

Felix Geiger

The Yield Curve and Financial Risk Premia

Implications for Monetary Policy

Lecture Notes in Economics and Mathematical Systems

654

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Felix Geiger

The Yield Curve and Financial Risk Premia

Implications for Monetary Policy

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ISSN 0075-8442

ISBN 978-3-642-21574-2

e-ISBN 978-3-642-21575-9

DOI 10.1007/978-3-642-21575-9

Springer Heidelberg Dordrecht London New York

Library of Congress Control Number: 2011935533

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Cover design: eStudio Calamar S.L.

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Preface

This book was written during my time as research and teaching assistant at the Department of Economics at the University of Hohenheim, Germany; it was accepted as dissertation at the University of Hohenheim in November 2010. I hereby would like to address my gratitude to a number of persons.

First and foremost, I would like to thank my supervisor Prof. Dr. Peter Spahn for his encouragement and support to pursue my research program. Offering me a fruitful and productive working environment throughout the years of completion of the dissertation at his chair, I have benefited tremendously from his personal guidance, advice and sharing of great ideas with me.

I would like to thank Prof. Dr. Harald Hagemann as my second supervisor. Not only did he support my work with valuable comments at various seminars at the University of Hohenheim, but he introduced me to the field of economic research at the end of my diploma studies. I would also like to express my gratitude towards Prof. Dr. Gerhard Wagenhals for joining the committee for my oral examination.

My colleagues and friends at the Department of Economics and Business Administration of the University of Hohenheim; Oliver Sauter, Lukas Scheffknecht, Arash Molavi Vasséi, Ulli Spankowski, Arne Breuer, Barbara Flaig, Dr. Constanze Dobler, Dr. Kai Schmid, Julian Christ, Patricia Hofmann, Ralf Rukwid, Dirk Sturz, Katharina Nau, Dr. Michael Knittel, Dr. Sybille Sobczak, Niels Geiger, Johannes Schwarzer, Larissa Talmon-Gros, Martin Lempe, Lenka Severova and Christine Eisenbraun helped me in various ways and created a very enjoyable atmosphere that contributed to the successful completion of my dissertation. I will treasure the memory of lively discussions with my friends at university, during evenings and at night. Big thanks also go to Luigi Giordano for producing such a delicious Barbera d'Alba.

Moreover, I would like to thank Dr. Wolfgang Lemke, Jesús Vázquez, Edward Nelson, Francisco Palomino, Michael Joyce, Oreste Tristani, Eric Swanson and Tobias Adrian for comments on parts of my research and for sharing selected data sets.

I am deeply grateful to my parents, Ulrich and Barbara, as well as my twin brother Moritz. They encouraged me in a lovely way throughout the years of working on this exciting project. Last but certainly not least, I would like to thank Martha for her wonderful support. In particular, as she did all the time consuming proof-reading for the last draft of this book.

Stuttgart-Hohenheim,
April 2011

Felix Geiger

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Chapter 1

Introduction

1.1 The Macro-Finance Approach to the Analysis of Monetary Policy and Financial Risk

One striking feature since the 1980s has been an observed trend decline in macroeconomic volatility for most industrialized countries. Standard deviations of output and inflation have significantly fallen with the timing of occurrence varying across countries. However, the phenomenon has been so pronounced that it has been labeled the “Great Moderation” (Blanchard and Simon 2001; Bernanke 2004c). Apart from the “good luck” hypothesis, the “good policy” approach, with an improved performance of macroeconomic policy, has been promoted as a major contributor to increased economic stability. This stylized fact has been accompanied by central banks becoming more independent and more transparent using the channel of communication and adopting some variants of the inflation targeting framework. The argument relies on the proposition that the commitment to deliver price stability in the medium term has promoted an environment of increased predictability of monetary policy thereby reducing overall macroeconomic uncertainty.

The process of enhanced central bank transparency has been spurred by advances in monetary policy modeling, with its focus on the role of private expectations for the effectiveness of monetary policy. The widely adopted New-Keynesian paradigm of describing the macroeconomy with forward-looking market participants reveals that the management of expectations matters, indeed “little *else* matters” (Woodford 2005a, 3). The ability of monetary policy to affect aggregate expenditures rests on the premise to influence market expectations regarding the future path of short-term interest rates. The extent to which a central bank can alter macroeconomic dynamics depends on its impact on financial market prices, in particular on its leverage effect on the long-term interest rate that determines the level of credit demand and, hence, expenditures. A credible central bank under inflation targeting supports the anchoring of long-term inflation expectations near the target level, if it reacts more than one-by-one to inflation dynamics so that it alters ex-ante real interest rates to

constrain aggregate demand. This “Taylor principle” delivers a nominal anchor for inflation and guarantees the uniqueness and the determinacy of the equilibrium on goods, labor and financial markets (Woodford 2003).

The arguments above rely on the assumption that the link between short-term interest rates and long-term interest rates is established through the *Expectations Hypothesis* of the term structure of interest rates. Long-term bond yields are determined by the weighted average of expected future short rates. Since monetary policy adjusts its short rate in response to macroeconomic shocks, the consistency between short- and long rates allows us to draw statements about how current and expected macroeconomic events should account for movements farther out the yield curve. The yield curve, thus, does not only represent the fulcrum between monetary policy and aggregate demand, it also provides a central bank with valuable information about private market expectations including the expected evolution of short rates, inflation and output dynamics. Monitoring expectations is essential for the conduct of monetary policy since expected changes in economic variables can have a significant impact on current market behavior. Along similar lines, monetary policy can extract information about inflation expectations from long-term bond yields for the purpose of assessing its own credibility towards maintaining price stability as perceived by market participants.

While it appears attractive to represent the main transmission mechanism of monetary impulses through this expectational channel, there is a serious shortcoming: its empirical failure. Froot (1989, 283) remarks that “if the attractiveness of an economic hypothesis is measured by the number of papers which statistically rejected it, the Expectations Hypothesis of the term structure is a knockout.” Against this background, a proper understanding of monetary policy effects on the long end of the yield curve is indispensable in order to draw statements on monetary policy effectiveness. Financial factor models of the yield curve arrive at the conclusion that the long-term interest rate is mainly driven by a time-varying risk premium demanded by risk-averse agents rather than by the expected path of the short rate (Dai and Singleton 2002) – a finding not captured at all within the prototype New-Keynesian economy. These models are successful in reproducing the properties of bond yields, i.e. an upward-sloping yield curve, high volatility of long-term bond yields and a time-varying risk compensation. For monetary policy, there is an inherent interest in disentangling interest-rate expectations from required excess returns along the term structure. The economic drivers of risk premia depend on the stance and on the strategy of monetary policy as long as the latter has an effect on risk perceptions (quantity of risk) and on risk tolerance (price of risk). Thus, for any part of the transmission mechanism that works via changes in asset prices, the effects from risk premia and the effects of monetary policy on risk valuation have to be taken into account.

Along with the “Great Moderation,” there is a further stylized fact for most countries, i.e. the frequency of periods of financial distress has intensified. Accompanied by the process of reduced regulatory requirements for financial markets and for the financial intermediary sector, both industrialized and emerging-market countries have suffered from multiple financial disruptions since the 1980s

(Borio and Lowe 2002; IMF 2003). Ultimately, the most severe financial crisis since the Great Depression of the 1930s, with the first turmoil starting in 2007 in money markets, has promoted a refocus on macroeconomic implications of developments in the financial sector. In the run-up to this crisis, there has been a serious underpricing of risk characterized by a flat term structure of credit spreads, high asset price inflation across the whole range of markets, unusual low financial market volatility and immense leverage dynamics of financial institutions. In part, these low risk premia have resulted from very low policy interest rates initiated by central bankers around the world after the Tech bubble in 2001 and have started overshooting in the opposite direction with the break out of the systemic liquidity crisis in 2008 (Goodhart 2008a).

The conventional view of dealing with evolving financial imbalances from a monetary policy perspective relies on the insight that it is hard to detect ex-ante asset price bubbles and that monetary policy is likely to be ineffective in addressing the bubble component with changes in interest rates. Consequently, the most promising strategy is based on a benign neglect vis-à-vis financial developments with monetary policy “cleaning-up” afterwards, once the unsustainable boom has turned to bust despite the recognition “that monetary policy has an important influence on asset prices – indeed, this influence is at the heart of the transmission of policy decisions to real activity and inflation“ (Kohn 2006). However, this view seems at least doubtful: “Whatever the merits of the above arguments, the fall-out in the real economy from the [recent] banking crisis seems to have made a policy of benign neglect towards potentially unsustainable credit/asset price booms untenable” (Bean et al. 2010, 19).

Considering these circumstances, the theory and practice of monetary policy must refocus on the above mentioned aspects. Closely related, Blinder (2006) posed sixteen, partly unresolved, questions and provided about twelve alleged accurate answers on these monetary policy issues. Among them, he dealt with the following questions: (1) What is the role of the term structure of interest rates in the monetary policy transmission? (2) Should monetary policy lead or follow financial markets? (3) How should central banks respond to asset price bubbles? (4) Should central banks also be bank supervisors?

The aim of this book is to address these questions within a joint *macro-finance approach*. This approach explicitly acknowledges the close feedback between the macroeconomy and financial conditions. In standard macroeconomic models, financial conditions are represented by a single interest rate with no reference to various forms of risk premia including compensation for interest-rate risk and default risk or liquidity risk. In contrast, finance models typically abstract from any macroeconomic content where latent factors drive asset prices and the primary focus lies in the consistent pricing of various assets across markets. By combining both modeling strategies, a proper understanding between macroeconomic dynamics, the conduct of monetary policy and asset price movements becomes possible. Along similar lines, financial frictions and financial intermediaries in standard policy analysis play hardly any role. Within the traditional credit view of monetary policy transmission, a case is made for financial frictions that give rise to a financial

accelerator stemming from balance sheet constraints in the borrowing sector. However, as the recent financial crises demonstrated, obstacles to credit supply dynamics likewise matter and they tend to dominate propagation effects of monetary policy impulses that work through the financial sector. The insight that investors and financial intermediaries actively manage their portfolios and their balance sheets significantly alters the logic of monetary transmission. This risk-taking view allows us to take on a different perspective about the role of financial constraints and it enables policymakers to develop guidelines for the conduct of monetary policy and for financial supervision of how to cope with financial imbalances.

1.2 Plan of the Book

Part I of this book, consisting of Chaps. 2, 3 and 4, develops the theoretical and empirical basis for analyzing term structure and financial risk aspects for monetary policy analysis. Chapter 2 introduces the no-arbitrage approach of asset pricing. The absence of arbitrage opportunities focuses on the relative asset pricing model. It implies that it is not possible to invest in a risky asset and to earn an expected return higher than the risk-free return without bearing the risk of capital losses. At the same time, the absence of arbitrage opportunities guarantees that assets are consistently priced relative to each other. If the model is augmented by an individual or a representative agent who optimally allocates wealth in different states of the world and in different periods to finance expenditures, an economic interpretation is possible. The optimality condition results in an Euler equation that equals the New-Keynesian model of describing aggregate demand; it is in essence an asset-pricing equation (Woodford 2003). Thereby, the risk premium depends on the specific risk characteristic of the asset and on the risk appetite of investors.

Chapter 3 applies the no-arbitrage model to the term structure of interest rates. The Expectations Hypothesis of the term structure of interest rates is one approximation within this framework but it ignores the importance of time-varying excess returns along the yield curve. It can be shown that risk-neutrality, the pure form of the Expectations Hypothesis and the no-arbitrage approach are different concepts for describing movements in the cross section of bond yields. Arbitrage opportunities are excluded if the long rate equals risk-adjusted expectations of the average future short rates. One increasingly used arbitrage-free term structure model in monetary policy analysis is the affine model according to which bond yields can be described as an affine function of some state variables within a financial factor model. Chapter 3 introduces a specific form of such a model in the spirit of Ang and Piazzesi (2003).

A systematic view on the sources of risk premia is the focus of Chap. 4. Risk premia can be decomposed into three components, i.e. risk originating from the uncertain future path of interest rates, default risk and liquidity risk. This Chapter applies the affine term structure representation for Germany, USA and UK on government bond securities to extract interest-rate risk from the evolution of the

yield curve. Verifying these premia to be highly time-varying, they show a secular decline starting at the beginning of the mid 1990s. Default risk is discussed with the help of a case study of government bond dynamics within the euro area. The main purpose of this case study is to disentangle the recent drivers of government bond spreads, whether they can be attributed to country-specific developments or whether global factors are the primary sources of observed bond differentials. The last part of Chap. 4 attends to the concept of liquidity risk. It deals with the difference between market liquidity and funding liquidity and how both concepts are related to each other. In this respect, a stylized model of liquidity crisis is presented in which a reinforcing loop process of falling market and funding liquidity promotes a market environment of suddenly disappearing liquidity. This endogenous nature of liquidity can be also regarded through the lens of financial intermediation and Keynes' concept of liquidity preference theory.

Part II of this book consists of Chap. 5 and 6 and it focuses on the macro-finance view of term structure transmission and monetary policy analysis. Chapter 5 introduces ways how monetary policy can use the yield curve for an assessment of its policy stance and how a joint modeling strategy of yield curve dynamics and the macroeconomy can be specified. The affine bond yield representation is applied to the New-Keynesian model economy so that statements about sources of the movements of long-term interest rates and about the shape of the yield curve becomes possible. Furthermore, this model is extended by parting with the rational expectations paradigm. Instead, with imperfect information, agents form expectations in an adaptive learning environment and the pricing of bond securities is based on subjective beliefs about the state of the economy. The learning approach, thus, allows different term structure dynamics.

Chapter 6 augments the analysis along several dimensions. Firstly, it provides a model analysis of determinacy of bond rate transmission and it sheds light on the question of whether monetary policy is advised to directly react on bond yield information in a rule-based, short-term interest rate setting. Secondly, the consistency between short- and long term interest rates allows to address the discussion whether central banks follow the gradualistic approach of setting their policy rates. Thirdly, since long-term bond yields heavily depend on market expectations, the effects of monetary policy communication on the dynamics of the yield curve are evaluated, in particular how the yield curve reacts to various communication channels and central bank announcements. Finally, Chap. 6 compares nominal and real interest rates and it provides insights to what extent the nominal term premium is related to its real counterpart and how much of nominal term premium variation is due to inflationary risk.

Part III augments the perspective by introducing financial intermediaries and financial constraints into the analysis. Chapter 7 develops the argument in favor of a broad risk-taking channel of monetary policy transmission that acknowledges active balance sheet management on part of financial market participants as a propagation effect. This Chapter analyzes how monetary policy influences risk perceptions and risk tolerance by applying different methodological tools, among them the New-Keynesian model augmented by risk premia, rules of risk management practices

and insights of the theory of financial intermediation. Financial frictions originating from the lending sector emphasize the endogenous nature of financial cycles and the high procyclicality of the financial system. The simultaneous build-up of financial imbalances are high in gear when balance sheet growth, leverage and the degree of maturity transformation pick up speed. They allow the proposition that the yield spread is an effective reference point for monetary policy. The spread gives limits on how fast and how deeply the financial sector “breathes” and is engaged in liquidity transformation. Against this background, new arguments in favor of a pre-emptive tightening in the course of the financial cycle are developed. The last part of the Chapter concentrates on the different instruments a central bank is equipped with in order to cope with evolving financial imbalances. It is also discussed whether macro-prudential regulation should be the task of a central bank or of another financial supervision body and whether the objective of financial stability should be directly assigned to the central bank.

Chapter 8 summarizes the main results of the book and it provides an outlook for future research efforts.

Part I
Theoretical Foundations for Policy
Analysis

Chapter 2

Financial Markets and Asset Pricing

2.1 Asset Pricing Theory

2.1.1 No-Arbitrage and the Stochastic Discount Factor

Essentially all modern asset pricing models rely on a single fundamental pricing equation according to which the price of an asset follows the relationship

$$P_{i,t} = E_t[M_{t+1}X_{i,t+1}] \quad (2.1)$$

where $P_{i,t}$ is the price of an asset i at time t , M_{t+1} is the stochastic discount factor (SDF) and $X_{i,t+1}$ represents the payoff of asset i at $t + 1$. Payoffs in general can be split up into a future price component ($P_{i,t+1}$) and an earning stream ($D_{i,t+1}$) such as coupon payments on coupon-bearing bonds or dividends on stocks. Since future payoffs are uncertain, the SDF is used to value the state-contingent possible payoffs of the asset in $t + 1$ so that the SDF takes the uncertain payoff back to present. But even if the cash flows generated by asset i may be known with certainty, the discount factors and future interest rates respectively are uncertain numbers depending on the future state of the economy.

Uncertainty in financial markets is mainly modeled by a probability space $(\Omega, \mathcal{F}, \mathcal{P})$ where Ω units all possible events of the state of the world and $\mathcal{F}$ is a collection of events of the sample space (σ -algebra). Every event in $\mathcal{F}$ has a certain numerical probability where the function $\mathcal{P} : \mathcal{F} \rightarrow [0, 1]$ assigns a probability between 0 and 1 to any event imbedded in $\mathcal{F}$. Since the valuation of assets has a time dimension, Ω can be interpreted as a set of increasing dimension of time. The filtration describes the evolution of information (events) through time with $\mathcal{IF} = \{\mathcal{F}_0, \dots, \mathcal{F}_T\}$. A filtration can be understood as an ever increasing stream of information (events) and agents do not forget implying that $\mathcal{F}_s \subset \mathcal{F}_T$ whenever $s \leq T$. Accordingly, if the process $X_t = \{X_0, X_1, \dots, X_T\}$ is a random variable, it is said to be $\mathcal{F}$ -measurable and adapted with respect to the measurable space $(\Omega, \mathcal{F})$. Such a stochastic process can be a simple random walk in discrete time

$$\begin{aligned}
W_t &= W_{t-1} + \epsilon_t \\
W_0 &= 0 \quad \epsilon_t \sim N(0, \sigma^2).
\end{aligned} \tag{2.2}$$

In order to introduce the concept of the SDF and its asset pricing implications, this section starts with two famous theorems which guarantee that, under weak conditions, a SDF (1) exists and (2) the SDF is strictly positive. The *law of one price* asserts that if two portfolios have the same payoffs in every state of nature, then both portfolios must have the same price. Otherwise, an immediate situation would arise in which the prospect of an arbitrage profit would lead investors to sell the expensive portfolio and to buy the cheaper portfolio instead. Cochrane (2001, 64) shows that the law of one price implies the existence of a discount factor. However, the law of one price is restrictive since it assumes that the two portfolios are perfect substitutes and trade at the same price.

The concept of *no-arbitrage* goes further and describes that it is not possible to come up with a portfolio with zero costs but whose payoff is positive (and never negative) with a positive probability. This line of argument is close to the concept of market efficiency. A market is said to be efficient if there are no unexploited arbitrage opportunities which requires that all new and relevant information is instantly imbedded in market prices (Fama 1970, 1991).¹ As it will be demonstrated later in this Section, market efficiency and the concept of no-arbitrage then imply that returns of any asset should be the same except for compensations of various forms of risk. The following baseline model illustrates the concept of no-arbitrage.²

The model consists of N assets, each of them adapted with a price P_{it} . All prices at time t are compressed in the vector $P_t = (P_{1t}, \dots, P_{Nt})^\top$. A *trading or portfolio strategy* is a process $H_t = (H_{1t}, \dots, H_{Nt})$ where H_{it} stands for the amount of asset i held in the portfolio within the time interval $(t, t + 1)$. The value of the portfolio at time t is defined as $V_t = H_t P_t = \sum_{i=1}^N H_{it} P_{it}$. The corresponding gain process to the trading strategy H satisfies

$$\begin{aligned}
\delta_t(H) &= H_{t-1} P_t - H_t P_t \\
H_{-1} &\quad \text{zero by convention}
\end{aligned} \tag{2.3}$$

The logic is that the investor holds a portfolio H_{t-1} from $t - 1$ and at time t prices are revealed and the investor reallocates her asset holdings. If $\delta_t < 0$, the investor has to finance the new portfolio with capital from outside; if $\delta_t > 0$, the new portfolio is cheaper than the one in time $t - 1$ and the difference is a gain and can be put aside. If $\delta_t = 0$, a trading or a portfolio strategy is said to be self-financing so that

¹For a survey on the efficient market hypothesis and its critics the reader might be referred to Malkiel (2003). Various forms of financial market efficiency are typically categorized according to the information set. Thereby, the literature distinguishes between the weak, the semi-strong and the strong form of market efficiency.

²See Irle (2003) and Duffie (2003) for a detailed analysis of arbitrage in a n-period model.

rebalancing the amount of assets held in the portfolio does not require or produce additional funds. A trading strategy is called *arbitrage strategy* if $\delta_t(H) \geq 0$ for $t = 0, 1, \dots, T$ and the gain process has a positive probability $\mathcal{P}(\delta_t(H) > 0) > 0$ for at least at one point in time t . To be concrete, a portfolio H constitutes an arbitrage strategy if it holds that the initial value of the portfolio in $t - 1$ is negative $V_{t-1} < 0$ but the value of the portfolio in t is always non-negative with probability $\mathcal{P}((V_t \geq 0) = 1)$. An arbitrage opportunity, thus, arrives when any zero-net-investment guarantees a positive payoff in some future state with no possibility of a negative payoff in all other future states. To put it in jargon, arbitrage implies the existence of a *free lunch*. However, there is no such thing as a free lunch! The model is said to be arbitrage-free whenever such an arbitrage strategy is not possible. Indeed, if a risk-free chance to come up with a zero-cost portfolio that does not permit a loss possibility existed, smart investors would pick this trading strategy to earn risk-free returns. However, at the same time, they would alter the asset prices held in the portfolio to bring about a no-arbitrage equilibrium.

The existence of a positive stochastic discount factor establishes a necessary and sufficient condition for a market to be arbitrage-free (Irle 2003, 114). Vice versa, Harrison and Kreps (1979) formally state that assuming away arbitrage opportunities is equivalent to the existence of a positive discount factor within a particular arbitrage-free market. In addition, Cochrane (2001, 51) demonstrates that with a complete set of contingent claims and state prices, a positive discount factor exists.³ The latter result is reviewed in the next paragraphs.

In a simple discrete state model, there are $s = 1 \dots S$ states of the world. A complete market implies that for each state s , there is a contingent claim with price $p_c(s)$ that pays off one unit in state s and 0 otherwise.⁴ In a next step, for $i = 1 \dots N$ assets in the economy, the price of an asset i is defined as $P(i)$ with payoff $X(si)$ in state s . In matrix and vector notation, the $N \times 1$ vector P collects the assets prices, X is the $S \times N$ matrix of payoffs and Q is the gross return on each asset for each state so that $Q = X^\top / P$ with element $Q_{si} = 1 + R_{si} = X(si)/P_i$. The state price vector then satisfies $X^\top p_c = P$ where p_c is the $S \times 1$ vector including the number of state prices $p_c(s)$. The price of asset i then follows

$$P(i) = \sum_{s=1}^S p_c(s)X(si). \quad (2.4)$$

This relation states that the price of an asset is simply equal to the sum of the state price in a given state multiplied by the amount of the payoff of the asset in state s . It can be expressed as a bundle of these contingent claims with $X(1i)$ contingent

³An excellent review on neoclassical finance, no-arbitrage and market efficiency is the Princeton Lecture Series on Finance by Ross (2005).

⁴The necessary condition of complete markets does not mean that investors have to trade explicit contingent claims. They rather need enough securities to span or to synthesize all contingent claims.

claims to state 1, $X(2i)$ claims to state 2 etc. Another way of looking at this relation is to replace the sum over states with the probabilities $\pi(s)$ of the states. To do so, the state-density function is defined as

$$M(s) = \frac{p_c(s)}{\pi(s)}$$

according to which $M(s)$ is the state price divided by the probability of state s occurring. An asset's price now can be written as

$$\begin{aligned} P(i) &= \sum_{s=1}^S \pi(s) M(s) X(s) \\ &= E[MX] \end{aligned} \quad (2.5)$$

The interpretation is straightforward. In states of small M , the state s is cheap in the sense that investors are reluctant to pay a high price to receive the payoff in this state. Consequently, for each $\{t\}_{t=0}^T$ and each asset $\{i\}_{i=1}^N$ with information $\mathcal{F}$ available at time $t-1$ and $\mathcal{P}(M_t > 0) = 1$ it holds that

$$P_{i,t-1} = E_{t-1}[M_t X_{i,t} | \mathcal{F}_{t-1}]. \quad (2.6)$$

M_t is the stochastic discount factor, a random variable. Applying (2.6) to above trading strategy, gives $V_{t-1} = E(M_t V_t)$. If it is assumed that V_t is strictly positive, ruling out arbitrage opportunities implies that M_t must be likewise positive so that V_{t-1} is positive, too – the condition of no-arbitrage. It also results in the proposition that a positive state price only exists if there are no arbitrage opportunities. Furthermore, the SDF and the state price vector are unique only in case of complete markets. In case of incomplete markets, many M 's may exist (Campbell et al. 1997).

Equation (2.4) can be expressed in terms of a gross return of the asset price. In a contingent claim market, (2.4) can be divided by the price of the asset $P(i)$ where the gross return is defined as $(1 + R(si)) = X(si)/P(i)$ for all s . This gross return can be interpreted as a payoff $X(s)$ with price one so that

$$1 = \sum_{s=1}^S p_c(s)(1 + R(si)). \quad (2.7)$$

Finally, from (2.5) it follows that

$$\begin{aligned} 1 &= \sum_{s=1}^S \pi(s) M(s)(1 + R(si)) \\ &= E[M(1 + R(i))] \end{aligned} \quad (2.8)$$

so that the product of the pricing kernel and the asset's expected return equals one.

Since all assets are priced according to the stochastic discount factor approach, it applies also to risk-free assets. An asset is said to be risk-free if it delivers the same payoff in all states of the world so that the payoff $X(s)$ is independent of s and one can write $X(s) = \bar{X}$ for all s . Often, instantaneous maturing assets are risk-free assets, since they have a payoff of 1 in the next period with certainty and no risk of price fluctuations are eminent. The price of such an asset is then

$$\begin{aligned}
 P(f) &= \sum_{s=1}^S p_c(s) \bar{X} = \bar{X} \sum_{s=1}^S p_c(s) \\
 &= \bar{X} \sum_{s=1}^S \pi_s M(s) \\
 &= \bar{X} E[M]
 \end{aligned} \tag{2.9}$$

which implies no-arbitrage. As the payoff of the risk-free asset can be characterized as a zero-coupon bond with payoff 1 in all states, the price equation collapses to

$$P(f) = E[M]$$

or in terms of its gross return

$$\begin{aligned}
 1 &= 1/P(f) E[M] \\
 1 &= (1 + R(f)) E[M] \\
 E[M] &= \frac{1}{1 + R(f)}.
 \end{aligned} \tag{2.10}$$

Having pinned down the risk-free interest rate, another convenient way of expressing asset prices is to use the risk-neutral valuation approach instead of using the subjective state probabilities in order to price assets. For that purpose, risk neutral probabilities for state s are defined as

$$\pi^Q(s) = (1 + R_f) M(s) \pi(s) \tag{2.11}$$

with $1 + R_f = 1/E[M]$ so that an asset's price stated in state probabilities can be converted into a price equation in terms of risk-neutral probabilities. The asset pricing formula can be rewritten as

$$P(i) = \sum_{s=1}^S \pi(s) M(s) X(s) = \frac{1}{1 + R_f} \sum_{s=1}^S \pi^Q(s) X(s) = \frac{E^Q[X]}{1 + R_f} = E[M] E^Q[X] \tag{2.12}$$

where asset i is priced as if investors are risk-neutral but apply π^Q instead of π to the states in the economy. Thus, risk-neutral probabilities align more weight on unpleasant states if investors are risk-averse (see on this account Chap. 3.5).

2.1.2 Individual Agent Optimality and Asset Pricing Equations

The stochastic discount factor so far offers no link to consumption decisions. Indeed, the use of the SDF to price all assets in an economy must not rely on an explicit model of intertemporal optimal consumption or the presence of a representative agent. Even the absence of complete markets would not alter the existence of a positive, but maybe not a unique SDF (Cochrane 2001). The pricing equation can hold individually for a single investor with a single portfolio. In the following, the model is extended by an investor who possess a utility function with utility derived from consumption. In this case, it can be modeled to describe the optimal allocation of wealth between individual current consumption and financial assets. This is reasonable to assume since typically an investor holds and reallocates the portfolio for some purpose, for instance to transfer wealth in different states and periods to finance expenditures in different states in the future.

This section starts with a simple evaluation of household optimality in a contingent claim model following the work of Duffie (2003).⁵ The single investor's expected utility is described by a strictly increasing utility function⁶ with preferences over consumption whereas utility takes an additive form. The investor tries to maximize current plus expected discounted utility where C is current consumption, e are current endowments and β is the subjective discount factor. The problem is to maximize

$$\max U(C) = u(C) + \beta \sum_{s=1}^S \pi_s u(C(s))$$

subject to

$$C + \sum_{s=1}^S p_c(s)C(s) = e + \sum_{s=1}^S p_c(s)e(s).$$

The budget constraint contains current consumption, current endowments, future consumption and endowments both unknown to the agent in the current period. State prices $p_c(s)$ are used to value future consumption and endowments. In the

⁵See also Semmler (2003) or Wickens (2008) for this result.

⁶It must hold that if $a > c$ then $u(a) > u(c)$.

second period, the investor may purchase contingent claims to each possible state. Using the Lagrange multiplier analysis and defining λ as the multiplier, the first-order conditions (FOCs) are

$$\begin{aligned} u'(C) - \lambda &= 0 \\ \beta \pi(s) u'(C(s)) - \lambda p_c(s) &= 0 \quad \forall s = 1 \dots S. \end{aligned}$$

If both conditions are combined, it can be shown that state prices follow

$$p_c(s) = \beta \pi(s) \frac{u'(C(s))}{u'(C)} \quad \forall s = 1 \dots S \quad (2.13)$$

or

$$M(s) = \frac{p_c(s)}{\pi(s)} = \beta \frac{u'(C(s))}{u'(C)} \quad (2.14)$$

so that state prices are determined by state probabilities and the intertemporal rate of substitution between current and future consumption. Alternatively, the first-order condition says that the marginal rate of substitution between states in the coming period equals the relevant price ratios. For instance, it represents the rate at which the investor is willing to give up consumption in state 1 for consumption in state 2 through purchases and sales of contingent claims

$$\frac{M(s1)}{M(s2)} = \frac{u'(C(s1))}{u'(C(s2))}.$$

The aim of the next step is to derive the optimal path of consumption of an investor for an infinite time horizon. In this respect, the investor can transfer wealth over time through asset holdings. In this setup, she has the opportunity to trade one asset i so that total portfolio holdings are the sum of the asset holding i . One can think of asset i as being a simple zero-coupon bond. At the beginning of each period, the investor arrives with an endowment stream y_t and portfolio holdings from last period H_{t-1} valued with the price P_t at time t for resale. Resources are then the sum of the endowment and the revenue of sales of the asset in the portfolio at the prevailing price. At the end of period t , the single investor can either reallocate her portfolio holdings at the cost of P_t or she can use them for consumption purposes. In a dynamic setting, the investor, therefore, wishes to choose a stream of consumption $\{C_t\}_{t=0}^{\infty}$ to maximize the expected discounted sum of utilities.

$$\max E_0 \left[\sum_{t=0}^{\infty} \beta^t u(C_t) \right]$$

subject to

$$\begin{aligned} H_t P_t + C_t &= y_t + H_{t-1} P_t \quad \forall t \geq 0 \\ H_{-1} &\text{ given.} \end{aligned} \tag{2.15}$$

The amount of assets held in the portfolio H_{t-1} from period $t-1$ is the state variable and the portfolio H_t chosen to hold in period t together with current endowments y_t are the control variables whose level optimally depicted by the utility-maximizing investor affects the resources available in the next period $t+1$. The consumption level C_t could also be treated as a control variable, yet as it will be demonstrated below, the use of the portfolio holdings as the control variable has the suitable advantage that it is a more simple task to find the Euler equation.⁷ The intertemporal separability of the objective function and the budget constraint allows to convert an infinite period problem into a two-period problem with the appropriate rewriting of the objective function (Adda and Cooper 2003). This is achieved by considering that the investor commits to optimality in a dynamically consistent way, i.e. today's optimal choice is made with the knowledge that it will likewise be optimal next period onwards.

As a first step, the objective function can be expressed in terms of a value function $V(H_{t-1}, y_t)$ which is the value of the objective function at the optimum when the control variable has been optimally chosen. By using the recursive structure of the value function, it is possible to derive the Bellman equation whose first derivative allows for the derivation of the Euler equation. Essentially, the optimization problem is to seek a policy function $H_t = h(H_{t-1}, y_t)$ which maps the state (H_{t-1}, y_t) into the control H_t . As soon as the household has chosen the portfolio (and thus the consumption) level, the transition function $H_t = (y_t + H_{t-1} P_t - C_t) / P_t$ determines next period's state H_t . Such a procedure is recursive. The value of the objective function then can be written according to

$$\begin{aligned} V(H_{t-1}, y_t) &= \max \left\{ \sum_{t=0}^{\infty} \beta^t u(C_t) \right\} \\ V(H_{t-1}, y_t) &= \max \left\{ u(C_t) + \beta E_t \left[\sum_{t=0}^{\infty} u(C_{t+1}) \right] \right\} \\ V(H_{t-1}, y_t) &= \max \{ u(C_t) + \beta E_t [V(H_t, y_{t+1})] \}. \end{aligned} \tag{2.16}$$

Equation (2.16) is called the Bellman equation. To solve this problem, the literature offers different solution methods. The most popular techniques are the application

⁷As it is shown in the Appendix A, if it is possible to write the transition equation in such a way so that next period's state does not depend upon last period's state, the first derivative of the Bellman equation is easy to solve.

of the Lagrangian or the Envelope theorem. Appendix A gives a more rigorous formal statement of the Envelope theorem which should be used in this Section. After writing down the Bellman equation, the budget constraint can be solved for C_t . Applying this to (2.16) gives

$$V(H_{t-1}, y_t) = \max_{H_t} \{u(y_t + H_{t-1}P_t - H_tP_t) + \beta E_t[V(H_t, y_{t+1})]\} \quad (2.17)$$

The first-order condition with respect to the control variable H_t set equal to zero is

$$\frac{\partial V(H_{t-1}, y_t)}{\partial H_t} = u'(C_t)(-P_t) + \beta E_t \frac{\partial V(H_t, y_{t+1})}{\partial H_t} = 0. \quad (2.18)$$

Ljungqvist and Sargent (2004) show that for an interior solution to hold, the value function $V(H_{t-1}, y_t)$ is also differentiable with respect to its state variable H_{t-1} . The solution is called Envelope Theorem or the Benveniste-Scheinkmann condition. Consequently, the first derivative of the Bellman equation (2.17) is

$$\frac{\partial V(H_{t-1}, y_t)}{\partial H_{t-1}} = u'(C_t)P_t \quad (2.19)$$

Taking (2.19) and moving it one period forward gives

$$\frac{\partial V(H_t, y_{t+1})}{\partial H_t} = u'(C_{t+1})P_{t+1} \quad (2.20)$$

In a last step, (2.18) and (2.20) are combined to get the Euler equation

$$u'(C_t)P_t = \beta E_t[u'(C_{t+1})P_{t+1}]. \quad (2.21)$$

The Euler equation – the first-order condition for optimal consumption and portfolio choices of an investor – can be used to link asset prices and consumption. The equation offers an intuitive economic interpretation. The left-hand side of (2.21) gives the marginal utility (here it is a loss in utility) of giving up a small amount of utility and using the additional resources to buy an amount of assets which is added on the portfolio holdings H_t . The right-hand side expresses the discounted marginal utility gain at time $t + 1$ when pursuing this strategy. It captures the increase of marginal utility due to an increase of prices. The household continues to buy or sell assets until the marginal loss equals the marginal gain. This process continues until the household is indifferent to consuming a small amount at date t or in transferring resources via asset allocation to $t + 1$ to gain discounted marginal utility due to price increases. To sum up, if the agent is indifferent about changing the amount of assets she demands, then she is demanding the optimal amount of assets.

So far, this section only analyzed the asset Euler equation for a single asset, i.e. a discount bond. The payoff is its price in the next period. It would be also interesting to see how a stock price in such a setting behaves. In this context, the

available assets which can be used to transfer wealth have to be modified. As in the previous example, the investor can trade a discount bond with holdings H and, in addition, she can trade equity shares which entitles her to a (stochastic) stream of equity dividends $\{D_t\}_{t=0}^{\infty}$. Let $P_{eq,t}$ be the stock price of the amount of shares L the investor chooses to hold. The budget constraint of (2.15) is modified according to

$$H_t P_t + L_t (P_{eq,t} + D_t) + C_t = y_t + H_{t-1} P_t + L_{t-1} P_{eq,t} \quad (2.22)$$

H_{-1}, N_{-1} given

The investor again maximizes the value function

$$V(H_{t-1}, L_{t-1}, y_t) = \max \{u(c_t) + \beta E_t[V(H_t, L_t, y_{t+1})]\} \quad (2.23)$$

At interior solutions, the first-order conditions are the Euler equations associated with the controls H_t and L_t .

$$u'(C_t) P_t = \beta E_t[u'(C_{t+1}) P_{t+1}] \quad (2.24)$$

$$u'(C_t) P_{eq,t} = \beta E_t[u'(C_{t+1}) (P_{eq,t+1} + D_{t+1})]. \quad (2.25)$$

The basic result is that for any asset $\{i\}_{i=1}^N$ in an economy which can be traded to transfer resources over time it holds that

$$u'(C_t) P_{i,t} = \beta E_t[u'(C_{t+1}) X_{i,t+1}] \quad \forall i = 1, \dots, N \quad (2.26)$$

where $X_{i,t+1}$ is the payoff of the asset i in $t + 1$.

2.1.3 Representative Agent and Equilibrium Asset Pricing

The preceding Sect. 2.1.2 discusses the condition for optimal consumption. Indeed, it represents any equilibrium condition. An equilibrium holds if all agents maximize and the market clears. The introduction of a representative household does not change the fundamental asset Euler equation at all. It just opens the door to model general equilibrium considerations. Following Cochrane (2001), firstly, the model can ask what determines consumption at given price sequences and preferences. Then the task is to treat the price sequences and the periodical payoffs as exogenous variables and agents as price-takers. In addition, one can specify the generation of payoffs but they are not determined within the model. The finance literature often takes some linear production technologies as given to specify the real physical rate of interest to which the consumption stream adjusts.

Secondly, following the work of Lucas (1978) and Breeden (1979), one can solve the model at a given equilibrium consumption stream. Then, the model is given by identical, infinitely lived agents (or a representative agent) each of whom

tries to maximize lifetime expected utility. Market clearing means that every agent who wants to buy one unit of assets at price $P_{i,t}$ must have a counterpart agent who wishes to sell the asset at price $P_{i,t}$. In a general equilibrium, this leads to a situation in which the sum of the total demand for assets must be zero. Essentially, any positive demand must be equalled by a corresponding negative demand. It holds that $\sum H_t = 0$.⁸ However, since all agents are identical with identical preferences, this may appear as a pitfall. This is because if one agent wishes to buy an asset, all others wish to do so, equally. Therefore, to clear markets, prices must be exactly such that agents are indifferent of buying or selling assets. Such a situation occurs if one substitutes $H_t = H_{t-1} = 0$ in the budget constraint of (2.15) to see immediately that zero asset demand implies $C_t = y_t$. In equilibrium, agents consume their exogenous endowments at all dates. In this respect, the scope is to search for precisely those asset prices which make it optimal for the representative investor to consume her periodical endowments in each period. Taking (2.26), in equilibrium it holds that

$$u'(y_t) P_{i,t} = \beta E_t[u'(y_{t+1}) X_{i,t+1}]. \quad (2.27)$$

It is also possible to rewrite the basic Euler equation to isolate the current price of an asset i

$$P_{i,t} = \beta E_t \left[\frac{u'(y_{t+1})}{u'(y_t)} X_{i,t+1} \right]. \quad (2.28)$$

The current equilibrium price depends positively on future expected payoffs. It is greater the higher time preference β appears to be. If β is high, then the representative agent is patient in consuming her endowments and is willing to transfer wealth via asset purchases to receive payoffs in $t + 1$. Today's price also depends on the ratio of marginal utilities (marginal rate of substitution). The price of the asset i will be higher when marginal utility of consumption and income is high. Since it is assumed that the utility function is concave, the representative agent tries to smoothen consumption. That is, if the agent expects future consumption to be lower than today's consumption, then marginal utility derived from future consumption is higher than today's marginal utility. If the agent expects that income is very low in $t + 1$, the agent tries to transfer wealth via asset purchases from time t to $t + 1$. As long as the utility function is concave, the ratio of marginal utilities is inversely related to the change in consumption from date t to $t + 1$ (i.e. the consumption growth rate). This is the explanation why a higher intertemporal rate of substitution evokes a high asset demand and pushes today's asset prices up.

It is also possible to inspect the dynamics of the fundamental asset pricing equation in a rational expectations equilibrium. Any rational asset pricing equilibrium is a pair of processes $\{P_t X_{i,t}\}$ that satisfy (2.28), $C_t = y_t$ and the transversality condition ($E_t[\lim_{T \rightarrow \infty} u'(C_{t+T}) X_{i,t+T}] = 0$) given the exogenous process $\{y_t\}$.

⁸For simplification, in this Section, the amount of asset holdings in the portfolio is denoted by H .

The latter transversality condition implies that it is ensured that the agent does not overaccumulate assets so that a higher expected overall lifetime utility can be achieved by, for example, increasing consumption today. In a finite horizon, this would imply that the agent dies with positive asset holdings which is not optimal.

2.1.4 Asset Returns and a First Look at Risk

2.1.4.1 Returns, Pricing Kernel and Risk

This section sets the stage for how asset returns behave in an asset pricing equilibrium. To shed light on asset returns, there is no need to make references to an explicit consumption-based equilibrium framework with an intertemporal utility-maximizing investor in the first place. The basic asset pricing formula sets the current asset price equal to the expected product of the pricing kernel and the future payoff. Absence of arbitrage opportunities guarantees the existence of a unique and positive SDF. However, for the sake of a rigorous economic understanding, this section uses the consumption Euler equation as basic equilibrium condition to draw return and risk implications. Initially, recall that for any asset, the first-order condition satisfies

$$P_{i,t} = E_t \left[\beta \frac{u'(C_{t+1})}{u'(C_t)} X_{i,t+1} \right] \quad (2.29)$$

which is nothing else than a slight modification of the no-arbitrage pricing formula of (2.6). The SDF in a multi-period consumption-based asset pricing model is given by the expected ratios of marginal utilities

$$E_t[M_{t+1}] = E_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right]. \quad (2.30)$$

One-period gross holding returns which throughout this section are simply called returns take the general form of

$$1 + R_{i,t+1} = \frac{X_{i,t+1}}{P_{i,t}} \quad \forall i = 1, \dots, N.$$

By dividing the left-hand side of (2.29) by the price and recalling that this operation yields an expression for a return, the most fundamental asset pricing equation takes the form of

$$1 = E_t \left[\beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{X_{i,t+1}}{P_{i,t}} \right] = E_t [M_{t+1}(1 + R_{i,t+1})]. \quad (2.31)$$

This way of writing the model in discrete time goes back to Rubinstein (1976). It states that asset prices obey in such a way that their returns multiplied with

the stochastic discount factor equal one. To understand the implications of (2.31), especially the joint dynamics of prices of asset i with the SDF, one can use the definition of the covariance to specify the expected product of the SDF and the return. For any independent random variables x, z , it holds that $E_t xz = E_t x E_t z + cov_t(x, z)$ where $cov_t = E_t(x - E_t x)(z - E_t z)$.⁹ Then, the expectation of the product can be written as the product of expectations plus the covariance term.

$$E_t [M_{t+1}(1 + R_{i,t+1})] = E_t [M_{t+1}] E_t [1 + R_{i,t+1}] + cov_t(M_{t+1} R_{i,t+1}) \quad (2.32)$$

Substituting this expression into (2.31) and rearranging gives

$$1 + E_t[R_{i,t+1}] = \frac{1 - cov_t(R_{i,t+1} M_{t+1})}{E_t[M_{t+1}]} \quad (2.33)$$

The conditional covariance is a measure for risk since it describes how random variables move together. Within the consumption-based setting, an asset's return and future consumption are random variables following stochastic processes. Equation (2.33) establishes a connection between the risk premium and a function for the consumption process. An asset that offers a low return when the stochastic discount factor moves in opposite direction, must bear a risk compensation. The economic intuition is as follows; a large positive pricing kernel corresponds to a state of low consumption in $t+1$ and high consumption in t . An asset characterized by a negative correlation with expected marginal utility is risky because it is not able to deliver wealth in a situation in which wealth is most valuable to investors. Consequently, risk-averse investors demand a risk compensation for holding this asset because it performs purely in a state where wealth is particularly important to investors (Campbell 2000). This means that the asset's return covaries positively with expected consumption growth but negatively with marginal utility. The covariance then can be interpreted as the risk premium of an asset. In contrast, if the return of an asset covaries positively with expected marginal utility and negatively with consumption growth, this will induce investors to demand a negative risk premium and lower returns. This asset precisely delivers wealth when high expected payoffs go hand in hand with expected states of high marginal utility.

In terms of prices, an asset characterized by a positive covariance of its payoff with the SDF ($cov_t(M_{t+1}, X_{i,t+1})$) must be traded at a higher price since the payoff is positively correlated with the SDF. This asset yields higher payoffs in a situation in which the investor has high marginal utility of transferring wealth into the next period. Buying this asset and the corresponding payoff may help the investor to smoothen consumption over time; it provides a hedge for the investor so that she is ready to pay a higher price. These kind of assets are countercyclical to current consumption and they are priced at a premium since the demand for these assets is high. In contrast, an asset whose payoff covaries negatively with the SDF is

⁹See, for instance, Wooldridge (2006).

valued at a discount which is nothing else than an additional risk compensation. It is negatively correlated with the marginal rate of substitution and it covaries positively with consumption growth. Buying this asset would even make the consumption stream more volatile. Therefore, the investor demands a risk premium which lowers the price of the asset and drives up required returns.

Since the realization of a risky asset return is uncertain in t , its realized return is not known until in period $t + 1$ so that the timing convention $t + 1$ as subscript for expected risky returns is appropriate. Moreover, (2.33) holds for any asset in the economy no matter whether it may be risky or risk-free. This link allows to derive a risk-free interest rate. A risk-free interest rate is the return of an asset f whose covariance with the pricing kernel is zero ($cov_t(R_{f,t} M_{t+1}) = 0$). It implies that the return is known in time t with certainty so that it can be denoted with the subscript t . The risk-free return obeys

$$1 + R_{f,t} = \frac{1}{E_t[M_{t+1}]} \quad (2.34)$$

The definition for the risk-free rate can be substituted into (2.33) to obtain an expression for the expected excess return (risk premium) of asset i over the risk-free asset f .

$$E_t[XR_{i,t+1}] \equiv E_t[R_{i,t+1} - R_{f,t}] = -(1 + R_{f,t})cov_t(R_{i,t+1} M_{t+1}). \quad (2.35)$$

The expected excess return of an asset i is positive if and only if its return is negatively correlated with the SDF – and in a consumption-based setting with the marginal rate of substitution.

Although the covariance between the SDF and the return of asset i determines its riskiness, it is straightforward to show that the volatility of the asset's underlying payoffs and the volatility of the SDF contribute to the nature of the risk premium since the covariation is implicitly driven by both volatility components. Thereby, the definition of the covariance can be refined as

$$\begin{aligned} cov_t(M_{t+1}, R_{i,t+1}) &= corr_t(M_{t+1}, R_{i,t+1}) \sqrt{var_t(M_{t+1})} \sqrt{var_t(R_{i,t+1})} \\ &= \rho_{Mi,t} \sigma_{M,t} \sigma_{i,t} \end{aligned} \quad (2.36)$$

with the volatilities σ_M, σ_i together with the correlation coefficient ρ_{Mi} determining the covariation.

Against these conditions, it is also possible to represent the basic no-arbitrage relation of (2.35) in an expected return-beta representation (Cochrane 2001). It holds that

$$\begin{aligned} E_t[1 + R_{i,t+1}] - (1 + R_{f,t}) &= - \left(\frac{cov_t(M_{t+1}, R_{i,t+1})}{std_t(M_{t+1})} \right) \left(\frac{std_t(M_{t+1})}{E_t[M_{t+1}]} \right) \\ &= \beta_{Mi,t} \lambda_{M,t}. \end{aligned} \quad (2.37)$$

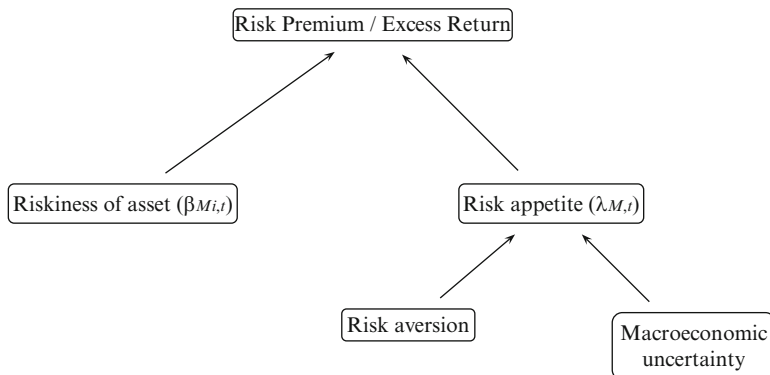


Fig. 2.1 Risk concepts (*Source: Gai and Vause (2006)*)

where $\beta_{Mi,t}$ is the quantity of asset-specific risk and $\lambda_{M,t}$ is the market price of risk per unit of risk which should be equal across all assets traded in the economy.¹⁰ The price of risk can therefore be defined as the reciprocal of investor's risk appetite; the latter measures the willingness of investors to bear units of risk. Thereby, the concepts of "risk premium," "risk appetite" and "risk aversion," though closely connected, are distinct concepts (Gai and Vause 2006). Figure 2.1 illustrates the risk premium decomposition of (2.37). First and foremost, risk appetite depends on investors' degree of reluctance towards uncertainty; risk aversion is determined by investors' preferences over uncertain outcomes which should not change rapidly and much over time. The second component of risk appetite is the level of uncertainty. It relies on the overall macroeconomic environment and moves periodically in response to factors such as unemployment prospects, the macroeconomic policy set-up and the volatility of consumption streams. Most generally, changing risk appetite or market sentiment over the course of business cycles should then reflect shifts in macroeconomic uncertainty rather than altering preferences towards risk. By taking the price of risk and the quantity of risk together, the excess return (risk premium) for each asset can be determined.

The implications of no-arbitrage asset pricing theory can be confirmed in stylized facts on international business cycles. Stock and Watson (1999, 2003a) identify procyclical effects of macroeconomic dynamics and asset returns on financial markets. For example, in a recession period, both returns and expected economic growth (consumption) are low, indicating positive risk premia on assets. Moreover, during booms the two variables are high and trigger excess returns as well. This attributes to the overall finding that in an economy with procyclical returns, assets

¹⁰In this respect, (2.35) together with (2.36) can be restated as the Sharpe ratio (SR) ($E_t[1 + R_{i,t+1}] - R_{f,t})/\sigma_{i,t} = -\rho_{M,i,t}\sigma_{M,t}E_t[M_{t+1}]^{-1}$. The highest possible Sharpe ratio applies to a return that is perfectly negatively correlated with the SDF so that $SR = \sigma_M E_t[M_{t+1}]^{-1}$. The highest Sharpe ratio, thus, coincides with the market price of risk $\lambda_{M,t}$.

should incorporate positive risk premia. Research also shows that risk premia are not constant over time but vary depending on business conditions and the state of the economy.¹¹ Typically, in a recession episode, risk premia tend to be higher than in a boom phase due to higher macroeconomic uncertainty and volatility (Cochrane and Piazzesi 2005). This is plausible since in “bad times” investors demand higher returns to transfer resources to the next period. When increased uncertainty is reflected by asset markets, agents heavily discount expected future events in current prices through the stochastic discount factor. By way of contrast, a boom is often associated with lower macroeconomic uncertainty and lower (but maybe still positive) risk premia.

2.1.4.2 A Log-Normal Representation

The log-normal model of representing returns and the SDF has become the workhorse for pricing financial assets (Campbell et al. 1997). It offers convenient features for modeling interest rates and it is a reasonable approximation to historical asset prices, at least in the long-run. For that purpose, it is assumed that the SDF and gross returns are jointly log-normal distributed. A random variable is said to be log-normally distributed if $\log(X)$ follows a normal distribution with mean μ and variance σ^2 . Conversely, if $\log(X) \sim N(\mu, \sigma^2)$, then the expected value of X is

$$E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$$

and

$$\log(E[X]) = \mu + \frac{\sigma^2}{2}.$$

Applying the joint distribution properties to (2.31), it follows

$$\begin{aligned} 1 &= E_t[M_{t+1}(1 + R_{i,t+1})] \\ &= \exp\left\{E_t[\log(M_{t+1}(1 + R_{i,t+1}))] + \frac{1}{2}\text{var}_t(\log(M_{t+1}(1 + R_{i,t+1})))\right\}. \end{aligned} \quad (2.38)$$

Taking the logarithm gives

¹¹See Pesando (1975), Fama (1984), Tzavalis and Wickens (1997), Hejazi and Li (2000) or Cochrane and Piazzesi (2005).

$$\begin{aligned}
0 &= \log(E_t[M_{t+1}(1 + R_{i,t+1})]) \\
&= E_t[\log(M_{t+1}(1 + R_{i,t+1}))] + \frac{1}{2}var_t(\log(M_{t+1}(1 + R_{i,t+1}))) \\
&= E_t[m_{t+1}] + E_t[r_{i,t+1}] + 0.5var_t(m_{t+1}) + 0.5var_t(r_{i,t+1}) + cov_t(m_{t+1}, r_{i,t+1})
\end{aligned} \tag{2.39}$$

where $\log M_{t+1} = m_{t+1}$ and $\log(1 + R_{i,t+1}) = r_{i,t+1}$.¹²

The risk-free rate with $cov_t(m_{t+1}, r_{f,t}) = 0$ and $var_t(r_{f,t}) = 0$ becomes

$$r_{f,t} = -E_t[m_{t+1}] - 0.5var_t(m_{t+1}). \tag{2.40}$$

Subtracting (2.40) from (2.39) gives the no-arbitrage condition for asset i

$$E_t[r_{i,t+1}] - r_{f,t} + 0.5var_t(r_{i,t+1}) = -cov_t(m_{t+1}, r_{i,t+1}) \tag{2.41}$$

where $0.5var_t(r_{i,t+1})$ stems from Jensen's Inequality. Finally, using the definition of the correlation coefficient the market price of risk takes the form of

$$\lambda_{M,t} = \sigma_{M,t}. \tag{2.42}$$

2.1.4.3 Pricing Nominal Returns

So far, the analysis above deals only with real assets according to which payoffs and asset prices are denominated in units of goods. It allows to derive a real risk-free interest rate which coincides with the reciprocal of the stochastic discount factor. Such a risk-free asset exactly delivers, say, one unit of good for the fixed delivery date. In practice, one hardly finds assets and returns denominated in consumption goods which may serve as a proxy for a real risk-free return. Index-linked bonds (TIPS) are the closest attempt to explicitly trade "real" assets on financial markets. They have part or all of their payoffs linked to a basket of weighted prices of consumption goods. However, even this type of asset may be not fully indexed since in the market for TIPS there are liquidity and technical factors that affect market prices (D'Amico et al. 2008a).

Still, on a modeling level, it is straightforward to introduce nominal assets and to adequately price these asset in nominal terms. Using the fact that assets and goods are priced in currency units, the budget constraint of an investor can be modified to incorporate nominal bonds in real terms. Let $P_{CPI,t}$ be the price index, then a nominal bond costs in nominal terms $P_{i,t}^{\$}$ and in units of goods $P_{i,t}^{\$}/P_{CPI,t}$;

¹²The variance term can be written as the sum of the variances of the random variables and two times their covariance since it must hold that $var_t(x + y) = var_t(x) + var_t(y) + 2cov_t(x, y)$ (Wooldridge 2006).

it pays \$1 or equivalently $\$1/P_{CPI,t+1}$ in units of goods. An investor is faced with a maximization problem according to (2.15) with a modified budget constraint

$$H_t P_{i,t}^{\$}/P_{CPI,t} + C_t = y_t + H_{t-1} P_{i,t}^{\$}/P_{CPI,t}.$$

By no-arbitrage, there exists a stochastic discount factor which discounts expected real cash flows, i.e. payoffs denominated in units of goods. The real SDF can be used to value nominal assets whereas the nominal payoffs can be denominated in real terms. The real price of any nominal asset then obeys

$$\frac{P_{i,t}^{\$}}{P_{CPI,t}} = E_t \left[M_{t+1} \frac{X_{i,t+1}^{\$}}{P_{CPI,t+1}} \right]$$

or in terms of returns

$$1 = E_t \left[M_{t+1} \frac{P_{CPI,t}}{P_{CPI,t+1}} (1 + R_{i,t+1}^n) \right] \quad (2.43)$$

$$1 = E_t \left[M_{t+1} \frac{1}{(1 + \pi_{t+1})} (1 + R_{i,t+1}^n) \right] \quad (2.44)$$

with $(1 + \pi_{t+1}) = P_{CPI,t+1}/P_{CPI,t}$ denoting the gross inflation rate and $R_{i,t+1}^n$ the nominal return of asset i . The product of the real stochastic discount factor M_{t+1} and the price deflators can now be interpreted as a nominal stochastic discount factor or nominal pricing kernel with $M_{t+1}^{\$} = M_{t+1}/(1 + \pi_{t+1})$.

The pricing of nominal bonds also allows a derivation of a “modern” version of the Fisher equation. Fisher’s theory of nominal interest rate determination (1986) implies that the gross nominal interest rate is a function of the gross real rate of return (the increase in real income and consumption, respectively) and the rate of appreciation or depreciation of one commodity standard in terms of another which is the expected gross inflation or deflation rate. In a no-arbitrage equilibrium, the goods-denominated return on a nominal bond and the return on a capital asset must be the same. Under perfect foresight, goods price changes are accurately predicted and fully incorporated in nominal returns. As a result, these returns fully adjust to inflation and leave the real rate unchanged at the constant level of the given return on the capital asset. Fisher was well aware that uncertainty or risk about the return on a real bond and expected inflation distorts the equilibrium condition; however, he stopped to give a rigorous geometric or algebraic interpretation. He, thus, assumed in a first step perfect foresight so that the investor is faced with perfectly known income streams, a given (ex-ante) real rate of return and goods prices.

When relaxing some of the restrictive assumptions, the Fisher equation has to be modified. If expected inflation is said to be random and perfect foresight is not imposed, price changes will not be fully reflected by one-for-one changes in nominal returns. Moreover, inflation risk will be translated into the pricing of nominal assets.

This is so because the real gross return of the nominal bond will as well become an uncertain number. Any risk-averse investor will demand a compensation for that inflation risk in terms of higher returns. To put it in a more formal statement, (2.43) can be modified to give a “modern” version of the Fisher no-arbitrage condition. If a one-period real bond exists which perfectly matches the properties of the real stochastic discount factor, risk-averse investors consider inflationary risk, so that the Fisher generalization for a one-period nominal bond can be written according to

$$(1 + R_{1,t+1}^n)^{-1} = (1 + R_{f,t})^{-1} E_t \left[\frac{P_{CPI,t}}{P_{CPI,t+1}} \right] + Cov_t \left(M_{t+1}, \frac{P_{CPI,t}}{P_{CPI,t+1}} \right) \quad (2.45)$$

or

$$(1 + R_{1,t+1}^n) = (1 + R_{f,t}) E_t [1 + \pi_{t+1}] \zeta^{-1} - 1 / Cov_t (M_{t+1}, (1 + \pi_{t+1})). \quad (2.46)$$

The nominal one-period interest rate equals the real one-period risk-free return adjusted for expected inflation and an inflation risk premium.¹³ The covariance term, thereby, captures the effect of inflationary risk on the nominal bond. As it was stated, with random inflation, the real return of the nominal asset remains uncertain. Inflation risk can result either in an increase or decrease of the nominal one-period bond’s real return depending on the sign of the conditional covariance between the real stochastic discount factor and the reciprocal of gross inflation. If the covariance term of (2.45) is negative, then investors demand a higher nominal return to compensate for inflationary risk. In particular, in a consumption-based setting, if high marginal utility tomorrow with low expected consumption growth coincides with high inflation, high inflation erodes the nominal bond’s real return in states in which the investor suffers from low expected consumption (Ireland 1996; Sarte 1998). This could be the case for a positive demand shock where inflation is pushed upwards due to high current consumption. Expected consumption growth then turns out to be weak (for given consumption expectations) so that high marginal utility of transferring wealth into the next period coincides with a reduction of the nominal bond’s real return. If, however, the covariance term is positive, low marginal utility and high expected consumption go hand in hand with high inflation delivering a lower real return in times when expected consumption is already high. This constellation would produce low or even negative inflation risk premia and it would reduce the one-period nominal return. For instance, a negative supply shock brings down current consumption but triggers an increase in inflation. Since the investor simultaneously experience a lower expected marginal rate of substitution, buying a nominal bond acts like a hedge to her.

¹³A n -period version for pricing nominal bonds is provided by Wolman (2006).

Another reason why the modern version of the Fisher equation does not correspond to the original one under perfect foresight is that even if the covariance term is zero, a Jensen's Inequality effect ζ exists. Equation (2.46) implies that in general $E_t[P_{CPI,t+1}/P_{CPI,t}]$ is not the same as $1/E_t[P_{CPI,t}/P_{CPI,t-1}]$ so that the modern version has to be augmented by the variability of inflation (see in a log-normal setting Benninga and Protopapadakis 1983; Sarte 1998). Greater variability in goods prices expressed in a higher ζ likely increases the expected real return of the nominal asset. Therefore, it rises the price of nominal bonds and makes them more attractive compared to inflation-indexed or real bonds. The higher the price of the nominal asset, the lower the (required) nominal return will tend to be. As inflation becomes more variable, the nominal bond's real value will increase since it is a convex function of the price level. Hence, to maintain a no-arbitrage equilibrium, the nominal return has to decline.

2.1.4.4 Valuation of Stock Prices

Traditional valuation principles for equities rely on the discounted cash-flow or the present value method. This model relates the price of a stock to its expected future payoffs discounted to the present using either constant or time-varying discount factors (Campbell et al. 1997). Since all expected future dividend streams enter the present value formula, temporary movements in expected cash-flows effect stock prices far less than persistent swings. Similarly, persistent changes in the discount rate have much greater influence on the valuation of stocks. Since the general no-arbitrage equilibrium condition of (2.31) holds for any asset, the stock price V_t satisfies

$$V_t = E_t[M_{t+1}(D_{t+1} + V_{t+1})] \quad (2.47)$$

where the next period's payoff X_{t+1} is determined by next period's dividend stream D_{t+1} and the stock price V_{t+1} . This expectational difference equation can be solved forward by repeatedly substituting out future prices. Using the law of iterated expectations, future-dated expectations can be eliminated. The price of a stock is given as

$$V_t = E_t \left[\sum_{j=1}^T M_{t,t+j} D_{t+j} \right] + E_t[M_{t,t+T} V_{t+T}] \quad (2.48)$$

$$M_{t,t+j} = \prod_{i=1}^j M_{t+i}$$

where $M_{t,t+j}$ compounds the one-period discount factors M_{t+i} .

The traditional present-value model (PVM) states that in terms of the no-arbitrage relation, there is a constant discount factor implying that $M_{t+i} = 1/(1+R)$ for all i . Taking the limit $T \rightarrow \infty$, the price of an equity is

$$V_t = E_t \left[\sum_{j=1}^{\infty} \left(\frac{1}{1+R} \right)^j D_{t+j} \right] + E_t \left[\lim_{T \rightarrow \infty} \left(\left(\frac{1}{1+R} \right)^T V_{t+T} \right) \right].$$

In the previous Sect. 2.1.3, the Lucas model (1978) was presented in which the representative agent in each period consumes her periodical endowments. To explore the dynamics of a rational expectations equilibrium in this asset pricing economy it holds that

$$V_t = E_t \left[\sum_{j=1}^{\infty} \beta^j \frac{u'(y_{t+j})}{u'(y_t)} D_{t+j} \right] + E_t \left[\lim_{T \rightarrow \infty} \frac{\beta^T u'(y_T)}{u'(y_t)} V_{t+T} \right].$$

Here, the constant risk free rate is replaced by a sequence of expected ratios of marginal utilities in equilibrium. The transversality condition which imposes a no-bubble solution in equilibrium $E_t [\lim_{T \rightarrow \infty} (\beta^T u'(y_T)) / u'(y_t) V_{t+T}] = 0$ finally breaks the last part of the solution down to the fundamental stock price equation where the price solely depends on expected future dividends

$$V_t = E_t \left[\sum_{j=1}^{\infty} \beta^j \frac{u'(y_{t+j})}{u'(y_t)} D_{t+j} \right]. \quad (2.49)$$

In this respect, the Gordon model is a specification of the fundamental no-arbitrage pricing equation in which the expected rate of growth of dividends g^D is assumed to be constant and nonzero over time (Gordon 1962). Applying the present value formula with a constant risk-free rate, stock prices evolve according to

$$V_t = D_t \sum_{j=1}^{\infty} \left(\frac{(1+g^D)}{1+R} \right)^j.$$

The fundamental value of an equity is made up of expected dividend streams. Tests on the present value, however, consistently arrive at the conclusion that, if at all, the link between prices and dividends appears to vary over time. For example, the dividend-price ratios and the dividend yield are not constants nor do they fluctuate around these constants. Indeed, there may be a dividend-based fair value for stock prices, but basically it means that over time, current prices diverge from the fundamental value, though dividend-price ratios can serve as predictors for excess returns on stock prices (Fama and French 1988). Other macro variables forecast stock returns as well including the investment/capital ratio and the

consumption/wealth ratio (Lettau and Ludvigson 2001). Still, the stylized relation between stock returns and macroeconomic forces is found to be unstable so that unambiguous links lack empirical support (Panetta 2001).

The fair value of a share is obtained by the expectational difference equation and solving this equation forward. This equation is stable if the expression $(1/1 + R)$ is smaller than one which is clearly the case; in addition, if the last term of the present value shrinks to zero with an increasing time horizon, then technically speaking the system has a unique solution. The market process guarantees that deviations from the present value are only a temporary event. Higher stock prices relative to the fundamental value reduce the dividend yield and trigger stock sales to finally reach the equilibrium solution again. However, rising stock prices and sudden crashes are the very nature of stock price dynamics. Ex-post, one would title such episodes as bubbles which have to burst some time in the future. The existence of a bubble is often associated with asset price dynamics that behave in a completely irrational way with what fundamentals would suggest.¹⁴ However, it can be shown that bubbles may reflect rational expectations on behalf of investors.¹⁵

In general, there is another solution for the stock price $V_{B,t}$ that likewise satisfies (2.47) with $M_{t+i} = 1/(1 + R)$ and that differs from the fundamental solution by the term B_t . Any solution including bubble solutions can be written as

$$\begin{aligned} V_{B,t} &= V_t + B_t \\ &= \frac{1}{1 + R} E_t(V_{B,t+1} + D_{t+1}) \end{aligned} \quad (2.50)$$

where B_t stands for the non-fundamental solution. V_t still serves as the fundamental value of the underlying stock price. It can be shown that by subtracting (2.47) from (2.50), the law of motion for the bubble term is given by

$$B_t = \frac{1}{1 + R} E_t B_{t+1}.$$

Following Brunnermeier (2008) and imposing rational expectations with $B_{t+1} = E_t[B_{t+1}] + \xi_{t+1}$ and $E_t[\xi_t] = 0$, the expression for the bubble term of the pricing equation becomes

$$B_{t+1} = (1 + R)B_t + \xi_{t+1}$$

which is an explosive process due to $R > 0$. Since bubbles burst and suddenly disappear, rational bubbles rely on a mechanism which prohibits an ever increasing or decreasing path of price misalignments when the system is hit by an innovation ξ . To do so, Blanchard and Watson (1982) suggest a bubble which takes the form

¹⁴See Kindleberger (1995) for a treatment on episodes of bubbles, maniacs and panics.

¹⁵DeBondt and Thaler (1985) point out that with the help of experimental psychology, irrational market behavior can be observed when investors overreact to unexpected news.

$$B_{t+1} = \begin{cases} \left(\frac{1+R}{\pi}\right)B_t + \xi_{t+1} & \text{with } \pi; \\ \xi_{t+1} & \text{with } 1 - \pi. \end{cases}$$

The bubble can burst with probability $1 - \pi$ in any period and a probability of growing further with π . Since investors know that the bubble may collapse with a positive constant probability, the bubble grows faster than R to compensate for the likelihood of bursting.

2.1.4.5 Bond Prices and the SDF

The SDF model and its fundamental asset pricing equation is mostly used to price bonds and the term structure of interest rates. It allows linking prices of bonds with different maturities in a no-arbitrage framework. Compared to stock prices whose periodical cash flows, i.e. dividends, are stochastic, fixed-income securities have the convenient property that the underlying periodical payoffs are deterministic. For example, zero-coupon instruments always deliver a specified final payoff at the maturity date without any payoff during the life of the bond. Only if the investor sells the bond before maturity, she is faced with an uncertain price. The obvious risk solely arises from time variation in discount rates which is the basis for the valuation of the bond price. The variation itself is driven by the stochastic behavior of the discount rates. Any co-movements between discount rates and the underlying bond consequently stem from these time-series dynamics.

The stochastic discount factor can be defined either in real (M_{t+1}) or nominal terms ($M_{t+1}^{\$}$) depending on what kind of zero bonds are priced. For the sake of simplicity, in this Section, M_{t+1} stands for both factors. A zero-coupon bond trades at $P_{n,t}$ in period t and pays 1 \$ at maturity $t + n$. An one-period gross return of this asset is related to its price in the way that $(1 + R_{n,t+1}) = P_{n-1,t+1}/P_{n,t}$. The payoff of the zero bond in period $t + 1$ is, thus, $P_{n-1,t+1}$ which can be substituted into the no-arbitrage Equation of (2.31). Then, the two prices with different maturities are related recursively according to

$$P_{n,t} = E_t[M_{t+1}P_{n-1,t+1}] \quad (2.51)$$

where n denotes time to maturity. The price of a bond with one period to maturity can be expressed directly in terms of the stochastic discount factor of the next period conditional on time t . For $P_{0,t+1} = 1$ at the maturity date, the one-period discount price is

$$P_{1,t} = E_t[M_{t+1}]$$

and given the relation between one-period prices, returns $(1 + R)$ and yields $(1 + Y)$, the one-period short rate is given by

$$(1 + R_{f,t}) := (1 + Y_{1,t}) = \frac{1}{E_t[M_{t+1}]}$$

as the reciprocal of the SDF. By means of repeated substitution and solving forward by the law of iterated expectations the basic pricing equation (2.51) can be used to describe the determination of any bond price with maturity $n > 1$. For the two-period bond price, it holds that

$$\begin{aligned} P_{2,t} &= E_t[M_{t+1}P_{1,t+1}] \\ &= E_t[M_{t+1}E_{t+1}[M_{t+2}]] \\ &= E_t[E_{t+1}[M_{t+1}M_{t+2}]] \\ &= E_t[M_{t+1}M_{t+2}]. \end{aligned}$$

Long-term bonds in general can be expressed as

$$P_{n,t} = E_t[M_{t+1}, \dots, M_{t+n}] = E_t \left[\prod_{i=1}^n M_{t+i} \right] = E_t M_{t,t+n} \quad (2.52)$$

where the price of a n -period bond is the product of successive expected one-period discount factors until maturity. In this context, pricing the term structure of interest rates is just a specification of a time series model for the expected dynamics of the stochastic discount factor. Indeed, as will be demonstrated later, most term structure models rely on a specification model of the stochastic properties of the discount factor which is firstly restricted by the absence of arbitrage opportunities in bonds markets and secondly, is used to price the whole spectrum of bond prices. Taking conditional expectations of the product of the SDFs given the information set at t is sufficient to describe the whole yield curve. A risk-adjustment to the price can be also included so that (2.52) becomes

$$P_{n,t} = \prod_{i=1}^n E_t[M_{t+i}] + \sum_{i=1}^{n-1} cov_t(M_{t+i+1}, M_{t+i}). \quad (2.53)$$

Furthermore, if the continuously compounded interest rate $i_{n,t}$ is defined as the negative of the logarithm of the pricing kernel, any bond price with maturity n can be written as

$$P_{n,t} = E_t \left[e^{-\sum_{i=0}^{n-1} i_{1,t+i}} \right] \quad (2.54)$$

so that the price is its expected present value discounted by the sum of one-period interest rates. The handling of risk components in the specification of (2.54) depends on the question whether actual probabilities $E^P[.]$ depart from risk-neutral probabilities $E^Q[.]$, i.e. whether the expectations operator $E[.]$ is specified as $E^P[.]$ or $E^Q[.]$. A formal representation of no-arbitrage term structure modeling is subject to Chap. 3.5.

2.2 Asset Pricing with Utility Specifications

2.2.1 Agents and Risk Aversion

Section 2.1.4 discussed asset pricing implications for a no-arbitrage equilibrium. In the consumption-based setting, in order to deduce empirical or dynamic results from these asset pricing relations, sufficient specifications on preferences and risk attitudes have to be made. Intertemporal asset pricing models rely on the assumptions about certain forms of utility functions. The von Neumann-Morgenstern utility function (vNM utility function) framework allows for the identification of expected utility and the identification of various forms of risk attitudes of inspected agents where the general probability space of Sect. 2.1.1 holds.

The probability space and the random variables makes it feasible to describe an agent's choices under risk.¹⁶ Let the random variables defined on a finite set of monetary values be expressed as $X = \{x_1, x_2, \dots, x_S\}$. Usually, x is regarded to be a monetary measure on income or wealth. Each of the random variables are combined with a probability $[\pi_1, \pi_2, \dots, \pi_S]$ whose combinations with the variables are called a gamble a . The probabilities add up to one

$$a \equiv [x_1, x_2, \dots, x_S; \pi_1, \pi_2, \dots, \pi_S] \quad (2.55)$$

$$\sum_{s=1}^S \pi_s = 1.$$

If the preference relation $\mathcal{R}$ of an agent satisfies the von Neumann-Morgenstern axiomatic approach to expected utility, then the preference relation over the lottery is

$$U(a) = \sum_{s=1}^S \pi_s u(x_s) = EU(x) \quad (2.56)$$

where u is the utility function that represents $\mathcal{R}$ for money obtained with certainty in the different states of the world s (Barucci 2003). $U(a)$ can be expressed as the weighted sum of a function u in the different states of the world. $U(a)$ is then linear in the probabilities and is called expected utility function or felicity function.

In this setting, the agent faces choices characterized by risk according to which not only the utility associated with an amount of money has to be considered but also the probabilities of receiving that amount of money. What is likewise important, is the question about certain risk attitudes which is captured by the preference relation $\mathcal{R}$. When the preference relation can be expressed through the expected

¹⁶For the derivation of the following propositions see for example Hirshleifer (1975), Hammond (1987), Pratt (1992), Machina and Rothschild (1992), Barucci (2003), Rubinstein (2006) or Cowell (2006).

utility $u(x_s)$, it is possible to show the agent's risk preference with the help of the shape of the utility function.

Let the random variables of a gamble only take on two values $\{\varepsilon_1, \varepsilon_2\}$ with probabilities $\{\pi, (1 - \pi)\}$. Thus, one can construct a lottery $\tilde{x}$ with two possible wealth outcomes: $x_1 = x + \varepsilon_1 < 0$ with probability π and $x_2 = x + \varepsilon_2 > 0$ with probability $1 - \pi$; hereby, initial wealth is x . Further it is assumed that $\tilde{\varepsilon}$ represents a fair gamble, i.e. $E[\tilde{\varepsilon}] = \pi\varepsilon_1 + (1 - \pi)\varepsilon_2 = 0$. Does a vNM-utility maximizer accept this lottery, i.e. does she put any positive value to this lottery? An agent is said to be risk averse if she is unwilling to accept this fair gamble but she strictly prefers another lottery that delivers the sure outcome x with certainty. Therefore, it must necessarily hold that she prefers x to $\tilde{x} = x + \tilde{\varepsilon}$. It represents the utility-decreasing aspect of pure risk-taking. If the first lottery is accepted, expected wealth is $E[\tilde{x}] = E[x + \tilde{\varepsilon}] = \pi x_1 + (1 - \pi)x_2 = x$ and expected utility takes the form of $E[U(x + \tilde{\varepsilon})] = \pi u(x_1) + (1 - \pi)u(x_2)$. If the first lottery is not accepted but instead the sure outcome is preferred, expected wealth is $E[x] = x$ with expected utility $E[U(x)] = U(x)$.

By exploiting the vNM utility function, an agent is said to be risk averse if $U(x) > E[U(\tilde{x})] = \pi(x + \varepsilon_1) + (1 - \pi)(x + \varepsilon_2)$. Since $E[\tilde{\varepsilon}] = 0$, denying the lottery yields an expected utility of $U(E[\tilde{x}]) = U(x)$. It implies that $U(E[\tilde{x}]) > E[U(\tilde{x})]$. This is equivalent to assume a concave utility function with Jensen's Inequality to hold.¹⁷ As Fig. 2.2 demonstrates, the utility function is concave, since drawing a line connecting the two points $u(x_1)$ and $u(x_2)$, the resulting utility $E[U(\tilde{x})]$ strictly lies below $U(E[\tilde{x}]) = U(x)$.¹⁸

Another way of looking at the lottery is to ask how much compensation a risk-averse agent would demand in order to accept taking the risky gamble. This compensation could be represented by an additional payment. A risk-averse agent is indifferent between the risky gamble $\tilde{x}$ and an amount of money CE which she can receive with certainty when both possibilities offer the same (expected) utility such that $U(CE) = E[U(\tilde{x})]$. This effect is called "certainty equivalence". It means that the expected wealth of the gamble $E[\tilde{x}] = x$ is always greater than the certainty equivalent CE . Stated differently, the certainty equivalent is the maximum price that she is willing to pay to receive the risk gamble. This is the "risk premium" associated with the risky outcome $\tilde{x}$, defined as

$$\rho_{\tilde{x}} = E[\tilde{x}] - CE = x - CE. \quad (2.57)$$

It is easy to consider the certainty equivalent as a risk-free asset which delivers a return R_f with certainty and yields wealth in case of investing an initial stock of

¹⁷Jensen Inequality states that for a concave function $h(x)$ where x is a random variable, it holds that $E[h(x)] \leq h(E[x])$ with $h''(x) < 0$.

¹⁸For the sake of completeness, an agent is said to be a risk lover, if the utility is convex so that $U(E[\tilde{x}]) < E[U(\tilde{x})]$ with $U''(\tilde{x}) > 0$. An agent is said to be risk-neutral if $U(E[\tilde{x}]) = E[U(\tilde{x})]$ with $U''(\tilde{x}) = 0$.

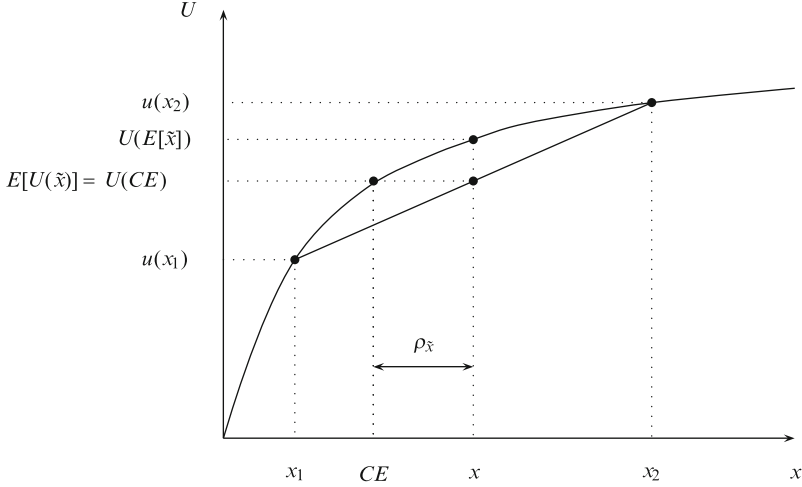


Fig. 2.2 Risk averse utility function

wealth in form of $x_{ce} = x_0(1 + R_f)$. In contrast, investing in a risky asset does not guarantee this wealth increase. Thus, a risk-averse investor demands the risk premium compensation $\rho_{\tilde{x}}$ in order to be indifferent between investing in the riskless asset with gross return R_f or in a stock of asset with expected gross return $E[R] = R_f + \rho_{\tilde{x}}$.

The magnitude of the risk aversion depends on the derivatives of the vNM utility function. By a second-order Taylor approximation of $U(\tilde{x})$ around x , one obtains

$$U(\tilde{x}) = U(x + \tilde{\varepsilon}) \approx U(x) + \tilde{\varepsilon}U'(x) + \frac{1}{2}U''(x)\tilde{\varepsilon}^2.$$

Expected utility then approximately takes the form of

$$E[U(\tilde{x})] \approx U(x) + \frac{1}{2}U''(x)\sigma^2$$

since $E[\tilde{\varepsilon}] = 0$ and $\sigma^2 = E[\tilde{\varepsilon}^2]$. Another Taylor expansion of the certainty equivalent centered at x up to the first order is

$$U(CE) = U(x - \rho_{\tilde{x}}) \approx U(x) - U'(x)\rho_{\tilde{x}}.$$

Since it must hold that $U(CE) = E[U(\tilde{x})]$ by definition, the risk premium can be calculated according to

$$\rho_{\tilde{x}} \approx -\frac{1}{2} \frac{U''(x)}{U'(x)} \sigma^2 \quad (2.58)$$

where

$$r_x^a = -\frac{U''(x)}{U'(x)}$$

is the Pratt-Arrow measure of absolute risk aversion (Barucci 2003). The risk premium is decomposed into two components, the variance of the lottery and the agent's risk aversion coefficient. If the curvature of an agent's utility function is greater when comparing it with another agent's utility function, then her risk coefficient is also greater which means that she demands a higher compensation for taking a risky lottery. By multiplying the coefficient of absolute risk aversion by x to get a percentage expression for the risk aversion, one finally obtains the coefficient of relative risk aversion.

$$r_x^r = \frac{U''(x)}{U'(x)}x. \quad (2.59)$$

2.2.2 Power Utility and General Equilibrium

Section 2.2.1 introduced some basic concepts for modeling expected utility. In order to shed further light on the dynamics of asset returns and consumption, one needs to define specific utility function. As a starting point, most of the finance and macroeconomic literature employs the easiest form of expected utility which is the time-separable power utility function. Following the work of Mehra and Prescott (1985), with power utility, the representative household maximizes the utility function of the form

$$U(C_t) = \frac{C_t^{1-\gamma} - 1}{1-\gamma} \quad (2.60)$$

where γ is the coefficient of relative risk aversion. As γ approaches one, the utility function approaches the log utility function $U(c_t) = \ln(C_t)$. As Campbell (2000, 2003) points out, the power utility function has several important properties. Firstly, it is scale-invariant to changes in aggregate wealth and consumption. This can be shown when calculating the first and second derivative to get a measure for the relative risk aversion. It holds that $-U''(C_t)/U'(C_t)C_t = C_t[-\gamma C_t^{-\gamma-1}]/[(1-\gamma)^{-1}(1-\gamma)C_t^{-\gamma}] = \gamma$ and therefore the power utility function has a constant relative risk aversion (CRRA) over all consumption levels. A second property of power utility is that the elasticity of intertemporal substitution ψ coincides with the reciprocal of the coefficient of risk aversion γ . This is in so far tricky as the elasticity of the intertemporal substitution is well defined in a world without uncertainty where a consumer moves consumption between time periods whereas the coefficient of relative risk aversion describes the willingness to transfer consumption between different states even in a one-period model. Finally, by taking the derivative of the

utility function, the basic asset pricing equation of (2.31) is

$$1 = E_t \left[\beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} (1 + R_{i,t+1}) \right] \quad (2.61)$$

Gross returns are high when investors are impatient, i.e. when β is low. Real returns also tend to be higher in times of high consumption growth and they are more sensitive to consumption if the risk aversion parameter is larger. A strongly shaped utility with high curvature reflects an agent's desire to smoothen consumption over time and states of the world and therefore, she is less willing to accept rearrangements of consumption streams in response to return changes.

To derive further insights into the dynamics of asset prices, the literature often specifies the time-series distribution of consumption and returns. This step can of course be heavily disputed since it is questionable that return and consumption really follow the assumed specifications. However, for the sake of completeness and further implications, the log-normal distribution of random variables is likewise applied to returns and the consumption process. Following Sect. 2.1.4.1 and the log-normal representation of the SDF and returns, the basic asset pricing equation with power utility (2.61) is

$$0 = E_t[r_{i,t+1}] + \ln \beta - \gamma E_t[\Delta c_{t+1}] + \frac{1}{2} (\sigma_{i,t}^2 + \gamma^2 \sigma_{c,t}^2 - 2\gamma \text{cov}_t(r_{i,t+1}, \Delta c_{t+1})) \quad (2.62)$$

where $r_{i,t+1} = \log(1 + R_{i,t+1})$, $\log C_{t+1} - \log C_t = \Delta c_{t+1}$, $\sigma_{i,t}^2$ denotes the conditional variance of the asset's return, $\sigma_{c,t}^2$ the conditional variance of consumption and $\text{cov}_t(r_{i,t+1}, \Delta c_{t+1})$ the conditional covariance of return i and the consumption process. Equation (2.62) was first derived by Hansen and Singleton (1983). It offers implications for both time-series and cross-section return dynamics. In this context, the risk-free return obeys

$$r_{f,t+1} = -\ln \beta - \frac{\gamma^2 \sigma_{c,t}^2}{2} + \gamma E_t(\Delta c_{t+1}). \quad (2.63)$$

The risk free rate is a linear function of expected consumption growth. The sensitivity of interest rate changes to consumption is captured by the slope parameter, i.e. the coefficient of relative risk aversion. The variance term reflects the precautionary savings motive. When consumption is uncertain and more volatile, investors with power utility value low consumption states higher than high consumption states (reflected by γ^2); this preference structure drives down the risk-free return. Cross-section implications are revealed by taking the difference of (2.62) and (2.63) to get a measure for the excess return and the risk premium of a risky asset i over the risk-free rate

$$E_t[r_{i,t+1} - r_{f,t+1}] + \frac{\sigma_{i,t}^2}{2} = \gamma \text{cov}_t[r_{i,t+1}, \Delta c_{t+1}]. \quad (2.64)$$

Finally, the market price of risk becomes

$$\lambda_{M,t} = \gamma \sigma_{c,t}. \quad (2.65)$$

2.2.3 Pitfalls and the CCAPM

Excess returns should obey in a way that the underlying asset i must be risky in the sense that it is positively correlated in terms of covariances with consumption growth and equivalently negatively correlated with the marginal rate of substitution. When taking the simple consumption-based asset pricing model to data, the outcome is both disappointing and puzzling. Table 2.1 reports summary statistics for US and German asset returns including the means and standard deviations of real stock returns and real short-term returns. Columns (1) to (4) report first and second moments for real stock and bond returns; columns (5) to (6) provide statistics on mean consumption and its standard deviation; finally, columns (7) and (8) display first and second moments of real dividend growth. The US stock market, on average, delivered real returns of 6.92% from 1970 to 1998 whereas German stock markets performed with an average real return of 9.84% over the sample period. In contrast, sample means for real bond returns ranged at 1.49% and 1.15% for the US and Germany, respectively. Volatility measured as the standard deviation of returns is

Table 2.1 Stylized facts on CCAPM data

Country	(1) $E[r_{eq}]$	(2) σ_{req}	(3) $E[r_f]$	(4) σ_{rf}	(5) $E[\Delta c]$	(6) $\sigma_{\Delta c}$	(7) $E[\Delta d_{eq}]$	(8) $\sigma_{\Delta d_{eq}}$
USA	6.92	17.56	1.49	1.69	1.81	0.91	0.61	16.80
GER	9.84	20.10	3.22	1.15	1.68	2.43	1.19	8.93
The equity premium puzzle								
			(9) $E[XR_{eq}] = E[r_{eq,t+1} - r_{f,t}] + \frac{\sigma_{eq}^2}{2}$			(10) $cov(\Delta c, r_{eq})$	(11) γ	
USA			6.35			4.23	150.10	
GER			8.67			1.45	599.47	
The risk-free rate puzzle								
			(12) $[r_f]$			(13) γ	(14) β	
USA			1.49			150.10	-175.92	
GER			3.22			599.47	9,757	

Note: Quarterly data are from Campbell (2003) and annualized accordingly. The U.S. sample period ranges from 1970:1 to 1998:4 and for Germany from 1978:4–1997:4. The calculated parameter of relative risk aversion is multiplied by 100

much higher for stock returns than for real bond returns. Average consumption growth over the sample period is curtly below 2.0% with low volatilities between 1.0 and 2.5%. Finally, dividend growth in Germany is slightly higher compared to US data; whereas US dividend growth volatility exceeded Germany's standard deviation.

Using (2.64) and interpreting asset i as a real stock price index, the model's implied parameters can be linked to historical stock market data. Calculating the equity premium as the difference between the average equity return and the short-term real interest rate, and taking the variance of stock returns and the covariance of stock returns and consumption growth as given, propositions on overall risk aversion become possible. Here γ is simply computed by dividing column (9) by column (10) since it holds that $E[XR] = \gamma \text{cov}(\Delta c, r_{eq})$. For instance, in the US, γ requires to take on the value of 150.10 and in Germany of 599.47 for (2.64) to fit. Thus, in order to explain the equity premium with the consumption-based model and power utility, for most countries the coefficient of relative risk aversion must be unreasonably high. This equity premium puzzle first described by Mehra and Prescott (1985) and heavily discussed in the literature is documented across most countries and sample periods.¹⁹ The problem with the consumption-based model is that although the data suggest a fairly stable consumption stream over the sample period as documented by its low volatility, only an extreme high risk aversion coefficient can explain why investors demand such high equity premia over time and across countries. Indeed, even in a moot scenario in which the correlation coefficient $\rho(r_{eq}, \Delta c)$ is set to one so that the product of the standard deviations of equity returns and consumption growth equals its covariance, the coefficient of relative risk aversion would still take on an implausible value of 41.18.²⁰ A high equity premium, thus, arises from the smoothness of consumption rather than from the low correlation of returns and consumption (Campbell 2000).

Another pitfall with the CCAPM and power utility is that it misses to meet the historical data of the real risk-free rate. To see this, consider (2.63) and take unconditional expectations to get a measure of the mean consumption growth rate. The average real risk-free interest rate is affected by three factors. Firstly, the real rate is high if time preference, i.e. if $-\log \beta$ is high. Secondly, the rate is high if the average consumption growth is high whose effect on the level of the real rate depends on γ ; finally, the rate tends to be lower with an increasing volatility of consumption due to precautionary savings. If the aim is to match the model's implied real return with the underlying sample and ignoring the precautionary savings motive, there are two ways to reconcile a positive consumption growth rate with a low real interest rate. Either agents prefer to spend resources for consumption later than sooner or they accept a high intertemporal substitution of consumption

¹⁹For excellent reviews of the consumption-based puzzles see Campbell (2003), Cochrane (2008a) and Mehra and Prescott (2003).

²⁰The correlation coefficient of any two random variables is defined as the covariance divided by the product of the standard deviation of the two variables. It holds that $\rho(x, y) = \text{cov}(x, y) / \sigma_x \sigma_y$.

between two periods without much compensation (Söderlind 2003). The former explanation is expressed in terms of a high β and the latter in terms of low risk aversion, i.e. a low γ . Surely, if one accounts for the quadratic term and the risk aversion coefficient is low enough, it presses the real rate down. For the sample period, with a given mean of the risk-free rate, the variance of consumption and the implied risk aversion coefficients of the equity premium calculation, the time preference parameters are reported in column (14). Obviously, they are at odds with sound economic reasoning which was firstly stressed by Weil (1989).

There are many suggested ways to cope with these puzzles within an equilibrium setting. One major strand of literature asks how to measure best implied consumption and return volatilities; while other approaches deal with changes in the specification of the utility function. Since one major explanation of the return puzzles stems from the low volatility of consumption growth, one resolution is the modeling of idiosyncratic shocks or time-variation in second moments (conditional variances) within GARCH models (Söderlind 2008). In addition, the composition of aggregate consumption might contribute to a better fit of the data with the consumption-based asset pricing model.

In continuing with the modification of the utility function, the goal of modeling utility functions that allow separating risk aversion from intertemporal substitution, delivers more promising results of replicating the stylized asset return facts. Such specification is important since in the power utility model both measures are jointly determined by the parameter γ though both concepts express different economic implications. Risk aversion, in general, represents a measure for an agent's sensitivity toward risk by means of substitution consumption across different states. Intertemporal substitution mirrors an agent's willingness to substitute consumption across time. The work of Epstein and Zin (1991) makes it possible to parameterize both dimensions separately though it covers many attractive features of a CES-based power utility function. Epstein-Zin preferences are said to be recursive, i.e. today's utility depends on tomorrow's expected utility. Thus, utility is – as opposed to the power utility framework – non-additive across states. In a log-normal representation, it also offers the convenient property that a high risk aversion coefficient does not imply a low average risk-free rate since the elasticity of intertemporal substitution and the coefficient of risk aversion may well diverge (Campbell et al. 1997). Another body of research focuses on non-separability in utility over time by allowing for habit formation, a positive effect of today's consumption on tomorrow's marginal utility (Constantinides 1990; Abel 1990; Campbell and Cochrane 1999).²¹ The basic idea behind this concept is that current utility is determined by current consumption relative to some reference/habit level. This could be either the agent's past individual consumption (internal habit) or past aggregate consumption (external habit). Habit-forming agents, thus, dislike large and rapid cuts in consumption. As a consequence,

²¹ When modeling the dynamics of inflation and output in macroeconomic models, the literature often refers to habit formation which allows for a lagged term in aggregate output in order to capture the empirical regularity of high persistence in output (Fuhrer 2000).

they demand a premium for holding risky assets that might force them to cut down rapidly on consumption. The result is a risk premium that is higher than the one implied by the time-separable power utility model.

Various other lines of explanations have been put forward to rationalize high equity returns relative to bond returns (DeLong and Magin 2009). Among them, the existence of transactions costs has been put forward that prevent investors to participate in equity markets in order to bear equity risk. Moreover, it has been argued that the data sample considered does not capture enough low-tail return risks coupled with consumption drops that rational investors should expect, either because the sample is too small or investors simply can not know the true lower-tail risk of equity returns and estimates are ad-hoc prior beliefs rather than derived from probability distributions. Finally, equipped with the behavioral finance approach to asset pricing, investors may pay too much attention to high short-term risks of equities and may value losses more seriously than gains so that they can not realize the very little long-term risk in equity returns relative to bond returns.

Though there are various attempts to cope with empirical asset-pricing regularities, it turns out to be difficult to approach these facts within a single theoretical framework. “Economists still do not have a complete explanation for the equity premium. Each of the explanations [...] has a well-developed research literature. Yet none of these explanations has achieved even a rough consensus: the plurality opinion is that the equity return premium remains a puzzle” (DeLong and Magin 2009, 203). To conclude, Mehra and Prescott (2003, p.982) claim that “over the long horizon the equity premium is likely to be similar to what it has been in the past and the returns to investment in equity will continue to substantially dominate that in T-bills for investors with a long planning horizon.”

Chapter 3

The Theory of the Term Structure of Interest Rates

3.1 Bond Pricing Representation and Yields

3.1.1 Notation and Pricing Relations

It is of essential interest to define a common financial arithmetic as to how to calculate the price of an asset.¹ Asset prices are typically calculated according to the *net present value approach* with market prices being valued by the expected discounted payoffs generated by the assets. Zero-coupon bonds should be used as a starting point. This Section only applies bond prices and yields on *default-risk-free* zero-coupon bonds. A zero-coupon bond is a bond that has a single fixed payment (“principal” and usual fixed to 1) at a given date (“maturity”). It guarantees the holder of the discount bond to get the principal at the maturity date. There are no intervening coupon payments and consequently, the bond sells less than the principal before the maturity date n . It holds that $n := T - t$ with T representing the maturity date. The total gross return at time t from investing in this bond maturing in T is $1/P_{n,t}$, i.e. the payoff divided by the current price. The discretely compounded return or spot rate is the number $R_{n,t}$ which satisfies

$$\frac{1}{P_{n,t}} = (1 + R_{n,t})^n, \text{ or} \quad (3.1)$$

$$\frac{1}{P_{n,t}} = (1 + Y_{n,t})^n \quad (3.2)$$

It is often convenient to define returns $R_{n,t}$ as the simple net return and $R_{n,t} + 1$ as the simple gross return from investing in an asset. For a zero-coupon bond, the spot rate corresponds to the yield to maturity $Y_{n,t}$ since there are no periodical coupon

¹For the subsequent explanations see Shiller (1990), Elton and Gruber (1991), Backus et al. (1998), Campbell (1995), Cochrane (2001), Hull (2003), Choudry (2004), Brandimarte (2006).

payments. It can be defined as the rate which makes the current bond price equal to the future cash-flow stream discounted at this rate.

The *term structure* of interest rates at time t is the mapping between time to maturity and the corresponding yields. The term structure can be written according to a function Θ_t with

$$\Theta_t : [0, Z], \quad n \mapsto \Theta_t(n) = Y_{n,t} \quad (3.3)$$

where Z is an upper bound on time to maturity n . Plotting this function against the time to maturity is called the (*spot*) *yield curve*.

A forward discount contract is an agreement of two counter parties today stipulating in time $t + m$ a discount bond that pays one unit of account (“numéraire”, principal) at time $t + n$ ($t < m < n$). The former date can be understood as the settlement date; whereas the latter can be called the maturity date. Forward rates can be synthesized by a set of zero-coupon spot rates. These rates are obtained by trading different discount bonds of different maturities available today. To do so, suppose that in period t the holder buys a discount bond maturing in $t + n$ at the price of $P_{n,t}$. At the same time, the holder of the bond sells $P_{n,t}/P_{m,t}$ of $t + m$ -maturing bonds at the value of $P_{n,t}$. Consequently the net investment (cash flow) in period t is zero. In period $t + m$, the holder pays the principal of the $t + m$ -maturing bond at the cost of $P_{n,t}/P_{m,t}$. Finally, at period $t + n$ the payoff is one. The forward agreement has the same payoff in $t + n$ and, thus, must specify the same net investment in $t + n$, i.e. the forward price $P_{n,t}/P_{m,t}$. The return on a forward contract $F_{m,n,t}$ must satisfy the equation

$$\begin{aligned} \frac{1}{P_{n,t}/P_{m,t}} &= (1 + F_{m,n,t})^{n-m}, \text{ or} \\ 1 + F_{m,n,t} &= \left(\frac{P_{m,t}}{P_{n,t}} \right)^{1/(n-m)} \quad \text{with } t < m < n. \end{aligned} \quad (3.4)$$

Similarly, the *forward rate curve* can be expressed according to

$$\Phi_t : [0, Z], \quad m, n \mapsto \Phi_t(m, n) = F_{m,n,t} \quad (3.5)$$

By using the basic relation between prices and yields, the (implied) forward rate can be written according to

$$1 + F_{m,n,t} = \frac{[1 + Y_{n,t}]^{n/(n-m)}}{[1 + Y_{m,t}]^{m/(n-m)}} \quad (3.6)$$

Forward rates are sometimes identified by the initial date m , and sometimes by the ending date n .² It is also possible to derive the yield to maturity by analyzing forward

²See for an example Elton and Gruber (1991) or Hull (2003).

rates of different maturities at time t . Using a recursion argument, the n -period yield equals the geometric average of h -period forward rates over t to $t + n$. If the time to maturity n can be split into n/h periods of h , then the following relationship holds

$$Y_{n,t} = [(1 + F_{0,h,t})^{h/n} \times (1 + F_{h,2h,t})^{(h/n)} \times \dots \times (1 + F_{n-h,n,t})^{(h/n)}] - 1$$

$$= \prod_{i=0}^{n/m-1} [1 + F_{ih,(i+1)h,t}]^{h/n} - 1. \quad (3.7)$$

If an investor sells the discount bond before it matures, she earns a holding-period return which is -contrary to the yield-to-maturity- not known in advance since the holding-period return primarily depends on the uncertain price in the future at which the holder can sell the bond. The total m -period return of an n -maturing bond is

$$1 + THPR_{n,t+m} = \left(\frac{P_{n-m,t+m}}{P_{n,t}} \right), \text{ or}$$

$$1 + THPR_{n,t+m} = \frac{(1 + Y_{n,t})^n}{(1 + Y_{n-m,t+m})^{(n-m)}}. \quad (3.8)$$

The per-period average return of holding a zero bond over m periods is then

$$1 + HPR_{n,t+m} = \frac{(1 + Y_{n,t})^{n/m}}{(1 + Y_{n-m,t+m})^{(n-m)/m}}. \quad (3.9)$$

For reasons of avoiding compounding horizons and of mathematical convenience, another fundamental idea of representing interest rates derives from the concept of continuously compounding. Continuously compounding uses the fact that the gross yield $(1 + Y_{n,t})$ can be approximated according to $\ln[1 + Y_{n,t}] \approx i_{n,t}$ whereas $i_{n,t}$ is the continuously compounded yield. It holds that

$$1/P_{n,t} = \exp(ni_{n,t}), \text{ or}$$

$$i_{n,t} = -\frac{\ln P_{n,t}}{n}$$

$$i_{n,t} = -n^{-1} p_{n,t} \quad (3.10)$$

with logs denoted by lowercase symbols.

Similarly, the continuously compounded forward rate becomes

$$f_{m,n,t} = (n - m)^{-1} [p_{m,t} - p_{n,t}], \text{ or}$$

$$= (n - m)^{-1} [ni_{n,t} - mi_{m,t}] \quad \text{with } t < m < n \quad (3.11)$$

and the n -period spot rate equals the average continuously compounded forward rates

$$i_{n,t} = \frac{h}{n} \sum_{i=0}^{n/h-1} f_{(ih,(i+1)h),t}. \quad (3.12)$$

Holding period returns can be expressed as

$$\begin{aligned} r_{n,t+m} &= p_{n-m,t+m} - p_{n,t}, \text{ or} \\ &= ni_{n,t} - (n-m)i_{n-m,t+m}. \end{aligned} \quad (3.13)$$

In order to reduce the complexity of representing forward contracts, forward rates are expressed as one-period rates so that with $n = m + 1$ and $h = 1$, implied one-period forward rates³ and spot rate can be expressed as

$$\begin{aligned} f_{m,m+1,t} &= -my_{m,t} + (m+1)y_{m+1,t} \\ &= i_{m+1,t} + m(i_{m+1,t} - y_{m,t}) \end{aligned} \quad (3.14)$$

$$i_{n,t} = \frac{1}{n} \sum_{i=0}^{n-1} f_{i,i+1,t}. \quad (3.15)$$

and one-period holding-returns ($m = 1$) become

$$\begin{aligned} r_{n,t+1} &= ni_{n,t} - (n-1)i_{n-1,t+1} \\ &= y_{n,t} - (n-1)(i_{n-1,t+1} - i_{n,t}). \end{aligned} \quad (3.16)$$

Equations (3.10), (3.14) and (3.16) describe the behavior and relationships between yields, one-period forward rates and one-period holding-period returns in a convenient way. For the one-period return of a one-period bond, the time subscript is changed to t , since the payoff in $t + 1$ is known to be one. It holds that $i_{1,t} = f_{0,1,t} = r_{1,t}$.

3.1.2 Coupon-Bearing Bonds and Duration

A default-free bond with face value (or principal) F paying a coupon c per period has a fair bond price by computing the present value of its cash-flow stream at a given gross discount rate $(1 + R)$. Time-variant discount rates imply that bond

³Note, that with this timing convention, implied forward rates are identified by the initial date.

pricing requires the knowledge of several discount rates. A coupon bond can be regarded as a portfolio of zero-coupon bonds with c_1 maturing in $t = 1$, c_2 maturing in $t = 2 \dots$ and $F(= 1)$ in $t = n$. Then, the price of the coupon bond with n periods equals the portfolio price

$$P_{c,n,t} = \sum_{i=1}^n P_{i,t} c_i + P_{n,t} (c_n + F)$$

$$P_{c,n,t} = \sum_{i=1}^n \frac{c_i}{(1 + R_{i,t})^i} + \frac{c_n + F}{(1 + R_{n,t})^n} \quad (3.17)$$

The *yield to maturity* is the constant interest rate which makes the current bond price equal to the future cash-flow streams discounted at this rate. The calculation assumes that the bond is held until maturity and that the holder of the bond is able to reinvest the coupon payments at the constant yield to maturity rate in all future periods over the life of the bond. The pricing equation solves

$$P_{c,n,t} = \sum_{i=1}^n \frac{c_i}{(1 + Y_{c,n,t})^i} + \frac{c_n + F}{(1 + Y_{c,n,t})^n}. \quad (3.18)$$

The yield to maturity is derived from the current market price with the periodical coupon streams and the redemption value taken as given.

When analyzing the basic pricing relation of (3.18), it becomes clear that an investor might want to know how sensitive bond prices and the future value react to small changes in the spot rate curve used to price the coupon bond. As a result, a long-term bond will gain in value in time t , if yields decline. However, at the maturity date, the future value of the investment has soared since a higher yield rate allows higher reinvestment opportunities of the periodical coupon payments. To shed light on the sensitivity of bond prices on small interest rate changes, one can take the derivative of a coupon bond pricing equation (the face value here is included in the last coupon payment; see Hull 2003).

$$\frac{dP_{c,n,t}}{dY_{c,n,t}} = -\frac{1}{1 + Y_{c,n,t}} \sum_{i=1}^n \frac{i c_i}{(1 + Y_{c,n,t})^i} \quad (3.19)$$

A general way to express the duration of a bond is to take the derivative of the bond price times minus one.

$$D = -\frac{dP_{c,n,t}}{dY_{c,n,t}}$$

$$= \frac{1}{1 + Y_{c,n,t}} \sum_{i=1}^n \frac{i c_i}{(1 + Y_{c,n,t})^i}. \quad (3.20)$$

Alternatively, the duration can be divided by the bond price $P_{c,n,t}$ to get the modified or adjusted duration

$$D_{adj} = D \frac{1}{P_{c,n,t}}. \quad (3.21)$$

The duration of a bond defined by Macaulay is thereby such a weighted average of the term of the discount bonds, where the weights correspond to the amount of the payments times a corresponding discount factor.⁴ When multiplying the duration by $(1 + Y_{c,n,t})/P_{c,n,t}$, the Macaulay's duration becomes

$$\begin{aligned} D_{mac} &= D \frac{1 + Y_{c,n,t}}{P_{c,n,t}} \\ &= \frac{\sum_{i=1}^n \frac{ic_i}{(1 + Y_{c,n,t})^i}}{\sum_{i=1}^n \frac{c_i}{(1 + Y_{c,n,t})^i}}. \end{aligned} \quad (3.22)$$

Macaulay's duration, thus, is a weighted average of the time to the coupon (and face) payments where the weights are the present values of the coupon payments. It signals how long an investor on average must wait before receiving payments (Campbell et al. 1997). For zero-coupon bonds, Macaulay's duration equals the time to maturity since $c = 0$. For coupon-bearing bonds, the duration is less than the time to maturity, since some of the cash payments are received prior to n . This effect is greater at high coupon rates and yield to maturities and it is increasing with time to maturity. In addition, the longer the terms of coupon payments, the smaller are their weights in the duration formula, thus, the duration between a 20-year and a 30-year coupon bond is relatively small.

The concept of duration can be re-expressed as the elasticity of the present value of a bond. To see this, the derivative of the bond price equation (3.19) can be modified to

$$\begin{aligned} \frac{dP_{c,n,t}}{dY_{c,n,t}} &= - \frac{1}{1 + Y_{c,n,t}} D_{mac} P_{c,n,t} \\ \frac{dP_{c,n,t}}{P_{c,n,t}} &= - D_{mac} \frac{dY_{c,n,t}}{1 + Y_{c,n,t}} \end{aligned} \quad (3.23)$$

which describes the relative percentage change of bond prices to relative small yield changes. It states that with increasing duration, the effects of small yield changes on bond price changes become larger.

⁴Shiller (1990) notes that Hicks (1946) defined independently the "average" period which coincides with Macaulay's concept of duration.

3.2 Stylized Facts on the Yield Curve

3.2.1 Moments of the US, German and UK Yield Curve

This Section starts with some of the typical characteristics of interest rates for government bond securities. Hence, a quarterly data set of annualized continuously compounded yields for the US, UK and Germany and a cross-section of yields with maturities from 1 up to 40 quarters is used. The data set is acquired from the Federal Reserve Board, the Bank of England and the Bundesbank, respectively. One exception is the German 1-quarter interest rate which is measured by the three-month interbanking rate LIBOR. The sample starts in 1970Q1 (UK), 1972Q1 (US) and 1973Q1 (GER) and it lasts until 2008Q3.

Table 3.1 reports descriptive statistics for yields, yield spreads and ex-post excess returns for the entire sample period. The first column on the left of the table shows that the average yield curve is an increasing function of time to maturity for all three countries.⁵ The curve is steepest for the United States with an absolute mean difference of 1.61%. The overall level of the yield curve for the UK is higher than for the other two countries and the unconditional mean of the curve is almost flat with a difference of 64 basis points between the 1-quarter and 40-quarter rate.

The average volatility curve as measured by the standard deviation is downward sloping on average for Germany, it exhibits a small hump at the short end for the US and a U-shaped pattern for the UK at the long end of the maturity range. For Germany, the volatility of long-term interest rates is lowest indicating relatively smooth dynamics at the long end of the term structure. Interest rate levels also exhibit high persistence as measured by the first autocorrelation of respective maturities with the persistence increasing slightly with time to maturity in the observed panel.

The moments of yield spreads partly behave different than its level counterparts. Average term spreads are positive, except for the 1 year spread of Germany and the UK. This fact does not accommodate selected periods of downward-sloping yield curves, among them the periods of 1973–1975, 1980–1981, 2000–2001 and 2006–2007 in the US. The standard deviations of yield spreads are increasing with time to maturity resembling the observation that most of the volatility stems from a high standard deviation of the one-quarter rate. As will be discussed throughout the next Sect. 3.2.2, interest rates are highly correlated across the yield curve but the tight link decreases over the maturity. Long-term yield spreads mirror this observation through higher volatility over the sample periods. Coevally, the autocorrelation of yield spreads rises for long-term interest rates.

⁵The German 3-month rate is not fitted from observable Government bond prices as it is done for the Nelson-Siegel interest rates for the maturities of 1 year up to 10 years. Therefore, the fact that the short rate is on average higher than the 1 year rate may stem from two different markets on which these securities are traded. For an introductory discussion of yield curve fitting, the reader is referred to the next Sect. 3.3.

Table 3.1 Descriptive statistics of the nominal yield curve

Quarter	Germany								
	Yield levels			Yield spread			Excess returns		
	Mean	Std	Auto	Mean	Std	Auto	Mean	Std	Auto
1	5.59	2.78	0.96	—	—	—	—	—	—
4	5.54	2.41	0.97	−0.06	0.76	0.82	0.01	0.48	0.41
8	5.77	2.27	0.97	0.18	1.02	0.85	0.09	0.97	0.34
12	6.00	2.18	0.97	0.40	1.21	0.87	0.17	1.40	0.32
20	6.33	2.01	0.98	0.73	1.46	0.90	0.32	2.21	0.33
40	6.75	1.86	0.98	1.15	1.76	0.92	0.56	3.59	0.36
US									
1	5.87	2.93	0.96	—	—	—	—	—	—
4	6.44	3.02	0.94	0.58	0.67	0.33	0.15	0.76	−0.25
8	6.67	2.91	0.95	0.81	0.83	0.55	0.23	1.56	−0.18
12	6.84	2.82	0.96	0.97	0.97	0.67	0.29	3.21	−0.14
20	7.09	2.68	0.97	1.22	1.16	0.77	0.38	3.29	−0.08
40	7.48	2.44	0.97	1.61	1.41	0.84	0.50	5.72	−0.05
UK									
1	8.19	3.31	0.95	—	—	—	—	—	—
4	8.12	2.98	0.96	−0.07	0.61	0.77	0.01	0.58	0.17
8	8.28	2.91	0.97	0.83	0.96	0.84	0.06	1.25	0.15
12	8.40	2.90	0.97	0.21	1.17	0.86	0.12	1.89	0.14
20	8.58	2.94	0.98	0.39	1.42	0.88	0.20	3.01	0.14
40	8.83	3.11	0.99	0.65	1.82	0.89	0.36	5.30	0.07

Ex-post quarterly excess returns are calculated accordingly, with the simplification that it is assumed that $i_{n-1,t+1} \approx i_{n,t+1}$. The right part of Table 3.1 reports statistics for ex-post quarterly excess returns with respect to the one-quarter short-term interest rate. Excess returns tend to rise in the maturity of the bonds. For instance, the average return earned from investing in a 1-year German bond for a quarter, in excess of the one-quarter return, is 1 basis point (4 basis points in annualized terms). Instead, an investment strategy of buying a 10-year German bond and holding it for one quarter, has yielded a return in excess of the one-quarter rate of 56 basis points (224 basis points in annualized terms). This fact should translate likewise in higher expected excess returns as predicted by the no-arbitrage approach of asset pricing. Moreover, what is typically found in a multitude of studies, is the high volatility of excess returns which increases with the life of the bond. Excess returns are, thus, considerably time-varying across the sample period with low persistence in terms of autocorrelation.

3.2.2 *Common Factors Driving the Yield Curve*

In finance theory as well as in the studies of monetary economics interest-rate models are based on a concept describing the behavior of interest rates. They can be either derived from theoretical reasoning, such as no-arbitrage considerations, or they can be specified empirically so as to fit historical shapes of yield curves across time. All model set-ups have in common that they try to identify the elements or factors that are believed to explain the dynamics among interest rates. It is likely that these factors are stochastic in nature so that they cannot be predicted with certainty. For instance, the Expectations Hypothesis of the term structure of interest rates might be interpreted as a factor model where the single factor is the short-term interest rate itself. The expected future path allows to price all bond yields along the yield curve.

For investors, the adequate availability of discount rates is essential to price and hedge the various forms of financial and physical assets, interest-rate products and derivatives. The branch of financial economics, thus, works likewise with the factor-model representation but with the adoption of much more sophisticated models of short-rate dynamics. The scope is to empirically discover the dynamics between yields across time and how they are interrelated to each other. The short-term interest rate may also be the single factor driving the other interest rates but the time-series process itself can be modified so as to fit the historical track much more appropriate. A multivariate factor extension allows for the improvement on the results; whereas the factors are usually unobservable so that they need to be estimated and derived from a historical set of yields. This strand of research tries to forecast interest rates with the underlying model specifications to put limits on the range of future prices that heavily affect future values of financial products (James and Webber 2000; Brigo and Mercurio 2006).

The following Sections aim at exploring various term-structure models designed for pricing and modeling fixed-income securities, especially zero-coupon bonds. They can be broadly categorized into theoretical and empirical based models. The choice of model can be characterized by a trade-off between historical fit or forecasting performance on the one hand and the ability to interpret the model in an economic sense on the other hand. For instance, sufficient restrictions on bond yield movements allow the separation of expected future interest rates from term premia which is not possible within a pure statistical interest-rate model. Any interest rate model must, however, have (at least) two ingredients. Firstly, it must provide a specification of the stochastic process of the underlying factors and secondly, it must provide a procedure for how to price zero-coupon bonds across maturities. The identification of factors which drive yields of different maturities, thus, seems to be crucial for explaining movements of interest rates; the model specification, in turn, imposes possible cross-restrictions on bond yields in order to describe specific yield curve models.

Most generally, a financial factor model can be described within a state-space representation where $i_{n,t}$ is the yield-to-maturity at time t (Favero et al. 2007).

Interest rates with different maturities are stacked into a vector $i_{n,t} = [i_{1,t}, i_{2,t}, \dots, i_{k,t}]$ with (3.24) describing the measurement equation. Yields are supposed to be driven by a maturity specific constant component a_n and a set of state variables (risk factors) collected in the vector X_t ; the latter are translated into yield movements via maturity-dependent factor loadings b_n . The dynamics of the state variables are described in the transition (3.25) and are assumed to follow a VAR process.

$$i_{n,t} = a_n + b_n^\top X_t + \varepsilon_{n,t} \quad \varepsilon_t \sim i.i.d.N(0, \sigma^2) \quad (3.24)$$

$$X_t = \mu + \phi X_{t-1} + v_t \quad v_t \sim i.i.d.N(0, \Omega). \quad (3.25)$$

As will be described extensively throughout the next Sections, this formulation allows for various fitting techniques of the yield curve including the Nelson-Siegel representation (Sect. 3.3), the Expectations Hypothesis of the term structure (Sect. 3.4.1), and the affine set-up of modeling bond yields (Sect. 3.5.2).

Principal component analysis (PCA) offers a starting point to investigate the volatility structure of interest rates. It does not rely on a specific interest-rate model which imposes cross-restrictions. Indeed, it identifies common factors driving the yield curve from a time series of historical term structures. Comovements among interest rates are often summarized by the covariance matrix which can be used to extract the risk factors that drive the term structure. PCA analysis has been a key technique in yield curve analysis including the work of Litterman and Scheinkmann (1991). They identified three factors and tried to interpret these factors according to their relative contribution to movements in bond returns. In this context, N yields at N different maturities over T time periods are stacked into the $N \times T$ vector Y that is generated by a linear factor structure with a fixed number of factors

$$Y_t = \bar{Y} + P^\top X_{n,t} + \varepsilon_t \quad \forall n = 1, \dots, N$$

where $\bar{Y}$ is the unconditional mean of the respective yields, $X_{n,t}$ is the $N \times T$ matrix of common *factors* affecting the yields, ε is a vector of idiosyncratic parameters and P is a $N \times N$ matrix containing the *factor loadings* depending on maturity.

A simple way to obtain these factors is to generate the principal components of the structure volatility of bond yields. The main idea is to transform the original N correlated yields into a set of new uncorrelated principal components which are linear combinations of the original yield set.⁶ With the principal components it becomes possible to explain the variance-covariance structure of random yields across different maturities. Principal components are arranged in order of decreased

⁶Geometrically, a principal component analysis is a rotation of axes in a multidimensional setting. The goal is to find linear combinations of the original variables that summarize as much information as possible. The new axes represent the directions with maximum variability. It provides a more parsimonious representation of the covariance matrix (Johnson and Wichern 2006).

importance so that the first principal component accounts for as much as possible of the variation in the original data set, the second accounts for the maximum possible variance among those components uncorrelated with the first, and so on. The analysis seeks to identify N linear combinations of demeaned yields

$$X_{n,t} = P_n^\top (Y_t - \bar{Y})$$

such that for the $N \times N$ matrix $P_n^\top$ the principal components X_n are uncorrelated and their variances are sorted in decreased order (Gentle 2007). The number of principal components can be theoretically as large as the number of underlying yields with different maturities N . If the first few components (factors) are able to account for the bulk of variability in the original yields set, then a reduction of dimension may be sufficient to reproduce the covariance structure of the data set. In particular, if the first k principal components amount for most of the total variation, we can approximate the original yields as

$$Y_t = \bar{Y} + P_k^\top X_{k,t}$$

with $P_k^\top$ denoting a $k \times N$ matrix and the principal components stacked into the $k \times T$ $X_{k,t}$ matrix. Finding the first principal component requires to maximize the variance of the first component subject to the normalization condition that the product of the first factor loading equals one

$$\max\{\text{var}(X_1)\} \quad \text{s.t. } P_1^\top P_1 = 1.$$

Since the variance of the first principal component is defined as $\text{var}(P_1^\top Y) = P_1^\top \Sigma P_1$ where

$$\Sigma = \text{Cov}(Y) = \begin{pmatrix} \sigma_1^2 & \cdots & \sigma_{1N}^2 \\ \vdots & \ddots & \vdots \\ \sigma_{N1}^2 & \cdots & \sigma_N^2 \end{pmatrix}$$

is the covariance matrix of the historical times series of term structures, the eigenvalue decomposition is the key to identify those components with maximal variance (Johnson and Wichern 2006). The symmetric matrix Σ can be factored as

$$P^\top \Sigma P = \lambda$$

with $P = [P_1, \dots, P_N]^\top$ and λ is a diagonal matrix with elements $\lambda_1, \dots, \lambda_N$ so that the variance of the first principal component is clearly maximized by the largest eigenvalue. The first eigenvalue is the solution to the equation system

$$(\Sigma - \lambda_1 I) P_1 = 0.$$

Therefore, the first principal component is $X_{1,t} = P_1^\top (Y_t - \bar{Y})$. The other principal components are calculated by the spectral decomposition where all eigenvectors

(factor loadings) are linear independent and the principal components are uncorrelated. Ultimately, to get a relative measure of the relative contribution of a principal component's variance to the total variance, the proportion of the variance by the n -th factor is

$$\frac{\lambda_n}{\sum_{n=1}^k \lambda_n}.$$

The same procedure can be repeated for yield changes rather than for yield levels by replacing Y_t with ΔY_t and $\bar{Y}$ with zero in the above derivation of the principal components.

Looking at principal components reveals that most of the variation in yield levels as well as in yield changes is explained by the first few principal components. As an illustration, $N = 11$ German zero-coupon yields (0.25, 1, 2, ..., 10 years) for the quarterly sample period 1973Q1 up to 2008Q3 are used. The baseline results are summarized in Table 3.2. For the sample period most of the variation of bond yields can be explained by a maximum amount of three latent factors.⁷ The principal components of yield levels offer a similar picture. In particular, 94.4% of the total variation is already covered by the first principal component. Together with the second and third factor, the variables account for almost 100%. For the comovements of interest rate changes, the first principal component describe 81.7% of the overall variation. Together, the first three factors account for over 99% of total comovements. Appending a further factor, thus, may be superfluous. Principal component analysis does not assume that the factors that best describe term structure movements over time must be observable. These factors are indeed latent factors since the selection of the components is made by the data set itself. Nonetheless, the latent factors and the corresponding factor loadings help to hedge against interest-rate risk. Litterman and Scheinkmann (1991, p. 57) conclude that “[f]or instance it is widely believed that changes in Federal Reserve policy are a major source of changes in the shape of the yield curve. If so – even though we have no clear idea of how to measure “policy” – we can insulate against Fed policy changes provided only that we can determine the *relative* effects of these changes on the returns to bonds of different maturities.”

Table 3.2 Proportion of variation explained by PCs

k th PC	% Explained in Y	% Explained in ΔY
1	94.4	81.7
2	99.4	97.0
3	99.9	99.1
4	100.0	99.9

Note: Principal component analysis for German yields for yield levels and yield changes.

⁷This number is similar for most studies. See Litterman and Scheinkmann (1991), James and Webber (2000), Piazzesi (2003) for the US and the UK.

The k principal components are linear combinations of nine yields. The factor loadings (eigenvectors) are the coefficients of the factors. A blip in the k th principal component X_k causes the term structure to alter by a multiple of the coefficient. In regression terms speaking, the factor loadings represent the regression coefficient with the vector of yields regressed on the principal components. Note, however, that in the principal component analysis there is no need to do such regressions since the data set in its covariance structure is sufficient to identify the factors and loadings. In order to give the single principal components a more profound interpretation, the loadings (coefficients) of the linear combinations of yield levels are plotted which are the $k = 3$ columns of P as function of the maturity of interest rates in quarters.⁸

Figure 3.1 implies that the loadings of the first principal component are horizontal and respectively roughly flat. This pattern basically indicates that a change in the first principal component likewise alters interest rates across all maturities. It corresponds to a parallel shift in the term structure and is therefore called the *level factor*. The loading of the second component is downward sloping and changes in the second factor triggers a tilt or rotation; it stands for a *slope factor*. In particular, if the second component rises then interest rates at the short end of the term structure will increase more than interest rate at the long end, thus, indication a flattening or inversion of the yield curve. The loading of the third principal component is humped-shaped (or U-shaped) causing the yield curve to flex. It occurs at intermediate maturities and the third factor is called the *curvature factor*. For the dynamics of the yield curve it is interesting that the level factor is highly persistent

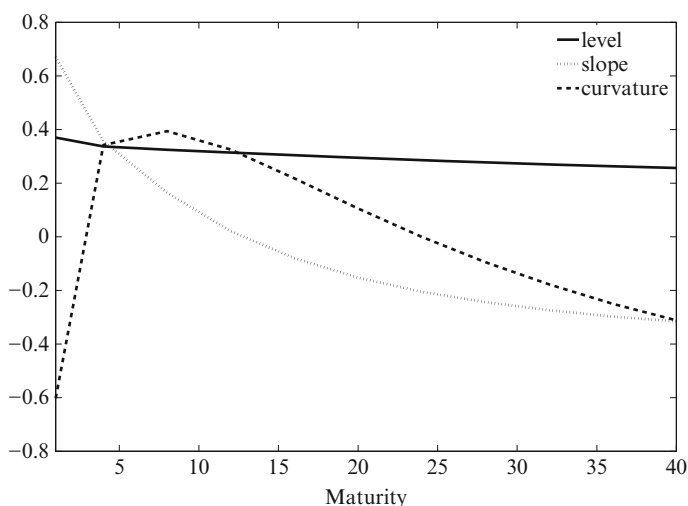


Fig. 3.1 Loadings of the German yield curve

⁸The coefficients on the principal components for interest rate changes follow a similar pattern; therefore, the focus in this section is on yield levels.

with a monthly autocorrelation of 0.98. The slope factor is slightly less persistent ranging at a value of 0.97 whereas the slope factor is autocorrelated with 0.87.

Although principal component analysis is a powerful tool for identifying the common factors of a times series of historical term structures, it lacks important prerequisites for a more fundamental macroeconomic and finance analysis. Extracted factors and its loadings derive from a simple analysis of volatility structures of bond yields. PCA just gives a statistical description of the time-series and cross-sectional behavior of observed interest rates. The model does not impose cross-restrictions on the co-movements of the whole yield curve nor does it rely on a theoretical model of asset-price relations which could be of the form presented in Sect. 2.1.1. Such assumptions are essential for the successful identification and interpretation of interconnections between the term structure of interest rates. Principal components do not offer an answer as to how investors form expectations about the future path of interest rates or how monetary policy affects the yield curve. Albeit most of the literature characterizes the first three components as level, slope and curvature factors, the question then remains what economic forces drive these factors. They are unobservable and need a further economic understanding. Hence, principal component analysis merely removes the economic problem to another level. Such a statistical model in general may fit the yield curve quite well but it fails in forecasting future yield dynamics unless some further model assumptions are imposed. For example, stacking the extracted factors in a VAR representation would allow to predict n -period ahead factors which in turn could be used to forecast bond yields. Yet, PCA has nothing to say about the relationship between observable economic variables and interest rates of different maturities. What it lacks is the appropriate study in economic terms.

3.3 Fitting Zero-Coupon Bonds

The term structure of interest rates is presented as a continuous function of interest rates differing in time to maturity. It implies that for any infinitesimal small step on the spot curve, there must exist a zero bond that is accurately priced in financial markets; or to put in mathematical jargon, the spot curve is continuously differentiable.⁹ In practice, however, prices and interest rates of zero-coupon bonds are only partly observable due to the presence of limited financial instruments across the maturity spectrum. Inconveniently, almost all bonds with time to maturity beyond 12 months bear coupon rates. Observing a sufficient amount of zero-coupon bonds in financial markets, thus, is not possible. Therefore, a spot curve usually must be fitted to (or backed out from) observed coupon-bearing bond prices. Fitting and estimating implicit spot rates and forward rates can rely either on a structural model

⁹For what follows the expressions term structure, yield curve and spot curve are used synonymously since the focus is on zero-coupon bonds.

or on smoothing techniques (James and Webber 2000; BIS 2005). When choosing the former, a parametric interest-rate model is calibrated against the instrument from which the yield curve derives. It relies on explicit assumptions about the evolution of state variables and asset pricing methods using arbitrage or equilibrium arguments. Conversely, smoothing models use financial instruments directly to statistically fit the yield curve as best as possible. Under this approach, the yield curve is derived without any reliance on asset pricing theories except that the bond's price represents the discounted cash flow stream.

The use of smoothed yield curves is very popular among market participants and central banks since it further allows the derivation of implicit forward rates which may serve as an indicator for monetary policy. In particular, spot rates are geometric averages of forward rates so that the shape of the yield curve reflects market expectations on the future evolution of interest rates under certain assumptions. The yield curve can be seen as the expected average of the relevant short rates, while the forward curve indicates the expected time path of the variables which is why forward rates are used to extract market information (Svensson 1994).

The literature lists several methods on how to fit the dynamics of interest rates. Each method has to fulfill at least two objectives. On the one hand, it should attempt to fit a zero-coupon curve that is accurate to market prices and on the other hand, the derived curve should be as smooth as possible. The McCulloch (1971), McCulloch (1975) method uses a polynomial approach and cubic splines to generate a smooth function of the discount function. Linear or non-linear regression techniques can be employed to extract the relevant factors to fit the function and to construct spot and forward rates. However, one of the main drawbacks of these methods is that they produce forward rates at the long end which exhibit unrealistic properties such as a steep fall or a rise in the curve (Anderson et al. 1996). Fama and Bliss (1987) construct yields not via an estimated discount curve, but rather with estimated forward rates at the observable maturities. These rates are subsequently fitted to the yield curve by various smoothing techniques. This could be done with cubic splines but it is exposed to the same critique as before.

Instead, a parsimonious, yet flexible functional form suggested by Nelson and Siegel (1987) is favored by many central banks.¹⁰ As previously stated, this fitting technique does not rely on a specific interest-rate model but captures the basic range of shapes generally associated with spot and forward curves. The Nelson-Siegel function starts with the identification of four parameters which specify a function for the instantaneous forward rate curve. The instantaneous forward rate is defined as the limit where the settlement date approaches the time to maturity (maturity date). Nelson and Siegel (1987) propose using a functional form that is associated with the solution of a difference or differential equation. Thereby, the Nelson-Siegel forward rate function is

¹⁰See the ECB's and BoE's webpage and for the Federal Reserve Board the staff estimations by Gürkaynak et al. (2006b).

$$f_{n,t} = \beta_1 + \beta_2 \exp(-\lambda n) + \beta_3 \lambda \exp(-\lambda n)$$

where $\beta = \{\beta_1, \beta_2, \beta_3, \lambda\}^\top$ is a vector of parameters. The curve allows for the family of forward rate curves that take on monotonic, humped or V-shapes depending on β . The corresponding yield curve is

$$i_{n,t} = \beta_1 + \beta_2 \left(\frac{1 - \exp(-\lambda n)}{\lambda n} \right) + \beta_3 \left(\frac{1 - \exp(-\lambda n)}{\lambda n} - \exp(\lambda n) \right). \quad (3.26)$$

An interpretation of the parameters reveals the basic dynamics of the forward and spot curve. The parameter β_1 is a constant and reflects the horizontal asymptote of the function. For n approaching ∞ , β_1 reflects the long end of the yield curve. The term $\beta_2 \exp(-\lambda n)$ is a decay term and monotonically decreasing (or increasing in the case of $\beta_2 < 0$) towards zero as a function of the time to settlement so that with the time to settlement approaching zero, the spot rate approaches $\lim_{n \rightarrow 0} = \beta_1 + \beta_2$ which corresponds to the current instantaneous one-period forward rate. The third component captures “hump-shaped” curve formations for $\beta_3 > 0$ (and a U-shape for $\beta_3 < 0$). Consequently, the Nelson-Siegel curve is constructed via short-term, long-term and mid-term factors. Svensson (1994) extended the original NS-function by adding a fourth term, i.e. $\beta_4 \exp(-\lambda n)$. Although it provides more flexible curve dynamics, empirical research shows that it is often sufficient restricting the functional form to the standard case (James and Webber 2000).

Recently, researchers have turned their attention to a time-series application of a Nelson-Siegel model. Diebold and Li (2006) made an important contribution in that field. They exploit the parsimony of this framework for the purpose of forecasting three Nelson-Siegel factors. In this respect, they reinterpret the β -coefficients as factors driving the yield curve over time. This is done by adding a time subscript to β_t . They show that the parameters are time-varying level, slope and curvature factors (henceforth NS-Factors) and the terms that multiply these factors are factor loadings. Principal component analysis has already revealed that such models typically can be described by dynamic latent factors and corresponding factor loadings. The difference between the NS-factors and principal components is the structure imposed on the loadings. The Nelson-Siegel approach restricts the loadings according to the functional forms as opposed to statistical PCA that extracts the eigenvectors from the covariance matrix of observed yields. Figure 3.2a plots the loadings of the NS-factors for quarterly maturities up to 10 years.¹¹ Again, the first factor is constant across all maturities similar to the first eigenvector in a PCA. It resembles the yield curve level and its empirical counterpart is the long-term interest rate. The second factor is monotonically decreasing. It corresponds to the negative of the slope as traditionally defined (“long minus short yields”). The empirical

¹¹Following Diebold and Li (2006), the parameter λ is set to a fixed value of 0.2391 which is the value that maximizes the term on $\beta_{3,t}$ for a maturity of 7.5 quarters. This corresponds to the Diebold/Li specification of a maximized third factor loading at a maturity of 30 months. For monthly yields, λ takes on a value of 0.0609.

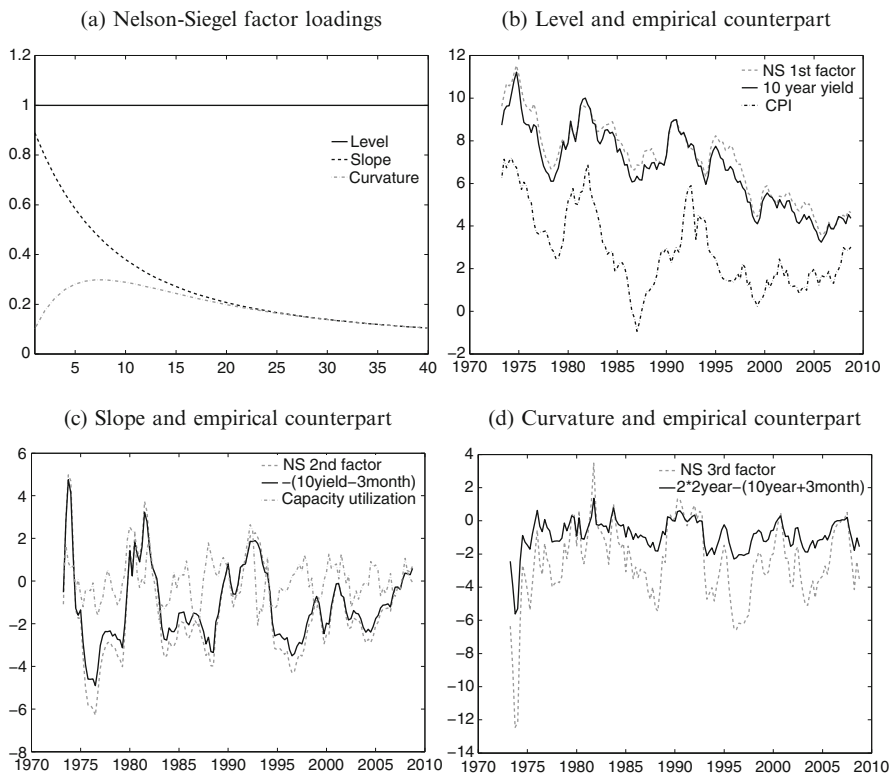


Fig. 3.2 Nelson-Siegel factors and empirical counterparts

counterpart for the curvature factor is approximated as twice the 2-year yield minus the sum of the 10-year and 3-month yields.

It is straightforward to use the dynamic interpretation of the Nelson-Siegel model to extract the driving factors and to use them for forecasting the yield curve. By fixing λ , it is possible to estimate the time-series process of the underlying factor dynamics in a convenient way. Following the work of Diebold and Li (2006), the factors are assumed to behave as independent first-order autoregressions

$$\begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix} = \begin{pmatrix} \mu_L \\ \mu_S \\ \mu_C \end{pmatrix} + \begin{pmatrix} \phi_{11} & 0 & 0 \\ 0 & \phi_{22} & 0 \\ 0 & 0 & \phi_{33} \end{pmatrix} \begin{pmatrix} L_{t-1} \\ S_{t-1} \\ C_{t-1} \end{pmatrix} + \begin{pmatrix} \eta_{L,t} \\ \eta_{S,t} \\ \eta_{C,t} \end{pmatrix}$$

or in matrix/vector form

$$X_t = \mu + \Phi X_{t-1} + \eta_t \quad \eta_t \sim i.i.d.N(0, \Omega). \quad (3.27)$$

with $X_t = [L_t, S_t, C_t]^\top$ and the variance term being a diagonal covariance matrix given by $\text{diag}(\sigma_{11}^2, \sigma_{22}^2, \sigma_{33}^2)^\top$.¹² The measurement equation can be specified according to (3.25) as

$$i_{n,t} = a_n + b_n^\top X_t + \varepsilon_t \quad \varepsilon_t \sim i.i.d.N(0, \sigma^2 I) \quad (3.28)$$

with

$$b_n^\top = \left[1, \frac{1 - \exp(-\lambda n)}{\lambda n}, \frac{1 - \exp(1 - \lambda n)}{\lambda n} - \exp(\lambda n) \right]$$

$$a_n = 0.$$

In general, there are two estimation strategies that can be applied, i.e. ordinary least squares or maximum likelihood estimation with the Kalman filter. The first way to estimate the factor process is a two-step estimation procedure. The factors for each point in time are estimated by simple OLS to extract the time-series dynamics; in a second step the parameters governing the AR(1)-processes are likewise estimated by standard OLS estimation. Alternatively, the factor dynamics and its Nelson-Siegel restrictions with respect to observed interest rates can be understood as a state-space system where the measurement errors are assumed to be the same for all interest rates and they are uncorrelated to the white noise transition Ω which is a condition for least square optimality of the Kalman filter (see Eqs. (3.24) and (3.25) for the most general set-up).

This state space set-up delivers maximum likelihood estimates with the Kalman filter algorithm for the parameter vector $\theta = \{\mu, \text{vec}(\phi), \text{vec}(\Omega), \sigma\}^\top$.¹³ It is an efficient and consistent estimator for the Nelson-Siegel model. The filter allows to extract and to estimate the parameters of the model simultaneously and it provides a correct inference about the estimation of parameters. Moreover, the method truly allows to treat the NS factors as latent factors driving the yield curve (Diebold et al. 2006).

The results may be sensitive to initial guesses about the unconditional mean and covariance matrix of the state vector. Therefore, the two-step OLS approach is used to extract the initial parameter values of the model and to get the unconditional moments. In a next step, the maximization of the log-likelihood function with respect to the estimated parameter vector is achieved by an unconstrained nonlinear numerical procedure with Matlab.¹⁴ Standard errors are calculated as the negative of the inverse Hessian of the log-likelihood evaluated at its maximum.

¹²Favero et al. (2007) also use a VAR model to estimate the Nelson-Siegel factor dynamics. However, in their work, state dynamics are restricted being independent from each other.

¹³For the Kalman filter see Appendix B and Hamilton (1994).

¹⁴Firstly, the Simplex routine *fminsearch* is applied using a maximum of 5000 iterations. Then, the derivative-based optimizer *fminunc* refines the parameter estimates.

The Nelson-Siegel model is examined covering the sample period 1973Q1–2008Q3 and interest rates with maturities of 1, 4, 8, 20, 28 and 40 quarters for Germany, 1972Q1–2008Q3 with maturities 1, 4, 8, 12, 20, 28 and 40 quarters for the US and 1970Q1–2008Q3 with maturities 1, 4, 8, 20, 28 and 40 for the UK. The results are similar for all the countries, thus, in the following, the focus is solely on Germany.¹⁵ Figures 3.2b–d give graphical illustrations of the extracted latent factors for Germany together with the empirical level, slope and curvature moments. Like for the PCA, the first NS-factor appears to be very persistent with an autocorrelation coefficient of $\phi_{11} = 0.98$ compared to the second and third NS-factors ($\phi_{22} = 0.92$, $\phi_{33} = 0.83$). Indeed, it almost perfectly matches its empirical proxy, i.e. the long-term yield earned on government bonds. The same results hold for the second and third NS-factors which reproduce their empirical counterparts accurately. The transition shock volatility as measured by the diagonal elements of Ω increases when moving from L_t , to S_t to C_t (Table 3.3).

Table 3.3 Estimated Nelson-Siegel factors

	GER	USA	UK
μ_L	0.103 (0.091)	0.163 (0.118)	0.132 (0.109)
μ_S	-0.110 (0.081)	-0.275** (0.104)	-0.074 (0.076)
μ_C	-0.441** (0.162)	-0.143 (0.59)	-0.176 (0.116)
ϕ_{11}	0.986** (0.012)	0.976** (0.015)	0.983** (0.012)
ϕ_{22}	0.915** (0.032)	0.860** (0.041)	0.903** (0.033)
ϕ_{33}	0.843** (0.048)	0.629** (0.069)	0.878** (0.043)
σ_{11}^2	0.129** (0.017)	0.246** (0.033)	0.336** (0.045)
σ_{22}^2	0.73** (0.089)	0.794** (0.099)	0.886** (0.111)
σ_{33}^2	1.805** (0.271)	5.567** (0.736)	1.399** (0.303)
σ	0.051** (0.004)	0.082** (0.003)	0.094** (0.006)

Note: Parameter estimates based on MLE and Kalman filter algorithm. Estimated standard errors are in brackets. An asterix (*) denotes significance at the 5% and two asteriks (**) denote significance at the 1% level.

¹⁵Using the same estimation strategy for US data, Diebold et al. (2006), Koopman et al. (2007), Nyholm and Vidova-Koleva (2010) derive very similar parameter estimates, standard errors and Nelson-Siegel factor dynamics as in this study for the US.

An interpretation of these latent variables is awkward because they do not derive from a model that includes economic variables. To overcome this problem, each subfigure of Fig. 3.2 plots a macroeconomic proxy that is correlated with the state variables. The level dynamics indicate an overall secular decline of the term structure with the first NS-factor affecting the whole spectrum of interest rates. In particular, many researchers attribute the level shift to the favorable macroeconomic conditions with low volatility of macro aggregates and a permanent decrease of market inflation expectations during the last decades that is translated into lower long-term interest rates as suggested by the Fisher equation (Backus and Wright 2007). Indeed, there is high correlation (0.78) between the level factor and inflation over the sample period.

Regarding the slope factor, it behaves procyclical with economic activity.¹⁶ Since the second NS-factor corresponds to the negative value of the term spread, a positive value should signal a macroeconomic environment of plumping economic growth and rising unemployment (Stock and Watson 2003a; Estrella and Mishkin 1997). The important point here is the lag structure of the slope factor and capacity utilization. Usually, one would expect the term spread to be negatively correlated with economic activity due to the presence of counter-cyclical monetary policy actions at the short end of the yield curve. Figure 3.2c does not reproduce this stylized fact with a correlation coefficient of 0.34. It rather mirrors the observation that periods of capacity constraints inveigle monetary authorities to let the policy rate hike which subsequently triggers a decline in overall economic activity. Critiques on monetary policy actions would claim that the central bank reacts to the macroeconomic environment too late to actively set interest rates precociously in a counter-cyclical way. Moreover, the length of the monetary transmission mechanism is crucial when interpreting term structure and macro effects. The longer it takes for a monetary impulse to spill over to the real economy, the more likely it is to find a positive correlation between economic activity and the term spread. Finally, the third factor in Fig. 3.2d hardly offers an obvious macroeconomic link (see also Diebold et al. (2006) for this result). This does not mean that it is not possible to interpret the curvature factor but the literature typically can not identify aggregate macro variables that significantly covary with it. One interpretation is the “flight to quality” aspect in times of crisis when economic agents increase their holdings of near monetary assets to make their portfolios more liquid (Duffee 2002).

Although the approach of modeling the yield curve by restricting the coefficients on the loadings offers significant progress for the identification of the driving state variables, the Nelson-Siegel representation (like the PCA) still lacks a well-founded economic interpretation. The model is not applicable to extract market expectations on the likely path of future risk-adjusted short-term interest rates nor is it possible to extract markets’ view on monetary policy actions. This comes from the omission of a theoretically-motivated model of interest rates in a partial equilibrium setting.

¹⁶Here, economic activity is measured as the cyclical component of real GDP using the Hodrick-Prescott filter.

What is presented in Sect. 3.4, is the most insightful way to think about the dynamics and cross-restrictions of interest rates along the yield curve.

3.4 Understanding the Term Structure of Interest Rates

3.4.1 *A Formal Representation of the Expectations Hypothesis and No-Arbitrage*

The *Expectations Hypothesis* of the term structure of interest rates relies on the general proposition that expectations about future interest rates affect the current level of long-term interest rates. It stands for numerous statements that link yields, returns and forward rates of different maturities and periods. In broadest terms, the slope of the yield curve reflects market expectations about future interest rates. From a historical perspective, the examination of the theory goes back at least to the work of Irving Fisher who elaborated on what would later be known as the famous Fisher equation according to which one-period real returns on nominal assets and goods-denominated assets under perfect foresight should be equalized. Fisher also augmented the investment period by allowing his equation to hold in a setting in which the present value of an investment is the discounted value of a given series of payments divided by the time-dependent geometric product of the series of one-period interest rates (Fisher 1896, 27).¹⁷ Initially, continuing with the contributions of Keynes (1930), Hicks (1946) and Lutz (1940), the theory did not originate from an equilibrium model or from rigorously formal statements. They were developed to interpret the interconnections between yields and forward rates as well as the time series property of the term structure of interest rates.

The Expectations Hypothesis heavily relies on the concept of the relative pricing approach. What is central to modern asset pricing theory is the concept of no-arbitrage as outlined in Chap. 2.1.1. It involves pricing one asset relative to another where the base price is an asset with risk-free returns. If expected returns between assets are not equalized, then investors realize profits through trading across these assets and across different time periods. This Section shows that the different equations surrounding the Expectations Hypothesis link unobservable expectations to a set of observable bond instruments. All forms of the Expectations

¹⁷Fisher writes that “interest realized on a very long bond, say 50 years is often lower than on a 25 years’ bond. This is explainable by the prevailing opinion that interest tends to fall, so that if the 50 years’ investment were in two successive bonds of 25 years each, the interest realized in the second would be lower than in the first. The “actuarial average” of the two is equal to the interest realized in the 50 years’ bond (p.29).” This is interesting since it implies that Fisher expected the 25 years’ bond to fall rather than to rise which corresponds to a downward sloping yield curve on average. This must come from either an expected appreciation of the monetary standard or likewise decreasing expected real returns over the investment horizon indicating to falling long-term nominal interest rates along the same line.

Hypothesis compare (at least) two different ways to transfer wealth across time. In essence, no matter how investors choose their portfolio, the returns should be identical. If, however, the evolution of interest rates is stochastic, these equations might be mutually inconsistent due to Jensen's Inequality. This comes from the fact that bond prices have positive convexity with respect to yields that drives a wedge between actual expectations of future short rates and implied forward rates derived from the yield curve. Most essentially, the presence of risk-averse market participants introduces a further variable driving current and expected rates, i.e. the term premium. Empirical tests on the Expectations Hypothesis show that expected returns on bonds are time-varying which mirrors a contraction of the theory (see Sect. 3.4.2).

A natural starting point to discuss the theory of the term structure is a world of certainty. In a certain economy, all future interest rates are known in advance so that any equilibrium must be satisfied by the no-arbitrage condition which equalize the forward rate $(1 + F_{n,t})$ and the short-term interest rate $(1 + Y_{1,t+n})$ at time $t + n$. If this condition does not hold, investors would realize arbitrage opportunities. A situation of higher forward rates could induce investors today to sell short one bond maturing at time $t + n$ and buy $(1 + F_{n,t})$ bonds maturing at $t + n + 1$. As already outlined in Sect. 3.1.1, the net cost of this portfolio is zero. At time $t + n$, investors would hold one unit of account on the maturing bond, but the bond maturing in the next period would be worth $1/(1 + Y_{1,t+n})$. The net position of the investment would be $[(1 + F_{n,t})/(1 + Y_{1,t+n})] > 0$, an obvious opportunity to arbitrage the profit away (see Ingersoll 1987, 388). Moreover, absence of arbitrage implies that the realized return on every bond over any period of time equals the prevailing riskless short-term return.

The concept of no-arbitrage can be translated into an economy in which future short rates are subject to unanticipated changes for each future date. As Cox et al. (1981) and Campbell et al. (1997) make clear, it is possible to distinguish between the "Pure Expectations Hypothesis" (PEH) which states that expected excess returns are zero over time and non-predictable and the "Expectations Hypothesis" (EH) which says that excess returns might exist, but they are non-zero and constant over time. This terminology is due to Lutz (1940). Moreover, the literature classifies different forms of the PEH since various ways exist to compare diverse investment strategies for bonds across maturity periods and across time. Their mathematical expressions also depend on the choice of using discrete time vs. continuous time and the choice of frequency of compounding – continuous (log returns) vs. discrete (simple/gross returns). Depending on the representation of the theory, inequivalence among them exist which is presented in the next paragraphs.

Starting with discrete compounding, a first form of the PEH equates expected one-period returns on one-period bonds and n -period bonds so that expected excess returns on long-term bonds over short-term bonds are zero. The *local expectations hypothesis* then states that

$$(1 + Y_{1,t}) \stackrel{!}{=} (1 + Y_{n,t})^n E_t [(1 + Y_{n-1,t+1})^{-(n-1)}]. \quad (3.29)$$

A second form augments (3.29) and equates the total n -period bond return with the return on one period bond and a $n-1$ period bond. The *return-to-maturity hypothesis*

$$(1 + Y_{n,t})^n \stackrel{!}{=} (1 + Y_{1,t}) E_t [(1 + Y_{n-1,t+1})^{n-1}] \quad (3.30)$$

claims that the total return from holding a zero-coupon bond to maturity is equal to the total return from continually rolling over a series of coupon bonds over the same period or by holding a short-term instrument for one period and subsequently holding a $n-1$ maturing bond until maturity. No matter how the investment strategy is chosen, the total return should be the same.

A related version to (3.30), the *yield-to-maturity hypothesis*, states that the periodic yield from holding a n -period bond is equal to the return from rolling over a series of one-period bonds but it refers to the annualized return rather than to the total return

$$(1 + Y_{n,t}) \stackrel{!}{=} E_t [(1 + Y_{1,t})(1 + Y_{1,t+1}) \cdots (1 + Y_{1,t+n-1})]^{1/n}. \quad (3.31)$$

Still, when solving forward, (3.30) implies (3.31). Finally, the last version of the PEH is the *unbiased expectations hypothesis* which shows that current forward rates are unbiased predictors of future spot rates.

$$(1 + F_{n,n+1,t}) = \frac{(1 + Y_{n+1,t})^{n+1}}{(1 + Y_{n,t})^n} \stackrel{!}{=} E_t (1 + Y_{1,t+n}). \quad (3.32)$$

This is the identity, early term structure theorists thought of, when linking current yields, forward rates and actual future spot rates.

Although the different forms describe the PEH and the basic equilibrium between bond prices and forwards across maturity and time, Cox et al. (1981) show that they are mutual inconsistent with each other if interest rates are stochastic in nature and if short-term interest rates can be described by an autocorrelated time series process. In particular, the one-period form (3.29) and the n -period form (3.30) of the return hypothesis are inconsistent because due to Jensen's Inequality the expected value of the inverse of a random variable is not in general equal to the inverse of its expected value ($1/(E(x)) \neq E(1/x)$). Moreover, the unbiased version conflicts with the local expectations version along the same argument. The same inequality holds for the yield-to-maturity and the unbiased expectations hypothesis. Equation (3.32) implies (3.31) but the opposite is not true because Jensen's Inequality forces the forward rate to be an upward biased predictor of the actual future short-term yield. When expressing the Expectations Hypothesis with discrete compounding yields, there are competing forms differing due to Jensen's Inequality which drives a wedge between the equilibrium concepts. It is crucial to recognize that general risk-neutrality is roughly the same as the PEH. Risk-neutral investors take the (constant) variance term stemming from Jensen's inequality into account when pricing bonds

and evaluating different investment strategies. For example, Piazzesi (2003) shows that, unconditionally, risk-neutrality implies a slightly downward sloping yield curve when the short rate follows a random walk; whereas under the PEH, according to which the yield of a long-term bond is the expected average of future spot rates, the yield curve is a flat function of time to maturity.

If discrete gross interest rates are lognormal and homoscedastic, the wedge can be quantified to be half of the variance between a one-period return on a n -period bond and the riskless interest rate. Campbell (1986) demonstrates that this effect is very small for short-term securities and becomes significant only at the very long end of the term structure. Most reduced-form empirical research uses a log form of the PEH and ignores the variance term imposed by the presence of stochastic and correlated interest rates when evaluating the PEH. This strategy can be justified as an approximation of the PEH in discrete time (McCulloch 1993). Then, all forms of the PEH are compatible to each other and equivalent so that Eqs. (3.29), (3.31) and (3.32) respectively can be expressed as

$$E_t[r_{n,t+1}] = i_{1,t} \quad (3.33)$$

$$i_{n,t} = \frac{1}{n} \sum_{j=0}^{n-1} E_t[i_{1,t+j}] \quad (3.34)$$

$$f_{n,n+1,t} = E_t[i_{1,t+n}]. \quad (3.35)$$

As introduced in Sect. 3.2.2, the EH can be restated within the factor model representation of Eqs. (3.24) and (3.25). For that purpose, it is assumed that the one-period interest rate follows a conventional AR(1) process of the form

$$i_{1,t} = \mu + \phi i_{1,t-1} + \varepsilon_t \quad (3.36)$$

with $\varepsilon_t \sim i.i.d.(0, \sigma^2)$. In this case, X_t is defined as the short rate $i_{1,t}$ and it is endogenous with respect to the measurement equation. It holds that

$$i_{n,t} = a_n + b_n i_{1,t} \quad (3.37)$$

with

$$b_n = \frac{1}{n} \sum_{i=0}^{n-1} \phi^i = \frac{1 - \phi^n}{n(1 - \phi)} \quad (3.38)$$

$$a_n = \frac{1}{n} \sum_{i=1}^{n-1} b_n^* \mu, \quad \text{where} \quad b_n^* = \sum_{i=0}^{n-1} \phi^i \quad (3.39)$$

for $n > 1$. A proof is reported in Appendix C.

The empirical literature on the theory of the yield curve is less restrictive than the pure theory. The *Expectations Hypothesis* modifies the PEH slightly by allowing the equilibrium conditions to be modified by introducing a constant term. It is constant over time but varies across the maturity spectrum of spot and forward rates as well as holding-period returns. Early theorists like Keynes (1930) and Hicks (1937) attribute a constant but maturity dependent term premium to capital risks which increase with the settlement date as interest-rate changes have a bigger impact on the present value of an investment strategy of holding and selling long-term bonds. Consequently, except if calculated until maturity, the return on a bond is uncertain and risk-averse investors may require a compensation for that risk. Closely related to this thought, Lutz (1940) points out that long-term securities are less liquid compared to short-term ones where the most liquid asset is money. The term premium is then a positive constant and increasing with time to maturity.¹⁸ Hicks (1946, p.146) made up his argument especially through a constitutional weakness on the demand side which offers “[...] an opportunity for speculation. If no extra return is offered for long lending, most people (and institutions) would prefer to lend short, at least in the sense that they would prefer to hold their money on deposit in some way or other. But this situation would leave a large excess demand to borrow long which would not be met. Borrowers would thus tend to offer better terms in order to persuade lenders to switch over into the long market.” The Keynes-Hicks view of term premia has been criticized by Modigliani and Sutch (1966) for its overemphasis on capital-value (present value) risk. They disputed that it is not clear why risk-averse investors are only concerned with short-term capital losses. Instead, “preferred habitats” may induce them to choose a bond with maturity n that coincides with the preferred investment horizon in order to minimize capital-income risk.¹⁹ A separate supply and demand in each habitat in principal allows for any term premium pattern. As a special case, the Keynes-Hicks approach could be represented as a preferred habit, namely that case in which all bond holders have a very short-term holding period preference.

Following Campbell (1986), the EH holds for each version including Jensen’s Inequality terms and constant term premia. In a continuously compounded setting, the EH can be formalized as

¹⁸In analogy to Keynes’ liquidity preference theory of interest, the risk premium has also come to be known as liquidity premium. Such a definition is, however, slightly imprecise since it does not separate interest-rate risk (due to uncertain future interest rates) from the characteristics of an asset to be liquid or illiquid (the ease of converting an asset into money as outlined in Keynes (1930)). Chapter 4.4 will further elaborate on this distinction.

¹⁹Kahn (1972) also draws attention to the investor’s sensitivity towards capital and income risk. In this respect, Kregel (1998) points to the different views of Keynes and Fisher on the Fisher relation that “[...] is based on the dominance of widows and orphans, investing on the basis of [...] expected constant real returns [...]. The Keynes-Kahn approach [...] emphasizes the impact of interest rates [...] and the cost of carry of bond portfolios [...]” which are absent in Fisher’s analysis. Capital value risk, thus, is the primary motive in the Keynes-Kahn tradition; whereas capital income risk takes center stage in the Fisher relation.

$$E_t[r_{n,t+1}] = i_{1,t} + xr_n \quad (3.40)$$

$$i_{n,t} = \frac{1}{n} \sum_{j=0}^{n-1} E_t[i_{1,t+j}] + \phi_n^y \quad (3.41)$$

$$f_{n,n+1,t} = E_t[i_{1,t+n}] + \phi_n^f \quad (3.42)$$

where xr is the one-period excess return for a bond with maturity n , ϕ_y is the yield term premium and ϕ_f the forward term premium, respectively.

As has been shown, the pure Expectations Hypothesis, the concept of no-arbitrage and general risk neutrality are different concepts. Under strict assumptions, they may be partly consistent. The pure form of the Expectations Hypothesis (PEH) only works if interest rates are non-stochastic and are known with certainty to all investors. Under this setting, Jensen's Inequality does not show up quantitatively and the PEH is the same as imposing the implications of absence of arbitrage opportunities. If interest rates follow a stochastic process and short-term yields are correlated, risk-neutral investors take Jensen's Inequality into account so that the PEH is a contradiction to the concept of no-arbitrage but the Expectations Hypothesis (EH) still may hold. In case of risk-averse investors, the EH is equivalent to no-arbitrage if and only if term premia are constant over time. Time-varying term premia are conform with no-arbitrage but conflict with the EH. Nevertheless, thinking in terms of the EH, allows to infer market views on expected interest rate dynamics and term premia where the difficult task is to distinguish between these determinants affecting the yield spread.

3.4.2 Empirical Tests on the Expectations Hypothesis

Most work on the empirical validity of the Expectations Hypothesis is conducted under the assumption of rational expectations. It states that all term premia depend only on maturity and not on time. Additionally, the EH implies that excess returns are not predictable over time. A changing slope of the term structure then reflects varying rational expectations on the future path of interest rates. One of the early empirical observations was made by Macaulay (1938, 33) who stressed that "the 'yields' of bonds of the highest grade should fall during a period in which short term rates are higher than the yields of the bonds and rise during a period in which short term rates are lower. Now experience is more nearly the opposite. The forecasting of short term interest rates by long term interest rates is, in general, so bad that the student may well begin to wonder whether, in fact, there really is any attempt to forecast." This result is a striking feature for what followed in the empirical research on the EH. The work mainly focussed on the spread of the yield curve as predictor for future interest rates.

It would be a merely unresolvable task to consider all work done in this research since so much was already done and is still continuing to grow as new approaches and methods to test the EH are evolving. The basic assertion of hundreds of studies is that – with few exceptions – the EH can be rejected no matter what testing equations, estimation techniques and information sets have been used. To this end, Froot (1989, p.283) summarized that “if the attractiveness of an economic hypothesis is measured by the number of papers which statistically rejected it, the expectations hypothesis of the term structure is a knockout.” Still, understanding the comovements of bond yields is important for both central banks and market practitioners. Extracting market’s interest rate expectations and the accurate modeling of spot rates is essential for forecasting, consistent pricing and hedging of fixed-income securities.

One of the most tested equations of the EH follow Campbell and Shiller (1991) who developed regression equations according to which the spread between the short- and the long-term interest rate is an efficient predictor of future long-term and short-term interest rate changes. The first approach can be determined from the local expectations hypothesis of (3.40) where the one-period return on a n -period bond equals the short rate. From this proposition, a testable equation can be derived

$$\begin{aligned}
 E_t[r_{n,t+1}] &= i_{1,t} + xr_n \\
 i_{n,t} - (n-1)(E_t[i_{n-1,t+1}] - i_{n,t}) &= i_{1,t} + xr_n \\
 E_t[i_{n-1,t+1}] - i_{n,t} &= (i_{n,t} - i_{1,t})/(n-1) - xr_n/(n-1) \\
 i_{n-1,t+1} - i_{n,t} &= \alpha_n + \beta_{1,n}(i_{n,t} - i_{1,t})/(n-1) + \varepsilon_{n,t} \quad (3.43)
 \end{aligned}$$

where the projection of changes in the long-term rate onto the slope of the yield curve should give a coefficient of unity for all n and α may reflect the constant term premium component.²⁰ Therefore, testing the EH implies the null hypothesis of $\beta = 1$ (CS-coefficient) for all n . The forecast error under rational expectations is orthogonal to the information set at t so that the term spread should be uncorrelated with the errors. Under the EH and a positive yield spread, long rates should rise so that expected capital losses of holding the long-term security offset the current positive yield difference.

Related to this type of regression, a second approach proposed by Campbell and Shiller (1991) rearranges (3.41), adds and subtracts the short rate on each side which gives

$$i_{n,t} = 1/n \sum_{j=0}^{n-1} E_t[i_{1,t+j}] + \phi_n^y$$

²⁰In the testable equations, rational expectations are assumed so that $E_t[x_{t+i}] = x_{t+i} + \varepsilon_t$ with ε_t being white noise.

$$\begin{aligned}
1/n \sum_{j=0}^{n-1} (E_t[i_{1,t+j}] - i_{1,t}) &= i_{n,t} - i_{1,t} - \phi_n^y \\
\frac{1}{n} \sum_{j=0}^{n-1} (i_{1,t+j} - i_{1,t}) &= \alpha_n + \beta_{2,n}(i_{n,t} - i_{1,t}) + \varepsilon_{n,t} \quad (3.44)
\end{aligned}$$

where the expression of the left-side is called the perfect foresight spread. It implies that the yield spread predicts the weighted cumulative expected change in the short rate over the life of the long-term bond. Consequently, the EH says that whenever long-term yields exceed the current short rate, expected future short rates tend to rise to equate the returns over the life of the long-term security. Similar to the first approach, if the EH holds, the projection coefficient β_2 should be unity so that an observed positive yield curve spread is associated with increasing future short-term rates.

Vast literature documents the failure of the EH since the estimated regression coefficients on the yield curve spread differ significantly from one. Moreover, Campbell et al. (1997) and Dai and Singleton (2002) among others²¹ find that the coefficients of the regression in (3.43) is additionally significant less than zero. Typically, the coefficients $\beta_{1,n}$ of regressions on the change in long rates fall monotonically with the maturity of the bond indicating a strong positive relationship between yield spreads and excess returns on long-term bonds (Campbell 1995). On the contrary, the coefficients $\beta_{2,n}$ for the short-rate change projections in (3.44) follow an U-shape (or smile respectively); they are smaller than unity for small n , declining up to a year and then increasing with values significantly greater than one for long-term bonds. Therefore, for short-term maturities, the EH cannot be rejected and at least some truth can be found in the hypothesis that observed bond yields can predict future short-term interest rates changes.

Another way of looking at testable equations of the EH is to apply the unbiased expectations hypothesis that states that current forward rates should appropriately predict and anticipate future short rates. Here, too, such regressions would imply that the forward premium is both constant and not forecastable. The seminal work on this approach was done by Fama and Bliss (1987). Instead of directly using the realized short-term interest rate in $t + n$ to test the EH, they show that (3.42) can equivalently tested by regressing either realized changes in the short rate over the life of the forward rate or the one-period excess return of an n -period bond on the forward-spot spread

²¹Campbell and Shiller (1984), Mankiw and Miron (1986), Campbell and Shiller (1991), Hardouvelis (1994), Rudebusch (1995), Bekaert et al. (1997), Tzavalis and Wickens (1997), Bekaert and Hodrick (2001), Bekaert et al. (2002), Cuthbertson and Nitzsche (2006), Rudebusch and Wu (2007).

$$i_{1,t+n} - i_{1,t} = \alpha_n + \beta_{n,3}(f_{n,n+1,t} - i_{1,t}) + \varepsilon_{n,t} \quad (3.45)$$

$$xr_{n,t+1} = \alpha_n + \beta_{n,4}(f_{n,n+1,t} - i_{1,t}) + \varepsilon_{n,t}. \quad (3.46)$$

The forward rate regression on changes in the short rate would imply a coefficient of unity and the excess return estimation calls for $\beta_{n,4}$ to equal zero if the EH is supposed to hold. Now, evidence points again towards the respective opposite result.²² In particular, excess returns on bonds are predictable via the forward-spot spread indicating to time-varying term premia. A paper that recently received a lot of attention is Cochrane and Piazzesi (2005) who re-examine the forecasting regressions of Fama and Bliss. They construct a single factor including the term structure of forward rates instead of a single forward rate. Their finding is that the coefficients from a regression of excess returns on this function of forward rates follow a tent-like shape and that this single common forward-rate factor forecasts excess returns at all maturities with a much higher goodness of fit compared to the single forward-rate expression.

Most of the results presented above have been applied on US data for various sample periods. Since the predictability of short-term interest rates may differ across countries, one might also expect varying results in favor or against the EH. For selected European countries, a number of authors find empirical support at the short end of the yield curve (<1 year) but generally reject the hypothesis of inferring interest-rate expectations of current observable long-term interest rates.²³ In particular for Germany, the predictive power of the term spread for ex-post future interest rate levels differs considerably from U.S. evidence with more promising results in favor of the EH. Consequently, monetary policy could be guided by the slope of the term structure since the latter signals future yield movements at least to some degree correctly (Boero and Torricelli 2002).

Empirical evidence overwhelmingly suggests that the EH in its constant term premium form can be rejected for the whole maturity spectrum. This does not imply that it is impossible to infer interest rate expectations from the term structure. It rather reflects the insight that the EH and its single-equation representation only predicts future yield levels to a limited extent. One way to improve regression tests is to model the Expectations Hypothesis in a bivariate stationary VAR in first difference of the short rate and the term spread. The validity of the EH imposes restrictions on the coefficients of the distributed lags by setting the null hypothesis according to the perfect-foresight spread. Campbell and Shiller (1987) find that although the EH likewise can be rejected, the theoretical spread derived from forecasting future short rates within the VAR framework are strongly correlated with the observed spread. They suggest that despite its

²²In a slightly modified regression, Backus et al. (2001) show that although their estimates seem to find weaker evidence against the EH, the construction of an one factor model of the short rate reveals that these results still translate into large negative values for the CS coefficients.

²³Hardouvelis (1994), Kugler (1997), Gerlach and Smets (1997), Guidolin and Thornton (2008).

poor statistical fit there is an important element of truth in the Expectations Hypothesis.²⁴

In order to understand the single-equation results, the literature elaborates different reasons for the failure of the EH. Numerous attempts have been made to attribute one or more paradoxical results to specific violations of the EH. In this respect, the important question is whether the failure of the EH can be explained within the rational expectations approach or whether it constitutes evidence on non-rational asset price anomalies. The overreaction idea asserts that long-term interest rates overreact to expected changes in short-term interest rates. For instance, if the market as a whole raises the expected path of future interest rates due to some event, long-term bond yields overreact causing the current spread to increase by an amount that is larger compared to the spread warranted by the expectations change. Over time, this overreaction is removed although current short-term rates rise. This creates a negative correlation between the spread and long-term interest rate movements and generates the negative bias in β_1 (Froot 1989; Hardouvelis 1994). Related to this hypothesis, Bekaert et al. (2001) interpret the overreaction not to non-rational anomalies but to a rational overreaction version in spirit of the “peso problem” in which high-interest rate regimes occur less frequent than rationally anticipated. A recent terminology that describes this hypothesis is the *Black Swan* in the short-term money market. From mid-August 2007 on, US interbanking rates and various yield spreads dramatically increased in anticipation of a deterioration of credit-risk quality for a number of financial institutions – albeit the market had no new information on increased counterparty risk between financial institutions borrowing and lending in the interbanking market. Clearly, declining asset prices and realized risks in derivative-based securities triggered doubts about the worthiness of market participants. Still, the day the turmoil started, there was no definite signal which could have induced traders to demand excess returns on LIBOR securities. Investors expected such an unfavorable event ex-ante but it did not occur during the particular short time frame (see for a fundamental analysis Taylor and Williams 2009b).

Regarding further explanation attempts for expectation errors, Campbell (1995) shows that changing views about excess long bond returns act like a measurement error. The failure of the EH can be explained by the fact that excess returns appear in (3.43) on the left-hand side as well as on the right-hand side. If expectations of excess returns change over the life of the bond, this biases β_1 down since they negatively affect the dependent variable and positively the regressor. In contrast, regressions of the yield spread on future interest rates as in (3.44) biases β_2 up since changing expectations of excess returns only affect the right-hand side positively.

Along similar lines, the explicit consideration of monetary policy can help in explaining to some part the low predictive power of the yield spread for future

²⁴See for the same results Bekaert and Hodrick (2001) or Favero (2006) who also apply a slightly modified VAR approach.

interest-rate movements. For example, Mankiw and Miron (1986) make clear that the predictions of the EH fit the data better in periods in which interest rate movements have been highly forecastable such as the period before the founding of the FED. This implies that firstly, the power of the EH changes across different monetary policy regimes and secondly, a central bank that heavily smoothes interest rates makes it harder for market participants to forecast future movements if the short rate follows a near random walk (Rudebusch 1995; Balduzzi et al. 1997; Thornton 2006). Consequently, in such an environment, any positive yield curve spread mirrors term premia and excess returns of long-term over short-term bonds.

Time-varying risk premia then indicate to risk-averse agents who are only willing to buy long-term bonds in an environment of a steep yield curve if they can earn an extra return to get compensated for taking the risk of interest changes. Indeed, there is ample evidence for time-varying excess returns (Fama 1984; Tzavalis and Wickens 1997; Duffee 2002; Dai and Singleton 2002; Cochrane and Piazzesi 2005). Accepting time-varying risk has the advantage that any yield curve model still relies on rational expectations where the danger of forecast errors are priced into the term structure via a risk correction. The omission of time-varying risks in traditional regressions of the EH is then the main candidate for why the yield spread is unable to forecast future interest rates correctly. To fully account for this explanation, one must identify the underlying sources of risk and how they are translated into changing term premia. The modern finance literature tries to build such term structure models in which required expected excess returns vary according to risk factors within a fully specified no-arbitrage equilibrium. The most popular yield curves model represents bond yields as an affine function of some state vector (latent factors) in which risk premia also depend linearly on these factors. It can be shown that such term structure models are highly flexible and that they can match yield movements, cross-restrictions and the evolution of term premia quite well (Duffee 2002). The following Sect. 3.5 describes the general formulation of such term structure models. In this respect, the reader should keep in mind that derived excess returns are solely due to interest-rate risk, i.e. the possibility of forecast errors when making projections for the future interest-rate process. Chapter 4 will fully account for various other forms of term premia which might be priced into bond yields such as default risk or liquidity risk.

3.5 Affine Term Structure Representations

3.5.1 General Setup

The basic problem in term structure modeling is that market expectations about the path of interest rates are not observable. The future market might be seen as a good proxy but term premia might distort the information content of asset

prices. Separating expected interest rates from term premia is the main requirement for any yield curve model that tries to impose economically important no-arbitrage restrictions on the cross-movements of interest rates. Other yield curve specifications such as the Nelson-Siegel model are able to deliver a good fit as well as promising forecasting performances but they lack sound economic restrictions.²⁵ One successful approach that stems from the finance literature and that has been growing very rapidly in recent years, are no-arbitrage term structure models. They rely on the general proposition that movements in the cross section of bonds yields are closely tied together.

The basic concept of no-arbitrage was already introduced in Chap. 2.1.1 of this work. The absence of arbitrage says that it is not possible to design a risk-free self-financing portfolio that yields more than the instantaneous return of the risk-free (short) rate within a time interval. Expected excess returns, then, are the result of explicit risk-taking. This means that arbitrage opportunities exist unless long-term bond yields are equal to risk-adjusted expectations of future short-term yields. The assumption of absence of arbitrage opportunities seem quite logical in bond markets in which arbitrage opportunities are traded away immediately and markets can be characterized as highly liquid. The so called affine dynamic term structure models (ATSM) are the most popular among the class of no-arbitrage term structure models. They are best tractable since they assume bond yields to be affine functions of a set of risk factors driving the whole yield curve. In a continuous time setting, the drift and the volatility of these factors are themselves affine functions. They enable getting closed-form solutions for interest rates and such models are maximally flexible to reproduce the moments of bond yields and excess returns. The pioneering works by Vasicek (1977) and Cox et al. (1985) consist of a particular simple form of an affine term structure model where the short-term interest rate is the single factor that drives the whole yield curve at one moment in time.

In this case, changes in the short-term rate follow a Markov process which means that the evolution of the rate is a function only of its current level and not of the path by which it arrived there. This property is consistent with the weak form of market efficiency which states that it is not possible to trade on the information set of historical asset price movements and to simultaneously generate excess returns. Consequently, the history of the short rate process contains no information about its expected outcome that cannot be extracted from the current spot price. Such a time-series process can be characterized by its drift and volatility (diffusion). It can be shown that bond prices are also an affine function of the underlying factors, respectively the short rate in the single factor model.²⁶ Vasicek (1977) models the short rate's volatility as a constant term; whereas Cox et al. (1985) allow the volatility to be proportional to the square root of the short rate itself. The latter

²⁵Only recently, Christensen et al. (2009) have shown how to construct a generalized Nelson-Siegel model that allows for arbitrage-free consistency over time.

²⁶See Duffie and Kan (1996) or Bolder (2001).

specification has the advantage that it rules out negative values of interest rates since the short-rate volatility approaches zeros when the short rate is zero. In addition, both models share the feature that the short rate is mean-reverting so that whenever it exceeds the long-run mean, it is pushed back to this mean. This process is hampered by the diffusion of shocks that hit the short-rate dynamics in each period of time.

An equilibrium in such an affine framework requires that bond yields equal the path of expected risk-adjusted short rates. The most important implication of absence of arbitrage is the existence of a positive stochastic process, i.e. the pricing kernel, with which all future payoffs are discounted. The main task of modeling the term structure of interest rates is to find the evolution of the pricing kernel (SDF) that allows to separate EH-consistent bond yields from term premia. Finance literature basically offers two ways on how to specify the evolution of the SDF over time. They give the same functional form of the pricing kernel and for bond yields. If constructed appropriately, they should be equivalent to each other. The first way involves a direct specification of the evolution of the pricing kernel over time. Put differently, having pinned down the short-rate process and the market price of risk, it becomes possible to solve for the whole set of zero-coupon bonds to construct the term structure of interest rates. The market price of risk describes the required excess return per unit of risk. It should be the same for all bonds and independent of time to maturity (Maes 2004).

Another well established equivalent to the explicit specification of the pricing kernel is the concept of *risk-neutral pricing*. Central to this approach is that an asset's payoffs over its life are discounted by the uncertain future path of the riskless rate (the numéraire) where expectations are built as if agents are neutral towards financial risk. This implies that under this risk-neutral probability measure $\mathcal{Q}$, discounted bond price processes follow a martingale so that they are not predictable over time.²⁷ The same is true for expected returns. What lies between expectations under the artificial, risk-neutral measure and the historical, data-generating measure²⁸ is again a specification for the market price of risk that captures agents' attitude towards risk. Dertermining the pricing kernel with the help of the risk-neutral measure reveals that risk preferences of agents are implicit embedded in the pricing kernel as a function of the state variables and in the change of the probability measure from the risk-neutral to the true measure (see for example Singleton 2006, 203).

The easiest and most intuitive way of thinking about risk-neutral valuation is to recall the basic no-arbitrage asset pricing equation

$$1 + E_t^{\mathcal{Q}}[R_{n,t+1}] = (1 + R_{f,t}) - \frac{cov_t(R_{n,t+1}, M_{t+1})}{E_t[M_{t+1}]}$$

²⁷A martingale is a stochastic process that satisfies $E_t[X_{t+1} \mid X_1, \dots, X_t] = X_t$ and $E_t[|X_t + 1|] < \infty$ so that the conditional expected value of the next realization given all information (past observations) is equal to the current realization (Bingham and Kiesel 2004).

²⁸This measure is also referred to as the physical, true or actual measure.

which states that for any risky asset the expected return under the historical $\mathcal{P}$ measure equals the short rate plus a term that captures the covariance between the asset's return and the SDF. The risk-adjusted return of the bond with maturity n can be written as

$$1 + E_t^{\mathcal{P}}[R_{n,t+1}] + \frac{\text{cov}_t(R_{n,t+1}, M_{t+1})}{E_t[M_{t+1}]} = (1 + R_{f,t})$$

$$1 + E_t^{\mathcal{Q}}[R_{n,t+1}] = (1 + R_{f,t}). \quad (3.47)$$

To obtain the risk-neutral distribution for one-period returns, the true distribution is shifted to the left by adjusting the expected return in terms of the risk premium. As a result, the risk-neutral pricing approach guarantees that all expected returns are equal to the risk-free return and agents price bonds as if they were risk-neutral due to zero expected excess returns. If so, risk-neutral pricing translates the distribution of the discounted asset price process to a martingale by removing the predictable drift.

The construction of a dynamic term structure model relies on several functional relations which allow to adequately price all bonds along the yield curve. These primary ingredients are (1) the *risk-neutral* time-series process of the state variables or risk factors, (2) the *historical* time-series process for the state variables and (3) the mapping between these risk factors and the short-term interest rate (Singleton 2006). Together with an affine functional relationship between bond prices for any maturity n and the state variables paired with maturity-dependent parameter restrictions guarantee the absence of arbitrage. The parameter restrictions comprehend the parameters governing the relations between the state variables under both the physical and risk-neutral measure as well as the short rate equation. If these restrictions can be chosen to fulfill the basic asset pricing (2.51), then the guess for the solution function of bond prices has been correct. In fact, the guess-and-verify strategy of the parameter restrictions for bond yields is actually the same method used in modeling the rational expectations equilibrium for difference equations in monetary economics. The method of undetermined coefficients suggests to guess a function of the state variables and to solve it according to the minimum state variable solution (McCallum 1983).

In discrete-time, the three ingredients can be summarized in three basic equations; first, a process describing the evolution of the risk factors and its innovations under the physical measure, second, a specification of the pricing kernel including the relationship between the risk factors, market prices of risk and the short-term interest rate and finally the fundamental pricing (2.51). Section 3.5.2 presents a general form of “essentially affine term structure models” which have the convenient property that bond prices are exponentially affine in the state vector.

3.5.2 An Essentially Affine Term Structure Model

The main challenging task in term structure modeling is the appropriate derivation of the pricing kernel. It does not only capture the short-term interest rate but also makes a statement on the market prices of risk. It is not the scope of this work to give a fully-fledged overview of yield curve models. Within the last decades, this field of research has been rapidly growing as the finance industry seeks for sophisticated spot rate models that allow to price new financial products, especially interest rate derivatives.²⁹ Instead, this Section introduces one model specification which becomes increasingly popular among monetary economists. They use the affine term structure setting to derive implications of general equilibrium models of the macroeconomy extended for asset prices and to extract valuable information from the yield curve which is of essential interest for monetary policy makers.³⁰

In order to produce time-varying term premia, the literature basically works with two strategies which result both in a heteroscedastic model of the pricing kernel. Without this requirement, e.g. in case of a constant variance of the pricing kernel, the term structure would only produce constant excess returns. The first strategy allows for heteroscedastic risk factors (stochastic volatility). Consequently, the conditional variance of these factors characterized by a square-root process of the factor themselves are translated into the pricing kernel. The second way is to generate time-varying term premia through state-dependent risk price parameters which are not driven by the conditional variance of the risk factors but by the state of the economy. The general formulation of Duffee (2002) and its division in a set of subfamilies introduced by Dai and Singleton (2003) include both strategies. It turns out that from an empirical perspective, constant volatility factors and stochastic prices of risk parameters perform best in fitting historical yield curve dynamics (Dai and Singleton 2002, 2003).

The basic question is how to convert the risk-neutral measure to the historical measure (et vice versa). The finance literature typically shows this with Girsanov's Theorem.³¹ Here, following the work of Ang and Piazzesi (2003) and Singleton (2006), the change of measure and its implication for the pricing kernel process can be specified in discrete time. The starting point is the fact that future payoffs follow a stochastic process. Investors must form expectations and assign probabilities to the set of all possible events. In general, the density functions of a random variable Z under the risk-neutral and historical measure are $f_t^{\mathcal{Q}}(Z_{t+k})$ and $f_t^{\mathcal{P}}(Z_{t+k})$ respectively. The *Radon-Nikodym derivative* of the $\mathcal{Q}$ measure with respect to the $\mathcal{P}$ measure satisfies

²⁹Excellent textbook treatments are James and Webber (2000), Bingham and Kiesel (2004), Brigo and Mercurio (2006).

³⁰See for instance, Ang and Piazzesi (2003), Bekaert et al. (2006), Hördahl et al. (2006), Rudebusch et al. (2007) among others.

³¹See for instance Baxter and Rennie (1996), Bingham and Kiesel (2004).

$$\frac{d\mathcal{Q}}{d\mathcal{P}} = \xi = \frac{f_t^{\mathcal{Q}}(Z_{t+T})}{f_t^{\mathcal{P}}(Z_{t+T})}.$$

Since the random variable is a stochastic process over time, so must be the Radon-Nikodym derivative a process over time. If the risk-neutral expectations at time t of a random variable in $t + 1$ is sought, then the amount of change of measure during this time interval is just ξ_{t+1}/ξ_t so that the transformation between $\mathcal{Q}$ and $\mathcal{P}$ can be written as

$$E_t^{\mathcal{Q}}(Z_{t+1}) = E_t^{\mathcal{P}}(\xi_{t+1}Z_{t+1})\xi_t^{-1}.$$

Thereby, ξ_{t+1} is assumed to follow a log-normal process

$$\xi_{t+1} = \xi_t \exp(-0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}). \quad (3.48)$$

where the same sources of randomness ε_{t+1} appear in the stochastic process of the Radon-Nikodym derivative and the risk factors driving the short-term interest rates (see below).

The *Novikov condition*³², implying that the variation in λ_t is finite, makes the derivative a strictly positive exponential martingale (see Appendix D Duffie 2003). $\mathcal{Q}$ is an equivalent martingale measure of $\mathcal{P}$ since ξ_{t+1} is a martingale and so is Z_{t+1} under $\mathcal{Q}$. The conversion from the physical to the risk-neutral measure with the Radon-Nikodym derivative guarantees that under the risk-neutral measure, all expected asset prices and returns are not predictable. This is the basic assertion if investors are risk-neutral.

With these expectational relations in mind, the fundamental asset pricing theory shows that any zero-coupon bond is the presented value, discounted by the expected path of the risk-free interest rate under the risk-neutral measure $\mathcal{Q}$. Recall the basic asset pricing equation for n -period bonds, especially for a one-period bond

$$P_{n,t} = E_t^{\mathcal{P}}[M_{t+1}P_{n-1,t+1}]$$

$$P_{1,t} = E_t^{\mathcal{P}}[M_{t+1}].$$

The pricing kernel is the essential variable for pricing the sequence of bonds along the yield curve. In particular, if investors are risk-neutral, the pricing kernel is simply the negative exponent of the continuously compounded risk-free interest rate

$$M_{t+1} = e^{-i_{1,t}}. \quad (3.49)$$

³²This condition formally states that $(0.5 \prod_{t=1}^T \lambda_t^\top \lambda_t < \infty)$.

For risk-averse investors, however, the pricing kernel is modified according to

$$M_{t+1} = e^{-i_{1,t}} \frac{\xi_{t+1}}{\xi_t}. \quad (3.50)$$

Substituting (3.48) in (3.50) gives

$$M_{t+1} = e^{-i_{1,t}} e^{-0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}}. \quad (3.51)$$

The pricing kernel is the negative of the short rate and its standard deviation is the negative of the market price of risk, i.e. $std_t(M_{t+1}) = -\lambda_t^\top$ (at least for the specification in 3.51). If λ_t is constant over time, the pricing kernel is homoscedastic and expected excess returns of n -period bonds are constant. Instead, if λ_t varies over time, bonds may exhibit changing expected excess returns.

Since the right-hand side of (3.51) is log-normally distributed with $\varepsilon_{t+1} \sim N(0, I)$, the conditional expectation of the pricing kernel always equals the short rate because

$$\begin{aligned} \log E_t^{\mathcal{P}}[M_{t+1}] &= E_t^{\mathcal{P}}[\log M_{t+1}] + 0.5\text{var}_t(\log M_{t+1}) \\ &= -i_{1,t} - 0.5\lambda_t^\top \lambda_t + 0.5\lambda_t^\top I \lambda_t \end{aligned}$$

and hence,

$$E_t^{\mathcal{P}}[M_{t+1}] = e^{-i_{1,t}}$$

and there is no source of uncertainty (risk) in the pricing kernel.³³ Again, the risk-free asset does not load any risk since it is perfectly correlated with the SDF and its covariance is zero. Typically, an instantaneously maturing bond carries such an interest since there are no uncertain (stochastic) payoffs in the immediate next time interval. However, note that short-term yields are random variables from the vantage point of date t ; expected short rates $i_{1,t+n}$ ($n > 0$) might be correlated with the payoff stream, an asset generates. Risk-averse agents want to be compensated via market prices of risk λ_t per unit of risk if there is covariation between the path of expected short rates and future prices of zero-coupon bonds.

Any term structure model under the historical measure can be then expressed as

$$P_{n,t} = E_t^{\mathcal{P}} \left[\frac{\xi_{t+1}}{\xi_t} \exp(-i_{1,t}) P_{n-1,t+1} \right]$$

³³If x is a normally distributed random variable, $Y = e^x$ is log-normal distributed with $E[Y] = e^{E(x) + 0.5\text{var}(x)}$.

and using the recursion argument as

$$P_{n,t} = E_t^{\mathcal{P}} \left[\frac{d\mathcal{Q}}{d\mathcal{P}} \exp \left(\sum_{i=0}^{n-1} -i_{1,t+i} \right) \right]. \quad (3.52)$$

Similarly, under the risk-neutral measure, bond prices follow

$$P_{n,t} = E_t^{\mathcal{Q}} [\exp(-i_{1,t}) P_{n-1,t+1}]$$

and

$$P_{n,t} = E_t^{\mathcal{Q}} \left[\exp \left(\sum_{i=0}^{n-1} -i_{1,t+i} \right) \right]. \quad (3.53)$$

The class of discrete affine term structure models (ATSMs) uses the above set-up; it typically starts with a parameterization of the factor dynamics under the historical probability measure. Thereby, N factors follow a VAR process with stochastic volatilities of innovations. State vector dynamics $X_t = [X_{1,t}, X_{2,t}, \dots, X_{N,t}]^\top$ can be expressed

$$X_t = \mu + \phi X_{t-1} + \Sigma S_t \varepsilon_t \quad (3.54)$$

where Σ is a $N \times N$ constant, S_t is a diagonal matrix (“volatility matrix”) describing the conditional variance of the factors and $\varepsilon_t \sim N(0, I)$ are sources of risk. This general formulation allows the state vector to be homoscedastic or heteroscedastic. For later reference, Dai and Singleton (2000) uniquely categorize the family of N -factor ATSMs into $N + 1$ subfamilies $A_M(N)$ with N factors and M number of factors that are present in the conditional factor variances. For instance, if the ATSM consists of $N = 3$ factors where one factor exhibits stochastic volatility and the other two have constant variance, the model would be written as an $A_1(3)$ model.

Evaluating (3.51), the law of motion of the short rate and market prices of risk specify the pricing kernel. In this respect, the short-term interest rate is given by

$$i_{1,t} = \delta_0 + \delta_1^\top X_t \quad (3.55)$$

so that it simply depends on a constant term and it is linear in the state variables. The parameter vector $\delta_1^\top$ with size $N \times 1$ represents the loadings on these unobservable factors X_t .

In what follows, the complexity of the model is highly reduced to an $A_0(N)$ ATSM version that is mostly applied in monetary economics (Ang and Piazzesi 2003; Rudebusch and Wu 2004; Dai and Philippon 2005; Ang et al. 2007). It has the convenient feature that it works with constant volatility of risk factor dynamics,

i.e. $M = 0$ and $S_t = I_{N \times N}$, but it introduces time-varying excess returns via changing market prices of risk.³⁴ Against this background, the λ parameter is simply specified as

$$\lambda_t = \lambda_0 + \lambda_1 X_t \quad (3.56)$$

so that it imbeds a constant (λ_0 is a $N \times 1$ vector) and a time-varying ($\lambda_1 X_t$ with λ_1 describing a $N \times N$ matrix or parameters) market price of risk component.

Taking (3.51) as the nominal pricing kernel which prices all bonds in the economy, the total gross return of any bond n satisfies $E_t^{\mathcal{P}}[(1 + R_{n,t+1})M_{t+1}] = 1$ so that bond prices can be derived recursively as

$$P_{n+1,t} = E_t^{\mathcal{P}}[M_{t+1} P_{n,t+1}]. \quad (3.57)$$

The state dynamics of X_t (3.48) together with (3.51) form an essentially affine Gaussian term structure model where bond prices are given by

$$P_{n,t} = \exp(A_n + B_n^\top X_t) \quad (3.58)$$

and the bond specific factor loadings follow the recursions

$$\begin{aligned} A_n &= A_{n-1} + B_{n-1}^\top (\mu - \Sigma \lambda_0) + \frac{1}{2} B_{n-1}^\top \Sigma \Sigma^\top B_{n-1} - \delta_0 \\ B_n^\top &= B_{n-1}^\top (\phi - \Sigma \lambda_1) - \delta_1 \end{aligned} \quad (3.59)$$

with $A_1 = -\delta_0$ and $B_1 = -\delta_1$. These difference equations can be derived by induction as described in Appendix D.

Since continuously-compounded interest rates are related to the logarithm of bond prices, $i_{n,t}$ is given by

$$\begin{aligned} i_{n,t} &= -n^{-1} \log(P_{n,t}) \\ &= n^{-1} (-A_n - B_n^\top X_t) \\ &= a_n + b_n^\top X_t \end{aligned} \quad (3.60)$$

with $a_n = -A_n/n$ and $b_n = -B_n/n$ (Ang and Piazzesi 2003). Interest rates are also affine functions of the state vector where the loadings b_n describe how much variation in the state dynamics is translated into the term structure of interest rates.

Since interest rates take an affine form, expected returns are also affine in the state variables. Recall, that the holding-period return on an n -period zero-coupon bond for τ periods in excess of the return on a τ -period bond is given by

³⁴For a more general form with both stochastic volatility and time-varying market prices of risk, the reader is referred to Duffee (2002).

$$xr_{n,t+\tau} = p_{n-\tau,t+\tau} - p_{\tau,t} - \tau i_{1,t} = A_{n-\tau} + B_{n-\tau}^\top X_{t+\tau} - A_n - B_n^\top + A_\tau + B_\tau^\top X_t.$$

Conditional expected returns can be computed using

$$E_t^{\mathcal{P}}[xr_{n,t+\tau}] = A_n^{xr} + B_n^{xr\top} X_t$$

where $A_n^{xr} = A_{n-\tau} - A_n + A_\tau$ and $B_n^{xr\top} = B_{n-\tau}^\top \phi^\tau + B_\tau^\top - B_n^\top$. The slope coefficients on expected excess returns can be written as

$$B_n^{xr\top} = B_{n-\tau}^\top [\phi^\tau - (\phi - \Sigma \lambda_1)^\top]$$

so that the one-period expected excess return follows

$$\begin{aligned} E_t^{\mathcal{P}}[xr_{n,t+1}] &= A_n^{xr} + B_n^{xr\top} X_t \\ &= B_{n-1}^\top \Sigma \lambda_0 - 0.5 B_{n-1}^\top \Sigma \Sigma^\top B_{n-1} + B_{n-1}^\top \Sigma \lambda_1 X_t. \end{aligned} \quad (3.61)$$

Expected excess returns comprises three terms (1) a Jensen's inequality term $-0.5 B_{n-1}^\top \Sigma \Sigma^\top B_{n-1}$, (2) a constant term premium $B_{n-1}^\top \Sigma \lambda_0$ and (3) a time-varying term premium $B_{n-1}^\top \Sigma \lambda_1 X_t$.³⁵ As previously discussed, the term premium is governed by the vector λ_t . A negative sign leads to positive expected excess returns. If a positive shock ε_{t+1} hits one of the state variables, according to (3.58), this lowers (expected) bond prices and triggers declining holding-period returns. When λ_t is negative, the shock drives up the logarithm of the pricing kernel so that the realized negative correlation between returns and the SDF leads to a positive pricing kernel. Such a dynamic is consistent with risk-averse investors. In order to afford a positively sloped yield curve, at least one parameter in the vector of market prices of risk must be significantly negative. In contrast, when λ_t is positive, this works like a hedge, since the shock drives down the pricing kernel so that bond returns are positively correlated with the pricing kernel; then, risk premia are negative.

The $A_0(N)$ model of the yield curve reveals that a non-zero λ_t vector affects the long-run mean of the term structure so that on average, it can be upward-sloping as confirmed by the stylized facts. The pure Expectations Hypothesis (PEH) is a contradiction to this observation as it would postulate an economic environment in which investors on average expect rising short-term interest rates. In fact, the PEH predicts a mean yield curve that is flat or even slightly falling due to Jensen's inequality. If $\lambda_0 \neq 0$ and $\lambda_1 = 0$, the expectations hypothesis (EH) holds. Time-varying market prices of risk $\lambda_1 \neq 0$ can be understood as a rejection of the EH so that yield spreads do not necessarily predict future changes in interest rates but rather time-varying dynamics of term premia.

³⁵Here again, the difference between the pure form of the Expectations Hypothesis and general risk-neutrality is revealed. If investors are risk-neutral, they still price Jensen's inequality as apposed to the PEH. To be concrete, Jensen's inequality need not to be interpreted as a "risk premium" since it does not compensate for explicit risk-taking.

Chapter 4

A Systematic View on Term Premia

4.1 Forms and Sources of Term Premia

In recent years, the concept of term premia has become a focus of attention for academics, policy makers as well as the investment community. This heightened attention was initially triggered by the puzzling behavior of long-term interest rates in the United States and in other industrialized countries (Greenspan 2005). The interest-rate conundrum manifested itself in stable and even falling long-term bond yields despite a reversal in the short-term FED funds cycle. Over the period between June 2004 and February 2005, the FED decided to increase the target rate by over 120 basis points. Over the same time, the 10-year treasury rate lost temporarily over 100 basis points. Among the global saving glut, declining inflation expectations, reduced global macroeconomic and financial uncertainty were cited as explanations. Attempts, shrinking bond term premia though were the most promising fact to capture the conundrum within a coherent macroeconomic framework (Kim and Wright 2005; Rudebusch et al. 2006; Backus and Wright 2007).

The financial crisis, starting in the middle of 2007 and evoking distressing parallels to the Great Depression in the 1930s, serves as a further indication for the increased sensitivity for overall risk attitudes. In particular, the explicit mentioning of risk and term premia has become commonplace in policy discussion, within central banks and their communication with the public.¹ In this respect, policy makers agree on an overall assessment of the causes of the crisis as being triggered by a heightened risk tolerance and appetite of market participants.² An environment of low interest rates promoted banks and other financial intermediaries to invest in more risky positions in a “search for yields” across a wider class of assets

¹See, for instance, Kohn (2005), Bernanke (2006), Plosser (2007) and Trichet (2008) among others for the United States and the euro area.

²For example, the Bundesbank identifies four main channels through which the financial crisis spread: (1) recklessness in securitization, (2) low risk perception, (3) slack lending standards and (4) high credit expansion in the aftermath of 2003 (Zeitler 2009).

by “making on the carry”. In particular, the whole financial industry imitated the traditional banking model of “maturity transformation” and expanded it to the “originate and distribute” version of banking in which loans are pooled, tranced and resold via securitization. The outcome was documented in highly leveraged borrowing, in soaring asset prices and diminishing required risk premia on part of the market participants across the whole set of asset classes (see Chap. 7.2).

The interest in term premia also coincides with developments in academia. By improving the estimation and modeling of bond yields and corresponding term premia in asset markets, progress has been made in combining methods in financial and monetary economics in order to disentangle the structural macro sources of time-varying term premia. Since term premia have a great deal to do with private-sector’s expectations of the future risk-neutral payoff of securities, identifying them relies on a model according to which agents can form these expectations. However, there is still no consensus on how to measure or even define term premia since the latter heavily relies on model specification criteria and estimation techniques (Swanson 2007). Nonetheless, for a given measure, term premia provide many macroeconomic linkages. For instance, they seem to be counter-cyclical so that they can be used to predict changes in real economic activity (Hamilton and Kim 2002; Favero et al. 2005; Ang et al. 2006).

The previous Chap. 3.5 already introduced the reader to one form of term premia, i.e. excess returns stemming from interest-rate risk. The uncertain path of future interest rates and, thus, payoffs provoke risk-averse investors to demand higher expected returns due to the danger of capital losses during the investment horizon. This sort of risk is the major concern when trading government securities. In general, a government bond market is said to be highly liquid and its default-risk is negligible. Whenever the investment period diverges with the maturity of the underlying asset, future changes in discount rates force investors to re-value their assets with the possibility of capital losses.

However, as the recent financial turmoil of 2007 arrestingly demonstrates, the abstraction of credit risk and liquidity issues in security markets may be only partly legitimated. Turning to government bond markets, yield spreads among the euro-area member countries, especially vis-à-vis Germany have risen dramatically since 2008. This observation is a contradiction to the assumption of a single homogenous capital market as interest-rate spreads mirror default and liquidity risk with exchange-rate risk being eliminated due to the single currency. More generally, two further factors make the future payoff stream of an asset an uncertain and risky number. Firstly, investors demand a compensation for credit default risk. It can be defined as the possibility that a bond issuer will fail to repay its principal and interest in a timely manner. In case of recovery, the lender receives just a fraction of the due payment. Modeling and analyzing credit risk has become very common in recent years, in particular for corporate bonds, driven by advances in pricing derivative securities and increasing attempts to transfer credit risk outside the institutions’ balance sheets (BIS 2003). Credit risk premia stand at the top of the risk management agenda for financial and non-financial institutions. Central banks as well monitor this risk since it heavily contributes to the stability of the financial

architecture. To this end, the ongoing debate on the likely break-up of the euro area causes the ECB to address scenarios in which member countries may default on their sovereign bonds. In order to capture these various aspects, this Chapter also gives an overview of the sources of risk premia originating from credit default risk.

The second factor that contributes to overall term premia is a liquidity risk premium. It is difficult to fully grasp the idea of liquidity within a single framework since it is eminent in every economic transaction, in portfolio theory, in banks's liquidity management or in the availability of funds to finance long-dated securities; to put it in Charles Goodhart's (2008b, 41) words: "liquidity has so many facets that it is often counter-productive to use it without further and closer definition." In general, the microeconomic literature on finance defines liquidity of a security as the ease of trading this security at short notice, at low cost and with little impact on its price (Amihud et al. 2005). This is what Keynes (1930) had in mind in his *Treaties*. It represents the degree to which a security can easily be transformed into money. In macroeconomics, liquidity refers to a generally accepted medium of exchange, i.e. money. It is most liquid since it does not need to be converted into anything else in order to make purchases of goods and assets. Besides the characteristics of assets being convertible into money, the word liquidity is inevitably anchored to Keynes' theory of interest (1936), i.e. the liquidity preference theory which gives money a store-of-value function within a portfolio decision problem. Due to Keynes' speculative motive, money is a part of investment balances due to expectations of fears of losses on other assets (Tobin 1958). If so, investors command a premium over non-monetary assets to get compensated for that fear. As pointed out by Hicks (1946), liquidity preference then reduces the term premium to an interest-rate risk factor that can be attributed to the future course of interest rates.³ The speculative motive of Keynes does *not* allow for a liquidity premium per se in the micro-finance sense but for an interest-rate risk premium. Only if money is discussed in terms of preserving the option of deferring decisions in an environment of uncertainty, the whole concept of a liquidity premium becomes eminent. The question then arises how the micro-approach of defining liquid assets relates to Keynes' liquidity preference and the idea of a liquidity premium.

Besides the theoretical discussion on the appropriate distinction between interest-rate risk premia and liquidity premia, the concept "liquidity," as used in common examination of financial market conditions, also serves as a catch-all framework when trying to grasp recent asset-price dynamics. *Excess liquidity* or *awash in liquidity* are synonyms on an aggregated level for bloating balance sheets of both the financial and non-financial sector. In this respect, funding liquidity, as defined by Brunnermeier (2009), captures the ability to raise funds at short notice and allows financial intermediaries to operate under a maturity-transformation strategy. With this definition, it becomes clear what is meant when money markets dry up due to lack of funding opportunities. It is for this reason why central banks operate as

³At least in Keynes (1936), Chap. 13.

lenders of last liquidity to provide liquidity services in case of squeezes in interbank markets (Goodfriend 2002; Taylor 2005; Allen and Carletti 2008). Former FED chairman Alan Greenspan rigorously used liquidity provisions in times of crisis such as in 1987, 1998 or 2001. A similar strategy has been applied by his successor Ben Bernanke in the aftermath of the bursting of the real-estate bubble in 2007/08 and by other major central banks around the world.

As it becomes clear from the above demonstration, term premia can be decomposed in three forms of risk, interest-rate risk, credit default risk and liquidity risk. Chapter 4 deals with these various aspects. The following Sect. 4.2 introduces an affine term structure model that helps to estimate interest-rate risk premia as well as to forecast future interest rates along the yield curve. The focus is placed on three countries, i.e. the United States, United Kingdom and Germany. The model is built in a way that only interest-risk premia can be identified. It, thus, abstracts from credit and liquidity risk. Section 4.3 introduces the concept of default risk and its measurability on financial markets. The last Sect. 4.4 discusses the finance and macro view on liquidity premia and bridges the gap between these two concepts by describing financial intermediaries' behavior towards liquidity preference.

4.2 Evidence on Interest-Rate Risk Premia

4.2.1 A Two-Factor Affine Term Structure Model

Estimating and measuring term premia is a challenging task. One major drawback is that it is awkward to simultaneously determine and distinguish between the various sources of term premia related to interest-rate, credit and liquidity risk. Even if one abstracts from the latter two risk components by analyzing highly liquid government bond markets, the adequate specification of the term premium in the term structure of interest rates relies on model assumptions that allow to extract interest rates expectations as well as term premia from the yield curve. Both variables are not directly observable on financial markets. The Expectations Hypothesis, for instance, takes the current forward rate as the expected future short rate, thereby, imposing rational expectations on the part of investors. Due to its empirical failure, models of the yield curve rely on the broader and more flexible no-arbitrage approach of asset pricing that explicitly contains time-varying term premia. Government bonds are then priced with the help of affine term structure models (ATSM) as discussed previously. Before estimating an ATSM, the empirical analysis is introduced by collecting a number of prominent term premia measures and by examining both their similarities and differences.

Most of the measures on term premia in bond markets derives from regression analysis. Starting point is usually some form of the EH from which risk premia can be quantified and predicted. The easiest way to detect term premia is the unbiased version of the EH according to which the future interest rate is an

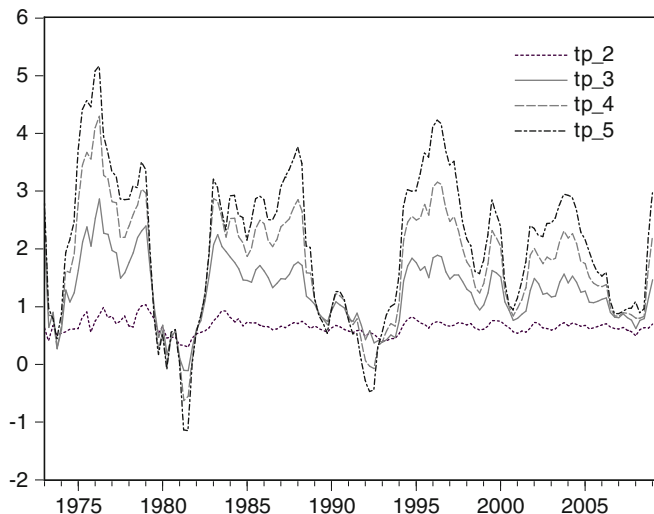


Fig. 4.1 Fama-Bliss regression for Germany (Note: Fama-Bliss regression of excess returns on the forward-spot spread $xr_{n,t+1} = \alpha_n + \beta_n(f_{n,t} - y_{1,t}) + \varepsilon_{n,t+1}$. The figure shows the forecast of excess returns from 2 years up to 5 years against the 1 year government bond. Forward rates are defined as a forward contract from $t + n - 1$ to $t + n$)

unbiased predictor of the future short-term interest rate. Deviations from this constant-form equilibrium condition is measured as residual and may be interpreted as term premium. This simple derivation is, however, inadequate due to its lack of consideration of measurement errors and expectations errors. The seminal work of Fama and Bliss (1987) shows that if it is possible to find ex-ante variables that help to predict ex-post excess returns or the difference between the future rate and the ex-post realized short rate, the estimation would imply the presence of a term premia. The predictable component of these explanatory variables may then be used to measure the premium. When the Fama-Bliss regression for Germany along (3.46) using quarterly forward rates ranging from 2 years to 5 years from 1973 through 2009 is estimated, annualized expected excess returns are presented in Fig. 4.1.

Recently, Piazzesi and Swanson (2008) extended this methodology by including macroeconomic variables such as year-on-year employment growth together with various other business cycle indicators as explanatory variables into an estimation of the spread between the forward rate and the ex-post short-term rate. They find that forward premia, i.e. the predictable part of the forward-spot spreads, are highly countercyclical peaking shortly after the beginning of recessions in the US. Similar results hold for the euro area and the UK (Ferrero and Nobili 2008; Joyce et al. 2009b) so that there are similarities across the set of countries. A disregard of predictable forward premia can lead to systematic forecast errors in using unadjusted forward contracts as a measure for monetary policy expectations.

Despite its attractiveness to extract term premia from a set of only a few financial market and macroeconomic variables, central banks until recently hesitated to use such measures. Instead, they applied a rule-of-thumb approach (!) and used a simple one basis point per month term premium correction on forward rates (at least Federal Reserve staff has followed this procedure according to Kim and Orphanides 2007). The main reason for such a reluctance for reduced-form results is that they heavily depend on sample choice. Expected excess returns markedly differ not only in their size but also in their time-varying characteristics that change their predictability. On this account, the inclusion of survey-based estimates may improve the fit since they already attain risk-adjusted market expectations obtained by interviews. Still, survey-based measures and reduced-form regression can be criticized for a number of reasons. Firstly, surveys on interest-rate forecasts may not be informative in a way consistent with rational expectations due to systematic forecast errors and due to the lack of exploitation of all available information on past interest-rate movements (Friedman 1980; Froot 1989). Secondly, at longer horizons, reduced-form models fail to adequately capture the concept of term premia. Estimating the forward premium for longer maturities is hampered because the size of the sample shrinks due to overlapping observations which reduces the effectiveness of term premia estimates considerably. For a sample of 20 years, deriving 10-year forward premia is only possible for the last 10 years. However, if one imposes a sufficient structure on the functional form of both yields and forward rates, this problem can be fully overcome. The affine class of yield curve models provides such a functional form where interest-rate risk imbedded in return-, yield- and forward premia are consistently derived from the underlying model.

In recent years, the vast majority of empirical research elaborated on affine term structure models. What they all have in common, is that they work with the three ingredients, i.e. (1) the *risk-neutral* time-series process of the state variables or risk factors, (2) the *historical* time-series process for the state variables and (3) the mapping between these risk factors and the short-term interest rate (Singleton 2006). They differ, however, in various aspects that can be basically split up in two main blocks, namely model specification and data considerations.

The first question in estimating an ATSM surrounds the choice on the general model – whether it is Gaussian with constant variances of the state variables, or whether it can be described as a stochastic volatility model in a CIR-style format. This basically questions whether observed excess returns and the failure of the EH is produced by stochastic market prices of risk or by stochastic interest rate volatility. Dai and Singleton (2002) clearly argue for the former force. Furthermore, the number of risk factors (state variables) driving the underlying interest rates must be determined. According to the level, slope and curvature evidence of Litterman and Scheinkmann (1991) it is reasonable to work with two or three factors in order to adequately fit bond yields and, consequently, most research follows this line. It must also be evaluated whether these factors are treated as latent or observable factors. Initially, ATSMs have been estimated with pure latent variables but the attempt to combine macroeconomic and finance topics in coherent models breed the inclusion of observable macro variables (Ang and Piazzesi 2003; Dai and Philippon 2005,

among many other work). This strand of literature helps in exploring the linkages between interest rate behavior and the business cycle. The choice of the number and form of risk factors is not the only preselection for the estimation. Any sensible parametrization of ATSMs must be theoretically admissible and econometrically identified. The theoretical model is *admissible* if it rules out negative conditional variances and negative short-term interest rates. From a statistical point of view, factor models need to be specified in a sense that one cannot find a “rotation” of the risk factors that leaves bond yields unchanged. An ATSM is said to be in its canonical form if finding restrictions on the state-space system make it possible to get it econometrically identified (Dai and Singleton 2000).⁴

A second estimation question deals mainly with the set of bond prices or yields with which one can estimate an ATSM. The frequency typically ranges from weekly up to quarterly observations depending on the aim which the yield curve model tries to capture. It may be reasonable to use CIR-models on a weekly basis since stochastic volatility emerges mainly at short frequencies. Quarterly data may be applied in a Gaussian setting. This is more likely in the macro-finance literature in which the macroeconomic model works with constant variances of its state variable anyway. The estimation can become quite difficult if the sample on interest rate data is too short to accurately provide reliable information about the data-generating dynamics of the risk factors. Due to highly persistent interest rates, the sample may suffer from a sufficient number of mean-reversion observations so that estimates on long-term expectations of the short rate might get distorted. To overcome this problem, some studies include survey information on the expected path of the short rate that support pinning down the estimated parameters of the data-generating drift of the state variables (Kim and Orphanides 2005).

The following model applies the sketch of Sect. 3.5.2 with the $A_0(N)$ workhorse among the no-arbitrage models as mostly adopted by monetary economists. The motivation of estimating the Gaussian model specification results from previous studies that document the overwhelming success in fitting historical behavior of bond yields and basic diagnostics on the observed empirical puzzles when testing the Expectations Hypothesis within single-equations regressions. Many studies on these countries differ in model specification, data selection and the use of survey data which make a comparison of cross-country results awkward.⁵ Moreover, some studies simplify the model so as to allow only for constant risk premia if estimated

⁴The problem is that specific numerical values of the underlying parameters give rise to the same term structure. In this respect, invariant transformations of the original ATSM, by restricting and normalizing specific parameter constellations before the estimation, is a procedure to guarantee identification of the model and to present it in its canonical form. Dai and Singleton (2000) show that if one restricts a specific set of parameters, this allows to treat the more “interesting” parameters to the econometrician as free parameters.

⁵Most work on affine term structure models rely on US data. See for instance, Dai and Singleton (2002), Duffee (2002), Dai and Philippon (2005), Kim and Orphanides (2005), Kim and Wright (2005), Lemke (2006), Bolder (2006), Rudebusch and Wu (2007), D’Amico et al. (2008b), Pericoli and Taboga (2008) and Adrian and Wu (2009). For Germany, studies have been carried out by Cassola and Lus (2003), Fendel (2005) and Mayer (2008) and for the UK by Bianchi et al. (2009).

jointly with a DSGE model whose solution only produces homoscedastic pricing kernels. Therefore, the estimations of the ATSMs for the US, UK and Germany are restricted to the two-factor Gaussian term structure model $A_0(2)$ for each country.

For estimation, the theoretical model is cast into state-space form where the transition equation and the measurement equation are built in accordance with the theoretical model. The transition equation consists of two latent factors which follow a simple VAR(1) process with $X_t = [X_{1,t}, X_{2,t}]^\top$ and

$$X_t = \phi X_{t-1} + \Sigma \varepsilon_t \quad (4.1)$$

where ϕ is upper-triangular, Σ is a 2×2 constant with diagonal elements and the distribution of the factor innovations is $\varepsilon_t \sim N(0, I_2)$. In addition, the means of the risk factors are normalized to zero ($\mu = [0, 0]^\top$) and the short rate is given by

$$i_{1,t} = \delta_0 + \delta_1^\top X_t \quad (4.2)$$

where δ_1 takes on the value $[1, 1]^\top$.⁶ To price all bonds, the stochastic discount factor takes the form

$$M_{t+1} = \exp(-i_{1,t} - 0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}) \quad (4.3)$$

and the vector of market prices is

$$\lambda_t = \lambda_0 + \lambda_1 X_t. \quad (4.4)$$

As already derived in Appendix D, bond yields satisfy

$$i_{n,t} = a_n + b_n^\top X_t \quad (4.5)$$

with initial conditions $a_0 = b_0 = 0$ and $a_1 = \delta_0$ and $b_1 = \delta_1$.⁷ Since (4.5) represents the measurement equation, we may add a measurement error so that we get

$$\tilde{i}_{n,t} = a_n + b_n^\top X_t + u_t \quad (4.6)$$

⁶See Singleton (2006) and Nyholm and Vidova-Koleva (2010) for the admissibility and identification conditions.

⁷The presence of latent factors make the model invariant to affine transformations. To identify the model, it is normalized by imposing the following restriction: (1) ϕ is upper-triangular, (2) Σ is diagonal, (3) the mean μ of the latent factors are zero and (4) the loadings δ_1 on the short rate are each set to one. Furthermore, the parameter δ_0 is set to the long-run mean of the short rate in order to reduce the number of parameters to be estimated. The a-priori restrictions follow Dai and Philippon (2005).

Table 4.1 Data for estimation $A_0(2)$ -model

	USA	GER	UK
Yields	Nelson-Siegel	Nelson-Siegel	Spline
Short Rate	LIBOR	FIBOR	Spline
Source	FED	BuBa	BoE
Sample	1972Q1:2008Q3	1973Q1:2008Q3	1970Q1:2008Q3
Measurement equation			
$n^\top$	$(1, 4, 12, 20, 28, 40)^\top$	$(1, 4, 8, 20, 28, 40)^\top$	$(1, 4, 8, 12, 20, 28, 40)^\top$

with the simple specification that

$$u_t \sim N(0, h^2 I_n).$$

This results in the empirical fact that the difference between the theoretical and observed yields has the same variance for all maturities.

The empirical analysis uses quarterly data for the US, UK and Germany (see Table 4.1 and the specification of the observation vector). Since interest rates are annualized but the model defines the length of a period as unit of time, the measurement equation has to be multiplied by 400. It implies that the model parameters are those corresponding to quarterly continuously-compounded yields and so are the time-series dynamics of the state variables but the measurement error is in annualized terms. As pointed out by Lemke (2006), this strategy circumvents possible numerical difficulties for the estimation of the measurement error that would be otherwise very low when working with quarterly and not with annualized yields.

Before estimating the $A_0(2)$ -model with maximum likelihood and the Kalman filter, some last technical comments need to be specified.⁸ The parameters to be estimated are stacked into the vector $\theta = \{\text{vec}(\phi), \text{vec}(\Sigma), \lambda_0^\top, \text{vec}(\lambda_1), h^2\}^\top$. To assure that some parameters fulfil admissability conditions in numerical optimization, the likelihood function is reparameterized with the help of auxiliary parameters.⁹ To begin the maximization procedure, the initial vector of starting values is based on a VAR(1) estimation with the 40-quarter yield and the difference between the 1-quarter yield and the 40-quarter yield taken as state variables. The VAR allows to load the initial matrices of the transition equation. Maximization of the likelihood is performed with Matlab in a two-stage maximization routine: firstly, the numerical Simplex routine *fminsearch* with a maximum of 3,000 iterations is carried out; after this round, taking the parameters of the Simplex as initial

⁸See Appendix B for details on the estimation.

⁹Especially, the diagonal elements of ϕ need to be smaller than 1 for stationarity. This is guaranteed by introducing $\phi_{ii}^{\text{aux}} = -\log(.999/\phi_{ii} - 1)$. In addition, the covariance matrix Σ needs to be strictly positive which is derived by setting $\Sigma_{ii}^{\text{aux}} = \log(\Sigma_{ii})$. Converting the two auxiliary matrices back in its original form reveals that the true values always fulfil the admissability restrictions.

vector, the derivative-based optimizer *fminunc* refines the parameters estimates.¹⁰ Following Lemke and Werner (2009), standard errors are computed in line with the quasi-maximum likelihood variance-covariance matrix of the estimated parameters as described in Hamilton (1994, Sect. 5.8).¹¹

4.2.2 *An International Comparison of Essentially Affine Risk Premia*

Table 4.2 contains the maximum likelihood estimates of the parameters and associated standard errors for the three countries subject to the cross-equations restrictions implied by the no-arbitrage assumption. An inspection of the state equation dynamics reveals that the first factor is highly persistent for all countries. In particular, in the UK, it nearly hits the boundary condition of 1 which might indicate a non-stationary time-series property. The persistence of the second risk factor is smaller, though the low mean-reverting feature is apparent, too. This observation is basically the main result of all factor-based yield curve models, e.g. models based on principal component analysis or the dynamic Nelson-Siegel version as presented previously. Moreover, the variance structure confirms that the second factor is more volatile than the first one, so that it is reasonable to characterize the first variable as describing level movements of bond yields; whereas the second factor causes the term structure to flip. The stationary assumption of the risk factors imposed in the underlying model may exhibit one decisive weakness. For a sufficiently long horizon, expected short-term interest rates must inevitably converge to its long-run mean, i.e. the constant part of the short rate equation δ_0 . Thus, movements of forward rates with long maturity are always due to time-varying forward premia. Some ATSMs modify the short rate process with the help of shifting endpoints so that long-run forecasts do not necessarily coincide with the constant term δ_0 (see Kozicki and Tinsley 2001). This is mainly justified by varying inflation perceptions on the part of market participants. The estimated models for Germany, US and UK in Sect. 3.2 reveal that the short rate does not asymptote at the considering maturities so that inspections of further model results are possible.¹²

Since affine yield curve models are able to extract term premia from observed bond prices, a special focus lies on the derived market prices of risk. As expected, the constant term λ_0 is negative at least for one entry in the USA, GER and UK so

¹⁰A similar, but much more intensive hands-on procedure is proposed by Duffee (2009).

¹¹The Matlab function to calculate the standard errors is partly provided by Piazzesi and Schneider (2007).

¹²As a test for robustness, the log-likelihood function also has been modified in the line of Chernov and Mueller (2008) by adding a term premium component to the log-likelihood function. It introduces an additional burden and uses term premia as a “last resort” in fitting yields. It basically means that the model first tries to fit yields via the expectations hypothesis. If this does not work, it allows term premia to do so. It turned out that the penalty did not alter results at all.

Table 4.2 Maximum likelihood parameter estimates for $A_0(2)$

	USA		GER		UK	
ϕ_{1n}	0.960*** (0.006)	0.069*** (0.023)	0.969*** (0.019)	0.037*** (0.021)	0.998*** (0.002)	-0.012 (0.013)
ϕ_{2n}	0 (-)	0.874*** (0.028)	0 (-)	0.920*** (0.029)	0 (-)	0.903*** (0.014)
$\Sigma_{1n} \times 10^2$	0.159*** (0.015)	0 (-)	0.103*** (0.010)	0 (-)	0.145*** (0.011)	0 (-)
$\Sigma_{2n} \times 10^2$	0 (-)	0.150*** (0.023)	0 (-)	0.152*** (0.026)	0 (-)	0.197*** (0.021)
δ_0	0.015 (-)		0.014 (-)		0.021 (-)	
λ_{01}	-0.125*** (0.046)		-0.144*** (0.032)		-0.079*** (0.021)	
λ_{02}	-0.261** (0.121)		0.014 (0.131)		-0.014 (0.104)	
$\lambda_{1(1n)}$	-22.821*** (3.317)	73.283*** (6.684)	-28.661 (19.142)	75.997*** (25.829)	-6.271 (0.973)	-0.460** (0.182)
$\lambda_{1(2n)}$	-26.129*** (4.031)	25.269*** (7.775)	-20.418*** (7.165)	65.072*** (5.546)	2.530** (1.271)	27.673*** (0.108)
h^2	0.269 (0.016)		0.260 (0.032)		0.231 (0.011)	
LR-Test:	36.63		2.158		2.003	
p-value:	0.00		0.14		0.16	

Note: Parameter estimates of the $A_0(2)$ model based on ML and Kalman filter with standard errors in parentheses. An asterisk (*) denotes significance at the 10%, two asterisks (**) at the 5% and three asterisks (***) at the 1% level. The parameters for which the standard errors are not reported have been either restricted for admissibility or estimated in advance to reduce the number of parameters to be estimated

that the yield curve is on average upward sloping. In contrast to the time-varying components of the state prices, estimated parameters of the constant price term are rather small. This suggests that risk premia are driven by an important time-varying component as displayed by the λ_1 values in absolute terms. This also leads to the finding that under the risk-neutral measure, the mean of the yield curve is higher than under the data-generating measure. However, estimated standard errors and corresponding t-statistics indicate to econometrically insignificance for some individual parameters so that inference based on individual estimates should be exercised with caution.¹³ It is for this reason that it is abstracted from a direct economic interpretation of single risk price parameters. However, as pointed out by many other studies, it is hard to pin down the single parameters of market prices risk so that lack of significance does not necessarily indicate to a poor model fit (Ang and Piazzesi 2003; Hördahl et al. 2006; Moench 2008).

¹³One line of technical defense is that standard errors calculated with Matlab tend to be higher than with other software programs such as Fortran (Duffee 2009).

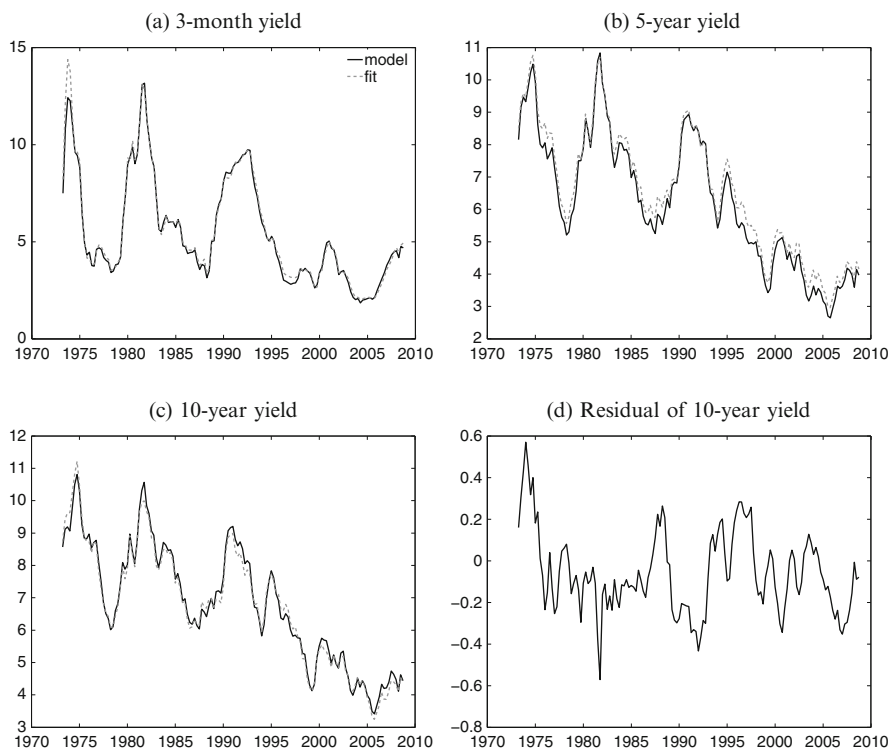


Fig. 4.2 Fitted and observed yields for Germany (*Note: Model-implied yields are calculated with the individual yield loadings of the measurement equation*)

To see this, Fig. 4.2 exemplary provides a plot of selected model-implied yields and its observed counterparts for Germany. The model fits the data well which should not come as a surprise due to its flexible specification. The measurement error is quite small and quantified with 26 basis points in annual terms for Germany. Similar results are obtained for the US and UK.

To further reveal the deep characteristics of the estimated model, fitted yields can be decomposed to ask at any point in time, how much of the bond yield and forward rate corresponds to expected future interest rates and how much to yield- and forward term premia, respectively. Appendix D describes how to extract the different term premia concepts, risk-neutral yields and forward rates as well as model-implied expected one-period short-term interest rates. Figure 4.3 pictures the loadings of the German yield curve to one-standard deviation of shocks to the factors in basis points. The first factor is the level factor that induces an equal change in yields of all maturities (b_{1n}); whereas the second factor induces the curve to rotate (b_{2n}). When decomposing these responses into the two components of interest rate determination, i.e. the average expected short rate and the yield term premium, interest rates at the short end are mainly driven by the expectational part b_{1n}^{rn} of the

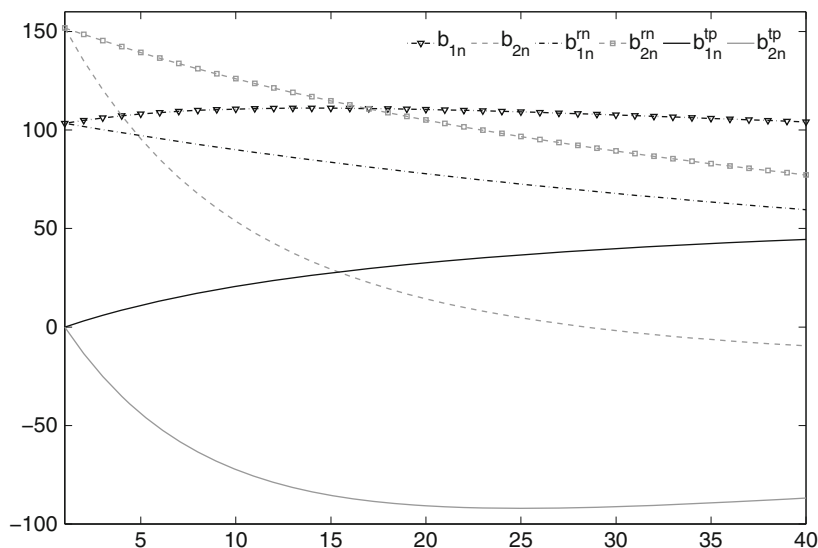


Fig. 4.3 Instantaneous yield curve response for Germany (Note: Factor loadings on yields $b_{1,2n}$, expected average short rates $b_{1,2n}^r$ and yield term premia $b_{1,2n}^{tp}$ to one standard deviation shock for the German $A_0(2)$ model. Bond yields are decomposed as $i_{n,t} = n^{-1} \sum_{i=0}^{n-1} E_t[l_{t+i}] + \phi_{in,t}$)

first factor. At longer maturities, yield changes are mainly dominated by the growing term premium component of the first factor (b_{1n}^{tp}). As regards changes in the second risk factor, rotating yield curve dynamics are mainly caused by the increasing term premium component and not by the expectational part. This means that the presence of a normally sloped term structure indicates only partly to increasing short-term interest rate expectations. The main effect comes from the high b_{2n}^{tp} component at the long end relatively to the short end. In times of a normal spread, the second risk factor takes on negative values which triggers bond yields to rise due to the term premium loading.

The decomposition can also be displayed from a time-series perspective. Figure 4.4a displays the 10-year yield and the decomposition into its risk-neutral level and the yield term premium. As can be seen, there is ample evidence for the time-varying component of the yield premium, although the latter only partly contributes to the run-up in bond yields through the early 1980s or 1990s for Germany. In the US, there is a rather mixed picture, where indeed both risk premia and risk-neutral yields account for the high interest rate levels.¹⁴ What is outstanding for all three countries is that after any one yield peak, short-rate expectations, as imbedded in risk-neutral yields, fell much faster than observed long-term yields pointing to an important premium component after the peaks.

¹⁴The US yield curve decomposition is not reported here but a similar result can be found in Rudebusch et al. (2007) for comparison.

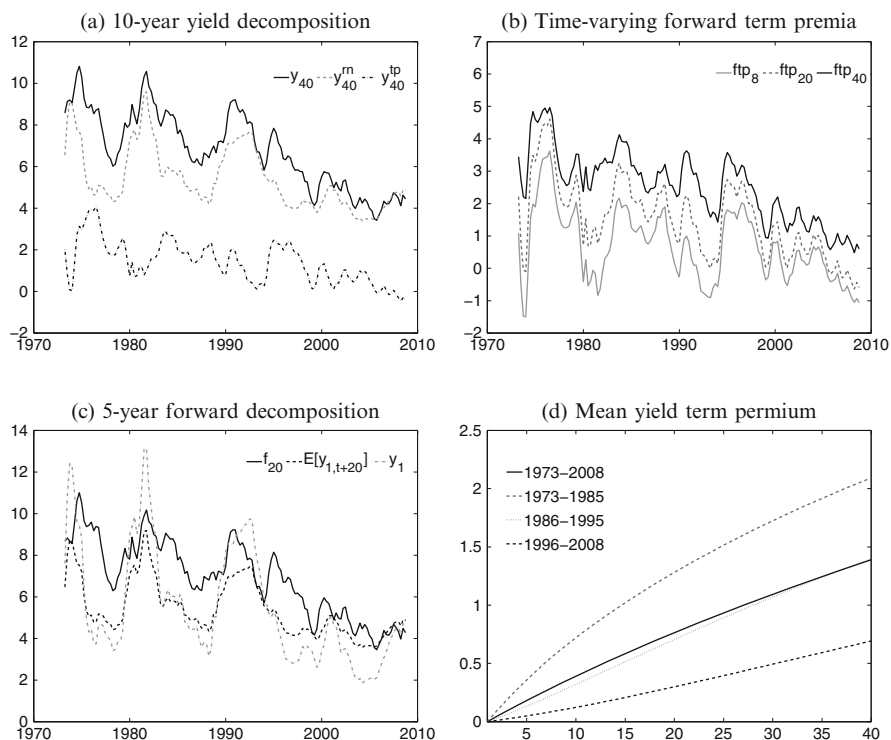


Fig. 4.4 Decomposing the German yield curve

Figure 4.4c presents a decomposition for the 5-year forward rate. There is a sharp divergence between the time series of forward rates and expected future interest rates. The expected one-period rate in 5 years tracks the current short rate through the four policy-induced rate declines much more accurately in terms of dynamics and time of reversals. It confirms the finding that the (expected) short rate process follows a “random-walk” pattern with the best estimate closely linked to the current level of the short rate, in particular for shorter maturities (Mankiw and Miron 1986). Moreover, Cochrane and Piazzesi (2008) find for the US that expected future rates decline even faster than current rates.¹⁵ Market participants know that in times of rate cuts, these will likely continue to fall. With forward rates remaining unchanged, extra returns can be earned on buying the forward contracts. The opposite holds in an increasing expected short-rate environment where forward premia are rapidly eliminated. The term structure models basically recommend to “get out” after the bottom of the short rate is reached.

¹⁵The authors build a modified affine term structure model based on monthly data. Their findings can be hardly captured in the quarterly frequency of the estimated model in this section.

To summarize, both forward premia and yield premia derived by the affine model show considerable dynamics with a secular decline starting at the beginning of the mid 1990s (Figs. 4.4b, d). The analysis clearly shows that interest-rate risk premia are positively correlated with the term spread and negatively with short-term interest rates. In an environment of rising (expected) short rates, there are two effects in opposite directions: first, with fixed risk premia, long-term bond prices may fall; and second, a falling risk premium induces bond prices to rise and long-term yields to decline. The evidence shows that the second effect tends to dominate the first. For that reason, the slope of the yield curve falls as interest rates are rising, leading to a positive correlation between the spread and expected excess returns. This is exactly the story as exemplified by the expectational puzzle, in reduced regression form, outlined by Campbell and Shiller (1991).

Further characteristics of international risk premia are sketched out in Fig. 4.5a. The mean of the term structure of yield premia is rising with time to maturity for all three countries, with the USA averaging at 1.7% for a 10-year bond, followed by Germany with 1.4 and UK with 1.0% over the sample period 1973Q1:2008Q3. As already observed for Germany, these have generally tended to a downward trend over time. Average forward term premia for 5 up to 10 year forward contracts for all three countries have been in a range between -1 to 2% points in the last 10 years of the sample. In addition, the panel clearly indicates co-movements in the same directions on an international level.¹⁶ Still, for the UK, the time-varying component of term premia are much smaller than for the US and Germany. What can also be read from the data is the considerable fall of UK term premia in 1997 which was accompanied by falling long-term bond yields. This fact coincides with two major decisions in UK monetary policy, i.e. the given operational independence

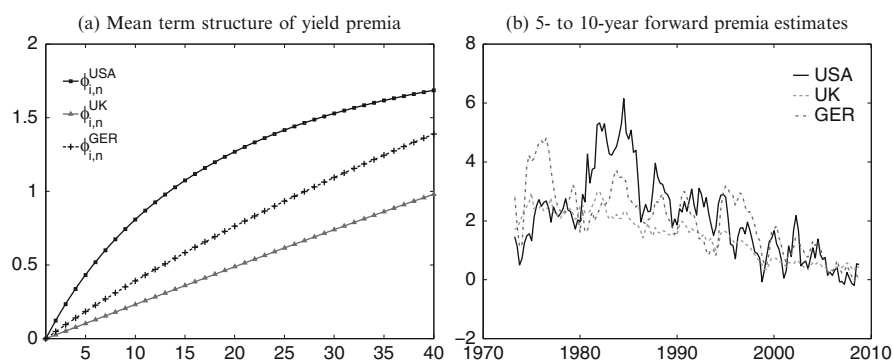


Fig. 4.5 International risk premia

¹⁶These findings are supported by Diebold et al. (2008). The authors identify a significant “global” yield curve factor that accounts for much of the variation in international yield curve dynamics.

of the Bank of England in May 1997 and the Monetary Policy Committee's announcement of a symmetric inflation target of 2.5% points for annual RPIX inflation in June 1997. These events have altered the shape of the yield curve considerably; they allow a straightforward economic interpretation: inflationary risk seems to drive risk premia at the long end of the yield curve. Piazzesi and Schneider (2006) point out that the UK real yield curve is on average downward sloped, while the nominal yield curve slopes up. In May and June 1997, the two curves behaved differently, with the real curve remaining unchanged and the nominal curve declining and flattening. This picture presumably reflects the effect of falling inflation expectations *and* lower inflation risk premia as estimated by the $A_0(2)$ for the UK.

According to the Expectations Hypothesis (EH), the yield on a n -period should increase one-to-one when the term spread widens. Evidence, however, reports the opposite with a negative relationship between bond yield changes and the slope of the yield curve in the presence of large and time-varying term premia. On this issue, Dai and Singleton (2002) have proposed to run two diagnostics for estimated ATSMs, to test the ability of the model at its maximized parameter values, to reproduce the stylized facts on the expectational puzzle. The first test asks whether the model shares the property that the pattern of the sample coefficient of fitted yields from a regression of yield changes onto the scaled yield spread matches the CS-regression of actual yield data – the authors call it the LPY(i) test. Matching LPY(i) means that the ATSM describes the historical behavior of yields under the $\mathcal{P}$ -measure. In addition, the authors suggest running a second kind of test -LPY(ii)- to concentrate on realized risk premia. If the model captures the risk-neutral dynamics well, a CS-regression with risk-adjusted yields changes onto the scaled yield spread should give a coefficient of unity.

In Fig. 4.6, the results are plotted for the three countries. As in Campbell (1995), the graphs confirm that the historical coefficients are significantly lower than one and decreasing with time to maturity. The results for the UK build an exception with a small hump at a maturity of 20 quarters. Moreover, for the US and Germany, the coefficients of the model-implied yields follow the coefficients closely of the original data set with downward sloping features. The results for the UK are somehow mixed where model-implied coefficients coincide with β only from 5-year bond yields onward to longer maturities.

Whether the model matches the LPY(ii) test mainly relies on the ability to generate highly persistent market risk premia. If this is so, risk-adjusted yield changes regressed on the scaled slope should result in a horizontal line with parameter values of 1. The grey lines β^{adj} display the model-implied coefficients. One observation of this exercise is that the risk premium always goes in the right direction, toward the EH-consistent line. Especially for Germany and UK, the test of LPY(ii) is positive and significant for longer maturities; whereas the opposite holds for the US where the risk-adjusted line converges to β^{EH} at shorter maturities.

To comprehend the findings, some further considerations are needed. The relatively high success of LPY(i) and LPY(ii) depends on the modeling assumptions,

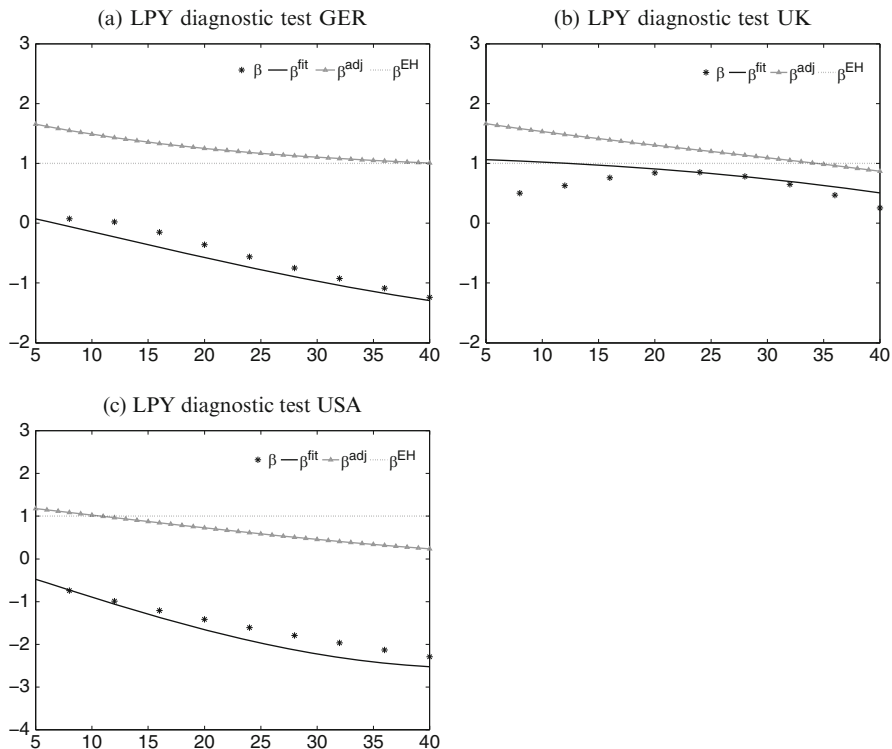


Fig. 4.6 Yield curve fitting diagnostics (*Note:* Regression coefficients of (adjusted) yield changes on the scaled yield spread. Model implied expected excess returns are calculated as $e_{n-h,t+h} = E_t^\mathcal{P} [p_{n-h,t+h} - p_{n,t} + p_{h,t}]$ with regression diagnostics LPY(i): $i_{n-h,t+h} - i_{n,t} = \alpha_n + \beta^{fit} h(i_{n,t} - i_{1,t})/(n-h)$ and LPY(ii): $i_{n-h,t+h} - i_{n,t} + 1/(n-h)e_{n-h,t+h} = \alpha_n + \beta^{adj} h(i_{n,t} - i_{1,t})/(n-h)$. The diagnostic tests are carried out for a holding period of one year ($h = 4$ in the quarterly model). The stars (β) are the coefficients of the original data, the solid line (β^{fit}) are the coefficients for model-implied yields, β^{adj} represent the results for risk-adjusted yield changes and β^{EH} is the EH-benchmark)

in particular on the flexible prices of risk that vary with the level of the sources of risk. If not modeled in this way, the test diagnostic would be rather disappointing.¹⁷ Moreover, the assumption of a zero factor correlation ($\phi = \text{diag}$) performs poorly in matching LPY(i) and LPY(ii), although a conventional likelihood ratio test would not reject the restriction of zero-factor correlation. It is therefore the combination of market prices of risk and factor correlation that allow to make LPY positive.¹⁸

¹⁷If the model is estimated with the assumption that λ_1 has zero entries in its off-diagonal elements or if it is even empty, the risk-adjusted coefficients would not lie near β^{EH} .

¹⁸The restricted model implies a diagonal matrix of ϕ . The LR statistic is χ^2 with 1 degree of freedom. The 5 and 1% critical values are 3.84 and 6.64.

Finally, the fact that only one end of the risk-adjusted coefficients along the yield curve successfully lies within the EH-range, may be due to the inability of the $A_0(2)$ to reproduce all dynamics within the yield curve. For instance, money markets are regularly exposed to non-continuous distortions stemming from “flight to quality” or regulatory issues. This lack can be rectified by the inclusion of a third or even a fourth risk factor as outlined by Dai and Singleton (2002) among others. It can be assumed that this modification might support the risk-adjusted coefficient line to fall within the EH-theoretical line across all maturities.

Affine term structure models are a powerful tool to extract bond premia from observable bond prices. The empirical analysis reveals that the Gaussian subclass of ATSMs with correlated factors and flexible risk parameters can help resolve the expectational puzzles surrounding the expectations hypothesis. The decomposition of international yield curve confirms the presence of a large time-varying component of term premia. The latter are positively correlated with the slope of the term structure and negatively with the short-term policy instrument of the central banks (provided that the one quarter rate is regarded as the policy rate).

4.3 Compensation for Default Risk

Fixed-income securities are, in addition to the risk of short-term interest rate changes, in general exposed to a further sort of risk, i.e. default (or credit) risk. It relates to the possibility that the issuer of a bond is unable to meet her contractual obligations, i.e. to pay the cash-flow stream generated by the underlying bond. Such an event is labeled as default. Default does not necessarily imply that the issuer of the security instrument is completely unable to honor her total debt contract which would result in bankruptcy. It can also mirror the possible partial failure to promptly pay interest and principal when one of both come due (failure to pay). If a default occurs, the recovery rate for a bond is defined as the market price of the underlying bond as a percentage of its face value. It reflects the value of the bond at the point in time when the original contract is surrendered. Default risk is then determined as the uncertainty of whether the issuer can make timely payments as fixed by the bond’s indenture (Fabozzi et al. 2005). The required compensation of a defaultable over a non-defaultable bond includes two components. First, it covers the expected loss from default calculated as the product of the probability of default (PD) times the loss-given default (face value minus the recovery value) so that the expected return of a defaultable bond equals the expected return of a default-free bond instrument with the same maturity. Moreover, it imbeds a credit risk premium associated with the fear that the realized loss from default may exceed the expected loss; it deals with the question about how risk-averse investors price this risk in bond valuation. Technically speaking, the risk premium reflects the fact that default probabilities calculated from historical data are generally smaller than default probabilities backed out from bond price data (Duffee 1999; [Duffie and](#)

Singleton 1999; Hull et al. 2005).¹⁹ Holders of defaultable bonds, thus, build in an extra expected return to compensate for default risk; they then expect to earn more than the default-free interest rate on average.²⁰

Credit spreads, the difference between yields on defaultable and non-defaultable bonds with the same maturity, can be decomposed in above two components: expected losses and credit risk premia. Amato and Remolona (2003) show that for US corporate bonds with different ratings, the historical levels of credit spreads appear to be significantly larger than the expected loss rates based on actual (historical) defaults mirroring the existence of large and positive risk premia priced in corporate bonds. Using a data set covering the period 1866–2006, Giesecke et al. (2010) find that corporate credit spreads on average have been twice as large as expected default losses. In this respect, default risk and risk premia are related to the business cycle. Expected losses are counter-cyclical so that an increase in output lowers the credit spread reflecting stronger balance sheets of the bond issuers. This pattern is relatively more distinct for lower-rated corporate bonds due to sharper financial frictions these borrowers are faced with. The same holds for required risk premia which fall whenever real activity picks up (Amato and Luisi 2006).

Another way to measure default compensation is to look at the credit-default-swap (CDS) market which has been rapidly growing in recent years. A CDS allows the protection buyer and seller to trade credit risk of an underlying reference entity, thereby providing insurance against a default event of the reference entity on part of the protection buyer. Both parties agree on a fee, called the CDS premium or the CDS spread, which the buyer of the contract pays the seller.²¹ If, over the time of the contract, no default event happens, the protection buyer is left empty-handed. If, however, a default event occurs, the buyer can either transfer the underlying obligation to the seller at face value (physical settlement) or she receives the difference between the face value and recovery value of the security (cash settlement). Based on arbitrage considerations, the CDS spread should equal the credit spread as measured by the difference between the yield of the respective defaultable bond and the risk-free benchmark security. If, for example, the CDS spread is smaller than the credit spread, an arbitrageur finds it profitable to buy both the CDS contract and the underlying defaultable bond; she will earn a rate of return that is higher than the default-free interest rate without being exposed to any

¹⁹Similar to the literature on default-free affine bond pricing of the previous Sect. 4.2, PDs from historical data refer to the physical probability measure; whereas PDs derived from bond prices are related to the risk-neutral measure. The difference results from the market price of credit risk per unit of this risk and it describes the systematic default risk which can not be diversified in portfolio optimization.

²⁰The credit-risk literature attributes excess returns on defaultable bonds to a number of factors, including systematic, non-diversifiable risk, risk due to an insufficient number of defaultable bonds in portfolio allocation, liquidity risk, tax effects, contagion or a misspecified risk-free interest rate. See among others Longstaff (2004), Hull et al. (2005) and Collin-Dufresne et al. (2010).

²¹Typically, this fee is quoted in basis points per annum and payments follow a quarterly or semi-annual payment basis.

credit risk. On average, the two prices are equal albeit there are discrepancies in the short run owed to technical, institutional and liquidity reasons (Hull et al. 2004; Zhu 2004). In particular, in times of market stress, the CDS market tends to lead the bond market because liquidity typically dries up faster in the latter and there is no funding restriction in the CDS market, in the sense that the protection seller needs to make initial payments that are used to settle any credit event; this makes price discovery in the CDS market more efficient.

Creditors of corporate debt have the possibility to take legal actions when a company defaults on its debt which, in general, is regulated through the liquidation of a company's assets or through assuming control. Turning to government debt, sovereign default is mainly a political decision – though influenced by macroeconomic factors – since sovereigns can not go bankrupt in the strict sense; holders of government debt may not have recourse to a bankruptcy code (Duffie et al. 2003). A government rather trades off the economic, political and reputations costs against the cost of honoring its debt. This trade-off includes the access to international capital markets, the hampering of international trade or losses of assets held abroad. Rather than defaulting outright, the sovereign has the leeway in renegotiating and restructuring its obligations. Still, pricing government securities is based on the same model set-up as for defaultable corporations, with the focus of implementing restructuring schemes in loss-given defaults and recovery rates.

Sovereign default risk is reflected in the difference between a country's bond yield against a benchmark default-free government bond yield. The history of debt crisis and currency crisis in emerging market countries highlights the surge of sovereign credit spreads in times of worsening economic conditions as measured by indicators of external debt, fiscal balances as well as balance-of-payment variables, inflation, real GDP growth, per capita incomes and overall macroeconomic volatility.²² Fundamental, local-economy variables are seen as important determinants of sovereign credit spread movements. But they reveal only one part of the picture since global factors also considerably contribute to the valuation of government default risk. These “push” factors are associated with economic developments on a global scale, including market sentiments and world interest rates (Calvo et al. 1993; Eichengreen and Mody 1998).²³ For emerging market economies, low global interest rates can thereby narrow government spreads with debtor countries being

²²For a review of recent sovereign lending episodes and default events see Andritzky (2006). The analysis of country risk, more generally, can be classified into (1) models of early-warning indicators, (2) studies on sovereign yield spreads and (3) studies on countries' credit ratings; see Edwards (1986), Kaminsky and Reinhart (1999), Reinhart (2002), Pan and Singleton (2008) and Hilscher and Nosbusch (2010).

²³Longstaff et al. (2007) find that two-third of sovereign credit risk can be linked to global factors whereby sovereigns spreads are related to US stock and corporate bond markets as well as global risk premia. Local economic measures matter less because there is little country-specific compensation for bearing local risk. Excess returns are earned for bearing global macroeconomic risk.

exposed to lower borrowing costs and with international investors searching for higher returns in a low interest-rate environment.²⁴

Taking on a European perspective, prior to and shortly after the introduction of the Single Currency in 1999, there has been a steady convergence of government interest rates within the euro area both unconditionally and conditionally on newly arriving market relevant information (Ehrmann et al. 2007). Before the monetary union, yield differentials were driven by expected exchange rate movements and exchange rate risk, liquidity, default risk and different tax treatments. With the start of the currency union, factors related to exchange rates and taxes have been eliminated so that default risk and liquidity factors and its pricing implications are seen as the only determinants of sovereign bond spreads in the euro area. Indeed, interest rate spreads have declined dramatically in the run-up of the currency union, with maximum yield spreads against the benchmark German Bund averaging at 50 basis points for 10-year government bonds until September 2008. There is evidence that the degree of liquidity and liquidity risk as well as default risk play a significant role in explaining this positive spread, though the absolute effect remains controversial depending on methodological tools and sample periods.²⁵ Moreover, a bulk of the analysis identifies a time-varying common factor that can explain the variation across issuers and maturities within the euro area. Yield differentials are to a large extent determined by international risk factors associated with international risk aversion and aggregate macroeconomic uncertainty.²⁶

One aspect in the early years of EMU has been the degree of market discipline in the context of euro area government bond markets. Following Manganelli and Wolswijk (2009, 196), “market discipline [...] may be broadly defined as the influence exerted by market participants on governments by pricing different risk of default.” As previously mentioned, national fiscal variables and the balance-of-payment position are seen as local factors driving default risk embedded in national bond yields, corresponding CDS spreads and country ratings. Bond spreads, then, signal market participants’ assessment of sustainability of national fiscal positions. The discrimination of the quality of national fiscal policies would make the issue of new debt more costly, and higher interest rates would punish high governments debts and would provoke a market-based adjustment mechanism towards sound fiscal policies (Lane 1993). However, after the introduction of the euro, it seems that markets did not adequately discipline governments via appropriate risk premia; for spreads continued to decline after 1999.

²⁴Remolona et al. (2007) decompose international sovereign bond spreads into expected losses and risk premia. They find that risk premia account for a larger part of spreads so that international risk aversion as imbedded in the price of risk of unexpected losses play an important force in valuing government debt.

²⁵See Codogno et al. (2003), Manganelli and Wolswijk (2009) and Favero et al. (2010) and the following Sect. 4.4 on liquidity risk.

²⁶See Bernoth et al. (2004), Sgherri and Zoli (2009) and Favero et al. (2010). International risk appetite is often proxied by the spread of US corporate bonds over US Treasury bonds.

There are several reasons for the lack of ability of market discipline. One is that there is little or no default risk since government debt throughout the euro area is simply too small to demand a risk premium, and the Stability and Growth Pact (SGP) represents an effective instrument to restrict government debt dynamics. This line of argumentation clearly misses the empirical evidence that many EMU countries have broken the fiscal rules and have been faced with excessive deficit procedures and increasing macroeconomic imbalances ever since 1999 (Geiger and Spahn 2007). A more convincing reason is grounded in the political and institutional setting of the Eurosystem. It has been argued that the credibility of political arrangements is not sufficiently high enough to prevent financial markets to expect the possibility of a EMU bail-out for a single member country, albeit, the no bail-out clause is constituted in the EU Treaty. This implicit commitment lets markets treat all government securities as being priced with the same default risk (von Hagen 2003). Moreover, as pointed by Buiter and Sibert (2005), the collateral framework of the Eurosystem's implementation of open market operations unintentionally provokes a distorted market equilibrium in which fundamentally justifiable default-risk differentials are suppressed. Government bonds serve as eligible assets that can be used as collateral in repurchase agreements of the Eurosystem. The operational practice of the ECB does not address the creditworthiness of sovereign issuers as the sovereign debt instruments can all be used on the same terms. Consequently, "[s]ince the Eurosystem is the dominant player in the euro Repo markets, its behaviour and the markets interpretation of its motivation, objectives, practices and actions will be the single most important determinant of the risk-spreads established by these markets" (Buiter and Sibert 2005, 8); it, thus, treats sovereign bonds as default-free securities. Finally, a last reason can be attributed to the previously mentioned global environment with low interest rates and high risk appetite of international investors that triggered a search for higher yields and compression of sovereign (and other) risk premia (Manganelli and Wolswijk 2009).²⁷

The picture considerably changes for the latest developments in financial markets since October 2008. As Fig. 4.7 shows, with the intensification of the financial crisis following the default of Lehman Brothers, cash credit spreads against the 10-year German Bund and corresponding CDS spreads have been dramatically widening mirroring a deterioration of euro area member countries' fiscal position accompanied by various downgrades of sovereign debt ratings in the subsequent period. The massive fiscal activities were due to financial-sector rescue packages and fiscal stimuli aimed at restoring financial market stability and promoting aggregate demand. The deterioration of public finances was particularly severe for those countries with large current account deficits and accumulated losses in competitiveness in the run-up of the financial crisis.²⁸ These countries are likely to face a prolonged period of sluggish growth and – perhaps, but hopefully

²⁷For a detailed approach to the risk-taking channel of monetary transmission see Chap. 7.2.

²⁸This holds especially for Greece, Portugal, Spain and to some extend to Italy (European Commission 2010).

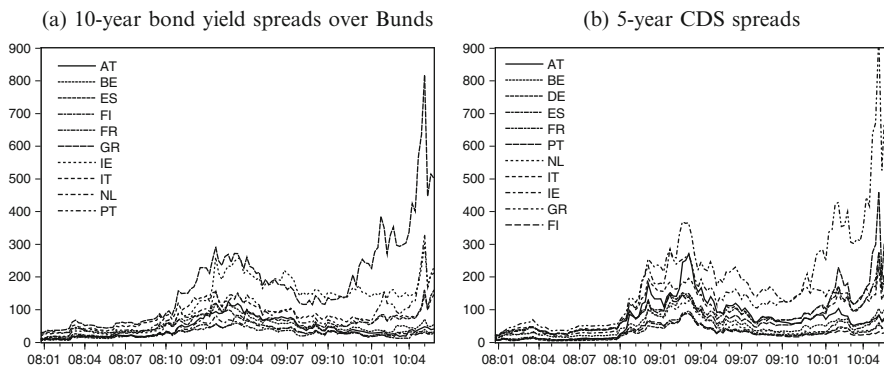


Fig. 4.7 Euro area sovereign spreads

not – unsuccessful fiscal consolidation attempts in the years to come. In this respect, the macroeconomic arrangement within the euro area will aggravate the necessary adjustment process towards a balanced intra euro area macroeconomic constellation.²⁹

Recent empirical research re-focused on the impact of financial conditions in explaining euro-area yield differentials. The overall finding is that markets have become much more concerned about past, current and expected fiscal balances since October 2008. Again, countries with large current account deficits experienced a strong linkage between financial sector vulnerability and sovereign spreads. This result was mainly due to the “credit risk transfer” from the financial to the public sector with jumping public debt levels. To some extent, the surge in bond spreads also can be attributed to the safe heaven status of the German bond market that is characterized as the most liquid market with little default risk (at least until this publication).³⁰ These insights suggest that euro area countries may have switched to a new regime with serious strains on public finances. It can turn out to be worrisome because sovereign bond spreads exhibit some degree of persistency and non-linearity so that positive spreads are likely to remain a fact in the European bond market. Against this background, Deutsche Bank (2010) remarks in its Default Study: “One of the problems [...] is that for the cash credit market we benchmark everything off the risk-free rate which has typically been government bonds. For us, credit started to be a stand-out buy after the Lehmans default because that event marked the acceleration of the deterioration in government balance sheets around the world. Most governments had to stress their own balance sheets to save the global economy and their financial sector from collapse. Given that spread is simply

²⁹For the problem of real and nominal divergence and adjustment failures within the euro area see Geiger and Spahn (2007) and Wickens (2007).

³⁰See Sgherri and Zoli (2009), Mody (2009), Haugh et al. (2009), Attinasi et al. (2009), Ejsing and Lemke (2009) and Schuknecht et al. (2010).

the difference between the supply and demand of one asset against the other, this period marked the period where credit spreads were set to rally hard. However, 18 months on and we are increasingly questioning what is the appropriate risk-free rate in an era of perilous government finances, especially in the developed world.” Indeed, an urgent challenge to be addressed by financial institutions, policymakers and academia.

4.4 Liquidity Risk and Asset Prices

4.4.1 *Micro-Finance Approach to Liquidity*

When attached to the characteristics of an asset, the concept of liquidity can be in a first attempt summarized as the ease of trading a particular security. Keynes (1930, 150) labeled “bills and call loans as more *liquid* than investments, i.e. more certainly realisable at short notice without loss [...]” It relates to the ability to sell an asset at a price which is the same independent of the length of notice agreed between seller and buyer. In this respect, an asset is characterized to be more liquid relative to another asset the more certain it is considered that no price losses occur when selling the asset. It thus incorporates elements of time but also transaction costs and the volume of transaction affect the degree of liquidity. A refined interpretation would define liquidity as the ability to trade an asset at short notice, at low cost and with little impact on its price (Nikolaou 2009). The market for a liquid asset is then determined by three factors. Firstly, it is deep so that in case of a large volume the transaction can take place without affecting the price, secondly it is tight in the sense that transaction prices do not diverge from mid-market prices (bid-ask spread) and finally a liquid market is resilient with price fluctuations resulting from trades and order imbalances dissipating quickly.

The above description is typically characterized as *market liquidity*. It measures the degree and the ability to realize value and cash flow respectively by selling owned assets without disturbing underlying market prices. Following Amihud et al. (2005), illiquidity of a particular asset depends on transactions costs including fees and transactions taxes. Another source of illiquidity comes from demand pressure and inventory risk when a buyer quickly wants to sell an asset but potential buyers are not present in the market all the time. Similarly, market makers pass over the risk of price changes to which they are exposed to when they buy in anticipation of laying off the position in the near future. More generally, the sale price can be depressed in case of search frictions and information asymmetries. The former typically holds in over-the-counter markets when the timing and the volume of transactions of the seller and the potential buyer do not coincide and searching a counterpart is associated with financing and opportunity costs. Moreover, if a potential buyer regards the seller to possess more reliable information about the “fundamental” value of an asset (or vice versa), trading would end up with a loss for the uninformed agent. All these costs should then be already incorporated in today’s asset price; it should

be lower the higher illiquidity costs turn out to be. Alternatively, by looking at the effect of liquidity on required returns, the latter should increase with the degree of illiquidity.³¹

Since liquidity varies over time, investors are uncertain about the degree of liquidity in the future when they (possibly) need to sell the asset. For agents having a preference over liquidity, liquidity fluctuations manifest themselves in liquidity risk beyond the uncertainty of interest-rate induced asset price changes. Liquidity risk is regarded as a premium in the asset pricing literature for which agents want compensation. Therefore, higher liquidity risk goes hand in hand with a larger liquidity risk premium which increases the expected and required return of investors. Liquidity risk can be derived and estimated within a liquidity-adjusted β -pricing model (Acharya and Pedersen 2005). According to the standard CAPM, an asset's expected return equals the risk-free return plus a risk compensation which is proportional to the asset's β , i.e. the quantity of risk determined by the covariance between the asset's return and the market return. The model yields three additional risk corrections that derive from liquidity risk. Firstly, the required return increases with the correlation between the asset's illiquidity and the market illiquidity. The intuition is that market participants are more likely to be reluctant in holding an illiquid asset when the market in general becomes illiquid; they rather prefer those securities in their portfolios which can still be easily traded when the market is illiquid. Indeed, the covariation between individual liquidity and market liquidity is mostly positive as found in many empirical studies, both within and across asset markets including both bond and stock markets (Chordia et al. 2005). Secondly, a positive correlation between expected return and market illiquidity tends to negatively contribute to liquidity risk compensation since investors are willing to accept a lower return but with a precisely high return in an environment of low market liquidity. However, empirical evidence shows that this covariation is mostly negative because a fall in market liquidity is typically associated with a fall in asset values.³² The last factor is due to covariation between an asset's illiquidity and the market return. An investor demands a liquidity risk premium if the asset's illiquidity is high and the market return is low. In this case, the general asset market performs poor and the possibility to trade easily is particularly valuable to the investor. If, however, the liquidity of the single asset is likewise low, a compensation for that risk is required. Acharya and Pedersen (2005) estimate that this latter effect appears to have the most pronounced impact on expected excess returns derived from liquidity risk.

Market liquidity is thereby correlated across markets; a negative information shock to stock markets may cause a shift from stock to bond market liquidity

³¹ Asset pricing deals with optimal pricing given a conditional information set. Agents require a higher return in case of increasing illiquidity costs. In contrast, realized (ex-post) returns depend positively on liquidity; they are greater if the liquidity of the asset is higher (than initially expected when the asset has been priced).

³² A similar implication has been found in a model version of Holmström and Tirole (2001) and it is documented by Pastor and Stambaugh (2003) in US stock markets.

as investors allocate their portfolios in favor of safe assets potentially triggering order outflows from the stock market. However, in most cases, a commonality is documented across stock and bond markets, i.e. market liquidity co-moves positively (Chordia et al. 2005). In order to explain such a pattern, it is worth introducing a further liquidity concept, namely *funding liquidity*. An entity is said to be liquid as long as cash inflows equal or exceed the amount of cash outflows. Typically, there are various sources of liquidity for investors. Most generally, they include new borrowing from financiers or the sale of (marketable) assets (Borio 2009). Notice that there is no clear cut between market and funding liquidity because, according to the above description, the latter points to the ability to sell an asset at short notice so that both liquidity concepts are interconnected. For this reason, sometimes funding liquidity is exclusively associated with liabilities and that funding through asset sales with market liquidity. Funding liquidity can then be understood like borrowing against a cash flow generated by an asset when borrowing takes place with secured lending; whereas market liquidity covers the transfer of the asset with its entire cash flow (Brunnermeier 2009; Tirole 2010).

Banks, for example, can get liquidity from the attraction of deposits, through the interbank market by engaging in secured and unsecured borrowing, or by bidding directly in the open market operations of the central bank. Moreover, they can sell part of their assets in terms of securitization in their role as “originator and distributor”. Funding liquidity is then characterized by the ability to raise funds in short notice through the liquidity sources described above in order to service liabilities as they fall due. The risk of funding liquidity arises from the possibility that over a given period investors are unable to settle their obligations with immediacy (Drehmann and Nikolaou 2009). Financial institutions are, in particular, exposed to such risk because their balance sheets have a maturity mismatch with both short-term nominal liability contracts and long-term, possibly illiquid, asset holdings. When they heavily rely on short-term debt, they face the risk of being unable to roll this debt over when funding liquidity is low and a funding shortage can thus arise. More generally, for financial institutions, funding liquidity risk arises along three dimensions, (1) haircut funding risk (see below), (2) debt roll-over risk and (3) redemption risk, i.e. the risk associated with withdrawal of demand deposits or even equity (Brunnermeier 2009).

The effect of haircut risk can be described in a repurchase agreement (repos) which can be best understood as a secured loan. The borrower sells an asset at a price lower than the current market price to borrow short-term against it. Since she cannot borrow against the total current market value of the underlying asset, the buyer of the repo requires a collateral buffer (haircut). The repurchase price then equals the current market value of the underlying asset plus an agreed repo interest rate. If the value of the asset shrinks during the maturity of the repo, the buyer may demand margin calls in order to get compensation for the losses associated with the price decline of the asset. For instance, if the haircut is 2%, then the repo seller can borrow €98 for €100 worth of the underlying asset. If the raised funds are used to buy assets worth €100, the seller must come up with €2 of equity. Thus, the reciprocal of the haircut simultaneously determines the maximum permissible

level of leverage, i.e. in the above example 50 (Adrian and Shin 2009b). Since financiers (buyers) can reset margins in every period, borrowers are faced with funding liquidity risk from existing positions on the asset side of balance sheets.

Strong linkages between market and funding liquidity arise not only because of their classification depending on the definition but because liquidity can be shown to be endogenous to the financial system where a squeeze in market liquidity triggers a shortage in funding liquidity (et vice versa) with positive feedback effects leading to a liquidity spiral. In normal times, with low liquidity risk, the central bank supplies the *aggregate liquidity* in order to cover the liquidity deficit of the financial system on aggregate. The amount of liquidity is then distributed among those financial institutions allowed to take actions along side with the central bank, and is redistributed to those institutions that are liquidity constrained and who ask for the amount of liquidity in order to cover their funding liquidity needs; finally, this liquidity is recycled within the financial system (Nikolaou 2009).

With increasing liquidity risk, the proper functioning of liquidity provision may be hindered in severe ways. A reinforcing loop process of falling market and funding liquidity promotes a market environment of disappearing liquidity which leads to a systemic liquidity crisis although the aggregate amount of liquidity provided by the central bank did not change at all. Individual funding liquidity risk is not a systemic problem on its own. It is only when funding and market liquidity simultaneously collapse that mutual reinforcing processes can aggravate and provoke a profound liquidity crisis. A funding shock to a (small) number of institutions can lower market liquidity along several dimensions. As banks are highly connected in interbank markets, through balance sheet linkages and through cross-holdings of liabilities, individual illiquidity risk can quickly translate into overall market liquidity risk. This holds true in markets with asymmetric information about the solvency and liquidity status of financial institutions; this condition may lead to higher counterparty risk which can produce a domino model of liquidity contagion (Brunnermeier et al. 2009). Enhanced default risk may lead to a reduction in market liquidity and falling assets prices. If market participants are forced to sell off their assets, possibly at fire-sale prices in order to avoid other costly project liquidations, market liquidity will further shrink.

Yet, increased counterparty risk is not even necessary to generate liquidity spirals. Since financial institutions mark their balance sheets to market, a negative shock to asset prices worsens funding conditions because assets are typically used in secured lending operations and institutions' capital and liquidity constraints may quickly bind. Consequently, they are forced to sell assets in order to meet their cash-flow commitments. Asset prices increasingly move away from fundamentals because financial institutions face similar budget constraints and market liquidity shrinks. This market pattern is further amplified within the margin spiral where creditors are faced with increased asset price volatility and uncertainty about the fundamental value of the underlying asset used in secured lending. It, thus, provides a feedback process from market liquidity risk to funding liquidity risk. A shock induced by asset price losses triggers higher haircuts; increased haircuts, in turn, worsen borrowers' funding conditions and increases funding liquidity

risk which forces investors to reduce positions (de-leveraging); as a consequence, market liquidity further shrinks which in turn gives rise to higher haircuts, and so forth (Brunnermeier and Pedersen 2009). Prices of assets then become heavily disconnected from their “fundamental” values as determined by expected cash flows; in times of liquidity distress, prices mirror solely the degree of market liquidity. Falling asset prices then trigger a shrinkage in capital positions and can manifest in a profound solvency crisis for the financial sector as a whole. It becomes clear that solvency and liquidity are not independent concepts. Liquidity problems, then, can trigger solvency problems even if there were none before (et vice versa).

In times of downward liquidity spirals, market liquidity evaporates predominantly toward high-risk securities which are exposed to most heightened liquidity risk. Investors re-balance their portfolios toward the most liquid assets, particularly towards cash positions, short-term overnight and government securities and, if offered, interest-rate bearing central bank deposits. Such a phenomenon is typically labeled as “flight-to-liquidity” and it aggravates market and funding illiquidity across a broad spectrum of financial assets. Despite aggregate liquidity, as offered by the central bank, not having been altered, the availability of liquid funds is rare to meet prospective cash flow commitments due to “liquidity hordings”.

Indeed, such a pattern could be documented in the latest turmoil in money markets starting with the financial crisis in 2007. One measure of liquidity tensions is the spread between the unsecured interbank deposit rate (i.e. the EURIBOR) and the overnight indexed swap (OIS) with the same time to maturity. The latter is the fixed rate that banks are ready to pay in exchange for getting the average overnight rate for the duration of the OIS arrangement. It reflects the average expected path of future overnight rates and embodies the same credit and liquidity risk characteristics as its overnight pendant so that it rarely incorporates any of these two risk components. The EURIBOR-OIS spread is then a reliable measure of perceived credit and funding liquidity risk (ECB 2009b). The left graph of Fig. 4.8 displays the massive tensions in the euro interbanking market starting in August 2007.³³ The 3-month EURIBOR-OIS spread reached its highest value after the bankruptcy of Lehman Brothers in September 2008. Eisenschmidt and Taping (2009) recently analyzed this spread dynamic to determine whether it is mainly a reflection of liquidity or credit risk. They find that liquidity hoardings have been in the center of financial institutions’ liquidity management. These institutions were eager to lend funds in unsecured money markets toward longer maturities because they faced high funding liquidity risk. If lenders lend out in the term money market, with a certain probability, they may be exposed to a liquidity shock before the loan matures so that they themselves are forced to raise funds at possibly higher costs when the shock arrives. Therefore, they did lend out repeatedly at overnight rates to reduce their own funding liquidity risk, thereby bringing overnight rates down and EURIBOR rates up.

³³The OIS spread is defined as the difference between the 3-month EURIBOR and the 3-month EONIA swap rate.

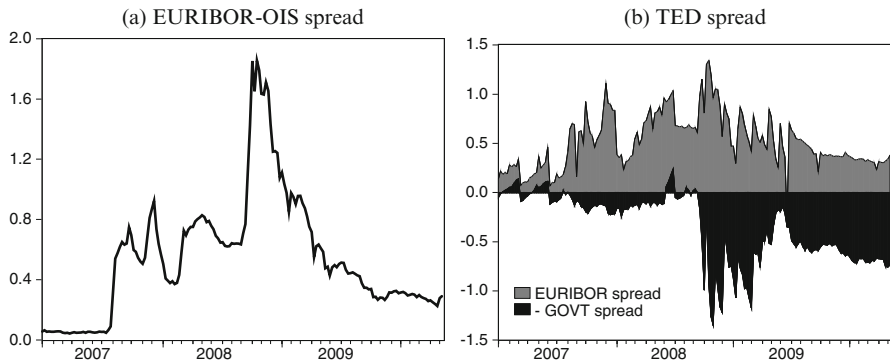


Fig. 4.8 Liquidity risk indicators euro area

A further indicator of euro money-market tensions is the TED spread, the gap between the unsecured 3-month EURIBOR rate and the 3-month government bond rate. In times of higher uncertainty and risk, lenders charge higher rates for unsecured loans which puts an upward pressure on long-term EURIBOR rates. At the same time, investors seek for first-rate collateral values for secured transactions in order to minimize the required haircuts demanded by financiers. Holding short-term government securities becomes a dominant investment strategy and it brings down government bond rates. The TED spread can be further decomposed into the EURIBOR spread, i.e. the difference between the 3-month EURIBOR and the central bank's target rate, and the government-bond spread, i.e. the difference between the target rate and 3-month government bond rate. The EURIBOR spread should be a rough indicator of credit risk given that target rate expectations about the short rate remain unchanged. The second spread measures the “collateral and liquidity effect” of holding government securities on financial institutions' balance sheets (Liu et al. 2006). Finally, the TED spread with its two spread components can be refined as

$$\begin{aligned} euribor_{3,t} - govtrate_{3,t} &= (euribor_{3,t} - ois_{0,t}) + (ois_{0,t} - target_t) \\ &\quad + (target_t - govtrate_{3,t}) \end{aligned} \quad (4.7)$$

in order to take into account the significant deviations of the overnight interest rate ($ois_{0,t}$) from the target rate set by the central bank. Indeed, this deviation turned negative in the last quarter of 2008, as the overnight rate hit the floor of the policy corridor and approached the deposit facility rate when the ECB (and many other central banks) started conducting massive liquidity injections into the financial system with the scope of increasing aggregate liquidity (stock of base money). The right graph of Fig. 4.8 shows the development of the TED spread split into the contribution of the EURIBOR spread and the government spread. Here, for a better visualization, the black shaded area represents the negative of the

government spread so that higher negative values contribute positively to the overall TED spread.³⁴ Similar to the EURIBOR-OIS spread, money market tensions, as measured by the EURIBOR spread, first showed up in August 2007 when newly arriving information about large exposures of European banks in the US subprime market triggered an increase in long-term money market interest rates.³⁵ The 2008 spike in the TED spread is the result of both a further rise in the EURIBOR spread and a dramatic increase of the government spread. The latter should be then seen as a reflection of “flight to liquidity” and “flight to quality” considerations by investors searching for the most liquid and best-rated security instruments in order to deal with market and funding liquidity risk.

4.4.2 Liquidity Preference and Uncertainty in Light of Financial Intermediation

The portfolio approach of Tobin (1958) and its extension of Acharya and Pedersen (2005) of dealing with liquidity rests on the proposition that investors possess probability distributions about an asset’s return and liquidity characteristics. It is conducive to discuss the concept of liquidity and money as a perfect store of value within a portfolio decision problem where an investor allocates her wealth among a set of (risky) assets.³⁶ In its simplest form, money is the only riskless asset so that maximizing expected utility implies a risk-return trade-off depending on the degree of risk aversion on the part of investors. In this respect, risk is understood as a dispersion around some expected value and is measured as the standard deviation of a probability distribution. The way of grasping the idea of uncertainty within the above context, however, does not fully deal with the theoretical concept of liquidity.

The conceptualization of uncertainty is based on risk versus fundamental uncertainty. Like Knight (1921) and Keynes (1921, 1936) sees a large part of the decision making process as one where it is not possible to come up with subjective, exact numerical probabilities for the states of the world in the future. He remarks that “by ‘uncertain knowledge’, [...] I do not mean merely to distinguish what is known for certain from what is only probable. The game of roulette is not subject, in this sense, to uncertainty; nor is the prospect of a Victory bond being drawn. [...] The sense in which I am using the term is that in which the prospect of a

³⁴The 3-month government interest rate is provided by ECB and it is the cross-country average of national government bonds with best credit-rating quality.

³⁵As stated, changing interest-rate expectations might also have a significant influence on the EURIBOR spread. A positive spread could then be interpreted within a rising short-rate environment. However, the period under consideration clearly does not allow for such considerations. If anything, markets were expecting falling policy rates which should have narrowed the spread between current overnight and 3-month rates rather than increased it.

³⁶For a theory of interest-rate determination within the portfolio and the stock of wealth setting see Spahn (1994).

European war is uncertain, or the price of copper and rate of interest 20 years, hence. [...] About these matters there is no scientific basis on which to form any calculable probability whatever. We simply do not know” (Keynes 1937, 213–4). Whereas Knight emphasizes the incapability of measuring such probabilities, Keynes highlights the problem of confidence in a probability statement. It heavily depends on the evidential weight and the weights of argument, respectively. These weights describe a measure of the degree of completeness of the relevant evidence for the judgement of probability. Forming an expectation, thus, does “not solely depend [...] on the most probable forecast we can make. It also depends on the *confidence* with which we make this forecast – on how highly we rate the likelihood of our best forecast turning out quite wrong. If we expect large changes but are very uncertain as to what precise form these changes will take, then our confidence will be weak (Keynes 1936, 148).”

A preference towards liquidity and holding money can be then seen as a “barometer of the degree of our distrust of our calculations and conventions (Keynes 1937, 215).” The less confident agents are in the validity of their judgement, the more likely they abstain from making commitments that constrain the degree of flexibility of reshuffling their portfolios. Staying liquid, thus, enables agents to react to new information at low costs since it avoids the sale of assets in times of unexpected liquidity needs. Therefore, Keynes makes a clear distinction between a pure risk premium either stemming from the expected path of future interest rates or from the possibility of default and the liquidity premium: “I am [...] inclined to associated the risk premium with probability strictly speaking, and liquidity premium with what [...] I called ‘weight’. [...] [t]he difference corresponding to the difference between the best estimates we can make of probabilities and the confidence with which we make them” (Keynes 1936, 240).³⁷

Liquidity preference does not only emerge on part of the non-financial sector. In a capitalist economy with a mature financial intermediary sector, the most essential liquidity preference is that of the banking system (Minsky 1982, 74). Liquidity preference manifests itself in the active management of banks’ balance sheets by trading-off maximum profit and the risk of illiquidity. Banks do not passively accommodate any demand for credit but make their decisions about the composition of the asset and liability side dependent on the degree of uncertainty towards the demand for liquidity of bank customers and the availability of (funding) liquidity

³⁷Many theories on liquidity preference build up on this theme. Hahn and Solow (1995, 144) point out that “there is thus a probability that a portfolio, once made, is not optimal in light of what will be learned. This consideration, when combined with transaction costs, leads to a premium on ‘liquid’ or low-transaction-cost assets. This premium is in nature of an option purchase.” Similarly, the user cost of money can be defined as the value of potential gains or losses that has been forgone or avoided by parting with money (Kregel 1998). Likewise, Hicks (1974, 57) gives liquidity a social function as “it gives time to think.” Finally, Jones and Ostroy (1984, 13) highlight the possible liquidation costs associated with re-allocating a portfolio, in particular out of non-monetary positions: “The more variable are a decision maker’s beliefs, the more flexible is the position he will choose.”

in case of unexpected liquidity shocks. The customers' demand for liquidity can occur on both sides of the balance sheet, in terms of granted credit lines (loan commitments, overdraft facilities) to borrowers and in terms of demand deposit withdraws that is to some extent unpredictable (Kashyap et al. 2002). Financial intermediaries need to come up with some overhead that represents a cost to them through foregone return opportunities and the threat of forced liquidation of existing projects. They hold "liquid" buffer stocks in order to cope with these uncertainties, in particular to the extent that they are exposed to funding liquidity risk, i.e. the inability to roll-over debt. Most essentially, the choice of the buffer is not a simple dual decision problem between reserves (cash) and long-term illiquid asset holdings. It is the degree of liquidity of a set of assets active balance sheet management is based upon. If uncertainty rises, so does the liquidity preference which can be documented by asset demands biased towards more liquid assets (with lower expected returns). Banks will then cut short long-term, illiquid projects, typically lending to non-financial corporations and households.³⁸ This process does not necessarily mean to substitute these projects by a corresponding increase in the amount of reserves. It implies that the asset side becomes, on average, more liquid and since the liquidity degree of an asset is associated with the underlying maturity, the average duration of the asset side should shrink.

Liquidity preference also has implications for the liability management of financial intermediaries. From this standpoint, a rise in liquidity preference is reflected in the desire to hold dated liabilities, possibly at higher re-financing costs but with a lower likelihood of being unable to roll-over debt, thereby increasing the average duration of liabilities. If this strategy is pursued by the banking sector as a whole, the degree of financial intermediation falls which is accompanied by shrinking balance sheets. Taken together, liquidity preference can be understood through the lens of financial intermediation. An increase in Keynesian uncertainty goes hand in hand with a process of disintermediation along the dimensions of reduced liquidity transformation and maturity transformation. A crisis in confidence and a deterioration of weights of arguments can then initiate profound negative liquidity spirals. The flight to liquidity is then a reflex of an increase in the precautionary motive of the demand for the most liquid assets (Bibow 2009).

³⁸This transmission mechanism is closely related to the risk-taking channel of Chap. 7.2.

Part II
The Term Structure of Interest Rates
and Monetary Policy Rules

Chapter 5

The Macro-Finance View of the Term Structure of Interest Rates

5.1 On the Use of the Yield Curve for Monetary Policy

It is a widely accepted consensus that the operating instrument with which a central bank conducts monetary policy is the short-term interest rate. Typically, it sets its policy rate conditional on the macroeconomic environment for the purpose of achieving its final goals of price stability as well as output stability. Managing aggregate demand operates through various transmission channels where interest rate moves affect the whole set of asset prices, the net worth of balance sheet positions and the lending behavior of banks. An important feature in the traditional interest-rate channel is the emphasis on real rather than nominal interest rates and the role of long-term interest rates.¹

It is not the short rate (alone) that influences the decision-making of firms, households and governments. The whole set of interest rates differing in their maturity affect spending and debt-management decisions. In particular, aggregate demand is mainly driven by movements in long-term real interest rates. One can easily think of purchases of durable goods such as housing and capital which should represent the most interest-rate sensitive elements of aggregate demand. With interest-rate driven monetary policy, the term structure of interest rates gains a prominent role for the maintenance of effective monetary policy. In particular, how are short rate changes translated into long-term bond yield movements? Monetary policy is heavily concerned with this “leverage effect”. A central bank’s leverage over longer-term interest rates comes from the idea that these rates are priced by market participants as the weighted sum of expected future short-term interest rates. Moreover, according to the Fisher equation, bond yields are determined by the inflation compensation component required by investors for holding long-term nominal securities. This effect can feed through the term structure from

¹See Allsopp and Vines (2000) for an early assessment of the consensus view and Woodford (2003) for a rigorous microeconomic approach to macroeconomic policy analysis.

long-term interest rates to the short end. Monetary policy is well advised to monitor such dynamics since they can signal necessary actions in order to keep inflation expectations on track.

Following Fendel (2007), for the conduct of monetary policy three closely inter-related fields of application arise in the presence of yield curve effects. Firstly, bond prices, like other financial assets, reflect the market's view about the future. They provide timely and forward-looking information about a number of macroeconomic and financial factors that are of pivotal relevance for policy makers. Secondly, given the importance of the long end of the yield curve in monetary transmission, the channels need to be pinned down through which changes of monetary policy affect the term structure of interest rates and, in turn, other aggregate variables. Finally, the implementation of a chosen monetary policy strategy can be mostly described by a short-term reaction function either derived from a simple rule as described by Taylor (1993) or from an optimized targeting rule within the flexible inflation targeting framework (Svensson and Woodford 2005). Little emphasis has been put on the question of whether economic outcomes can be improved by explicitly reacting to term structure dynamics such as the term spread or the long-term interest rate. Each field of application will be discussed in turn and further insights are provided in subsequent Sections.

5.1.1 The Information Content and Its Interpretation

Bond prices mirror market views that are inherently forward-looking in nature. From the theory of asset pricing it is known that the current price of an asset can be described by the expected discounted value of its stream of future pay-offs. The value is linked to future developments in the economy and the way it interacts with the evolution of (stochastic) discount rates. Discount rates, in turn, can be decomposed into two parts: the compensation in terms of postponing current expenditures and parting with money respectively, and the compensation for bearing risk associated with the uncertainty of the future pay-offs. The no-arbitrage approach of thinking about the yield curve represents the most powerful tool in explaining the interconnection between short- and long-term interest rates. Long-term interest rates are described by the weighted average of expected future short rates adjusted for risk compensation. In order to extract market expectations about the future path of short-term interest rates, a model to estimate term premia is necessary.²

For monetary policy, interpreting spot yields and implied forward rates reveals how an initial policy rate change is translated into a sequence of expected rate changes as priced in by market participants. Such information is important because

²See Chap. 3.5.2 in this work. As described there, term premia estimates are subject to considerable uncertainties surrounding the estimation procedure and the structural specifications of term structure models.

it is difficult to adequately assess the amount of rate changes necessary in order to fight inflation or to prevent a recession. These rate changes usually follow the “Brainard principle” that calls for gradual short rate moves so that monetary policy can observe the sensitivity of the relevant bond yields to its action. For instance, a subdued increase in the policy instrument might be accompanied by a rise in interest rates with higher maturities if the public knows that monetary policy credibly operates within adjusted policy cycles in the presence of disequilibria and shocks. Viewed from the perspective of the role of long-term interest rates for aggregate demand, the degree of persistence on developments of the short-term interest rate becomes crucial.³ Against this background, it can be absolutely possible that markets begin to expect a sequence of policy changes in one direction in advance of any policy actions. In this way, policy works via the expectations channel of monetary transmission. Consequently, term structure information reveals whether the policy intention is understood correctly. At the same time, it gives an idea of how “much policy is in the pipeline” through active action and consequential expectational impulses (Goodfriend 1998).

This *steady-handed* approach makes it possible for changes in policy rates to have much larger effects on aggregate demand and price dynamics than would otherwise be; it could indeed be possible that less activist policy actions are necessary to achieve the goals of price and output stabilization.⁴ Such a fashion can become very important if policy rates hit a lower bound and were to be constrained. A clear commitment to gradualism and its communication to markets might support a flattening of the yield curve initiated by falling long-term interest rates (Krugman 1998; Reifschneider and Williams 2000). Still, an overreliance on the gradual approach could turn out to be ineffective when a central bank does not prove its commitment to it. Market participants will form adequate interest rate expectations if and only if they have the reason to do so as learned and experienced by past and current policy rate cycles. For monetary policy, an intrinsic tension emerges between using the pipeline effect with less pronounced short-term rate changes and the need to push strong policy signals, especially when prospects are vague and uncertain (Spahn 2001b; Dennis and Sderström 2006). In an environment of overall uncertainty, strong signals and out-of-rule based interest rate decisions are appropriate measures, as there is a need to maintain flexibility in order to act in response to changing conditions in the future. This means that a commitment to persistence can only be effective if it is conditional to current available information about the overall economic outlook; it can not hold from an unconditional perspective.⁵

³Extensive literature elaborates on the effectiveness of monetary policy and the role of expected future interest rates. A selected collection is Goodfriend (1991), Rudebusch (1995) and Woodford (1999b).

⁴See Trichet (2009) for a treatment from a policymaker’s view.

⁵It is in this context why central bankers do not stop highlighting that the decision-making body never precommits.

To sum up, the yield curve is an instrument for both policy makers and market participants with which the stance of monetary policy can be evaluated. If successful measures of term premia exist, risk compensation can be backed out so that risk-adjusted forward rates and risk-neutral bond yields can be used to filter out market expectations about future policy rates.⁶

Additionally, besides its use as indicator of expected short-term interest rates, the medium- and long end of the term structure provides valuable information about future economic activity. Leading indicators as derived from the term structure can be analyzed in order to forecast future dynamics of macroeconomic fundamentals.⁷ The term spread contains information about future output growth independent of that contained in various other macro variables. Ang et al. (2006) report that for the US, every recession after the mid-1960s has been predicted by an inverted yield curve within six quarters of the impending recession. The main explanation for this pattern is that a temporary tightening of monetary policy can be expected to trigger two effects. First, it produces a real economic downturn and second, it generates a fall in inflation with a sufficiently lag. For the shape of the yield curve it implies that it flattens or even gets inverted since short-term interest rates increase relatively to long-term rates. In the way that a monetary contraction is associated with an economic slump, markets may expect future lower policy rates that might support the economy in rebounding. In contrast, a monetary expansion causes both a steepening of the yield curve and an improvement of economic activity. Equipped with the Expectations Hypothesis, markets anticipate an increase in the policy rate due to a more positive outlook for output growth; consequentially, long-term interest rates pick up, too. Higher inflation dynamics can aggravate the findings at the long end of the yield curve. A point to stress is that such explanations for the co-movements of the term spread and future output growth can be attributed entirely to the informational role of (expected) monetary policy actions.

An alternative explanation finds its origin in the “deep” structure of the economy. Within this approach, the predictive content of the slope refers to the real rather than to the nominal term structure based on consumption smoothing motives (Benati and Goodhart 2008). Whether this intrinsic approach translates from the real slope to the nominal one, depends on the stochastic properties of inflation. If inflation is a random walk and innovations have a permanent impetus, then a shock to inflation alters inflation expectations at all maturities and the whole nominal yield curve shifts proportionally thereby leaving the slope of the nominal curve for a given real curve unaffected; changes in the real slope one-by-one move the observed nominal slope. If inflation, however, is less persistent and inflation expectations change only at short horizon due to its mean-reverting character, then the predictive content of

⁶For an empirical application on the futures market see Piazzesi and Swanson (2008) for federal funds futures and Joyce et al. (2009b) for the BoE’s bank rate

⁷See Estrella and Mishkin (1997), Hamilton and Kim (2002), Stock and Watson (2003a), Favero et al. (2005) or Ang et al. (2006) among many others.

the nominal spread may differ from the real one and it may blur the informational content of the nominal curve.

Although the predictive content has been well documented for more than two decades, recent evidence comes to the conclusion that it has somewhat weakened (Estrella et al. 2003). In this respect, monetary policy plays an important role by altering the dynamics of inflation and output through its respective reaction coefficients to macroeconomic events. The stronger a central bank smoothes the policy rate, the lower the predictable part of the yield curve slope becomes for future output. Along the same lines, if monetary policy only reacts to inflation, the correlation between (future) short-term interest rates and (future) output shrinks. The latter supports the view that the switch to the inflation targeting framework by many central banks has weakened the slope as leading indicator for future output growth.

In the presence of time-varying risk premia, the possibility can occur that risk attitudes blunt the yield curve's usefulness as an indicator. The term spread may only partly mirror the expected path of future interest rates. When measuring the spread in conventional ways, there is no distinction between the correlation between future output and expectations of future short rates and the required term premium. To improve the forecasting model, it is therefore suggested that risk-adjusted term spreads should be applied when forecasting the likely outcome of economic activity in upcoming periods (ECB 2006; Ang et al. 2006). Estimations with the risk-adjusted slope factor give much larger and significant regression coefficients.

Yet term premia measures themselves may present reliable benchmark variables that can be used for forecasting purposes. Recent research stresses that the future change of output is also correlated with the change in term premia as calculated by affine term structure models (Rudebusch et al. 2007). This observation has come to be known as the "practitioner's view," which is prevalent among central banks and market analysts: a fall in the term premium *ceteris paribus* triggers a decline in long-term interest rates, thereby providing expansive stimulus to future economic growth.⁸

Beyond its implication for predicting future output, the medium and long end of the yield curve provides valuable information about inflation expectations. According to the modified Fisher Hypothesis, any nominal interest rate can be decomposed into its real counterpart, expected inflation and an inflation risk premium. On part of monetary policy, inflation expectations are used to forecast current inflation dynamics and they are applied to measure credibility towards a well-defined inflation target. If the central bank credibly commits in achieving its target, expected inflation in the distant future should remain well anchored.

⁸See e.g. Kohn (2005) or Bernanke (2006). Kohn notes that "[...] the decline in term premiums in the Treasury market of late may have contributed to keeping long-term interest rates relatively low and, consequently, may have supported the housing sector and consumer spending more generally." On the contrary, a bulk of evidence suggest that term premia and output growth are positively correlated (Hamilton and Kim 2002; Favero et al. 2005).

Though real interest rates and expected inflation are two key economic variables, their dynamics are essentially unobserved. In order to disentangle real rates from inflation expectations, a measure of proper ex-ante real bond yields or a measure of markets' inflation expectations are necessary. On this account, an increasing number of fiscal authorities have started to issue inflation-indexed bonds that provide information about the term structure of real rates. The difference of nominal yields and real yields can serve as a measure of the term structure of inflation expectations (Christensen 2008). However, expected inflation derived from financial markets should still be interpreted with caution, since various other factors such as liquidity issues or inflation risk premia can distort a “clean” measure of break-even rates. Usually, inflation expectations tend to be overstated by the presence of inflationary risk and understated by liquidity effects.⁹ A cross-checking with survey-based data can help to reduce such biases.

5.1.2 Term Structure Reaction to Monetary Policy Events

The reaction of long-term interest rates to monetary policy changes is inextricably connected to expectations dynamics of markets concerning future macroeconomic conditions and, thus, future monetary policy. The “pipeline” effect of the preceding Sect. 5.1.1 provides a first mechanism of how the long-end of the yield curve responds to short-rate changes. Obviously, the Expectations Hypothesis describes that long-term bond yields should respond positively to current short-rate moves, provided that expectations remain constant or follow the same direction of the initial change. Cook and Hahn (1989) were one of the first to empirically investigate the effects of monetary policy actions on the term structure of interest rates for the US economy. They identified a strong positive relationship between target rates and long rates, although the correlation diminishes with increasing maturities of long-term bonds. They found that a 100-basis point increase in the FED's nominal target rate has increased the nominal 30-year rate by 13 points. Many other studies confirm positive comovements between short-term and long-term interest rates.¹⁰ They support the view that, on average, the yield curve shifts rather than rotates.

However, there are many exceptions of this empirical finding. In May 1994, after having induced a tightening cycle in January, the FED increased its policy rate by a quarter point up to 4.25% points whereas long-term interest rates fell. One description of events could have been that the last policy decision did not put much new policy into the pipeline since those moves had been already anticipated by markets. Indeed, markets could have even thought of an easing monetary

⁹See Chap. 6.4 on the decomposition of the nominal yield curve.

¹⁰See for instance for the US Gürkaynak et al. (2005b), Peersman (2002) or Beechey (2006). For the euro area, results are rather mixed. Empirical evidence suggests that far distant forward rates do not respond much to macroeconomic events (Ehrmann et al. 2007).

policy in light of worsening economic conditions (Goodfriend 1998). Another well-documented case is the period of 2004–2005 in which the FED triggered a restrictive FED funds cycle but long-term interest rates did not rise at all – they actually fell once the policy change was initiated.

The post-2000 experience is supported by the view that positive movements in the long rate in an environment of rising short-term rates is inconsistent with monetary theory. The reason is that a short-term hike should reduce inflation and hence, should support falling long rates (Romer and Romer 2000; Cochrane 2007b). On similar grounds, Gürkaynak et al. (2005b) report that long-term interest rates should respond only to a limited extent to policy changes, since in macroeconomic models short-term interest rates tend to return quickly to equilibrium after a shock hits the economy. On forward markets, the expected outcome should result in virtually no reaction of forward rates in the distant future.

Seemingly, partly antithetic approaches exist that try to give answers to how and why long-term bond securities respond to monetary policy. The Expectations Hypothesis predicts that, with short-rate expectations following the same direction of the initial policy move, bond yields move together; whereas the Fisher Hypothesis supports the view that a restrictive monetary policy should reduce inflation (expectations) and long-term interest rates. In order to adequately describe the links, one must allow for considerable refinements. This pertains to whether market participants react to monetary policy actions that have been anticipated or not and whether or not the information set on part of central banks and the private sector coincides or differs.

Interest rate policy actions that have been anticipated should likely not affect long-term rates much, if at all. The reason is simply that markets probably consider the whole sequence of the expected policy cycle. Depending on the monetary policy stance in place during the cycle, long-term yields would react sharply, not at all, or even in the opposite direction. Policy leverage depends on the expectations of what a given policy move implies for future short-term interest rates. An unanticipated central bank move should, however, have an immediate and pronounced impact on long-term rates. This “excess sensitivity” behavior can be further specified by arguing that when a central bank tightens monetary policy and market participants have not anticipated this move, the latter may infer that they are equipped with unfavorable information about the future dynamics of inflation; accordingly, they revise their expectations for inflation upward and provoke an increase in long-term interest rates. A monetary policy that responds to new and possibly private information about the state of the economy, triggers a shift in the yield curve in the same direction as the policy innovation. On the contrary, if a central bank changes its weight in the policy objective in favor of inflation stabilization, the yield curve rather rotates since economic agents adjust their inflation expectations downward (Romer and Romer 2000; Ellingsen and Söderström 2001; Ellingsen and Sderstrm 2004). In a nutshell, the positive response of long-term interest rates relies on the assumption that economic variables need to exhibit strong persistence to which monetary policy reacts. This persistence can be either unanticipated, for instance through time-varying inflation targets or through asymmetric information between policymakers

and market participants. Both approaches can contribute to the observed pattern of long-term interest rates and will be further discussed in subsequent Sections.

5.1.3 Implementation of Monetary Policy and the Yield Curve

Optimal interest rate rules or simple Taylor-style rules regard the short-term interest rate as the instrument with which a central bank conducts monetary policy. Although their aim is to alter the bond spectrum of longer-term maturities, policymakers and academics resile to directly use a selected long-term interest rate as instrument. If monetary policy directly influences a specific long-term rate, it inevitably takes action in the allocation of resources for durable consumption and investment decisions and may alter supply and demand schedules of the private sector, in particular between various capital market segments. Moreover, from the point of view of political economy and political independence of central banking, a trade-off emerges between the effectiveness of monetary policy and fiscal policy goals. Consider the case in which a monetary authority decides to use the 10-year government bond as instrument. With this knowledge, fiscal policy can supply debt instruments with this maturity structure without bearing the risk of sharp interest-rate fluctuations. In the short-term, monetary policy monetizes government debt but it may be forced in the medium run to take restrictive measures to restore macroeconomic stability according to its legislative commitment to price stability. In the end, such a strategy considerably soils the informational content of asset prices if a central bank acts as supplier and buyer of bond securities on capital markets.

Despite justifiable critics in giving long-term interest rates an instrument role in monetary policy, the experiences of Japan in the 1990s and the fears of US deflation in 2003 and 2009 have let both policymakers and academics rethink their view on the field of application of the term structure of interest rates. Bernanke (2002) remarks that in the face of the zero lower-bound of the short-term policy rate “a [...] method [...] which I personally prefer, would be for the Fed to begin announcing explicit ceilings for yields on longer-maturity Treasury debt (say, bonds maturing within the next two years). The Fed could enforce these interest-rate ceilings by committing to make unlimited purchases of securities up to two years from maturity at prices consistent with the targeted yields.” Indeed, one way to overcome the zero-bound is to carry out open market operations that can in turn provide signals that allow market participants to expect a particular short-rate policy path (Clouse et al. 2003). Long-term interest rates fall if these purchases lead to a market view where expected short rates are revised downwards – compared to the initial date before open market operations have been conducted. Notice that the long-term interest rate could also be translated into a short-term policy reaction function given the theory of the Expectations Hypothesis. When solved for the short-term interest rate, the typical reaction function including inflation and output is augmented by a term structure effect where the short rate is a function of its own expected path.

This approach of “targeting” the long rate is a type of targeting rule put forward by Svensson (2001) where monetary policy is committed to restore a link between short-term and long-term interest rates (McGough et al. 2005).

Besides the strategy of including long-term interest rates as instruments in times of deflation, another argument in favor of a more prominent role of the yield curve comes from informational requirements. The simple Taylor rule suggests that the short rate should react to current inflation and to the current output gap. Such a rule can hardly be feasible from an operational point of view. The period a central bank sets its interest rate, current inflation and output data are heavily measured with noise and they are subject to sampling errors and revisions, whereas the output gap is not observable at all. The term structure of interest rates provides high-quality financial data in real time while the variables in the Taylor rule are based on more slowly gathered macroeconomic information. Given its informational content, term structure measures can be explicitly incorporated in a modified reaction function. In this case, the policy rate responds to term-structure information that are in nature forward-looking. Against this background, monetary policy itself can be forward-looking in the sense that it reacts to expected inflation and output (Clarida et al. 1999). This specification allows a central bank to consider a broad array of information to form expectations about the future state of the economy. Under a forward-looking regime, interest rate decisions are based upon a broad set of conditioning information in a “data-rich environment” rather than on a few key aggregate variables (Bernanke and Boivin 2003).

The yield curve reflects market’s view on future inflation and near-time business cycle prospects expressed in terms of the long rate and the slope of the curve. Moreover, if one accepts the argument pursued in this Section that aggregate demand is mainly affected by interest rates of longer maturities, the logical implication for policy analysis is to evaluate whether a more direct response to these rates help in achieving the goals of price and output stability. McCallum (2005) sets up an interest rate rule in which a central bank reacts to the lagged policy rate (interest rate smoothing) and to the current term spread (yield curve smoothing). The rationale for this term-structure augmented rule is that monetary policy starts tightening whenever the yield spread is large. The latter serves as indicator of the expansiveness of monetary policy and represents a reliable indicator from a cyclical perspective provided that market participants value bond prices correctly according to expected output and a possible risk premium correction. At the same time, the long-rate hints to “inflation scares” that might call for policy moves if the long rate is unusual high (Goodfriend 1993). Empirical evidence suggests that selected term structure variables play a significant part in describing actual central bank behavior. Term-structure based reaction functions lead toward more stability in explaining the dynamics of policy rates across different monetary policy regimes and countries.¹¹ Even if one accounts for short-term inflation expectations through the inclusion of survey forecasts or its rational expectations forecast in the spirit of Clarida et al.

¹¹ See Mehra (2001), Gerlach-Kristen (2003), Fendel and Frenkel (2005) and Vázquez (2009).

(2000), long-term interest rates dramatically improve the track of actual policy moves, especially in times in which standard rules have produced large predictions errors.¹² Along these lines, Mehra (2001) shows that the latter are strongly positively correlated with the evolution of the 10-year bond rate indicating to the active role of the long-term rate in the short-term reaction function.

5.2 Joint Modeling Strategies of Interest Rates and the Macroeconomy

5.2.1 *The Macro-Finance View of the Term Structure of Interest Rates*

Although the ways of thinking about the economy substantially differ, two masterminds of their respective economic school come to one similar conclusion. The advocate of the Keynesian view, Paul Krugman (2009), recently stresses that “[...]economists will] have to do their best to incorporate the realities of finance into macroeconomics.” His opponent, Chicago economist John Cochrane (2008a, 242) finds that “[...] clearly, finance has a lot to say about macroeconomics, and it says that something is desperately wrong with most macroeconomic models. In response to this challenge, many macroeconomists simply dismiss asset market data. Somethings wacky with stocks, they say, or perhaps stocks are driven by fads and fashions disconnected from the real economy. It makes no sense to say markets are crazy and then go right back to market-clearing models with wildly counterfactual asset pricing implications. If asset markets are screwed up, so are macroeconomic models, so are those models predictions for quantities, and so are their policy and welfare implications. Many financial economists return the compliment, and dismiss macroeconomic approaches to asset pricing because portfolio-based models “work better” — they provide smaller pricing errors. This dismissal of macroeconomics by financial economists is just as misguided as the dismissal of finance by macroeconomists.”

The criticism is caused by the different methodical tools that are applied by the macro and finance literature. Indeed, research in macro and finance is disconnected as demonstrated by the differing approaches of modeling interest rates. In monetary economics, a major theme has been to use the term structure of interest rates in order to understand the relationship between interest rates, monetary policy and the macroeconomy. Modeling the yield curve typically relies on the Expectations Hypothesis despite its poor empirical fit as documented across sample periods and

¹²Clarida et al. (2000) estimate forward-looking Taylor rules that incorporate Feds expectations of these variables as rational ones and hence, they reflect $t - 1$ period information known to the Fed. Expected variables are replaced by their realized counterparts.

across countries. Financial economists, instead, focus on pricing and forecasting interest-rate instruments. These models are built on the premise of absence of arbitrage opportunities; they make it possible to extract risk-neutral expectations from observable security prices and they overwhelmingly proof the existence of risk premia being priced by investors. In this respect, risk premia account for most of the variation in long-term bond prices. Interest rates in such finance models are driven by a set of few unobservable factors extracted from the yield curve. These factors, in turn, imply no economic interpretation since they are not related to the macroeconomy.

A fruitful modeling strategy emerges if it becomes possible to combine powerful finance models with macroeconomic models. It allows us to understand, interpret and forecast term structure dynamics with macroeconomic fundamentals. In reverse, using a yield curve that is priced with macroeconomic variables, may augment the possibilities to extract valuable information of market expectations on key macro variables.

The central element of a joint modeling strategy is the short-term interest rate. From a macroeconomic perspective, the short rate is the policy rate of the central bank that sets the rate in accordance to its policy goals. It is typically modeled as a reaction function to inflation and output variables. Such interest-rate rules have been proven successful in explaining the actual dynamics of the short rate, although macro factors have been poor variables for fitting the long end of the yield curve. The short rate exhibits one characteristic, i.e. it is determined as an additive combination of multiple macroeconomic variables. Within the finance view, the short rate shares the same functional form; it is likewise a vector-based combination of risk (latent) factors. It provides the building block for interest rates of other maturities which can be represented as averages of risk-adjusted future short rates. Equipped with the no-arbitrage approach of asset pricing and its implied cross restrictions between short and long rates, macroeconomic dynamics should find their way through other interest rates and should amount for variations farther out in the yield curve.

The seminal work by Ang and Piazzesi (2003) on macro-finance modeling clarifies the joint macro-finance view: state dynamics of both risk factors and macro factors can be represented within the same state space form. To see this, state dynamics in general can be written as

$$X_t = \mu + \phi X_{t-1} + \Sigma \varepsilon_t \quad \text{with} \quad \varepsilon_t \sim i.i.d. N(0, I).$$

Bond yields are driven by either latent factors or observable macro factors collected in the vector X_t with covariance matrix $\Omega = E[\varepsilon_t \varepsilon_t^\top] = \Sigma I \Sigma^\top$. From a macro modeling perspective, this is the most general law of motion of the economy including equilibrium solutions to an economy with forward-looking agent behavior as well as simple reduced form representations. The short-term interest rate, in turn, evolves according to

$$i_t = \delta_0 + \delta_1^\top X_t$$

and may be interpreted from either a macro or finance view. Firstly, the short-term rate may be set according to a Taylor rule where δ_1 is parameterized appropriately to choose inflation and output from the state vector. Secondly, the loadings on the short rate may be simply specified in a most flexible way as it is commonly applied in finance models of the yield curve.

The last building block of a joint macro-finance model describes the mapping from state dynamics to the term structure of interest rates. In a general form, any yield curve model can be described as a financial factor model (see Chap. 3.4.1) in which bond yields are determined by a constant term and factor loadings which translate factor dynamics into bond yield movements

$$i_{n,t} = a_n + b_n^\top X_t.$$

Following Diebold et al. (2005), key questions emerge for jointly modeling interest rates and the macroeconomy. Firstly, there are various methods in the literature that specify the factor loadings for the yield curve model. The main advantage of describing the macro view and the finance view as above is that factor loadings can be parameterized to nest various classes of yield curve models including the Nelson-Siegel representation (Chap. 3.3), the essentially affine set-up (Chap. 3.5.2) or the simple Expectations Hypothesis version (Chap. 3.4.1). For instance, the EH version would imply that $i_{n,t} = n^{-1} \sum_{i=0}^{n-1} i_{1,t+i}$ which translates into a specific form of the factor loading, i.e. $b_n = n^{-1} \delta_1^\top (I - \phi)^{-1} (I - \phi^n)$.¹³ Consequently, the above set-up still allows to analyze macro-finance linkages in the “traditional” macro way in which the EH holds (Fuhrer and Moore 1995). On the contrary, it could be used as a framework for a fully-fledged finance model.

Secondly, although the Nelson-Siegel and affine latent factor models provide an accurate fit to the yield curve, little can be said about the nature of the underlying forces that drive its movements. Therefore, a significant research challenge would be to determine a link between macroeconomic fundamentals and the yield curve. A starting point is, in general, a specification of the state dynamics that can include both observable macro variables and latent factors. Ang and Piazzesi (2003) build up a VAR with measures of observable inflation, economic activity and additional latent factors. In this respect, a crucial modeling point is whether there are unidirectional links from macro factors to yield curve factors or whether a bidirectional link between the factors is favored. The choice of factors and the exact specification of the parameters of the macro-finance model define the degree of macro structure and the possible interlinkages between the driving variables of the economy and the term structure.

Finally, depending on the research question to be solved, the need to incorporate the concept of term premia may lead to demanding challenges. As suggested by many statistical tests, term premia tend to vary over time and may significantly alter macroeconomic dynamics. The contradiction of risk-neutrality may be seen as

¹³See Appendix C for a derivation.

important evidence for explicitly modeling risk premia within a joint macro-finance model. However, it turns out that finding a reliable modeling strategy is the “holy grail” in linking the macroeconomy and financial markets. Structural macro models are hardly capable of producing time-varying risk premia that can be documented in an estimated financial factor model.

In what follows, the most recent contributions to the macro-finance literature will be presented. Three broad research directions have emerged that all rely on above model set-up. However, they differ in specification and modeling issues regarding the underlying yield curve model, the structural form of macro and latent factors and, lastly, the inclusion of term premia.

5.2.2 VAR-Based Models

Examining the empirical relationship between the nominal term structure and the business cycle has been a major theme addressed by policy institutions and the academic community. The pioneering work of Sims (1980, 1986) on estimating (structural) vector-autoregressive systems initiated a bulk of macro-finance studies that have tried to detect the interconnections between monetary policy, interest rates and the macroeconomy. Early research has tried to identify various macroeconomic shocks that contribute to shifts and rotations of the term structure. These empirical studies had the aim of describing the stylized facts by directly modeling the links between interest rates and macro variables.¹⁴ They inferred the links through shocks to macro variables and monetary policy using impulse responses implied from the VARs. These “unrestricted” VAR studies share in common that only statements for those maturities can be made whose yields have been directly included in the VAR so that only their behavior can be inferred from macro dynamics.¹⁵ Insights into other bond prices with other maturities have been missing. This deficiency stems from the modeling assumption that cross-restrictions play no role in depicting connections between observable interest rates with different maturities. These unrestricted macro-finance models give only an insufficient picture of the yield curve. It is provided even more as latent factors that are essential in standard finance models can not enter the VAR.

Related to unrestricted VAR literature, attempts have been made to estimate restricted macro-finance VAR systems of yields under the validity of a financial factor model that imposes the Expectations Hypothesis (Campbell and Shiller 1991; Bekaert and Hodrick 2001). These systems do not include macroeconomic variables

¹⁴See for example Estrella and Mishkin (1997), Evans and Marshall (1998), Kozicki and Tinsley (2001), Wu (2003), Marzo et al. (2008).

¹⁵Notice, that in the following Section, the term “structural” and/or “restricted” does not refer to the econometric use of the words concerning model identification. They are mainly used to describe macro-finance models in terms of their structure and their cross-restriction properties between different bond prices.

but have only been applied to test the null of the EH. Within this set-up, an important step forward has been made by Favero (2006) who modeled the short rate in line with a Taylor rule. The macro-finance based VAR is used to construct projections of policy rates in which monetary policy is determined by inflation and output. The latter affect the long end of the yield curve by means of policy forecasts that themselves rely on forecasts of the relevant macro factors.

Although the Taylor-augmented VAR approach of representing EH-consistent yield dynamics fits the historical yield curve record “quite well”, the embedment of latent factors typically improves the description of bond prices. The use of the Nelson-Siegel curve is popular among market participants and central banks since it is relatively easy to estimate. Diebold et al. (2006) provide a macroeconomic interpretation of the dynamic three factor model by combining it with VAR dynamics for the macroeconomy including measures for inflation, output and monetary policy. They allow a bidirectional feedback from the yield curve to the economy and back again. The approach nests special classes of unrestricted VAR studies in which macro variables are either determined independently from the shape of the term structure; or it is assumed that there is only a unidirectional link from the yield curve to the macroeconomy.¹⁶ Most importantly, for the US, Diebold et al. (2006) find that a change in the slope factor is followed by a one-to-one response in the policy instrument. This comes either because the central bank reacts to yield dynamics or, given the frictions of monetary policy decision-making, interest rates react to macroeconomic information in anticipation of central bank actions. Upward changes in the level factor, i.e. changes in perceived long-term inflation expectations, are associated with positive impulses to all macro factors. An increase in the level factor lowers ex-ante real rates interest rates which is associated with a positive economic stimulus, i.e. an increase in capacity utilization. In turn, the slope factor responds directly to all macroeconomic shocks. This observation is consistent with the idea that the central bank increases the policy rate in times of inflationary and output shocks. The level factor is only influenced by inflation surprises which comes close to the assumption that long-term inflation expectations in the US are not firmly anchored (see Chaps. 6.3.2 and 6.4).

A rigorous no-arbitrage perspective is taken by Ang and Piazzesi (2003) who set up a model that allows to jointly describe the dynamics of the macroeconomy, the term structure *and* bond risk premia. They describe the state dynamics of the economy with observable macro and three latent yield factors whereas there is only a unidirectional link from the macro measures to the term structure. The short-term interest rate is determined by a Taylor rule augmented by factor loadings on the unobservable risk factors. Single reduced-form estimates on Taylor rules typically reveal that either persistent monetary policy shocks (correlated residuals and omitted variables, see Chap. 6.3.1) or interest-rate smoothing is necessary to fit central bank behavior. The macro-finance model takes this fact into account by including the set of latent factors into the short-rate equation. Risk premia are derived according to

¹⁶See Evans and Marshall (1998) or Estrella and Hardouvelis (1991).

the essentially affine representation of bond yields through a flexible market price of risk specification and constant state volatilities.¹⁷ As in standard latent-factor finance models, the stochastic discount factor has the functional form

$$M_{t+1} = \exp\left(-i_{1,t} - \frac{1}{2}\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}\right),$$

where prices of risk are assumed to be affine in the state vector of the VAR

$$\lambda_t = \lambda_0 + \lambda_1 X_t.$$

The relevant stochastic discount factor is derived in an ad-hoc version with no economic interpretation. For reasons of tractability, the authors assume that the unobserved factors do not interact at all with the observed macroeconomic variables (output and inflation). Yield curve factors do not affect macroeconomic outcomes and risk premia have no influence either. Still, the authors find that the macro factors mainly describe movements at the short end of the yield curve; whereas, the most persistent latent factor is the main driving force at the long end of the yield curve.

Building on the seminal work of Ang and Piazzesi (2003), a bulk of research has studied yield curve implications within the restricted macro-finance framework.¹⁸ They mainly differ in the way VAR dynamics are built, the number of included latent factors and the choice of constant or time-varying risk premia. All studies work out a common theme that emphasizes the importance of a long-term attractor for long-term interest rates. Whereas the short end of the yield curve mainly depends on macro factors, a macro-only model produces poor results for yields with longer maturities. The statistical description of the yields-only affine model in Chap. 4.2.1 has revealed that shifting endpoints of the yield curve are important features to describe the persistent time-series properties at the long end of the yield curve. Kozicki and Tinsley (2001) suggest that the missing factor is related to long-run inflation expectations which can be traced back to a central bank's inflation target. Against this background, a more structural interpretation of the VAR representation can give new insights into the logic of yield curve shifts.

5.2.3 *Semi-Structural Macro-Finance Models*

Restricted VAR-based macro-finance models lack sufficient economic structure in order to draw conclusions for policy analysis. A second strand of literature has incorporated more economic structure into the models by building structural

¹⁷See Chap. 3.5.2 for a discussion on the stochastic discount factor and market prices of risk.

¹⁸Bernanke and Reinhart (2004), Fendel (2004), Bundesbank (2006), Dewachter and Lyrio (2006), Bolder (2006), Ang et al. (2007), Lildholdt et al. (2007), Chernov and Mueller (2008), Pericoli and Taboga (2008).

VARs consistent with economic theory. Given its (still) workhorse model feature, a structural linearized New-Keynesian economy is typically assumed to represent the relationships between the macroeconomic variables. These semi-structural macro-finance models skip the latent factor approach and instead rely on aggregate demand and Phillips curve equations with forward- and backward looking expectations. Moreover, time-varying inflation targets or time-varying natural real rates should provide sufficiently persistence to fit the long end of the yield curve. In contrast to standard New-Keynesian studies, the use of an extended information set including interest rates along the yield curve makes the estimation and extraction of the perceived inflation target from bond yields much more tractable. The rational expectations solution of the economy is used to determine the law of motion for the term structure of interest rates. The latter is not solved within the model except for the short-term policy rate since there is no bidirectional link between the macro and finance block. Long-term interest rates have no effect on inflation or output, nor do term premia; financial market prices do not feedback to the macroeconomy. As it will be discussed later, this is exactly how the New-Keynesian economy is supposed to work.

Only the macroeconomic solution is necessary to pin down term structure implications. It can be calculated by standard techniques for solving and estimating (log-) linearized multivariate rational expectations models. Therefore, the stochastic discount factor follows the exogenous ad-hoc affine structure as in the VAR-based models above. Market prices of risk are kept flexible to produce time-varying risk premia in a manner determined by macroeconomic conditions that are derived structurally. The main advantage of this modeling strategy is that it is parsimonious and simple where expectations are allowed to be influenced by economic dynamics and term premia are associated with macroeconomic events (Rudebusch et al. 2007). New-Keynesian models with non structurally identified market prices of risk have been estimated for the US, the euro area and Germany.¹⁹ As suspected, perceived inflation targets have long lasting effects at the end of respective yield curves; whereas cyclical components trigger yield curve rotations.

5.2.4 Asset Pricing in a DSGE Model

VAR-based macro-finance models, as derived either in its unrestricted or its more structural form, can all be characterized by putting macroeconomic content into an otherwise standard finance model. Pricing the term structure of interest rates relies on a pricing kernel that is determined outside the macroeconomy. The sole link to macroeconomic factors is achieved through the short-term interest rate. From a theoretical point of view, setting up a general equilibrium model in which a set

¹⁹See Hördahl et al. (2006), Hördahl and Tristani (2007), Lemke (2008), Rudebusch and Wu (2008).

of asset prices does not appear, is clearly an insufficient way of representing an economy. Early attempts have been made to incorporate equities and bond securities but drawing implications for bond yields across the term structure or term premia within a fully-fledged general equilibrium model were missing.²⁰

Dynamic Stochastic General Equilibrium (DSGE) models have emerged as a triad of New Classical Macroeconomics, Real Business Cycle Theory and New-Keynesian economics. Its key features are the emphasis on rational expectations, the presence of intertemporal optimizing agents who adapt their paths of employment and consumption to sequences of stochastic shocks with micro-founded inflexibilities in prices and wages (Spahn 2009). These three components create an economy in which a representative household and a representative firm perform an optimization calculus integrating consumption, investment, labor supply and portfolio choices.²¹ Consequently, the economy is described by a system of non-linear first-order conditions that exhibit a deterministic steady state.

From an asset pricing perspective, the stochastic discount factor is determined within the model and directly derived from the utility function of the representative household. It is incorporated in the household's Euler equation, essentially consisting of the discounted ratios of marginal utility between two periods and possibly scaled by the expected change in the price level in case nominal bonds are traded. The DSGE model imposes cross restrictions between the macroeconomy and the stochastic discount factor where the driving factor dynamics and market prices of risk depend on the utility function arguments and on the degree of risk aversion – a simple example of such an economy will be laid out in the next Sect. 5.3. Statements on asset prices can then be made in an environment of “micro-founded” macroeconomics.²² If one sticks to the DSGE model, the

²⁰The general equilibrium models in the spirit of Lucas (1978) usually have simplified the economy by assuming exogenous processes for consumption, dividend growth or trivial production sectors. For an overview see Jermann (1998).

²¹In contrast to the standard New-Keynesian textbook model, DSGE models cover a more detailed view on the economy. Following Smets and Wouters (2003), households consume, decide how much to invest (asset purchases to transfer wealth) and are monopolistic suppliers of differentiated types of labor, which allow them to set wages. In turn, firms hire labor, rent capital and are monopolistic suppliers of differentiated goods which allows them to set prices. Both agents are confronted by a large number of nominal frictions which constrain their ability to reset prices or wages. On the real side, capital is accumulated in an endogenous manner and there are real rigidities arising from adjustment costs to investment or fixed costs. Households preferences display habit persistence in consumption, and the utility function is separable in terms of consumption, leisure and real money balances. Fiscal policy is usually supposed to work in a Ricardian setting, while monetary policy is conducted through a Taylor-type reaction function, in which the interest rate is set in response to deviations from an inflation target and some measure of economic activity. More recently, these models have been extended by including financial markets in particular financial frictions into the set up (Goodfriend and McCallum 2007; Canzoneri et al. 2008; Cúrdia and Woodford 2008).

²²Whether DSGE models are truly micro-founded or not is a matter of current debate. Critiques claim that these foundations are rather ad-hoc than economically justified neither from an empirical

micro derivation of the stochastic discount factor is essential; otherwise detecting market prices of risk from first principles would not be possible analytically and numerically.

Solving a non-linear macroeconomic model usually implies an approximation of the law of motion for the state variables around the non-stochastic steady state. Typically, the equations of a model are firstly (log-) linearized around the steady state. In a next step, the economy is represented as a linear difference model. Finally, its equilibrium paths and local stability are analyzed by standard techniques (Blanchard and Kahn 1980; Söderlind 1999). A major drawback of this strategy is that the non-linear system is approximated up to the first order which implies a manifestation of the well-known property of certainty equivalence in log-linearized models. Since economic agents act as if they were risk-neutral, the variability of the state variables do not alter the decision rules of the economy. Log-linearized methods do not take into account the information contained in the distribution of the shocks and therefore *expected* returns of any asset are identical. Though conclusions about term structure dynamics can be drawn in a first-order approximation which implies the pure expectations hypothesis, risk premia can not pop up. A second-order approximation to the solution produces a term premium that is nonzero but constant in which the variances of the shocks enter the decision rule of the economy; but if the aim is to generate time-varying premia, the model must be solved with a third-order approximation.

The use of a structural DSGE model for the investigation of relationships between risk premia, the term structure and the economy seems coherent from a general equilibrium perspective. However, there are also profound limitations. The structural approach suffers from substantial computational challenges since closed-form solutions do not exist. As reported by Rudebusch et al. (2007), numerical solutions are intractable except for very simple models that cannot match the data. In order to produce sizeable term premia, models need to assume very large values for the curvature of the utility function, large and highly persistent stochastic shocks or a high degree of backward-looking behavior of agents including interest-rate smoothing on part of the central bank. A general finding in DSGE models with term premium effects is that an appropriate parametrization to fit key macroeconomic variables such as inflation, output or wages produces average term premia that stand in sharp contrast to finance-based premia measures in terms of level and volatility; they tend to be far too low and stable. In contrast, even if one adjusts the models to match basic asset price moments, the implied macroeconomic moments

nor from an intuitive plausible perspective. The assumption of the representative agent who acts as a superior statistician defines away important challenges in macro theory such as coordination problems, interactions between heterogenous agents, learning aspects and imperfect markets (Spahn 2009; Colander et al. 2008). A defence in favor of DSGE models is that research has begun to incorporate the presence of heterogenous agents in a learning environment, imperfect credit markets, income constrained households in the spirit of Keynes, non-Ricardian fiscal policy regimes or the possibility of default (Goodhart et al. 2006; DeGraeve et al. 2008; Ratto et al. 2008; Annicchiarico et al. 2009; Fiore and Tristani 2009). Obviously, this comes at the cost of larger models.

are seriously distorted.²³ Still, the main advantage of considering nominal as well as real bond securities in a structural model is to assess how nominal rigidities are translated into asset price dynamics. Most recently, DSGE models which incorporate finance-based utility specifications with long-run economic risks, i.e. inflation and technology risks, are more successful in jointly describing the yield curve and the macroeconomy. Moreover, an heterogeneous agents environment, where only a limited number of agents can participate in financial markets to smooth consumption, prevents the possibilities of large risk sharing so that model-implied risk premium dynamics are more in line with actual data (Rudebusch and Swanson 2008a; DeGraeve et al. 2008).

By reason of computational burdens of asset pricing implications in fully structural DSGE models, a different approach has been taken up by the hybrid structural macro-finance literature. Like in typical New-Keynesian models, this strand sticks to the log-linear way of representing the economy's rational expectations equilibrium. The approach exploits the recursive nature of bond prices by first linearizing the equilibrium conditions of the macroeconomic model and then by assuming that the relevant arguments in the pricing equations are jointly log-normal distributed. As a result, the stochastic discount factor is entirely defined within the macro model including market prices of risk for each unit of shock. However, derived term premia are constant across time, unless explicit conditional heteroscedastic shock processes are assumed to hit the economy.²⁴

5.3 Term Structure Implications of New-Keynesian Macroeconomics

5.3.1 *Stylized Facts and Benchmark Results*

The modern approach of monetary policy highlights the importance of guiding and shaping private expectations about future policy actions. This forward-looking view has its root in the New-Keynesian framework in which aggregate demand is mainly determined by forward-looking expectations. The current short-term interest rate is, on its own, of little importance for spending decisions. Instead, aggregate expenditures are driven by the path of expected future short-term interest rates, especially as incorporated in long-term interest rates (along with adjustments for risk if needed). The practical implication for the effectiveness of monetary policy is how well a central bank is able to manage private expectations about

²³See Ravenna and Sepp (2006); DePaoli et al. (2007); Rudebusch and Swanson (2008a,b); Hrdahl et al. (2008).

²⁴This approach was suggested by Jermann (1998) and applied by Wu (2006); Emiris (2006); DeGraeve et al. (2009); Bekaert et al. (2010).

its future policy setting (Eggertsson and Woodford 2003; Gali and Gertler 2007). In order to extract the role of long-term interest rates in a baseline New-Keynesian framework, consider the log-linearized version of the aggregate demand relationship as measured by the output gap x_t with

$$x_t = E_t x_{t+1} - \sigma^{-1} (i_{1,t} - E_t \pi_{t+1} - r_{1,t}^{nr}) \quad (5.1)$$

where $i_{1,t}$ is the one-period, riskless nominal interest rate, π_t is the inflation rate and $r_{1,t}^{nr}$ is the one-period natural rate of interest which is assumed to follow a known stochastic stationary process. The output gap moves inversely to the short-term real interest rate gap with its sensitivity given by the intertemporal elasticity of substitution (σ^{-1}). Solving this equation $n - 1$ periods forward, the output gap can be represented in terms of the expected short-term real interest rate gaps over the relevant time horizon

$$x_t = -\sigma^{-1} E_t \sum_{j=0}^{n-1} (i_{1,t+j} - \pi_{t+1+j} - r_{1,t+j}^{nr}) + E_t x_{t+n}. \quad (5.2)$$

To give further intuition of the role of long-term bond yields, the pure version of the Expectations Hypothesis of the term structure is supposed to hold according to which the ex-ante real n -period bond yield equals the expected average real interest rate over the next n periods. The same holds for the expected average natural rate:

$$r_{n,t} = \frac{1}{n} \sum_{j=0}^{n-1} E_t [i_{1,t+j} - \pi_{t+1+j}]$$

$$r_{n,t}^{nr} = \frac{1}{n} \sum_{j=0}^{n-1} E_t [r_{1,t+j}^{nr}].$$

Aggregate demand can be represented in a modified way by the following equation

$$x_t = -n\sigma^{-1} (r_{n,t} - r_{n,t}^{nr}) + E_t x_{t+n} \quad (5.3)$$

which makes the output gap a variable depending on the difference between the n -period real long-term bond yield and its corresponding natural rate (the expected output gap for n periods in future should approximately get zero for large values of n , see Rudebusch and Williams 2006). Altering the real long-term interest rate relative to its natural rate counterpart, makes the management of expectations central for the conduct of monetary policy.

The inflation rate in the New-Keynesian model is determined by the following equation

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t + u_t$$

which can be transformed to

$$\pi_t = -\kappa\sigma^{-1}E_t \sum_{j=0}^{\infty} \beta^j \left(r_{n,t+j} - r_{n,t+j}^{nr} + x_{t+n} - \sigma/\kappa u_{t+j} \right). \quad (5.4)$$

The reformulated Phillips curve centers the pivotal role of the long-term real rate gap for inflation determination and the need to ensure macroeconomic stabilization through the whole *sequence* of expected future interest rates. The concept of the real rate gap makes clear that long-term interest rates play implicitly a central role in structural New-Keynesian (NK-) models because the expectations channel is fundamental to the monetary policy transmission mechanism.

As described extensively, a joint view on the macroeconomy and the yield curve enables to bring long-term interest rates into a macroeconomic analysis. Valuable information about macroeconomic expectations can be extracted from the yield curve; simultaneously, interpreting level and slope shifts of the term structure improves the understanding how to use the yield curve for policy analysis. For the conduct of monetary policy, the incorporation of the yield curve augments the otherwise parsimony of a standard model in which only a very limited information set is available (output, inflation, short-term interest rate). It is common sense, however, that policy operates in a data-rich environment (Bernanke and Boivin 2003); the yield curve efficiently augments the information set for policy makers and might support the correct identification of the sources of shocks and the optimal reaction to it. For that reason, a reduced-form New-Keynesian model in the spirit of Bekaert et al. (2010) is built and where the effects of macroeconomic shocks to the yield curve can be explored.

Although the following analysis relies on a reduced-form approach, the pricing kernel must be derived consistently with the underlying first-order conditions to fully understand the relationship between asset prices and macro dynamics. Since the simple NK-model abstracts from any investment, the derivation of the pricing kernel relies on the optimality properties of the representative household (more on that later).²⁵ The model contains three core equations: the intertemporal IS-curve, the Phillips curve and a monetary policy rule. These equations can be formulated with explicit micro-foundations as a dynamic general equilibrium model which exhibits endogenous persistence, sometimes characterized as a hybrid New-Keynesian setup.

The demand equation is based on forward- and backward-looking behavior where a representative household maximizes intertemporal utility with external habit persistence (Fuhrer 2000). The log-linear version (as deviation around the steady state) yields an aggregate demand function

²⁵ A textbook treatment of a reduced-form NK-model with investment as represented by Tobin's Q in the aggregate demand equation can be found in Gali and Gertler (2007).

$$y_t = (1 - \alpha)E_t y_{t+1} + \alpha y_{t-1} - \phi^{-1}(i_t - E_t \pi_{t+1}) + \varepsilon_{IS,t} \quad (5.5)$$

where y_t denotes log output, i_t is the one-period nominal interest rate and ϕ^{-1} captures the sensitivity of output to changes in the ex-ante real interest rate. Following Woodford (2003) and building on a modified version of staggered price-setting as in Calvo (1983), the hybrid version of the Phillips curve can be written as

$$\pi_t = (1 - \delta)E_t \pi_{t+1} + \delta \pi_{t-1} + \kappa(y_t - y_t^n) + \varepsilon_{AS,t}. \quad (5.6)$$

Inflation π_t depends on expected and past inflation as well as on the output gap $x_t = y_t - y_t^n$. Here, the natural rate of output y_t^n is the output that would prevail in the case of perfectly flexible prices. In this respect, the natural output process is characterized by an autoregressive process of order one

$$y_t^n = \tau_{y^n} y_{t-1}^n + \varepsilon_{y_t^n}. \quad (5.7)$$

Monetary policy is supposed to specify a monetary policy reaction function according to which it sets its current policy rate i_t in line with macroeconomic conditions, but with a tendency to follow a smoothing strategy

$$i_t = \rho i_{t-1} + (1 - \rho)[\rho(\pi_t - \bar{\pi}_t) + \chi(y_t - y_t^n)] + \varepsilon_{i,t}. \quad (5.8)$$

Finally, following Gürkaynak et al. (2005b), the inflation target $\bar{\pi}_t$ follows the law of motion

$$\bar{\pi}_t = \tau_{\bar{\pi}} \bar{\pi}_{t-1} + \nu(\pi_t - \bar{\pi}_t) + \varepsilon_{\bar{\pi},t} \quad (5.9)$$

so that it loads to some extent on the most recent history of inflation.

The macroeconomic model can be expressed in state space form where $X_t = [\bar{\pi}_t, y_t^n, \pi_t, y_t, i_t]$, $\varepsilon_t = [\varepsilon_{j,t}]$ with $j = \{\bar{\pi}, y^n, AS, IS, i\}$ and the matrices $\mathcal{E}$, $\tilde{\gamma}$, $\tilde{\psi}$, $\tilde{\Lambda}$ are defined so as to fit the structural model

$$\begin{aligned} \mathcal{E} X_t &= \tilde{\gamma} X_{t+1} + \tilde{\psi} X_{t-1} + \tilde{\Lambda} \varepsilon_t \\ X_t &= \gamma X_{t+1} + \psi X_{t-1} + \Lambda \varepsilon_t \end{aligned} \quad (5.10)$$

with $\gamma = \mathcal{E}^{-1} \tilde{\gamma}$, $\psi = \mathcal{E}^{-1} \tilde{\psi}$ and $\Lambda = \mathcal{E}^{-1} \tilde{\Lambda}$. Shock processes follow $\varepsilon_t \sim i.i.d.N(0, \Omega)$ with $\Omega = \text{diag}(\sigma_j^2)$. Since the model is already log-linearized, the rational expectations equilibrium can be expressed as

$$X_t = \Theta X_{t-1} + \Gamma \varepsilon_t. \quad (5.11)$$

What are the term structure implications of above New-Keynesian model setup? To shed light on this question, the essentially affine factor representation is applied.

Against this background, term structure effects are reproduced within the log-linear/log-normal strategy where the macroeconomic system is solved with standard techniques and the stochastic discount factor is explicitly derived from the household's first-order condition. To get closed-form solutions, the utility function and the budget constraint must be further specified. In order to derive a hybrid aggregate demand equation, consumer preferences must be exploited in an environment of habits. Bekaert et al. (2010) assume that the household maximizes

$$E_t \sum_{t=0}^{\infty} \beta^t \left[\frac{F_t C_t^{1-\sigma}}{1-\sigma} \right]$$

where C_t is the consumption index, F_t represents a shifting factor and β is the discount rate. To introduce habit formation, F_t is specified as $F_t = H_t$ where H_t is an external habit level that depends on past consumption so that $H_t = C_{t-1}^\eta$ and η measures the degree of habit persistence. The maximization of the objective function is subject to the following budget constraint

$$P_t C_t + \sum_{n=1}^N P_{n,t} B_{n,t} = Y_t + \sum_{n=0}^{N-1} P_{n,t} B_{n+1,t-1}$$

where households can carry over a portfolio of nominal zero-coupon bonds with maximal maturity N . They use current income Y_t and financial wealth brought over from the previous period to consume $P_t C_t$ or to alter their portfolio by purchasing or selling nominal bonds. The variable $B_{n,t}$ is the weight given to each asset in the portfolio with price $P_{n,t}$.²⁶

The intertemporal marginal rate of substitution in the model is defined as

$$M_{t+1} = \beta \frac{\vartheta_{t+1}}{\vartheta_t} \frac{P_t}{P_{t+1}}$$

where ϑ_{t+1} equals

$$\vartheta_{t+1} = \beta C_{t+1}^{-\sigma} C_t^\eta$$

from the first-order condition with respect to C_t in the household's maximization problem. The logarithm of the pricing kernel with the resource constraint $Y_t = C_t$ gives

$$m_{t+1} = \log \beta - \sigma y_{t+1} + (\sigma + \eta) y_t - \eta y_{t-1} - \pi_{t+1} \quad (5.12)$$

²⁶The notation of nominal bond holdings is such that each period a bond becomes a one-period bond and is, thus, not available anymore in the next period to form the portfolio.

which is consistent with the aggregate demand (5.5). The pricing kernel is used to price all nominal bonds in the economy through the no-arbitrage equation

$$P_{n,t} = E_t [M_{t+1} P_{n-1,t+1}]$$

so that for the one-period interest rate it must hold

$$E_t[M_{t+1}(1 + R_{1,t})] = 1. \quad (5.13)$$

With the short-term interest rate and the pricing kernel jointly log-normal distributed, the pricing kernel can be expressed as

$$\begin{aligned} E_t[m_{t+1}] + i_{1,t} + \frac{1}{2} \text{var}_t[m_{t+1}] &= 0 \\ m_{t+1} &= -i_{t,1} - \frac{1}{2} \lambda_0^\top \Omega \lambda_0 - \lambda_0^\top \varepsilon_{t+1}. \end{aligned} \quad (5.14)$$

In a New-Keynesian macro-finance model, risks are evaluated in marginal utility ϑ_t terms. Notice that the stochastic discount factor can be represented within the reduced-form (5.11), leading to a vector of prices of risk entirely restricted by the structural parameters

$$\lambda_0^\top = [0 \ 0 \ 1 \ \sigma \ 0] \Gamma - [0 \ 0 \ 0 \ \sigma + \eta \ 0]. \quad (5.15)$$

Following the recursive solution algorithm described in Appendix D and using the rational expectations VAR in 5.11), interest rates can be derived as

$$\begin{aligned} A_n &= A_{n-1} - B_{n-1}^\top \Gamma \Omega \lambda_0 + \frac{1}{2} B_{n-1}^\top \lambda_0 \Omega \lambda_0^\top B_{n-1} - \delta_0 \\ B_n^\top &= B_{n-1}^\top \Theta - \delta_1^\top \\ a_n &= -n^{-1} A_n \\ b_n^\top &= -n^{-1} B_n^\top \\ i_{n,t} &= a_n + b_n^\top X_t \end{aligned}$$

where the short-term interest rate is given within the recursive solution of (5.11) so that $\delta_0 = 0$ and $\delta_1 = [0, 0, 0, 0, 1]^\top$ picks $i_{1,t}$ from X_t .

In the model, the term premium is constant because the second moments are time-invariant. It does *not* interact with the economy, as the one-period interest rate is the central variable that equates current and future output. Along the same lines, required risk compensation is consistent with the first-order conditions in every case though it is a completely endogenous variable. It cannot affect the output level and inflationary dynamics since there is no way in which an “exogenous shock” to the term premium feeds through the model’s logic. This insight holds in each prototype

Table 5.1 Baseline parameter values for the NK-Benchmark

Calibration values					
α	0.4	τ_{y^n}	0.90	σ_{AS}	0.003
δ	0.4	$\tau_{\bar{\pi}}$	0.88	σ_{IS}	0.003
σ	3.15	ν	0.02	σ_i	0.003
η	4.29	ρ	1.5	$\sigma_{\bar{\pi}}$	0.001
ϕ	$\sigma + \eta$	χ	0.5	σ_{y^n}	0.001
κ	0.064	ρ	0.7		

DSGE model, no matter how specified it is (Rudebusch et al. 2007). Long-term interest rates only matter for aggregate demand as they imbed information of the expected path of one-period interest rates that equalize current and expected output in the first-order condition for optimal consumption; term premia do not play any role in altering economic outcomes.

In what follows, the dynamics implied by the structural model are described using standard impulse response analysis. The baseline parameter values are reported in Table 5.1; they are mainly taken from Bekaert et al. (2010) and are standard in literature. The backward-looking components α, δ of the aggregate demand and supply equation are each set to 0.4 implying a slight dominance of forward-looking behavior. The sensitivity of inflation to the output gap κ is set 0.064 mirroring the success of empirical estimates to get significant results for it. The curvature parameter is set to $\sigma = 3.51$ that is common in macro and finance models. Lucas (2003, 4) suggests that the coefficient of risk aversion should generally lie between 1 and 4. Habit persistence is calibrated as $\eta = 4.29$. Notice that although the parameter α strictly depends on the latter two, it is specified in an ad-hoc way since the structural parameters are only needed to represent the term premium. Furthermore, Bekaert et al. (2010) show that a structurally derived α is close to the value reported in the table. The monetary policy reaction function is standard according to which the monetary authority responds to the inflation gap and the output gap ($\rho = 1.5$ and $\chi = 0.5$). Interest-rate smoothing is conducted with the parameter ρ set to 0.7. Finally, the processes for the inflation target and the natural output are supposed to be very persistent with $\tau_{y^n}, \tau_{\bar{\pi}} = 0.90$ and 0.88 respectively. In contrast to the estimated model of Bekaert et al. (2010), the model is specified for a quarterly frequency; time-series variables are continuously compounded to match the affine factor setting so that annualized shock variances are divided by 400. An annualized standard deviation of 1.2% points would yield a quarterly value of 0.003.

To understand the response of the term structure to the underlying structural shocks, it is useful to start analyzing the dynamics in terms of output, inflation and the short-term policy rate. Figure 5.1 reports the dynamic responses of the key macroeconomic variables for the five exogenous stochastic shocks. A shock to aggregate supply shows a strong initial effect on current inflation simultaneously followed by an increase in the short-term interest rate. Since inflation expectations are firmly anchored in the Taylor-based policy regime, the resulting increase in the ex-ante real interest rate pushes output down but effects cancel out quite quickly – a

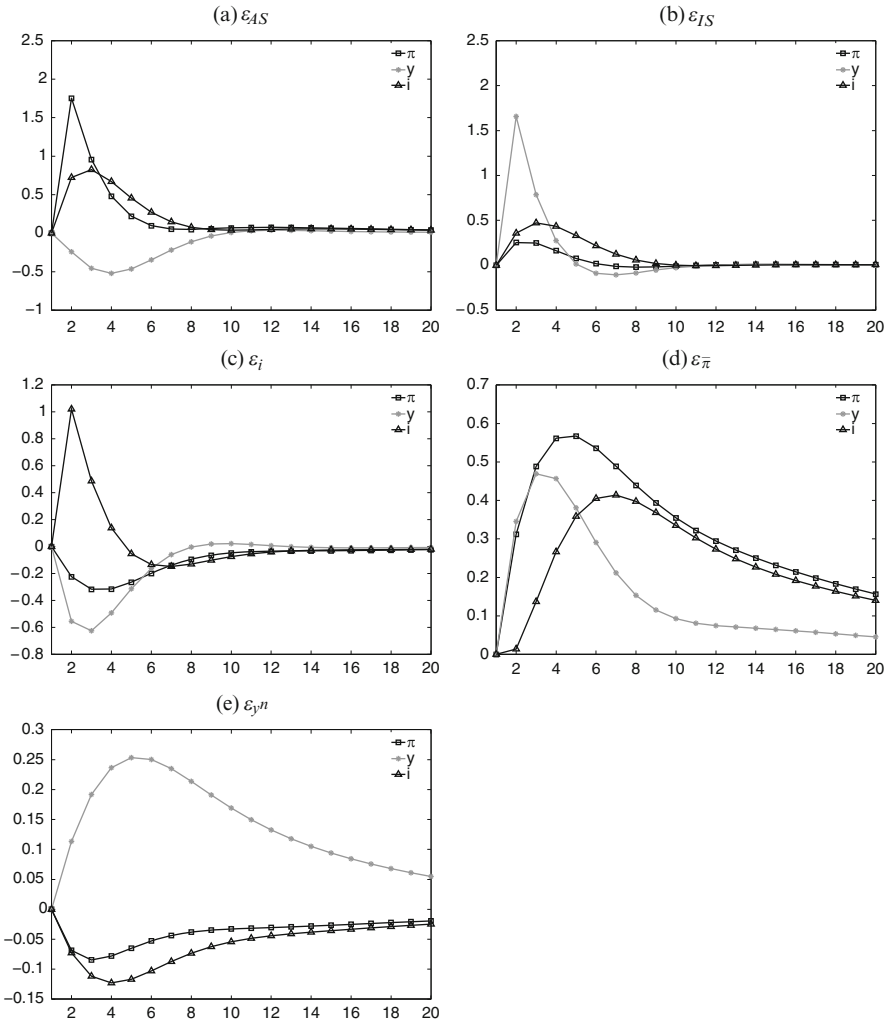


Fig. 5.1 Impulse response analysis NK-Model

result implied by the forward-looking nature of market participants. Similarly, a positive demand shock leads to a strong initial rise in output, inflation and the policy rate. A shock to the monetary policy rule triggers a simultaneous decline in inflation and output; this shock may be interpreted either as a discretionary impulse or as a central bank response to a variable set not being considered in the macro model. Obviously, the most persistent effects within the monetary transmission channel stems from stochastic variations in the inflation target and in the change of the natural output. A temporary rise in the inflation target is associated with a transitory increase in inflation and output. Due to its re-enforcing feedback-loop from current

inflation dynamics to the target rate, this process is rolled over for a significant period of time. Even after 20 periods, there are lasting amplitudes. The same holds for a shock to natural output whose impact on inflation and interest rates is slightly smaller compared to target moves.

Figure 5.2 displays the initial effects of the structural shocks to the yield curve. Figure 5.2a shows the factor loadings b_n for time to maturity given the state dynamics of the macro model. The factor decomposition is not attributed to the respective shocks, but to the state variables that can be hit by the shocks. At the short end of the yield curve, most of the contribution of yield curve movements is

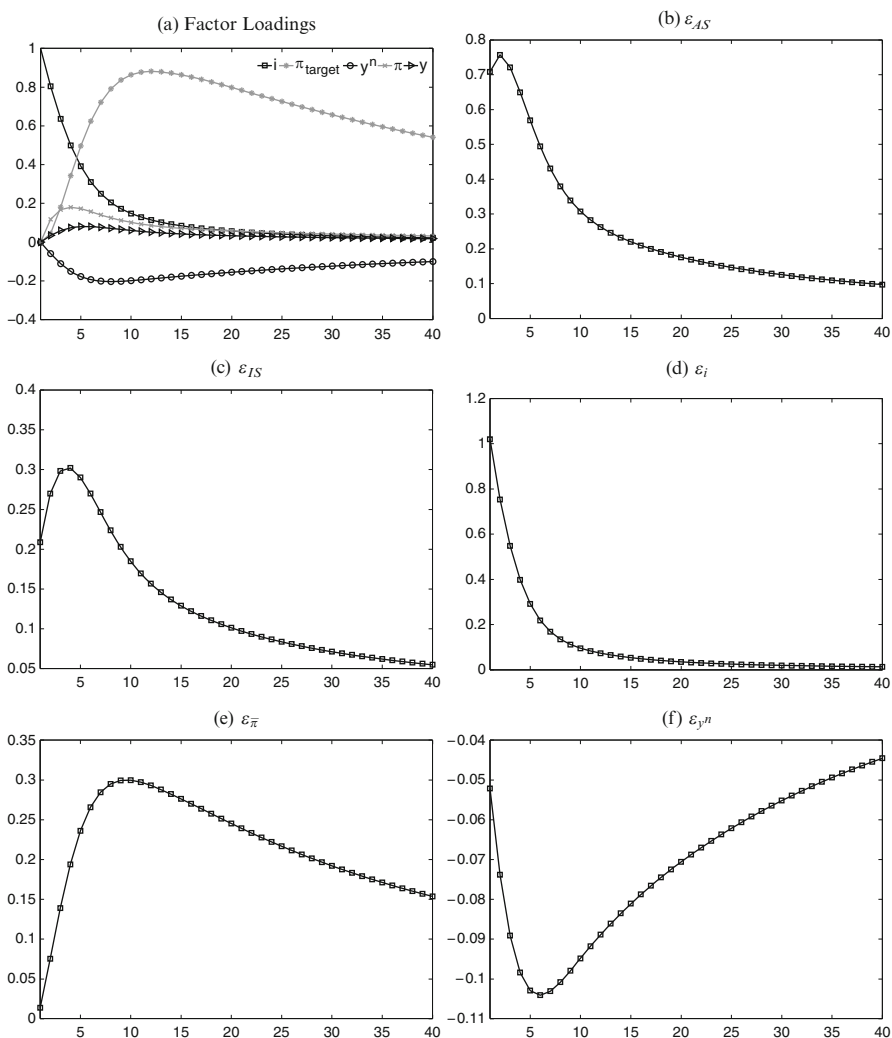


Fig. 5.2 Initial yield curve effects

associated with changes in the policy rate. This policy effect slowly decays in favor of the more persistent variables of the macroeconomy. The more persistent a state variable tends to be, the higher is its impact at the long end of yield curve. The most pronounced effect comes from the inflation target whose associated loading can be clearly understood as a positive level shifter. Similarly, movements in the natural output trigger a negative level move across the whole maturity spectrum. Transitory inflation and output dynamics are reflected in a humped-shaped loading on the yield curve and exhibit virtually no impact on the long end.

The remaining subfigures of Fig. 5.2 show the initial move of the yield curve in the period a single structural shocks hits the economy. To start with, a shock to aggregate supply (cost-push shock) provokes a sharp inverted character of bond yields with small impacts on longer-term interest rates. Notice, that the initial response of the short-term rate is less than one in case of interest-rate smoothing which is a well-known feature when studying dynamic macro models with such a policy rule (Woodford 2003). A positive aggregate demand shock ε_{IS} likewise leads to a negatively sloped term structure that shows negligible effects at the long end. In both cases, there is a hump-shape effect at around five quarters which can be also attributed to an interest-rate smoothing central bank. A monetary policy shock is associated with a strong response at the short end of the yield curve. Since the expectations hypothesis holds, a period of decreasing short-term rates for stabilization purposes is expected so that the impact on the yield curve cancels out quickly. This finding stands in contrast to the results of Ellingsen and Sderstrm (2004) who apply an empirical New-Keynesian model with multi-period lags and leads. They find that a monetary policy shock has a significant effect for all interest rates if the central bank possesses private information for the future macroeconomic environment.

In case of a transitory but highly persistent positive inflation target shock, monetary policy needs to shift its policy rate to negative values compared to the steady state equilibrium. Here, this policy is smoothed due to the policy parameter ρ . Still, the yield curve steepens since market participants adjust their inflation expectations upwards so that expected short-term interest rates pick up, and contemporaneous long-term interest rates move in opposite direction to the short-term interest rate – or at least they react more sensitive to the underlying inflation target shock. As a positive shock to the natural output triggers a monetary policy easing, the whole yield curve shifts down. Markets infer a sufficiently long period of below steady-state interest rates and translate this view into initial higher long-term bond prices and lower interest rates.

A general finding of the analysis is that the response of the n -period rate to a shock decays with increasing maturity. Moreover, a high autocorrelation of shock processes induces an environment of lasting dynamics around the steady state which in turn provokes yield curve effects even at the very long end. This becomes in particular clear when inspecting the factor loadings in b_n for the state variables. The contemporaneous term spread effects should be then smaller in case of the presence of persistent macro shocks. Figures 5.3a–b plot the time-series properties of the level (20 quarter yield) and the slope (20–1 quarter spread) component. As suspected, the

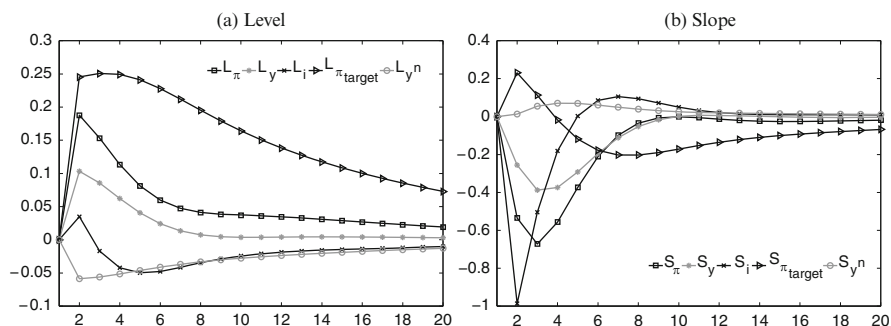


Fig. 5.3 Impulse responses of term structure factors

most persistent shocks produce long lasting responses of the level factor; whereas a monetary policy shock virtually does not alter long-term interest rates. In contrast, the term spread widens the strongest for the inflation, output and monetary policy shock with zero autocorrelations – albeit the inflation target contributes to sustained non-zero spreads after eight quarters.

In order to keep matters straight, implications for the average term premium will be discussed in Chap. 7.3.3 in which the effects of different monetary policy regimes for financial risk premia are discussed.

5.3.2 An Extension: Learning, Volatility and Persistence

The New-Keynesian model of the Sect. 5.3.1 assumes the standard methodology for studying macroeconomic dynamics, i.e. the dogma of rational expectations (RE). It is the central idea of how agents in the economy, including policy makers, form expectations about the future state of the macro variables. This hypothesis is based on the assumption that agents know the true model of the model except for unforecastable shocks that are not correlated with the information set available to agents. Working with the concept of rational expectations implies that conditional expectations are formed with the efficient use of all current information dealing with the structure, the quantification of parameters, the kind of policy formation and various stochastic processes. In other words, RE coincides with the case of perfect knowledge.

Given the pivotal role of expectations in standard macro models, it is doubtful whether RE equilibria on the part of agents can be computed in advance. In this respect, Spahn (2004, 1) points out that “in order to [compute this kind of equilibrium] and to act in accordance with [its] interest, one group of agents needs information on the expectations of a second group in the first place – which is not available because these other agents react to the behavior of the first group and, of course, to shocks.” In light of these strategic complementarities, the only feasible

way of forming expectations is that of *adaptive learning*. Within this framework it is acknowledged that agents possess only imperfect information. Typically, this means that agents in the model are characterized by bounded rationality since they formulate and estimate their models and update them as new information becomes available. They act like econometricians when they make forecasts of future prices and quantities of economic variables needed in their decision making. They make supply-demand choices deriving from these forecasts and interact with other agents on markets. The resulting outcome, in turn, generates new prices and quantities which enter the decision rules for the next period. Over time, “these estimates may become progressively nearer to being correct” (McCallum 2009, 3) since a rather strong assumption remains in the adaptive learning literature: it is assumed that agents *know* the true structure of the economy and agents (just) need to regularly update their beliefs about the sign of the structural parameters through a learning algorithm. Model uncertainty is ruled out; whereas uncertainty about the underlying parameters set takes center stage in modeling the economy – a rather stark restriction for a permanently changing economy.

In an environment of learning, the crucial question is whether the market process can be understood by agents so that a unique stable equilibrium can be reached based on past market experience. If this process leads in the limit to a particular RE solution path, then the solution is said to be learnable and “E-stable”. If, instead, a small displacement from it triggers a process that diverges from the fixed RE-point, then the market process is not learnable (Evans and Honkapohja 2001). When inspecting a model economy with learning, a first task to be done is to check for the existence of a unique stationary RE-equilibrium so that the model is determinate.²⁷ Secondly, the presence of learning agents requires a second task, i.e. to evaluate whether the E-stability principle holds under an appropriate statistical learning rule. In this respect, any RE solution can be described as a stochastic process with a particular parameter set $\bar{\phi}$. Under adaptive learning, agents do not know the true parameter set $\bar{\phi}$ but they estimate it as time passes using all available information. The issue is whether the estimated parameter set $\phi_t \rightarrow \bar{\phi}$ as $t \rightarrow \infty$ (Evans and Honkapohja 2001, 40, 237–238). Stability under learning can be determined by the E-stability equation

$$\frac{d\phi}{d\tau} = T(\phi) - \phi \quad (5.16)$$

in the neighborhood of the RE-solution where $T(\phi)$ is the mapping from the perceived law of motion ϕ to the true one and τ is notional time. The T-map, thus,

²⁷Most recently, McCallum (2009) elaborates the interconnection between determinacy and learnability of economic systems. Determinacy, as defined by the existence of a single RE solution, is neither sufficient nor necessary for the stable evolution of market processes. The central task to work on is the examination of whether the model is learnable. Though a model may possess more than one stable solution, the criteria of learnability allows to pin down the only one solution that “[...] could prevail in practice” McCallum (2009, 10).

takes perceived parameter coefficients to actual coefficients. Stability is ensured if all eigenvalues of the system of differential equations have real parts less than one.

In recent years, the E-stability principle for standard monetary models has been applied in many studies.²⁸ The central message of this work is that monetary policy should ensure and facilitate learning by private agents so that expectational dynamics do not result in instability of the economy. Simple monetary policy rules which react to current values or to contemporaneous expectations of macroeconomic variables are most desirable in terms of generating both determinacy and learnability. Some degree of history-dependence on part of policy-making is essential in order to ensure an equilibrium path that converges to the unique RE solution. Bullard and Mitra (2002) emphasize the importance of the “Taylor principle” in generating stable and determinate interest rate rules. Forward-looking policy rules, instead, do not always provide these desirable properties; in particular, when monetary policy strongly reacts to forward-looking expectations of the output gap, sunspot equilibria could be the outcome.²⁹ The same holds for optimal targeting rules which call for a timeless perspective so that monetary policy targets lagged macroeconomic variables. The feedback from inflation and output expectations to their current counterparts is less than one because deviations from the RE equilibrium are offset by policy in a way that market participants are guided to build expectations in line with the optimal RE.

There arises a further crucial aspect with respect to the conduct of monetary policy. It deals with the choice of explicitly incorporating the learning process itself into the optimal policy problem. Either the central bank sets interest rates under the assumption of perfect knowledge even when the private sector has only imperfect knowledge (naïve policy), or it is fully aware of the learning mechanism and adjusts policy accordingly (efficient policy). Comparing welfare losses within each policy regime reveals that a more vigilant response to inflation reduces inflation persistence and makes the economy less prone to costly stagflationary episodes (Orphanides and Williams 2005a; Gaspar et al. 2006). A clear communication strategy of a numerical inflation target also mitigates the influence of learning agents on macroeconomic outcomes since it helps to minimize confusion about the central bank’s objectives and supports an anchoring of long-term inflation expectations.

A number of empirical applications of the learning environment has recently developed in order to analyze the past evolution of inflation dynamics and monetary policy for selected countries.³⁰ The main finding is that the learning approach plays a central role in the historical explanation of inflation. In particular, Milani

²⁸See Sack (1998), Bullard and Mitra (2002), Locarno (2006), Bullard (2006), Evans and Honkapohja (2003), Evans and Honkapohja (2006), Evans and Honkapohja (2008).

²⁹Sunspots occur whenever the market process depends on extraneous random variables that influence the economy solely through the expectations of the agents.

³⁰See Bullard and Eusepi (2005), Sargent et al. (2006), Sargent et al. (2009), Orphanides and Williams (2005b), Milani (2007), Eusepi and Preston (2008).

(2007) shows that learning can help to endogenously generate persistence in key macroeconomic variables which is only feasible to reproduce in standard RE models if one allows for mechanical a priori, but rather ad-hoc micro foundations that result in persistence of both inflation and output. Moreover, misspecifications about potential output or the natural rate of unemployment on part of a learning monetary authority can explain historical episodes such as the high inflation periods in the 1970s in the US and in other industrialized countries.

Against this background, the concept of adaptive learning can be applied to the New-Keynesian model set-up, whose RE properties have been already presented in the Sect. 5.3.1. To do so, one needs to re-write the model in reduced form with a slightly different representation of the shock processes as

$$X_t = AX_{t+1} + BX_{t-1} + Cu_t \quad (5.17)$$

$$u_t = \rho u_{t-1} + \varepsilon_t \quad (5.18)$$

where $X_t = [\bar{\pi}_t, i_t, \pi_t, y_t]$, $u_t = [o_t, p_t, q_t, r_t, y_t^n]$, $\varepsilon_t = [\varepsilon_{\bar{\pi},t}, \varepsilon_{i,t}, \varepsilon_{IS,t}, \varepsilon_{AS,t}]$ and $\rho_{5,5} = \tau_{yn}$ with zero entries elsewhere. Following McCallum (1983, 2007), the model is solved for the RE equilibrium following the Minimum State Variables (MSV) solution to the system; it takes the form

$$X_t = \bar{a}X_{t-1} + \tilde{c}u_t$$

or

$$X_t = \bar{a}X_{t-1} + \bar{c}u_{t-1} + \bar{d}\varepsilon_t \quad (5.19)$$

with $\bar{c} = \tilde{c}\rho$ where the matrices can be solved by the method of undetermined coefficients. Under this RE assumption, agents know the true structure of the economy, its relevant parameters $\bar{\phi} = (\bar{a}, \bar{c}, \bar{d})$ and make use of this knowledge to form expectations of the future state of the economy.

Under adaptive learning, each period agents have in mind a perceived law of motion (PLM) which has the same functional form as the MSV solution. Since they do not know the true parameter set associated with the RE equilibrium, they use past data and a learning algorithm to obtain parameter estimates. The PLM follows

$$X_t = a_{t-1}X_{t-1} + c_{t-1}u_{t-1} + d_{t-1}\varepsilon_t. \quad (5.20)$$

It is important to note that due to its forward-looking nature, there is the problem of simultaneity, in the sense that the solution to X_t and the estimated coefficients ϕ_t have to be solved simultaneously, if agents use ϕ_t to form expectations in t . Therefore, following Carceles-Poveda and Giannitsarou (2007), it is

assumed that agents forecast X_{t+1} taking their estimates from the previous period ϕ_{t-1} .³¹

Agents use the PLM to make forecasts for $t + 1$

$$E_t^* X_{t+1} = a_{t-1} X_t + c_{t-1} u_t \quad (5.21)$$

where the operator E_t^* denotes non-rational expectations; moreover, it is assumed that the VAR process for the exogenous shocks ρ is known to all agents as is commonly taught within the learning literature.³² To this end, a constant term is not included in the expectations formation; it relies on the assumption that agents know that they are estimating deviations from a steady state. By inserting expectations into the reduced form (5.17) and substituting the exogenous shock process, one obtains

$$\begin{aligned} X_t &= A(a_{t-1} E_t X_t + c_{t-1} u_t) + B X_{t-1} + C u_t \\ &\Leftrightarrow (I - A a_{t-1}) X_t = B X_{t-1} + (A c_{t-1} + C) u_t \\ X_t &= \frac{B}{I - A a_{t-1}} X_{t-1} + \frac{(A c_{t-1} + C) \rho}{I - A a_{t-1}} u_{t-1} + \frac{A c_{t-1} + C}{I - A a_{t-1}} \varepsilon_t. \end{aligned}$$

Thus, an associated mapping from the PLM to the actual law of motion (ALM) is given by

$$\begin{aligned} T_1(\phi_{t-1}) &= \frac{B}{I - A a_{t-1}} \\ T_2(\phi_{t-1}) &= \frac{(A c_{t-1} + C) \rho}{I - A a_{t-1}} \\ V(\phi_{t-1}) &= \frac{A c_{t-1} + C}{I - A a_{t-1}} \end{aligned}$$

and $\{\bar{a}, \bar{c}, \bar{d}\}$ is a fixed point of this map. The ALM can be then described within a state space representation as

$$Y_t : \begin{pmatrix} u_t \\ X_t \end{pmatrix} = \begin{pmatrix} \rho & 0_{n \times k} \\ T_2(\phi_{t-1}) & T_1(\phi_{t-1}) \end{pmatrix} \begin{pmatrix} u_{t-1} \\ X_{t-1} \end{pmatrix} + \begin{pmatrix} I \\ V(\phi_{t-1}) \end{pmatrix} \varepsilon_t \quad (5.22)$$

with $Y_t = [u_t, X_t]^\top$. This means that at the beginning of a period, the current variables are realized as implied by the ALM which are then used to update the

³¹Alternatively, one could assume that the state variables in X_t are not included in the information set at period t so that expectations are formed with an information set up to $t - 1$.

³²This stark assumption can be macerated by treating the times series process of the shock process as unknown, too. This would end up in a unrestricted forecasting VAR with little economic structure.

coefficient estimates stacked into the vector $\phi = [\text{vec}(a_t, b_t)^\top]^\top$. The actual state is also used in combination with previous estimates to form expectations about the state variables in the next period. This outcome repeats itself at the next period $t + 1$ and so on.

More formally, using the realized observation of X_t , agents run the regression

$$X_t = Y_{t-1}^\top \phi + \psi_t$$

to get new estimates of ϕ . Typically, the literature assumes that agents perform learning under least squares estimates which yields

$$\phi_t = \left(\sum_{i=1}^t Y_{i-1} Y_{i-1}^\top \right)^{-1} \sum_{i=1}^t Y_{i-1} X_i$$

for the parameter estimates. The first term in brackets resembles the variance of the regressors and the second term describes the covariance between the vectors Y_{t-1} and X_t . The main advantage of least squares is that the estimator can be written in recursive form so that

$$\phi_t = \phi_{t-1} + g R_t^{-1} Y_{t-1} (X_t - Y_{t-1}^\top \phi_{t-1}), \quad (5.23)$$

$$R_t = R_{t-1} + g (Y_{t-1} Y_{t-1}^\top - R_{t-1}). \quad (5.24)$$

where R_t is the matrix of second moments of the regressors and g is the gain resembling the degree of memory of forecasting agents.

The specification of the gain allows the implementation of two learning algorithms, i.e. the recursive least square algorithm associated with decreasing gain learning (RLS) and a logarithm consistent with constant gain learning (CG). Under RLS, the gain takes on the value $1/t$, so that agents use all available past data to extract efficient estimates of the relevant parameter set. In this respect, each data point counts equally and the coefficients are updated in response to the most recent data point with a weight proportional to the sample size $1/t$. The weight on each new data point declines and approaches 0 for $t \rightarrow \infty$. The relative importance of newly information for affecting expectations, thus, diminishes over time and the recursive algorithm converges to the rational expectations solution (Evans and Honkapohja 2001, 34:6).

With CG, g is set to a small constant so that most recent observations used to reformulate the estimates enter the recursive formulation with greater weight and past data are discounted. This algorithm is equivalent to some form of rolling regressions with a fixed-width but rolling time window of past data. For instance, setting $g = 0.02$ implies that agents form their forecasts using 25 years (100 quarters) of historical data. With $g \rightarrow 0$, the time window increases towards the infinite memory recursion. The main advantage of the CG algorithm is that it can capture the degree of rationality on part of agents. A high gain allows agents to

expect less stability in the economy and it accounts for situations where agents are alert to structural shifts in the economy. Most recent data may be then regarded as carrying more valuable information. In contrast, a small gain is typically associated with a high degree of rationality since it resembles the decreasing learning algorithm that allows for convergence to the RE solution. Indeed, with a constant gain, Evans and Honkapohja (2001) show that the algorithm converges to a distribution rather than to the constant fixed point $\bar{\phi}$.³³

In order to rule out explosive dynamics of the forecasting model, simulation exercises often incorporate a “projection facility” which prevents the estimates from leaving a predetermined region. Following selected studies, at least for the RLS algorithm explosive solutions can be ruled out if the E-stability criterion holds.³⁴ To prevent unstable dynamics in the forecasting VAR (PLM) also under CG learning, the roots in the modulus (absolute value of the real part of the eigenvalues) should always take on values below one. If this does not hold in a simulation round, the forecasting model is not updated and the parameters are held at their respective previous period values. This can be justified by the view that, in practice, private agents would reject unstable forecasting models and stick to parameter values in line with stationary conditions and a well-defined model corridor (Leijonhufvud 2009). The disregard of “unusual” parameter estimates actually help to stabilize the system as a whole if market participants keep working with the same model structure and previously stable parameter guesses.

In what follows, learning under action is presented with the RE solution as benchmark scenario. Different expectations formations are evaluated in terms of persistence and volatility of the relevant macroeconomic variables. In order to generate unconditional first and second moments in the stochastic simulation, the initial conditions are given by the RE equilibrium parameter values $\bar{\phi}$. Moreover, all state variables are set to their steady state values and the initial values for the covariance matrix R_t are calculated according to their respective values corresponding to the reduced form of the RE equilibrium. Simulations are carried out for a total length of 5,000 periods which is sufficient for adequately calculating standard deviations of the macro variables. As just described, the projection facility is employed as well.

Evans and Honkapohja (2001, 238) show the propositions under least squares learning for E-stability of a multivariate difference model such as the hybrid New-Keynesian model of the Sect. 5.3.1. In simulations, it is checked for E-stability in each period; a general finding is that for the selected parameter values, the model is learnable, E-stable and has a unique RE solution. Moreover, since all eigenvalues of the matrix $\bar{a}$ lie within the unit circle, the recursive least squares algorithm locally converges to the rational expectations equilibrium. Figure 5.4a exemplary shows that the first-order autoregressive coefficient of output converges to the rational

³³For that reason, Orphanides and Williams (2005a) characterize the learning algorithm as perpetual.

³⁴Evans and Honkapohja (2001), Orphanides and Williams (2006), Gaspar et al. (2006), Carceles-Poveda and Giannitsarou (2007).

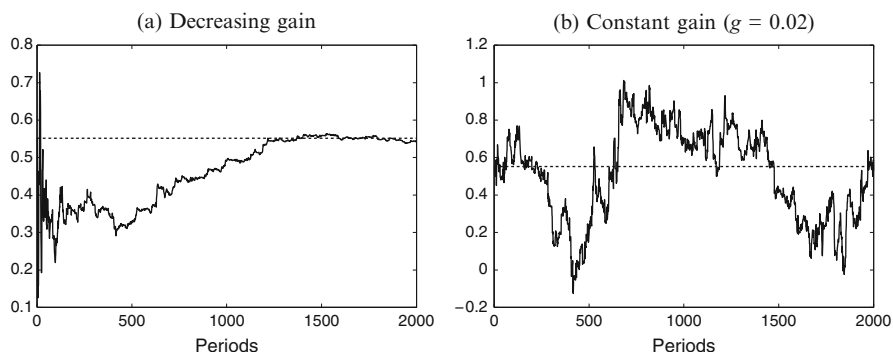


Fig. 5.4 Parameter estimates for the perceived law of motion

Table 5.2 Simulation of NK-Model with different expectations formations

Benchmark	Standard deviation			Autocorrelation		
	i	π	y	i	π	y
Expectations						
RE	2.45	2.75	2.64	0.80	0.72	0.67
RLS	1.84	2.16	2.86	0.69	0.61	0.70
CG $g = 0.02$	2.78	2.99	2.82	0.83	0.74	0.70
CG $g = 0.03$	2.93	3.19	2.81	0.84	0.76	0.71
CG $g = 0.04$	2.93	3.10	2.86	0.86	0.78	0.73
CG $g = 0.05$	3.23	3.34	2.98	0.88	0.80	0.87
Low mechanical inertia						
RE	2.23	1.49	1.58	0.26	0.29	0.39
CG $g = 0.04$	3.07	2.41	1.73	0.63	0.71	0.49

expectations solution represented by the dashed horizontal line. In contrast, under CG learning, the evolution of agents' beliefs about the persistence of output is much more volatile moving perpetual around the RE value during the simulation exercise (Fig. 5.4b).³⁵

A look at the moments of the macroeconomic variables sheds light on the question, how does learning alter overall dynamics of the system? The upper part of Table 5.2 reports standard deviations in annualized terms and the first-order autoregressive component of the key state variables, i.e. the short-term policy rate, inflation and output. The benchmark case of rational expectations (RE) produces second moments that are broadly in line with the respective empirical counterparts – though the numerical values should not be taken literally. Similarly, when inspecting the first-order autocorrelation of the variables, a considerable persistence can be documented for the simulated time series processes. This observation is the outcome

³⁵In simulation, the projection facility constraint applies in less than 2% of total simulation periods, so it is rather rare.

of the strong a-priori structural feature which produces forward- and backward-looking behavior of agents for both demand and supply decisions. The simulated persistence is supported by a policy-smoothing central bank which triggers adjusted policy responses to macroeconomic shocks. Against this background, the intention to build a New-Keynesian economy with a high degree of habit formation and price indexation to lagged inflation becomes understandable (for recent estimated models see Giannoni and Woodford 2003b; Smets and Wouters 2003; Christiano *et al.* 2005). Indeed, if the degree of mechanical persistence is reduced, both second moments and the autocorrelations of the state variables considerably decline. The lower part of Table 5.2 simulates the economy in which the parameters α, δ, ρ are all set equal to 0.1 so that the model is mainly forward-looking, accompanied by a low degree of interest-rate inertia. In particular, the model with low inertia fails to match the observed behavior of high persistence in the data, because inflation and output become pure jump variables (at least in the limiting case of $\delta, \alpha = 0$) so that both can quickly adjust to various shocks (Mankiw 2001). The low persistence can be documented in low autoregressive coefficients of the state variables.

Learning provides a strong transmission mechanism for macroeconomic shocks resulting in greater volatility and persistence. Taking the hybrid New-Keynesian model, learning agents with infinite memory (RLS) produce time variation in the estimates of the reduced form VAR for forecasting. Time-varying coefficients contribute to both increased volatility and persistence of the state variables. In case of constant-gain learning, the effects are considerably larger and they increase with a higher gain since the sensitivity of the coefficients to newly available information is greater. The revisiting of expectations formation in terms of learning enables to endogenously generate the persistence of inflation, output and the short-term policy rate. The re-calibration of the model with a strong emphasis on forward-looking behavior and low mechanical inertia reveals that constant gain learning helps to produce higher volatility and persistence, in particular of the short-term interest rate and inflation (see Table 5.2 and Milani 2007).

What are the term structure implications of the macro-finance model, if one allows for learning dynamics? First of all, standard rational expectations models (without time-varying risk premia) typically fail to generate yield curve dynamics consistent with the data. All movements in longer-term interest rates are driven by expectations of the future short rates over the relevant maturities; expected short rates, in turn, depend on expectations of the remaining state variables. Generally, the short-term interest rate is far less persistent and it converges too fast to its equilibrium values to account for the observed variability of long-term interest rates. To see this, consider the recursive bond pricing equation of (5.16). To obtain the variability of long-term rates, the solution³⁶ of the factor sensitivities is

$$B_n = -\frac{I - \Theta^n}{I - \Theta} \delta_1.$$

³⁶See Appendix C for a derivation of this result.

Following Gallmeyer et al. (2008) and in case of one state variable describing the whole yield curve, the relation between the standard deviations of bond yields and the short-term interest simplifies to

$$\sigma_{i_{n,t}} = \frac{1}{n} \frac{1 - \Theta^n}{1 - \Theta} \sigma_{i_{1,t}}.$$

For such a model, the volatility of interest rates depends entirely on the autocorrelation Θ of the relevant state variable. Atkeson and Kehoe (2008) show that the standard deviation of a representative long-term interest rate in proportion to the standard deviation of the short-term interest rates is considerably smaller than the corresponding ratio reported in selected sample periods across industrialized countries (see also the stylized results presented in Chap. 3.2).

To overcome this caveat, the macroeconomic model must generate high persistence. As explained, this can be achieved mechanically through the inclusion of adequate structural parameters or by introducing structural shocks characterized by large size and high autocorrelation (Ravenna and Sepl 2006; Rudebusch and Swanson 2008b). Another way is to build an interconnection between the yield curve and learning dynamics that likewise allow persistence in the state variables. Cogley (2005), Dewachter and Lyrío (2006) and Dewachter (2008) incorporate time variation in the intercepts as sources for adequately explaining term structure dynamics. As described earlier in this Section, bond yields tend to exhibit excess sensitivity to macroeconomic shocks which can hardly be captured by standard macroeconomic models (Ellingsen and Söderström 2001; Ellingsen and Sderstrm 2004; Gürkaynak et al. 2005b). Introducing an environment in which agents learn about long-term attractors, such as the inflation target or the natural rate, has been proved to be a promising extension to fit the entire yield curve and not only the short end.

In this respect, the approach in this Section of using the learning framework differs from the previous studies, since it explicitly allows for updating beliefs on all structural parameters based on the learning algorithm.³⁷ Agents use the same affine bond yield representation of (5.16) which presents interest rates as a factor model in which the Expectations Hypothesis holds. They use their PLM to make forecasts of the future state of the economy in order to price long-term bond yields. Consequently, the term structure of interest rates is built on the basis of subjective beliefs rather than on the equilibrium outcomes of the actual law of motion. Imposing no-arbitrage under the PLM reflects the view that asset prices are set by the private sector; they should be consistent with the available information set of these agents. For this purpose, the perceived dynamics of (5.20) are redefined in an extended state-space as

³⁷The closest work is Laubach et al. (2007) who allows time variation of the whole parameter set, but they do not impose any economic structure.

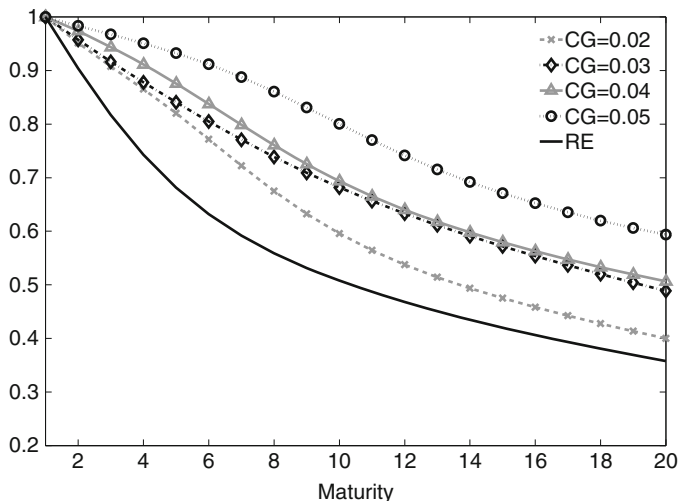


Fig. 5.5 Long-term volatility as proportion of short-term volatility

$$Y_t^P = \Theta_{t-1}^P Y_{t-1}^P + \Gamma_{t-1}^P \varepsilon_t \quad (5.25)$$

with $Y_t^P = [u_t, X_t^P]$ where the subscript P denotes the perceived dynamics of the state variables. As mentioned, bond securities and, thus, interest rates are then priced according to (5.16) with risk prices λ_0 set equal to zero in order to reduce the sources of volatility and possible explosive dynamics for long-term interest rates.³⁸ Only time variation in the estimated coefficients account for additional volatility of interest rates.

First of all, one can ask whether active learning under action can help to explain the volatility of the yield curve within an otherwise standard macro model. Figure 5.5 reports evidence for the benchmark case and for various learning algorithms differing from each other by the size of the gain. The black solid line mirrors the relationship between the volatility of interest rates solved under rational expectations. Long-term volatilities are considerably much lower than the short-term counterpart which stands in sharp contrast to empirical evidence. For instance, the proportion of the 5 year to the 3-month standard deviation in Germany takes on the value of 0.7 for the period 1960:1–2008:3; whereas the simulation of the yield curve in the benchmark case allows only for a value of 0.35.

Incorporating constant gain learning increases the volatility ratios across the selected maturity spectrum. As a result, a higher gain is associated with a tendency that is able to fit the moments within the yield curve. A crucial point is that the strategy of jointly modeling interest rates and the macroeconomy typically

³⁸Moreover, notice that we describe de-meaned dynamics and since the NK-Model only works with constant risk premia they should stay constant over time.

fails to track the volatility ratios even when taking time-varying risk premia into account. Either the macro-finance models can reproduce the basic properties of macroeconomic variables at the cost of screwing up financial outcomes, or it can track the yield curve with counterfactual macro effects (Rudebusch and Swanson 2008b). With the learning framework, it becomes possible to improve a joint macro-finance view with little misalignments on the macro side. Indeed, for selected gains, standard deviations of inflation or output do not rocket high; meanwhile the volatility ratios can be considerably improved without the inclusion of large and persistent shocks.³⁹ The reason for this observation is that high time variation in the loadings to factor sensitivities $b_{n,t}$ translate into higher volatility in long-term interest rates.

For the purpose of inspecting the yield curve response to macroeconomic shocks, 100 draws from the distribution of the coefficients of the simulation are randomly picked; they are used as initial conditions for generating dynamic impulse responses for the set of interest rates. This is like asking how sensitive bond yields react to shocks for different periods in time. Under learning, the impulse responses vary with the state of the economy and the beliefs determining expectations formations. With perfect knowledge, the bond loadings to factor changes are constant and exhibit identical dynamic properties across time. To get a broad picture of possible outcomes of yield curve effects, Fig. 5.6a plots the median and the 80% range of the distribution of the initial impulse response of the term structure following a monetary policy shock. Taking the RE response as benchmark, a shock to the policy rate has virtually no effect for longer-term maturities – a contradiction to empirical

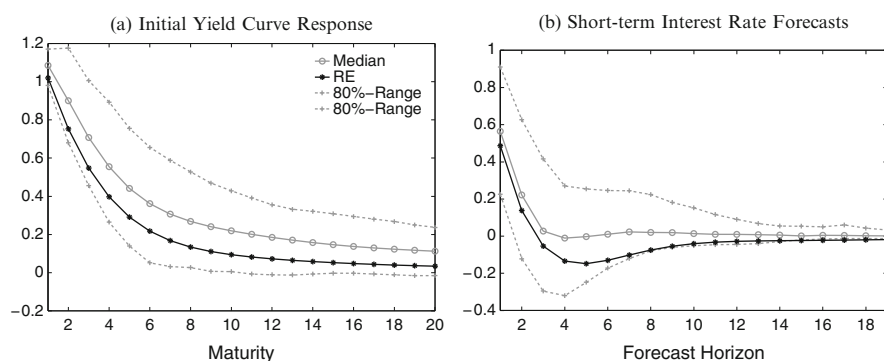


Fig. 5.6 Yield curve response under learning to monetary policy shock

³⁹In the macro-finance literature, shocks to natural output are often quantitatively captured by a quarterly standard deviation of over 2% points or the natural rate process and the inflation target are near-random walks that mechanically produce high persistence. In the model above, standard deviations of shocks are in the range found in empirical estimates and the processes of natural output and the inflation target are persistent, though they do not follow near random walks (Ravenna and Sepp 2006; Rudebusch and Swanson 2008b; Hördahl and Tristani 2007).

evidence. An inspection under constant gain learning ($g = 0.03$) reveals that long-term interest rates indeed react. The median move is larger compared to the RE case and it is skewed upward, which is an indication for a strong sensitivity of long-term rates following an unanticipated change in monetary policy.

Imperfect information about the parameters governing the state variables translates, on average, into higher short-term interest rate expectations. This comes as no surprise since a monetary policy reacting to more persistent and volatile inflation and output gaps according to the Taylor, produces a more prolonged response of its policy rate. The expected path of the short-term interest rate is documented in Fig. 5.6b. Here, starting from initial conditions again drawn from the simulated distribution, agents form expectations from their perceived law of motion by solving the VAR-process n -periods forward. Again, the median path and the 80% range mirror the persistency and the decayed effects of expected short rates. As the initial response of a long-term bond is captured by the weighted average of the future path of short-term interest rate expectations, it becomes clear why long-term rates react more sensitive to monetary policy shocks.

The rational expectations paradigm in which market participants make supply and demand decisions under perfect knowledge, does a poor job in matching basic yield curve dynamics. This also holds under a scenario in which time-varying risk premia are derived from a full-fledged DSGE model. By extending the basic macro-finance model through the inclusion of learning agents, yield curve puzzles can be partly resolved. Long-term interest rates sufficiently react to unanticipated macroeconomic news as documented in the history of term structure dynamics (Ellingsen and Sderstrm 2004). This holds even under the assumption of constant risk premia. Laubach et al. (2007) point out that the deviation of the Expectations Hypothesis is much smaller on average with a learning model for the term structure of interest rates. In particular, unlike in many other studies that imply a sizable downward trend in risk premia over the last decade, they find a much reduced trend in term premia. Learning dynamics largely account for the observed swings in long-term interest rates.

Chapter 6

Monetary Policy in the Presence of Term Structure Effects

6.1 The Term Structure of Taylor Coefficients

In macroeconomics, the short-term interest rate is usually modeled as a monetary-policy reaction function to a set of macroeconomic variables. A deliberately simple functional form is the Taylor rule; it is the answer to a perennial question in monetary economics of how the monetary authority should implement policy in a systematic manner. Taylor (1993, 214) developed a “hypothetical but representative” rule by using the gap concept of inflation and output. The Taylor rate is 1.5 times inflation plus 0.5 times the output gap, plus 1. For the period between 1982 and 1992, the rule was successful in capturing the actual behavior of the US federal funds rate. As this period is widely regarded as a shift from a “passive” to an “active” stabilizing monetary policy regime¹, the rule has been used ever since to ask where monetary policy *should* head in response to fluctuations in inflation and real activity.

Such a simple characterization of monetary policy has been favored by many researchers since Taylor rules perform well in a class of models that are in use in policy research (Davig and Leeper 2007). Most importantly, the proposition that central banks can stabilize the economy by adjusting their interest-rate instrument by more than one-by-one with inflation is an essential feature of the modern view of successful central banking. The size of the inflation coefficient assures that the nominal interest rate is raised enough to actually increase the ex-ante real interest rate. In many monetary models, the “Taylor principle” is necessary and sufficient for a unique expectations equilibrium, either derived from agents equipped with rational expectations or an adaptive learning scheme (Woodford 2003; Bullard and Mitra 2002). A failure to produce such a disproportionate reaction results in large movements of the fundamental variables as a reaction to otherwise standard shocks and brings about multiple equilibria or sunspot dynamics.

¹Sargent et al. (2009) and DiCecio and Nelson (2009).

Estimated extensions of Taylor-inspired reaction functions, in turn, allow to describe the actual interest-rate setting behavior on part of central banks over selected sample periods. Empirical reaction functions typically differ from the standard Taylor in order to fit the historical track of the short-term interest rate in a better way. The modifications typically include the assumption of partial adjustment of the policy rate, forward- and backward-looking specifications of the arguments, the use of survey or real-time data or the inclusion of various asset prices and monetary aggregates.²

In its original version, the Taylor rule postulates a stable, systematic reaction to deviations of inflation with respect to a constant inflation target rate. This stability of coefficients and the target rate has been recently challenged for the US by the empirical literature.³ It appears that during the last 50 years, the FED's reaction to inflation changed in favor of a more aggressive monetary conduct at the beginning of the 1980s. This changed policy view is closely connected to the term "Great Moderation" which indicates to the generalized phenomenon of reduced volatility of key macroeconomic variables.⁴ Monetary policy, thus, delivered price stability more effectively as it followed the Taylor principle in the Volcker/Greenspan/Bernanke administration. Moreover, there is evidence of a "time-varying" inflation objective expressed in a slow moving trend inflation. The explicit use of term structure information to extract the trend inflation component in a model of inflation supports the perception of regime shifts in monetary policy in the post WWII-period (Kozicki and Tinsley 2001; Ireland 2007).

The macro-finance view impressively clarifies the link between the yield curve, monetary policy and macroeconomic developments. All interest rates can be described by a constant term and maturity-specific factor loadings that translate factor dynamics into yield curve movements. At the heart of these financial factor models is the short-term interest rate that bridges the gap between the "pure finance" and "pure macro" approach. The dynamics of this riskless rate together with the stochastic characteristics of the discount factor determines long-term interest rates farther out the yield curve. Shifts in the behavior of monetary policy must likewise be translated into a changing dynamic of the term structure of interest rates. Analyzing the impact of such shifts from the perspective of policy rules offers a simple and straightforward way for term structure implications since long-term interest rates are determined by the response of the central bank to future economic conditions.

²Clarida et al. (2000), Rudebusch (2002, 2009), Gerlach and Schnabel (2000), Gerlach-Kristen (2003), Sauer and Sturm (2003), Orphanides (2003) and Fernandez et al. (2008).

³Clarida et al. (2000), Lubik and Schorfheide (2004), Cogley and Sargent (2005), Boivin (2006), Kim and Nelson (2006) and Castelnuovo et al. (2008).

⁴An alternative view of the Great Moderation is taken by Sims and Zha (2006). They identify the reduced volatility of less frequent and lower shocks to inflation and output as the main driver of smoothed macro dynamics during the relevant period beginning in the mid 1980s. To them, reduced inflation persistence was a matter of "good luck" rather than sound monetary policy.

Smith and Taylor (2009) combine the no-arbitrage theory of the term structure of interest rates with a Taylor rule to get closed form solutions for the *term structure of policy rules*. The yield curve can be represented as a series of implied policy rules – with one policy rule for each maturity. In this respect, the responsiveness of long-term bond rates to macroeconomic variables depend on the reaction coefficients of the one-period policy rule. Shifts in the Taylor coefficients trigger shifts in the responsiveness of the yield curve to macroeconomic events. To get further insights into the workings of the model, the model version of Smith and Taylor (2009) is reviewed where the central bank responds to inflation but not independently to output. The setup has the following equations

$$i_{1,t} = \delta \pi_t \quad (6.1)$$

$$i_{n,t} = -n^{-1} \log(P_{n,t}) \quad (6.2)$$

$$P_{n+1,t} = E_t[M_{t+1} P_{n,t+1}] \quad (6.3)$$

$$M_{t+1} = \exp(-i_{1,t} - 0.5\lambda_t^2 - \lambda_t \varepsilon_{t+1}) \quad (6.4)$$

$$\lambda_t = \lambda_0 + \lambda_1 \pi_t \quad (6.5)$$

$$\pi_t = \pi_{t-1} - \phi(i_{1,t-1} - \pi_{t-1}) + \sigma \varepsilon_t \quad (6.6)$$

with $\varepsilon_t \sim N(0, 1)$; $i_{n,t}$ is the nominal n -period interest rate with bond price $P_{n,t}$ and π_t is the inflation rate. It is a simple structural model for the term structure of interest rates where monetary policy responds to inflation and bond rates are related through the no-arbitrage relation in (6.3) with M_{t+1} representing the stochastic discount factor and market prices of risk determined by λ_t .⁵ Market prices of risk are specified to have a constant term λ_0 and a state-dependent term λ_1 . Inflation depends on the lagged real interest rate and its own lag.⁶

It can be shown that interest rates of maturity n are linear functions of the inflation rate

$$i_{n,t} = a_n + b_n \pi_t \quad (6.7)$$

where the solution is achieved through the method of undetermined coefficients and its representation follows the characteristics of a typical financial factor model. The exact derivation of the constant part a_n and the reaction coefficient b_n follows the logic of Appendix D and Smith and Taylor (2009); their recursive closed-form representation is not reported here for the sake of simplicity because the focus is on the basic mechanism of the policy coefficient on the bond rate loading. Therefore,

⁵See Chap. 3.5.2 for a detailed treatment on affine term structure models.

⁶The inflation equation can be derived from a modified version of Svensson (1997) where output is replaced by the lagged real interest rate which monetary policy can affect through its policy rule.

a 2-period bond rate is sufficient to describe this effect. The reaction coefficient of the 2-period bond is determined by

$$b_2 = \frac{\delta(2 - \phi(\delta - 1) - \sigma\lambda_1)}{2}. \quad (6.8)$$

What becomes clear for b_2 is that the inflation coefficient in the Taylor rule has a direct effect on the response coefficient to inflation for the longer maturity. A larger reaction coefficient δ – the direct impact is measured by the presence of δ outside the parentheses – produces an expectational environment of rising future short-term interest rates when future inflation rises. However, there are also countervailing effects. Firstly, a larger δ implies that inflation is expected to rise by a smaller amount in the future because the persistence of inflation falls as measured by the term $\phi(\delta - 1)$. Secondly, the net effect depends on the stochastic risk price and, thus, the risk premium λ_1 . Provided that λ_1 is negative, an increase in current inflation pushes the response coefficient b_2 additionally up.⁷ The impact of monetary policy shifts can be clarified by taking the first derivative with respect to the reaction coefficient δ

$$\frac{\partial b_2}{\partial \delta} = \frac{2 + \phi - \sigma\lambda_1}{2} - \delta\phi > 0. \quad (6.9)$$

As long as the reaction is not too large, the direct policy effect dominates the reduced persistence component of inflation so that the reaction of the 2-period bond rate to inflation is positive. A more aggressive monetary policy implies that bond yields likewise react more sensitive to inflation. Against this background, the empirical evidence on term structure effects on inflationary impulses highlights that the response coefficients decline with increasing maturity (see for instance Kozicki and Tinsley (2008) and the further analysis below). In the model, such a pattern is ensured if a modified Taylor principle holds, i.e.

$$\delta > 1 - \frac{\sigma\lambda_1}{\phi}. \quad (6.10)$$

The condition is sufficient to (1) prevent bond rates from exploding and to (2) replicate the empirical findings.

As shown by Smith and Taylor (2009), the core mechanism at work does not change if monetary policy reacts to both inflation and output in a model economy with an explicit output equation. The affine term structure of policy rules can be described by

$$i_{n,t} = a_n + b_{\pi,n}\pi_t + b_{y,n}y_t \quad (6.11)$$

⁷Ang and Piazzesi (2003) and the estimation results of Chap. 4.2.1 estimate risk prices to be negative for most sample periods.

where the maturity-dependent loadings each imbed a direct policy effect (δ_π , δ_y) and a countervailing force through reduced persistence of inflation and output with increasing reaction parameters in the Taylor rule.

The simple model impressively demonstrates the impact of monetary policy shifts on the term structure of interest rates. The appealing way of describing yield curve effects to Taylor-style reaction functions can considerably support the understanding of monetary policy responsiveness and macroeconomic dynamics. The informational advantage of extracting information from the whole term structure of interest rates about monetary policy and vice versa, allows to efficiently measure yield curve responses to changes in systematic monetary policy and to monetary policy shocks (Ang et al. 2007). If there are policy shifts in the description of central bank behavior, the joint model of policy rules and yield curve interaction predicts that changes in the coefficients in the policy rule will likewise lead to changes in the coefficients (loadings) for selected yield curve data.

Smith and Taylor (2009) estimate the term structure of policy rules for selected sample periods according to (6.11) with single equation regressions by OLS. They find that in the pre-Volcker area 1960q1–1979q4 the policy coefficients had been significantly lower than afterwards (1984q1–2006q4) indicating a preference toward a shift in terms of a strict and low inflation target and stability-based monetary policy (Bullard and Eusepi 2005). In addition, the policy findings are reflected in higher coefficients of the affine equations for long-term interest rates so that the heightened sensitivity of the yield curve to the policy rule variables is a clear sign of that policy shift. Moreover, the same estimation procedure is conducted for the “interest-rate conundrum” period of 2004–2005 when the FED started raising the federal funds rate which was not accompanied by appropriate increases in long-term bond yields. According to the outlined model above, a policy-related explanation of the unusual behavior of long-term interest rates would be that the policy rate significantly deviated from what estimated policy rules and the conventional Taylor rule would have predicted. Indeed, Smith and Taylor (2009) show that the inflation coefficient dropped during this period and the perception of this lower policy responsiveness induced market participants to expect lower interest-rate moves to future inflation and, thus, a lower reaction coefficient of the long-end of the yield curve to inflation dynamics.

Such policy behavior can be further evaluated when jointly estimating policy rules and yield curve movements under cross-equation restrictions implied by the no-arbitrage framework. As already mentioned, it augments the applied information set and allows to estimate policy coefficients and monetary policy shocks more efficiently compared to single equation OLS evidence. Interestingly, it seems that the extended model produces Taylor coefficients that are significantly smaller ($b_{\pi,1} < 1$) than the OLS estimates, due to an omitted latent variable in the conventional OLS estimate (Ang et al. 2007). Most importantly, the omitted variable in the policy rule is central for the fit of long-term interest rates and it is correlated with output gap and inflation. Further empirical evidence on regime shifts supports the view of changing coefficients in interest-rate reaction functions. Ang et al. (2009) find that monetary policy changed in important ways that are mostly captured

by the FED's action to inflation developments: there are clear signs of a time-varying inflation coefficient. It has been below 1 during the 1970s, increased during the 1980s and 1990s to almost 2 and decreased again below 1 during the early 2000s.

6.2 Incorporating Long-Term Interest Rates into Monetary Policy Analysis

6.2.1 *Determinacy with Bond Rate Transmission*

Most studies on monetary policy specify the policy interest rate to be a very short-term such as the overnight rate. However, as documented in Chap. 5, long-term interest rates play a pivotal role for the transmission of monetary policy. Woodford (2005a, 3) notes that “[...] the current level of overnight rates *as such* is of negligible importance of economic decisionsmaking; if a change in the overnight rate were thought to imply only a change in the cost of overnight borrowing for that one night, then even a large change would make little difference to anyone's spending decisions. The effectiveness of changes in central-bank targets [...] in spending decisions is wholly dependent upon the impact of such actions upon other financial-market prices, such as longer-term interest rates [...]” Bond rates mirror this expectational channel since they contain bond traders' expectations of future policy rates. Therefore, policy effectiveness highly depends on the policy perception of the bond market; it makes long-term interest rates an explanatory variable in aggregate demand as illustrated in the basic New-Keynesian model economy of Chap. 5.3.

There are at least two more interrelated mechanisms that give long-term interest rates a separate role in determining aggregate demand. Firstly, as already discussed, time-varying interest-rate risk premia may create a wedge between the relation of expected short-term interest rates and long-term bond rates as expressed in the pure version of the Expectations Hypothesis. However, though a long-term rate appears in the standard model, it does so only as a stand-in for the expectation of the path of the current short rate. Deviations of the long-term interest rate from the expectations theory of the term structure are recognized but the focus remains largely one of describing yield-curve dynamics for given macroeconomic data and due to the lack of a structural link between term premia and macroeconomic behavior (Rudebusch *et al.* 2007). Secondly, the absence of any direct feedback from long-term bonds rates not being generated by the expected future path of short-term interest rates can be overcome by the acknowledgment of imperfect substitutability between money and non-monetary assets. Following the work of Tobin (1969, 1982), a direct feedback from bond rates to the macroeconomy can be established through the portfolio approach which allows different returns of assets through changes

of a bond security's supply relative to other assets in the presence of transaction costs.⁸

Given the prominent role of the long-term interest rate in the monetary transmission, it is straightforward to ask whether a rationale exists to explicitly incorporate term structure information into the conduct of monetary policy.⁹ According to the history on the search for simple monetary policy rules, there would be at least some misgivings. Monetary policy rules, in general, differ with respect to the choice of instrument, the derivation within an explicit optimization framework or in a rather "heuristic", but economically comprehensible way, and in the choice of a set of variables to which monetary policy responds.

Research has impressively shown that the short-term interest rate as policy instrument performs better than monetary targeting rules. The insight does not only hold for practical central banking but in particular for monetary theory (Poole 1970). Moreover, the debate on simple vs. optimal rules and the inclusion of additional variables such as oil prices, equity and real estate prices, exchange rates or foreign output measures into reaction functions has demonstrated that the most robust policy rule is the standard Taylor reaction function.¹⁰ Since optimal targeting rules rely on the assumption about the truth of the underlying macro model, they suffer from the possibility of model uncertainty. If optimal rules and simple rules are evaluated in different model economies, it turns out that, on average, the Taylor rule generates the lowest losses in terms of inflation and output variability.

Still, McCallum (2005) analyzes an interest-rate reaction function with the lagged interest rate and the 2-period yield spread as variables to which monetary policy responds. He shows that the resulting equilibrium interest rate process is able to capture empirical dynamics of the yield curve better, in particular Expectations Hypothesis tests: the observed downward bias of the regression coefficient can be attributed to a combination of the persistence of a risk premium and the central bank's action of yield-curve smoothing through the reaction to the yield spread.¹¹ In addition, most of the empirical literature hints to the fact that central banks respond to long-term interest rates and to the slope of the yield curve.¹² This does not mean that they mechanically react to term structure information as it is the case for inflation and output in a standard Taylor reaction function. Since equilibrium interest rates are functions of the current state of the economy, the empirical observation

⁸A general equilibrium approach with imperfect asset substitution has been analyzed by Andres et al. (2004), Marzo et al. (2008) and Zagaglia (2009).

⁹The role of term spreads, financial intermediation, financial stability and implications for the conduct of monetary policy is discussed in Chap. 7.4.2. Here, the focus is on bond rate transmission in the presence of the traditional expectations channel of monetary policy.

¹⁰See Taylor and Wieland (2009) or Taylor and Williams (2009a). An excellent classification of monetary policy rules is provided by Fendel (2004).

¹¹See Chap. 3.4.2 for the empirical rejection of the Expectations Hypothesis as well as Kugler (1997) and Romhányi (2002) for an application to the n -period case.

¹²Mehra (2001), Cochrane and Piazzesi (2002), Gerlach-Kristen (2003), Fendel and Frenkel (2005) and Vázquez (2009).

may mirror macroeconomic fundamentals, such as expected inflation and expected real cyclical activity to which monetary policy reacts beyond current inflation and output.

Along similar lines, a number of studies have either looked at optimal and simple rules in a macroeconomic environment with bond rate transmission, or at term-structure augmented monetary policy in an otherwise standard model in which only the short-term interest rate affects aggregate demand.¹³ None of these papers, however, has tried to jointly evaluate monetary policy with term-structure effects in both aggregate demand and in policy rules. The idea that a central bank might wish to target a long rate and to intervene in bond markets has been on top of the agenda in 2003 when short-term interest rates were supposed to hit the zero-lower bound. McGough et al. (2005), among others, have studied the equilibrium outcomes for inflation and output that would not be attainable using the short-rate as policy rate. However, as pointed out by Woodford (2005b), there are serious shortcomings of such strategy in terms of implications for both money markets and bond markets such as high volatility and the imperfect controllability of the target bond rate. Against this background, long-term bond rates should be used as indicators of whether policy is on track. Monitoring long rates makes sense in order to check whether policy-intended changes of the expected future path of short rates are translated into long-term interest rates.¹⁴ If this is not the case, there is a clear rationale for monetary policy to react with its short-rate instrument to term structure information for the purpose of bringing about a desired equilibrium.¹⁵

The following paragraphs do not capture the discussion of the role of the long rate as monetary policy instrument, i.e. as operating target of a central bank. However, they explicitly work out the implications for equilibrium dynamics and welfare effects for term-structure augmented interest-rate reaction functions.

For analyzing monetary policy transmission in a simple model structure, the basic purely-forward looking aggregate demand and supply relations of Sect. 5.3 are introduced where a long-term ex-ante real interest rate determines output and inflation dynamics. For illustrative purposes, the equations are kept as simple as possible in order to gauge the mechanisms derived from bond-rate transmission. The model takes the form

$$\tilde{y}_t = -\sigma^{-1} \left(i_{n,t} - n^{-1} \sum_{j=1}^n E_t \pi_{t+j} \right) + v_{y,t}$$

¹³See Gallmeyer et al. (2005), McGough et al. (2005), Kulish (2006) or Fendel (2007).

¹⁴Goodfriend (1998, 16) remarks that “there is evidence that the long-term nominal bond rate moves primarily as a result of inflation expectations. Sharp bond rate movements ought to be taken as evidence of worsening or improving credibility on inflation, as the case may be, and taken into account in making decisions on short-term policy.”

¹⁵To be sure, such actions can be supported by active engagements in bond markets through open-market operations but it does not mean that a long rate might be used as instrument of policy as that term is generally understood.

$$\begin{aligned}
i_{n,t} &= \frac{1}{n}i_{1,t} + \frac{n-1}{n}E_t i_{n-1,t+1} + \frac{n-1}{n}v_{i_n,t} \\
\pi_t &= E_t \pi_{t+1} + \kappa \tilde{y}_t + v_{\pi,t} \\
i_{1,t} &= \alpha \pi_t + \gamma^\top \chi_t + v_{i_1,t}
\end{aligned} \tag{6.12}$$

where all variables are expressed in terms of deviations from the steady-state equilibrium. The first equation states that the output gap $\tilde{y}_t$ is determined by the n -period ex-ante real bond rate and by a stochastic disturbance term comprising stochastic changes in the natural rate of interest of relevant maturity n . According to the second equation, the n -period interest rate is priced due to the Expectations Hypothesis and an exogenous disturbance term which captures deviations from the EH with time-varying interest-rate risk.¹⁶ It is important to note that with such a specification, aggregate demand is not exchangeable with the one fully derived by microeconomic foundations since the term premium cannot be traced back to first principles in the rigorous sense. For the purpose of analyzing stability of monetary policy rules, this should not represent a major shortcoming. The third equation is a standard forward-looking inflation equation depending on expected inflation, the output gap and a stochastic cost-push shock. Finally, the model is closed via an interest-rate rule for the short-term policy rate where monetary policy reacts to current inflation (with the inflation target normalized to zero) and a vector χ that may include the output gap, term structure information or other policy-relevant variables. All shocks are assumed to follow an AR(1)-process $v_{j,t} = 0.5v_{j,t-1} + \varepsilon_{t,j}$ with $\varepsilon_{t,j} \sim N(0, \sigma_j^2)$ for $j = \{y, i_n, \pi, i_1\}$.

In order to facilitate the analysis of determinacy of the system, the case with a 2-period long-term rate is inspected with the monetary authority reacting only to current inflation so that $\gamma = 0$.¹⁷ For $n = 2$, the 4th, 2nd and 1st equation is inserted into the New-Keynesian Phillips curve to write down a single dynamic equation in inflation. This gives a second order representation with

$$\pi_t = \left(\frac{2\sigma + \kappa - \alpha\kappa}{2\sigma + \alpha\kappa} \right) E_t \pi_{t+1} + \left(\frac{\kappa}{2\sigma + \kappa} \right) E_t \pi_{t+2}$$

¹⁶The asset pricing relation can be derived by the pure form of the local return hypothesis according to which $ni_{n,t} - (n-1)E_t i_{n-1,t+1} = i_{1,t} + (n-1)v_{i_n,t}$. The left-hand side of this equation is the one-period return which should equal the short-term interest rate plus the risk premium which arises due to the uncertainty about the one-period interest rate. In the simple two-period case, the local hypothesis becomes $2i_t - E_t i_{1,t+1} = i_{1,t} + v_{i_n,t}$. The n -period case is thereby derived by recursion. For reasons of computational tractability, in the numerical simulations, $i_{n-1,t+1}$ is approximated by $i_{n,t+1}$ in above model set-up (see also McCallum (2005) and Romhanyi (2002) for this modeling strategy).

¹⁷Moreover, in the 2-period case, shock processes are set equal to zero in order to clarify results for determinacy.

which can be restated in first-order form

$$\begin{aligned} A_1 x_{t+1} &= A_0 x_t \\ x_{t+1} &= B x_t \end{aligned} \quad (6.13)$$

where $x_{t+1}^\top = [\pi_{t+1}, E_t \pi_{t+2}]$ and

$$B = A_1^{-1} A_0 = \begin{bmatrix} 0 & 1 \\ \frac{2\sigma}{\kappa} + \alpha - \frac{2\sigma}{\kappa} - 1 + \alpha \end{bmatrix}. \quad (6.14)$$

Woodford (2003, Appendix C, Case II) provides the general results regarding determinacy of the system. He shows that for a 2-dimensional vector of non-predetermined endogenous state variables with a 2×2 coefficient matrix B , a rational-expectations equilibrium is determinate, if and only if both eigenvalues of B lie outside the unit circle (see also Blanchard and Kahn 1980). For the above setup, the condition is satisfied if

$$\det(B) - \text{tr}(B) < -1 \quad \text{and} \quad \det(B) + \text{tr}(B) < -1.$$

It holds for

$$\begin{aligned} \frac{4\sigma}{\kappa} &> 0 \\ \alpha &> 1, \end{aligned}$$

where the latter resembles the well-known Taylor principle. The resulting “leaning against the wind” policy rule calls for an one-period nominal interest rate which is adjusted more than one-for-one in response to shocks that trigger a 1% point increase in one-period inflation. The intuition is that under such a rule, a rise in inflation is accompanied by an increase in the ex-ante real interest rate which dampens demand and inflationary pressure. If the principle does not hold, the economy is pushed away from the local equilibrium and it ends in either explosive or indeterminate equilibria.

Since the interest-rate channel in the aggregate demand specification of the model works through the 2-period bond rate, monetary policy must ensure that the 2-period ex-ante real rate must likewise react to alter aggregate demand. Clearly, such an environment depends on the effects of monetary policy on the long-term interest rate, in particular on the way how financial market participants form interest-rate expectations.¹⁸ With rational expectations, the Taylor principle is sufficient to guarantee uniqueness of the system. Monetary policy ensures that it moves current

¹⁸See Sect. 6.1 for the affine term structure of policy responses.

and future policy rates disproportionate to current and expected inflation. A shock to current inflation raises the current policy rate as well as expected inflation and the expected short rate. What matters is the net outcome of these effects. If monetary policy sticks to the Taylor rule with $\alpha > 1$, agents always perceive a more than one-for-one reaction. Although the 2-period nominal rate move is less than the policy rate reaction in absolute terms, the ex-ante 2-period real interest rate rises due to lower average expected inflation over the maturity of the bond.

Kozicki and Tinsley (2008) recently modified the model framework by building a short-rate equation in line with horizon-dependent policy perceptions on part of bond traders. They highlight that what is crucial for determinacy of inflation is the anticipated path of future policy rates. For a 2-period bond rate transmission, they specify the short-rate to follow $i_{1,t} = c_1 E_t \pi_{t+1} + c_2 E_t \pi_{t+2}$ where the perception of the future policy response c_2 does not need to coincide with the response perceived in the current period c_1 . For a unique equilibrium to exist, they show that the Taylor principle can be generalized to be $0.5(c_1 + c_2) > 1$. It implies that determinacy requires that the average anticipated response to expected inflation over the life of the bond must exceed unity. The condition is the same for the standard case as discussed above if the inflation coefficients in the policy rule are identical, i.e. $\alpha = c_1 = c_2$. If, however, the perceived responsiveness differs, even a passive policy with $c_1 < 1$ in the initial period of a shock could be in line with determinacy, as long as the *average* response satisfies the uniqueness condition. The essential point is that the “convolution” of policy parameters over the maturity of the long-term interest rate matters, not just the policy coefficient in the near-term interest-rate reaction function.

In the proceeding discussion, term structure information in monetary policy rules are considered in two ways. First, it is assumed that the short rate responds to the long rate with $\chi_t = i_{n,t}$ beyond current inflation. Second, the central bank reacts to the term spread with $\chi_t = i_{n,t} - i_{1,t}$. The analysis asks which policy parameter constellations yield a unique rational expectations equilibrium for bond rates with different maturities.¹⁹ The alternative reaction functions are studied numerically since analytical results are hardly tractable, especially as relevant interest rates move towards larger maturities. To this end, the remaining parameters are set to $\sigma = 4$, $\kappa = 0.17$, $\sigma_{y,\pi} = 0.4$ and $\sigma_{i_1,i_n} = 0.1$.

To get an intuition on the logic of central bank action, consider a 2-period bond $i_{2,t}$ in the policy rule – for the purpose of clarity, it is abstracted from any specification of shocks. Using the 2nd and 4th equation of system (6.12) with $\chi_t = i_{2,t}$, the short rate follows

$$i_{1,t} = \frac{\alpha}{1 - 0.5\gamma} \pi_t + \frac{\gamma}{2 - \gamma} E_t i_{1,t+1}. \quad (6.15)$$

¹⁹The system being analyzed is said to be determinate if the rational expectations solution is unique and dynamically stable. It is indeterminate in the case of more than one possible stable solution and explosive if none of the solutions are stable (McCallum 2009).

The rule differs from the simple Taylor rule as policy directly responds to $E_t i_{1,t+1}$, i.e. the expected short rate.²⁰ The derivation of a 2-period term-spread rule results in the same law of motion for the short rate except that the sign in both denominators changes. It becomes immediately clear that the short rate is a non-predetermined variable and no restriction pins down the current short rate by construction. Indeed, there is a continuum of paths for the short rate that is consistent with the term-structure augmented Taylor rule. Determinacy of the macroeconomy then depends on the joint setting of the Taylor coefficient α and the value of the reaction parameter γ for the long rate. Both the standard Taylor rule and the term-structure augmented rules belong to the class of implicit instrument rules where the formula for setting the policy instrument is a function of other variables which must be projected by the central bank with the projections themselves being conditional upon the instrument setting. In other words, there is no exogeneity of arguments in the policy rule (Giannoni and Woodford 2003a).

For $0 < \gamma < 2$, if the private sector expects rates to go up, the short rate is increased today. This constellation mirrors the view that the central bank follows the bond market where shocks to the long rate might call for subsequent moves in the short rate. For instance, if there is a shock to inflation expectations and the long end of the yield curve jumps up, the central bank might follow with higher short rates.

In contrast, if $\gamma < 0$, the monetary authority reacts inversely to long-rate movements. The fact that if the private sector expects future rates to go down the central bank raises the short rate, is a reflection of a joke about the Bundesbank which says, “[t]he BuBa is just like cream, the more you stir it, the thicker it gets.”²¹ This inverse relationship can be explained by the long-rate as being the main determinant of aggregate demand. Any given change in interest-rate expectations might imply an opposite move in the current short rate to maintain maximal price (and output) stability. Moreover, though not sketched out in (6.15) above, a decline in risk premia might be associated with an economic stimulus and, thus, might provide the case of reacting with a monetary constraint. At an intuitive level, if the monetary authority is able to detect the origin of bond rate movements, the appropriate response of the short rate differs. As pointed out by Bernanke (2006), the optimal interest-rate setting with the long rate as an argument heavily depends on the identification of the source of movements at the long end of the yield curve.

From the perspective of dynamic stability, what is of essential interest is the existence of interest-rate expectations in the policy rule. Bernanke and Woodford (1997, 669) emphasize that the general problem of a forward-looking policy rule is such that it “need not imply a determinate rational expectations equilibrium and so may permit fluctuations arising purely from self-fulfilling expectations. Not only may the response of the economy to exogenous [...] disturbances be indeterminate

²⁰With a n -period bond, the policy rule would include either the expected path of short rates or the expected long-term interest rate. If the Expectations Hypothesis holds, the relevant expected spot rate could be substituted by the implied forward rate.

²¹Cited from Eijffinger et al. (2006, 10).

but there may also exist “sunspot equilibria” in which the endogenous variables respond to random variables unrelated to the structure of the model simply because they are expected to.” For instance, on the one hand consider the case where monetary policy reacts negatively to the long rate. If, however, a higher long rate can also indicate to higher inflation, the policy advice of reducing the short rate would be counterintuitive. On the other hand, expectations on higher future short rates may become self-fulfilling in the case of a positive γ because higher expected short rates in the future may cause the long rate to rise, leading the central bank to increase its policy rate. Such variation can result in instability of inflation, output and interest rates.

Figure 6.1 shows the regions of uniqueness (white area) in the space α and γ for two bond rates with maturities of $n = 4$ and $n = 20$ periods. Starting with the long-rate analysis, there is a downward and an upward sloping part for uniqueness. The first part reveals that for maturities under inspection, the necessary condition for determinacy can be modified in a way that “the sum of response coefficients”²² must be greater than one. In contrast to the Taylor principle where the short rate must rise disproportionate ($\alpha > 1$) in order to alter ex-ante real rates in the right direction, reacting to long-term rates at least relaxes that requirement.²³ To see the logic of the stability condition, aggregate demand can be re-written in terms of the long rate so that for a 2-period bond the 1st equation of system (6.12) becomes

$$i_{2,t} = -\sigma \tilde{y}_t + 0.5 \sum_{j=1}^2 \pi_{t+j} + \sigma v_t.$$

By responding to two period bonds, the central bank implicitly reacts to the average expected path of future inflation, to aggregate demand shocks and to the current output gap.²⁴ The horizon-dependent policy, similar to the Kozicki and Tinsley (2008) model, allows to alter the policy rate by less than the Taylor principle as long as the coefficient γ ensures an appropriate real rate change. If this is not the case, the model is indeterminate which is denoted by the black dotted space.

For large values of γ , Fig. 6.1 highlights the possibility of sunspot equilibria and indeterminate equilibrium outcomes when monetary policy responds too aggressively to the long-term interest rate. It can be attributed to self-fulfilling expectations because expectations of high future short-rates causes long-term interest rates to rise triggering a further increase in the current short-rate. In the upward-sloping part,

²²This statement should not be taken literally. It rather signifies that the necessary condition of the ex-ante real rate to change more than one-by-one can be assured by a combination of the coefficients α and γ .

²³In the limit (not reported here), with a 40-period bond as relevant variable, it has been also checked for determinacy of the system. It is found that a small positive value of γ with $\alpha < 1$ still yields a unique RE equilibrium.

²⁴See for similar results in a different setup Kulish (2006) and Vázquez et al. (2009).

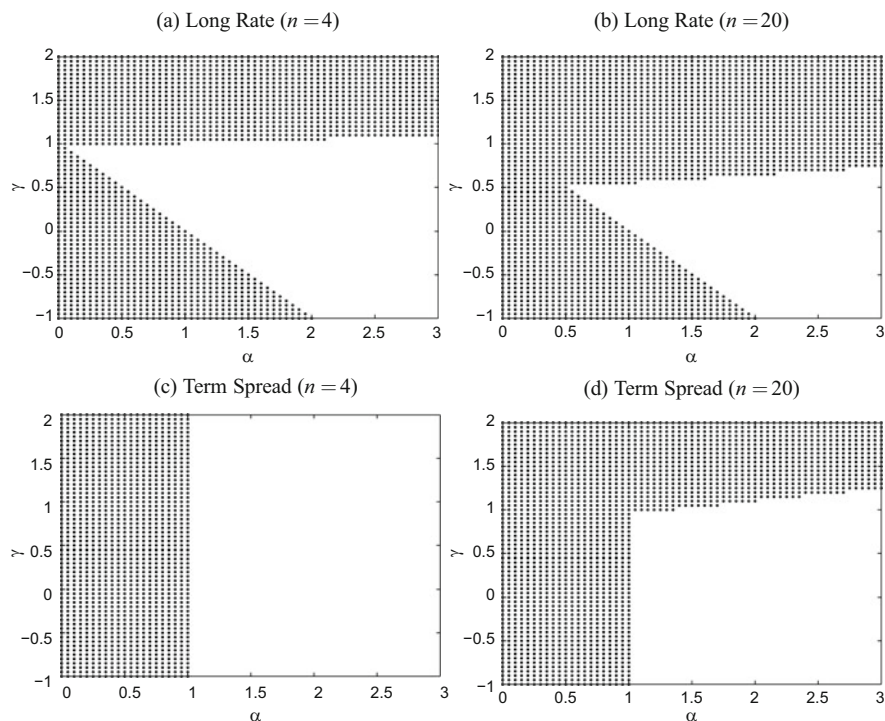


Fig. 6.1 Region of uniqueness for term-structure augmented Taylor rules

a strong reaction to the long rate must be accompanied by a larger inflation rate response in order to avoid bloating expectations. A high α allows bond traders to “learn” that the short-rate is a stationary process just because the solution to inflation is a stable process.²⁵

Turning to the term-spread augmented interest-rate rule, the region of determinacy changes considerably. For $\chi_t = i_{n,t} - i_{1,t}$, the parameter choices in space α and γ are greater compared to the long-rate rule, in particular for term spreads with shorter maturities. With a 4-period bond, each specification of γ leads to a unique rational expectations equilibrium as long as the Taylor principle is satisfied, i.e. $\alpha > 1$. To see this, consider a positive shock to aggregate demand which induces monetary policy to increase its short rate through inflationary pressure. Simultaneously, the yield curve flattens or even inverts invoking a relief in

²⁵For the sake of completeness, the reaction to the output gap in the rule would result in a slightly higher slope of the upward-sloping part of the uniqueness region. Since responding to $i_{n,t}$ implicitly means that monetary policy reacts negatively to current output, a higher value of the output coefficient allows for a higher value of γ since the latter is partly compensated by the positive output coefficient in the reaction function.

terms of monetary tightening. The net effect on the short rate depends on the reaction to inflation. As long as monetary policy disproportionately responds to inflation, bond traders form expectations of a rising short-rate environment so that long-term interest rates increase and the ex-ante real interest rate rise. When studying longer maturities, there is again an upward-sloping part. With longer maturities and high short rate sensitivity to the term spread, expectations can become self-fulfilling as explained previously; only a high α prevents expectations from exploding.

The analysis can be summarized with three main findings. Firstly, monetary policy must be horizon-dependent. It must sufficiently react to the average expected path of inflation in the presence of term-structure effects in aggregate demand. The explicit incorporation of yield-curve variables can ensure the condition for determinacy and may allow monetary policy to react to inflation by less than what the Taylor principle suggests. Secondly, there is an inherent danger of explosive dynamics and indeterminate system solutions if a central bank attributes a too large coefficient to the long rate or the term spread in the reaction function. Finally, the region of determinacy is larger for the term-spread augmented Taylor compared to the long-rate rule. It implies that the reaction to the slope of the yield curve likely ensures an appropriate change in the ex-ante real interest rate of relevant maturity to stabilize aggregate demand and inflation dynamics.

6.2.2 *Optimal Simple Rules with Term Structure Information*

Checking for the uniqueness of the underlying macroeconomic model reveals important insights into the area of opportunity of using term-structure based monetary policy rules. Little can be said, however, about the choice of rules from a welfare perspective. The standard approach to evaluate monetary policy performance is to calculate average losses for a specified periodical loss function which the monetary authority tries to minimize. Such optimal policy can be categorized according to two broad classes. The first class belongs to the parametric family of conditional simple instrument rules. Usually, in the best case, is optimized over the coefficients embedded in the interest-rate rule using an economic model to compute relevant welfare criteria with each set of possible parameters. In contrast to these optimized simple rules (OSR), the second class of rules derive from an optimal linear regulator problem (OLR). The monetary authority solves an optimization problem in which the structural equations of the model are taken as constraints in order to characterize the best possible pattern of responses to economic disturbances. In a next step, an evaluation of the type of policy rule takes place that can bring about the desired equilibrium.²⁶

²⁶See Giannoni and Woodford (2003a,b,c).

In what follows, the OSR approach is analyzed, thereby asking what the preferred monetary policy rule looks like for minimizing the objective function; in this respect, the expected value of periodical losses is given by

$$L = \frac{1}{2} E_o \sum_{t=0}^{\infty} \beta^t (\pi_t^2 + \lambda_y y_t^2 + \lambda_i i_{1,t}^2)$$

where β is the central bank's discount rate and the periodical loss is defined as the weighted sum of variances of inflation, output and the short-term interest rate. For welfare examination, minimizing L is equivalent to minimizing the weighted average of variances as approximated by their respective unconditional variances.²⁷ Computations are carried out through a numerical optimization routine where parameters are found by means of a grid search over those parameter values that ensure a unique solution. For each parameter combination, the average loss is calculated and the optimal simple rule is identified as the one for which the computed loss is smallest.²⁸ Moreover, optimal policy is evaluated against two preference specifications with differing weights λ_y, λ_i in the loss function.

Table 6.1 reports results for selected instrument rules. It includes the normative rule proposed by Taylor (1993) with given coefficients (I), an optimized Taylor rule (II), and the two term-structure rules with the long-rate (III) and the term spread (IV) as additional arguments.²⁹

Predictably, the pre-specified Taylor rule (I) performs worst in terms of volatility of the target variables. Since the rule is not tailored to the model setup, the result should not come as a surprise. By allowing the coefficients to vary, the average loss can be reduced by almost 30 basis points for the cases $\lambda_y = 0.5$ and for $\lambda_y = 1$. What is interesting is that the optimal reaction to inflation is unity – the condition for determinacy. Even with $\lambda_y = 1$ in the objective function, monetary policy performs best with $\alpha = 1$ and a small reaction to output.³⁰ When turning to the long-rate rule, losses can be reduced where optimal values of γ are of similar magnitude than α for $n = 20$. Notice that the sum of the two coefficients add up to unity. This pattern has been characterized as a sufficient condition for determinacy. It implies that, on average, monetary policy must react disproportionate to inflation dynamics in order to alter real rates. The numerical values for γ are all positive so that the

²⁷See Clarida et al. (1999) or Woodford (2003).

²⁸For that purpose, the OSR routine of DYNARE is applied but the procedure is modified in the following way; numerical optimization may be sensitive to the initial condition of the optimization routine. Therefore, a grid search over the parameter region is applied in the first place. Then, the optimized coefficients are used as starting values for the OSR routine. Finally, it is checked whether the generated loss can be further improved by using the OSR command.

²⁹For the sake of completeness, the term-structure rules can be augmented by an output coefficient. However, the losses remain basically unchanged compared to rule (III) and (IV).

³⁰This finding is also obtained for optimal simple rules based on an estimated euro area model (Stracca 2006).

Table 6.1 Optimal simple rules with term structure effects

	$\lambda_y = 0.5, \lambda_i = 0.25$		$\lambda_y = 1, \lambda_i = 0.25$	
	$n = 4$	$n = 20$	$n = 4$	$n = 20$
(I) $i_{1,t} = 1.5\pi_t + 0.5y_t$				
<i>Loss</i>	1.40	1.49	1.49	1.60
(II) $i_{1,t} = \alpha\pi_t + \beta y_t$				
<i>Loss</i>	1.17	1.20	1.28	1.31
α	1.0	1.0	1.0	1.0
β	0.01	−.01	0.30	0.01
(III) $i_{1,t} = \alpha\pi_t + \gamma i_{n,t}$				
<i>Loss</i>	1.04	1.07	1.15	1.17
α	0.17	0.56	0.16	0.56
γ	0.83	0.44	0.84	0.44
(IV) $i_{1,t} = \alpha\pi_t + \gamma(i_{n,t} - i_{1,t})$				
<i>Loss</i>	1.04	1.05	1.15	1.17
α	1.0	1.0	1.0	1.0
γ	3.01	0.79	3.00	0.79

rationale can be confirmed according to which monetary policy incorporates long-term interest rates as the latter reflect long-term inflation expectations. The term spread rule offers basically the same improvement as rule (III) where in all cases α takes on the value of unity. Moreover, there is clear tendency of yield-curve smoothing. If the spread widens, the short-term interest rate counteracts by adjusted policy moves. A common theme in the welfare analysis is that for central bank preferences welfare losses can be reduced when compared with the the optimized Taylor rule (II).

The discussion of term-structure augmented policy rules has shown that there are plausible parameter regions for a unique rational expectations equilibrium when using yield curve indicators in the conduct of monetary policy. Indeed, such rules may be the appropriate choice if there are strong, perhaps temporary deviations from the Expectations Hypothesis and long-term interest rates are not in line with what monetary policy intends to achieve through the expected path of future short-term interest rates. However, a crucial caveat remains, i.e. the use of such rules in a systematic, day-to-day way. The optimal parameter values are critical next to the region of indeterminacy. Consequently, monetary policy must be highly alert and it is well advised to monitor a variety of forecast measures for the purpose of preventing self-fulfilling expectations. This holds all the more if policymakers face uncertainty in terms of appropriate model specifications. It is not clear whether term-structure rules are robust to changing parameter values or even to different models. The standard Taylor rule proves it does, so a more rigorous analysis in different models would be in order for term-structure rules (Taylor and Williams 2009a).

6.3 Selected Further Issues on Interest Rates and the Conduct of Monetary Policy

6.3.1 Policy Inertia: What Does the Term Structure have to Say?

The dynamic adjustment process of monetary policy to the macroeconomic environment is a key feature for the effectiveness of central bank actions. It focusses on the persistent cyclical fluctuations in policy rates and asks how quickly central banks adjust monetary policy in response to developments in the economy. It is widely accepted that at high frequencies, central banks typically “smoothen” their policy rates over the course of several days and weeks. Changes in these rates are undertaken at discrete intervals and in discrete amounts (Rudebusch 1995; Balduzzi et al. 1997; Cochrane and Piazzesi 2002).³¹

It is also a common view that there is a smooth dynamic of short-term interest rates over the course of several quarters, which is the relevant frequency band for most macroeconomic analysis (Rudebusch 2002). Such adjustment includes, for instance, the increase of the target rate by two 25-basis-point movements in two proceeding decision meetings instead of raising the rate once by 50 basis points. Figure 6.2 illustrates the persistence in average quarterly policy rates for the United States and the euro area.³² It seems that the two central banks adjust interest rates only gradually. Indeed, the key rates mirror a tendency of moves of small step moves in the same direction and with few reversals, although there is a trend of faster policy moves in times of monetary easing than in times of monetary tightening – at least within the observed sample period. Former Vice-Chairman Alan Blinder (1998) confirms that “[...] in most situations the central bank will take far more political heat when it tightens pre-emptively to avoid higher inflation than when it eases pre-emptively to avoid higher unemployment.”³³ The documented asymmetry still allows an interpretation of interest-rate dynamics as being highly persistent. In this respect, the issue of the sources of such interest-rate behavior emerges, in particular, from a quarterly perspective.

The *intrinsic* view of partial adjustment highlights the explicit use of such behavior as part of an extended monetary policy strategy (Rudebusch 2006).

³¹Following Amato and Laubach (1999), such smoothing can be categorized as seasonal smoothing, event smoothing or day-to-day smoothing. It supports the elimination of calendar effects, the erratic behavior of the overnight rate in times of sudden events and it tries to keep the average level of the policy instrument as close as possible to the target level of the central bank.

³²For the US, the effective federal funds rate is the policy rate and for the euro area it is the main refinancing rate. Data are taken from Datastream.

³³This view can be justified by the presence of an inflation bias on part of a central bank’s loss function (Cukierman and Gerlach 2003); for a central banker’s perspective see Bini-Smaghi (2009). However, there is also evidence that the FED has exhibited such asymmetric preference only until the pre-Volcker area (Surico 2007).

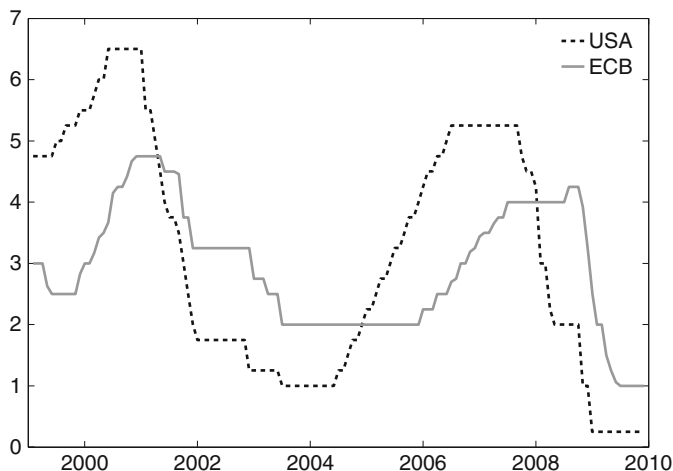


Fig. 6.2 Key interest rates USA vs. Euro area

Adjusted policy moves denote that the central bank knowingly distributes a desired change over the course of several quarters. This relatively slow adjustment has been referred to variously as a partial adjustment, as monetary policy inertia, interest-rate smoothing or gradualism. Its rationale originates from both empirical and theoretical considerations, such as the aim of low interest-rate volatility or the role of expectations in the transmission process. The opposing view is the *extrinsic* approach to partial adjustments, because it reflects the fact that monetary policy responds relatively promptly to changes in the economy. The reason for the alleged gradualism simply comes from a sluggishly evolving economy where relevant information for setting the target rate is accessible only slowly. It appears that in a smoothing pattern, the policy instrument moves just because the relevant arguments in the policy reaction function change slowly. Each view will be presented in turn with discussion on what the term structure has to say about gradualism of monetary policy.

Starting with the intrinsic approach, most research in empirical dynamic Taylor rules find a large coefficient on the lagged interest rate with quarterly data.³⁴ The finding implies that the monetary authority adjusts the actual policy rate only partially to the “desired” target level, where the latter typically resembles the standard Taylor specification with measures of inflation and the output gap as arguments. Generally, monetary policy inertia can be formalized to

$$i_{1,t} = \rho i_{1,t-1} + (1 - \rho) \beta^T X_t \quad (6.16)$$

³⁴See for instance Clarida et al. (2000), Rudebusch (2002), Sauer and Sturm (2003) or Castelnuovo (2007).

where the target rate is a linear function of a set of macroeconomic variables $i_{1,t}^* = \beta^\top X_t$ with X_t representing a vector of variables influencing policy (Rudebusch 2006). In literature, the smoothing parameter ρ takes on values between 0.7 and 0.9 so that the inertia approach seems to be confirmed by the empirical single-equation evidence.

From a normative perspective, there are rationales for explicitly conducting monetary policy through interest-rate smoothing; they include (1) the policy maker's uncertainty about the economy, (2) the leverage effect of shaping long-term interest rates and (3) the reduction of financial risk through gradualism.

Uncertainty is an elementary feature of central banking; it applies to the state of the economy, the economy's structure and the inference of the market participants to current policy actions. Bernanke (2004b) remarks that "[b]ecause policymakers cannot be sure about the underlying structure of the economy or the effects that their actions will have on economic outcomes, and because new information about the economic situation arrives continually, the case for policymakers to move slowly and cautiously when changing rates seems intuitive." This view, by a practical central banker, is supported by the literature on parameter and model and data uncertainty which calls for less aggressive responses to a changing and evolving economy (Brainard 1967). The small step strategy in an environment of imperfect knowledge can help to minimize the dangers of overshooting, to either the inflationary upside or the downside risk of a recession if data are measured with noise or the elasticities of the variables in the applied models can be hardly approximated numerically.³⁵ Credibility losses could be the outcome if frequent alterations in the path of policy rates could produce doubts about the ability of the central bank to understand the economy. In this respect, many interest-rate reversals could be interpreted *ex-post* as mistakes and could undermine the central bank's credibility (Williams 2003).

A complementary argument for monetary policy inertia emerges from the role of forward-looking behavior and the role of expectations for the effectiveness of monetary policy. It has been stressed throughout the Chap. 5.3 that if market participants' behavior is largely forward-looking, monetary policy should realize that maintaining price and output stability depends less upon current actions but, rather, upon how the private sector expects policy to act in the future. This fundamental insight plays a crucial role in the determination of long-term interest rates. Since, under the assumption of the Expectations Hypothesis of the yield curve, bond rates depend heavily on the anticipation of future policy actions, a policy that follows initial small interest-rate changes by further changes in the same direction will allow a moderate change in the current short-term interest rate to have a significant impact at the long end of the yield curve (Goodfriend 1991).

³⁵See Orphanides (2003) or Bernanke (2007). It should be noted that the gradualistic approach may not always be a valid advice under uncertainty, in particular when there is uncertainty surrounding the structure of the economy. Robust control methods rather advocate a more aggressive response, for example if the structural degree of the inflation process is not known (Söderström 2002).

Monetary policy gradualism, thus, can be a lever to stabilize and direct private expectations. In addition, such behavior can be optimal if the central bank credibly commits to policy inertia. One implication of such policy is that although the private sector is assumed to be forward-looking, optimal targeting rules produce a sufficient degree of a backward-looking monetary authority that can be translated into an optimized simple interest-rate rule with a high degree of smoothing (Woodford 1999b). Welfare analysis, in calibrated hybrid New-Keynesian versions, confirm the view that monetary policy should gradually adjust its policy rates even if there is no preference for interest-rate stabilization in the intertemporal loss function (Levin *et al.* 1999; Rudebusch 2002).

The case for explicitly considering interest-rate volatility in the central bank's loss function represents the third rationale for partial adjustment. It can be justified by the characteristics of the financial market and banking system. Commercial banks and leveraged financial intuitions face the cost of short-term funding. Since the asset side can be typically characterized by long-term maturities and the liability side by short-term, interest-rate sensitive debt positions (deposits and commercial papers), maturity transformation implies the danger of sudden cash-flow shocks to the financial system (Adrian and Shin 2008b). Policy inertia can support the maintenance of financial stability; financial institutions have more time to adjust the maturity spectrum of their balance sheets so that fire sales of assets can be prevented, resulting from direct monetary policy decisions. Closely related is the idea that gradual short-rate moves can minimize the risk of shocks to debt markets, which would create large capital gains and losses, and make the profitability of bond holders highly time-varying (Bernanke 2004b).

Although convincing arguments can be brought forward in favor of policy inertia, the extrinsic view denies a deliberate role of monetary policy in the smooth behavior of policy rates. Instead, the central bank sets the short rate precisely as needed to offset shocks to the economy. If the shocks cancel out very sluggishly or the adjustment mechanism of the key macro variables lead to a gradual policy response, the empirical observation of interest-rate smoothing is not triggered by the central bank but by a slowly evolving economy and possible persistent omitted variables in a Taylor-style reaction function (Rudebusch 2006). A non-inertial rule is then not misspecified in terms of dynamics but in terms of arguments. It holds that

$$i_{1,t} = \beta^T X_t + \vartheta^T Z_t \quad (6.17)$$

$$Z_t = \phi Z_{t-1} + \varepsilon_t \quad (6.18)$$

where Z_t is a vector of persistent latent factors to which monetary policy responds. Such incomplete description of monetary policy by the Taylor rule can be observed when plotting the deviations of the actual policy rate from the fitted, non-inertial target rate ($i_{1,t} - i_{1,t}^*$). Rudebusch (2006) impressively cites historical episodes where the gap has been significantly large. They include the US stock market crash in 1987, inflationary scares in 1994 reflected by a hike in long-term interest rates or the fear of deflation after the dotcom bubble in 2003/2004. The FED reacted to

information in an appropriate way beyond what is captured by the Taylor rule in order to fight the specific threats to price and output stability. As noted by Meyer (2002), “[a]lthough the Taylor rule has been a useful benchmark for policymakers, my experience during the last 5-1/2 years on the FOMC has been that considerations that are not explicit in the Taylor rule have played an important role in the policy deliberations.”

Both opposing views of observed interest-rate smoothing are based upon firm explanations from a theoretical and a policy-making perspective. In order to further assess the validity of arguments, one can evaluate the implications of the two approaches for macroeconomic and financial market dynamics. A straightforward way is to use term structure information since it imbeds expectations about future short-term interest rates. Market participants use a model of the short rate in order to forecast future interest rates (Ang et al. 2007). A high smoothing parameter ρ will clearly have a direct autoregressive effect on the predictability of the short rate and it will produce an indirect effect through an increased predictability of inflation and the output gap. Using a New-Keynesian model with lags and leads in the inflation and output gap equation, Rudebusch (2002) and Söderlind et al. (2005) find a high correlation between the size of ρ and R^2 from a regression of the realized short-term interest rate change on the model-consistent rational expectations of the interest-rate change.³⁶ More formally, the regression is performed as

$$\Delta i_{t+n} = \alpha + \gamma E_t \Delta i_{t+n} + \varepsilon_{t+n}. \quad (6.19)$$

In the limit, the coefficients should take on values of $\alpha = 0$ and $\gamma = 1$. From a modeling point of view, high monetary policy inertia must translate into forecastable movements in future policy rates. Only if ρ approaches unity the interest rate level becomes a near unit-root process with unpredictable changes (Mankiw and Miron 1986; Söderlind et al. 2005).

When testing such term-structure implications on historical data, the rational expectations version of the expected interest-rate change is replaced by corresponding forward rates, such as eurodollar or federal funds futures since they should be a good proxy for interest-rate predictability (Piazzesi and Swanson 2008; Ferrero and Nobili 2008). The empirical results are more than troublesome because the horizon-dependent regressor cannot at all explain the actual variation in actual short-term rate changes beyond the first quarter. Indeed, the empirical R^2 takes on values of zero when setting the forecasts horizon to more than three quarters. If monetary policy smooths the short rate and bond traders appraise this behavior, the latter should be able to predict future interest-rate changes better. This should also hold under the assumption of time-varying risk premia since the standard deviation of risk premia is typically smaller than the total standard deviation of the regression (Rudebusch 2002). The empirical insight,

³⁶The R^2 measures the forecastability of future interest rate changes; it reports the ratio of the explained variation to total variation.

thus, stands in stark contrast to the theoretical view that by credibly committing to future policy moves (possibly through the inertia term), the stabilization of expectations and the determination of long-term interest rates could be successfully managed.

As a consequence, the data cannot confirm the intrinsic view of a monetary policy actively distributing desired target changes over the course of several quarters, and communicating this strategy to the public through a comprehensible smoothing parameter.³⁷ Interest rate changes are then hardly predictable. However, these findings do *not* imply that monetary policy acts in a completely erratic, unsystematic way, entirely driven by monetary policy shocks. It is rather that the conventional Taylor rule is misspecified due to the absence of additional variables influencing policy decisions. Empirical macro-finance models of the yield curve confirm the single-equation evidence as the short rate quickly responds to shocks in inflation and output; they reflect the absence of partial adjustment and rather indicate to persistent (latent) factors extracted from yield-curve information to which monetary policy systematically responds. Indeed, in such models the smoothing parameter is often found to lie below 0.2 (Rudebusch and Wu 2008).

Recently, attempts have been made to detect such persistent factors through observable macro-related variables. Motivated by the factor-model view of representing empirical reaction functions, empirical research has confirmed that monetary policy decisions are based on large sets of macroeconomic information rather than on individual measures of inflation or output (Bernanke and Boivin 2003; Belviso and Milani 2006; Moench 2008). In this respect, key common factors are extracted from a variety of data sets through a dimension reduction (principal component analysis), used to study the interaction of monetary policy and key macroeconomic aggregates. Data can be compressed into a real activity factor, an inflation factor, a long-term interest-rate factor, a financial market factor or a money and credit factor. Empirical short-term policy rules from the augmented data set signal that there is indeed omitted information in the standard Taylor rule with spurious persistence in the short rate. Policy inertia, thus, seems to be less important where the excluded variables often mirror financial-market conditions. Such findings are consistent with the single-equation evidence where credit spreads or long-term interest rates (mirroring inflation scares in the bond market) are used to consistently describe central bank behavior (Gerlach-Kristen 2004).

³⁷Further evidence on interest-rate forecastability from survey data or from Taylor-augmented interest rate rules with serial correlated shocks largely seem to confirm the view that policy persistence may not (solely) originate from inertia (Söderlind et al. 2005; Gerlach-Kristen 2003; Castelnovo et al. 2003; Castelnovo 2007; Consolo and Favero 2009). Based on a simple analysis of a Taylor rule with a serially correlated disturbance term and the Fisher relation, Cochrane (2010) finds that a typical Taylor-rule regression of the policy rate on the inflation rate estimates the disturbance serial correlation parameter rather than the smoothing parameter.

6.3.2 *Monetary Policy Communication and Yield Curve Reflections*

The process of enhanced central-bank transparency has been steadily pushed forward during the last 25 years. Issing (2005a) documents the shift from central banks, characterized by secrecy and opacity, to an independent central-bank regime of greater accountability and transparency.³⁸ Such developments can be explained considering the circumstances of increased monetary-policy effectiveness when monetary policy uses the main instrument of transparency, i.e. its communication with the public. Managing expectations is at the heart of the modern view of central banking, in particular managing inflation expectations and the expected future path of monetary policy. Considering the fact that it is generally accepted that agents are uncertain regarding the economic environment, communication helps to clarify the systematic response pattern of monetary policy to economic developments. Steering market expectations can create an environment of reduced uncertainty, improved planning of market participants, lower interest-rate volatility and more effective monetary policy (Eijffinger and van der Cruijsen 2007); the latter highly depends on the perceived structure of the economy, possibly non-rational expectations and asymmetric information between market participants and the central bank (Blinder et al. 2008).

Interacting with the general public and financial markets has evolved and become an integral part of modern monetary policy. The content of communication can be summarized by at least two aspects: (1) the monetary policy strategy and (2) the monetary policy stance (ECB 2007). A firm guidance on the enunciated (numerical) objects fosters the anchoring of long-term expectations and separates them from current circumstances. In turn, expectations can influence current economic outcomes in terms of quantities and prices so that well-anchored expectations help to stabilize current inflation and output dynamics. Due to this reason, many central banks communicate quantitative targets given either by its legislative mandate or through its own initiative.

Asymmetric information and diverging model views about the economy may produce discrepancies between expectations of market participants and those of the central bank. In the past, it has been widely accepted that the historical record of policy actions would be the best way to align private sector expectations. In recent years, however, central banks consider the contingency of misinterpretations by explaining their policy decisions and providing extensive background information about the current state of the economy and the most likely outcome of the economy in the future. The main intention is to improve central bank watchers' understanding of the systematic behavior of monetary policy. Therefore, any forward-looking information with respect to future policy inclinations may allow for smoother

³⁸An excellent confrontation from a policy perspective can be found in Goodfriend (1986) who makes the case in favor of maintaining secrecy and Poole (2005) who outlines the benefits of increased transparency for the predictability of monetary policy.

dynamics on labor, goods and financial markets. Following ECB (2007), forward guidance can be achieved by three types: (1) the central bank may use indirect signals, often in the form of code words (language), (2) it may issue risk assessments that may hint indirectly to the future policy direction conditional on current information, (3) or it may explicitly communicate a quantified path of future policy rates.

Computations of projected paths for future interest rates are relevant in order to communicate forecasts on key macroeconomic variables such as inflation and output. A policy path can be described by “the sequence of current and expected future settings of the policy rate that central bankers believe will be consistent with achieving their goals” (Kahn 2007, 27). This communicated policy path may be based on constant interest-rate projects where a today’s constant policy rate may satisfy a specific target criterion in the medium-term future. For instance, if monetary policy publishes an economic projection based on the current short rate, private agents can compare the projection with a target criterion set by the central bank. In case of inflation, if this projection is higher than the target, the market infers that short-term rates will likely rise over the projection horizon. Such a procedure, though applied by various central banks, has been criticized for a number of reasons, among them is the danger of cumulative instability in the spirit of Wickseil when interest-rates are held constant by the central bank.³⁹ Alternatively, the policy path can be based on the assumption that policy includes a model of the central bank’s own future behavior which implies publishing some information about the time-varying policy path for the forecast horizon (Woodford 2005a). This projection exercise should be even more appealing since it allows market participants to directly use the announcements of the policy path without relying on further extraction techniques. However, in practice, the main concern against this projection is that such communication might be mistaken for commitment by agents failing to detect the conditionality of the sequence of future short-term interest rates (Kahn 2007). If the central bank is obliged to act inversely to the previously projected policy path in response to new arriving information, its credibility and reputation could be damaged even though the central bank credible makes the point of a changing macroeconomic environment.⁴⁰

Still, announcing a policy path, providing direct signals in form of statements or publishing implicit projections imbedded in releases of macroeconomic forecasts can help financial markets to price assets more efficiently by reducing near-term uncertainty concerning interest-rate volatility stemming from policy decisions. For the effects on the term structure, the question of what kind of assumption to make about the future policy path is of pivotal importance given the no-arbitrage

³⁹See Goodhart (2001) for an extensive discussion and critique of the constant-rate approach.

⁴⁰Another pitfall is grounded in the information content of asset prices. Too much communication might distort the use of asset prices for the purpose of extracting market expectations if, as a results of public information, not enough private information is priced into asset prices. On account of this “a central bank may face a trade-off between managing market expectations and learning from them” (ECB (2007, 65); Morris and Shin (2002)).

assumption in asset pricing models. A communicated path gives policy makers a greater leverage over long-term interest rates in such a way that private short-term interest-rate expectations are more in line with what the central banks intends to bring about. Significant deviations from the Expectations Hypothesis (EH) may then be at least reduced. As stated by Bernanke (2004a), “FOMC communication can help inform the public’s expectations of the future course of short-term interest rates, providing the Committee with increased influence over longer-term rates and hence a greater ability to achieve its macroeconomic objectives.” If private agents have imperfect and biased knowledge about central bank’s preferences or the parameters of the policy rule, direct signalling of future short rates in a theoretical New-Keynesian model reduces the forecast errors on part of private agents and welfare losses can be considerably minimized (Rudebusch and Williams 2008).

Evidence on the effects of central bank transparency and communication confirm the above remarks. Inflation expectations are heavily affected by an announced quantitative inflation target. Central banks that follow this strategy are far more successful in decoupling long-term expectations from short-term fluctuations. Short-term deviations are mainly driven by the business cycle and do not alter the long-term inflation outlook. This can be seen when inspecting the effects of macroeconomic releases on long-term inflation expectations extracted from the yield curve. Euro area bond rates are barely sensitive to such releases attributed to the numerical target of below, but close to, 2% over the medium term (Ehrmann et al. 2007). In contrast, the empirical evidence for the US differs where macroeconomic releases are often found to be statistically significant (Gürkaynak et al. 2006a; Beechey et al. 2008).⁴¹ Consequently, long-term inflation expectations seem to be more firmly anchored in the euro area than in the US as documented by financial-market based measures.⁴²

Turning to the communication of the monetary policy stance, financial market reacting to various communication channels can highlight the importance of “words” rather than of “deeds” for shaping asset prices. Stock and bond prices are a particularly accurate object of investigation since traders price assets by continually fine-tuning the views on the likely future policy path. Financial markets are quick to identify reactions where market prices are quoted with intra-day high frequency.

Consistent with the evidence on macroeconomic news, the analysis of intra-day market activity reveals that the volatility of bond markets is higher for the US than for the euro area. This fact indicates more expectational reversals regarding the expected level of the short-term interest rate in the US. It implies that monetary policy decisions in the euro area are more predictable given the conditional information for the private sector (Andersson 2007). Yet what is more interesting, is that central banks typically explain their monetary policy decisions as soon as they have been taken. For instance, the ECB provides a timely press conference with an introductory statement and a questions and answers session with the

⁴¹The finding is also supported by the analysis of survey expectations (Geiger and Sauter 2009).

⁴²See also Sect. 6.4 on the nominal yield curve decomposition.

financial-market press. From the perspective of the effects of such communication, ECB (2009a) reports trading activities on the days of the monthly press conferences for a selected number of future bond-market instruments with different maturities. The descriptive analysis clearly shows that trading activity not only increases in response to the announcement of policy decisions. Rather, the trade volume during press conferences on average exceeds the one during the actual monetary policy decision. This finding is robust in particular for medium- and long-term bond instruments – an indication that statements matter most for the implied policy path translating into yield curve effects (Gürkaynak et al. 2005a; Brand et al. 2006). Similar results can be obtained when testing the effects of speeches, interviews, periodical reports on the term structure of interest rates. The results however are country-specific and they depend on the channels of communication.⁴³

Transparency and communication contributes considerably to the effectiveness of monetary policy. If market participants form expectations that are not based on perfect information, but are built upon uncertainty, they need to learn from market experience. A central bank can align this process by providing in-depth analysis about the conditional macroeconomic environment surrounding its policy decisions. A firm forward guidance, in terms of (in-)direct signalling of likely developments of the policy rate in the medium-term future, can put a frame around financial markets without committing to it in the case of changing conditions. In this respect, given the pivotal role of the future path of short-term interest rates for the term structure, communication in form of statements and interviews is essential to offer “path news” that affect the medium- and even long-end of the yield curve, thereby preventing long-term interest rates from deviating too much from the level the central bank wants to implement in the bond market in order to achieve maximal price- and output stability.

6.4 Decomposition of the Nominal Yield Curve – BEIRs and Inflation Risk

Only during the last years have inflation-linked instruments become an established asset class in financial markets. There have been rare examples of issuances in the post-war era before 1990s. They include countries with high and volatile inflation in

⁴³See for instance Ehrmann and Fratzscher (2005), Andersson et al. (2006), Siklos and Bohl (2007). Recent research highlights the interaction of public and private information and its flows over time on money markets. Ehrmann and Sondermann (2009, 1) find that “due to the quarterly frequency at which the Bank of England releases its [inflation report, the reaction to news] can become stable over time. In the course of this process, financial market participants need to rely more on private information. The more time has elapsed since the latest release of an inflation report, market volatility increases, the price response to macroeconomic announcements is more pronounced, and macroeconomic announcements play a more important role in aligning agents’ information set, thus leading to a stronger volatility reduction.”

the 1970s which provided indexed bonds as the only possibility to raise capital for the purpose of financing large debt burdens (Colombia, 1967, Argentina, 1973).⁴⁴ In the 1980s, few governments started issuing indexed debt as a mean of credible disinflationary policies and of lowering the cost of borrowing (UK, 1981, Australia, 1985). Such indexing should tie the government's hands in using the incentives of inflating debt burdens and in taking the revenues that acquire from such inflating. Since the mid-1990, the latest development in inflation-protected bond markets has been initiated by major industrialized countries in an era of low and stable inflation dynamics. Outstanding bond values have grown substantially, with the US Treasury Inflation Indexed Securities (TIPS) market now representing the largest bond market. The rationale for the recent emphasis on indexed-bond markets has been to take a step forward towards the appealing idea of having complete markets. It fills the gap for both governments and bond holders by providing real security instruments (Garcia and van Rixtel 2007).

In contrast to nominal securities, inflation-protected bonds are adjusted according to the periodical change of a specific consumer price index so that the interest rate is tied to a price basket where bond holders are compensated for inflation by adjusting both the principal and the periodical coupon payments one-by-one to the consumer price index. For issuers, the ability to raise capital through indexed-bonds offers the benefit of reducing the cost of borrowing because investors are willing to pay a premium in order to be protected against inflation. This willingness can reduce the real costs of financing though the net effect might be unclear.

For (long-term) investors, inflation-linked bonds provide the only hedge against the risk of inflation since they provide a riskless real income stream up to the residual maturity. From the perspective of asset allocation, at first sight this view seems ambiguous and skin-deep. Conventionally, with inflation, the one-period real bond is regarded as a riskless investment opportunity where the one-period nominal bond stands in as a reliable proxy for the one-period real bond since it permanently adjusts its one-period returns to a changing price index. Short-term security instruments are therefore said to be safe investments in times of inflation. However, if one-period real and nominal interest rates are time-varying, a long-term investor is exposed to interest-rate risk when rolling over one-period securities for the preferred investment horizon. Campbell and Viceira (2001) prove within an optimal portfolio choice problem of an infinite-lived investor that when a long-term inflation-indexed bond is available, this bond represents the riskless asset since it finances a constant consumption stream. In case only nominal long-term bonds are available, the risk-averse investor shortens nominal long-term bonds due to inflation risk and holds them only for speculative purposes in the presence of a positive expected term premium.

⁴⁴ An excellent review about the development of indexed-bond markets is Garcia and van Rixtel (2007). They document an extensive economic literature dating back to W. S. Jevons, I. Fisher, J. M. Keynes, R. Musgrave and M. Fiedman who all favored indexing debt in general, and public debt in particular. For a discussion on the costs and benefits of indexation for macroeconomic stability, the reader is referred to Humphrey (1974).

Against this background, valuable measures can be extracted from inflation-indexed interest rates and they can provide an important source of information for financial market participants and central banks. With mature TIPS markets, it is possible to derive a direct market measure on the term structure of (ex-ante) real interest rates. The latter can be used to give appraisals of growth prospects and might mirror real rates of return on investment as perceived by financial markets. Equipped with the neoclassical growth-model apparatus, movements to the real rate of return, i.e. the marginal product of capital, can be attributed to demographic and productivity dynamics as well as the intertemporal time-preference of savers (Acemoglu 2009).

From the perspective of a central bank's mandate of maintaining price stability and the need of finding reliable inflation indicators, a large number of available linked bonds make it possible to decompose the nominal yield curve by comparing nominal and inflation-linked interest rates with comparable maturities. The difference between both curves gives a measure of inflation compensation, i.e. the inflation rate at which the expected real return from any two nominal and real bonds with identical maturity would be the same. These rates are typically labeled as spot break-even inflation rates (BEIRs) and are often viewed as a proxy for investors' average expected expectations of future inflation over the residual maturity of the bonds. The break-even inflation rate for maturity n can be calculated according to

$$BEIR_{n,t} = i_{n,t} - r_{n,t}. \quad (6.20)$$

The computation of implied forward break-even inflation rates typically offers more precise information about inflation compensation due to its flexible forward construction. The difference between the n -period nominal and TIPS forward rate maturing in m periods measures average expected inflation for the next m periods beginning in n . Consequently, spot break-even rates can be further decomposed into forward-break even rates with varying maturities. For instance, the 10-year instantaneous forward break-even rate displays today's market view about the annualized inflation in 10 years. Alternatively, the 5-year forward BEIR five years ahead measures average expected inflation as of today over the next five to ten years. Many central banks use these forward break-even rates to make an assessment about expected inflation in the distant future.⁴⁵

Despite its attractiveness in terms of timely availability and means-tested flexibility, BEIRs generally do not reflect expected inflation alone. First and foremost, nominal bond holders are faced with unanticipated changes in inflation which can not be forecasted. Since future inflation erodes the real payoffs of a nominal bond but not those on an indexed-bond security, the real payoff of the nominal

⁴⁵For example, the ECB's Monthly Bulletin regularly reports 5-year forward break-even inflation rates five years ahead to check whether medium-term inflation expectations are at a level consistent with the central bank's inflation objective; see for instance the December Monthly Bulletin 2009a.

bond is an uncertain number. Risk-averse investors, thus, require a premium to get compensated for inflation risk. Recalling from Chap. 2.1.4.3 that the inflation risk premium can be consistently derived from the no-arbitrage assumption, it is defined as the covariance between the real stochastic discount factor and the change in the price index. In a consumption-based setting, if output growth tends to be low when inflation is high, holders of nominal bonds will perceive nominal assets as more risky and demand an excess return to hold them. With rising inflation uncertainty over the horizon, risk premia on longer-term bonds should be higher than those on shorter maturities though the sign of the premium is not defined a-priori from economic reasoning. It depends on the sign of the covariation of the real stochastic discount factor and inflation. All in all, inflation risk premia tend to bias inflation compensation as measured by conventional BEIRs upwards.

Besides inflation risk, liquidity issues may further distort the information content of break-even inflation rates as the indexed-linked bond is characterized to be less liquid than its nominal counterpart (Campbell et al. 2009). As a reflex of low bond supply on part of governments, index-linked interest rates therefore embed a higher liquidity premium biasing break-even inflation rates downwards. Finally, technical and institutional factors may occasionally account for movements in BEIRs originating from tax distortions or changes in portfolio management strategies of large institutional investors with significant effects on bond yields (Garcia and van Rixtel 2007).

Abstracting from liquidity premia and technical factors, a representative nominal bond yield with maturity n can be expressed as⁴⁶

$$i_{n,t} = i_{n,t}^{rn} + \phi_{n,t}^{nom} \quad (6.21)$$

where $i_{n,t}^{rn}$ is the risk-neutral n -period nominal interest rate consistent with the Expectations Hypothesis (EH) and $\phi_{n,t}^{nom}$ stands for the total nominal term premium. Introducing an inflation-indexed bond with the same residual maturity, the nominal bond rate can be decomposed into

$$i_{n,t} = r_{n,t} + E_t \pi_{t,t+n} + \zeta_{n,t}. \quad (6.22)$$

It is the sum of the real interest rate $r_{n,t}$, the average expected inflation rate over the life of the bond denoted by $E_t \pi_{t,t+n}$ and an additional compensation for the risk of realized inflation turning out to differ from expected inflation, i.e. the inflation risk premium $\zeta_{n,t}$. For the sake of completeness, the real interest rate itself can be further specified in terms of its risk-neutral rate $r_{n,t}^{rn}$ and its real risk premium $\phi_{n,t}^{real}$ so that the nominal interest rate consists of four variables

$$i_{n,t} = r_{n,t}^{rn} + \phi_{n,t}^{real} + E_t \pi_{t,t+n} + \zeta_{n,t} \quad (6.23)$$

⁴⁶The equation also ignores the implications of Jensen's inequality terms.

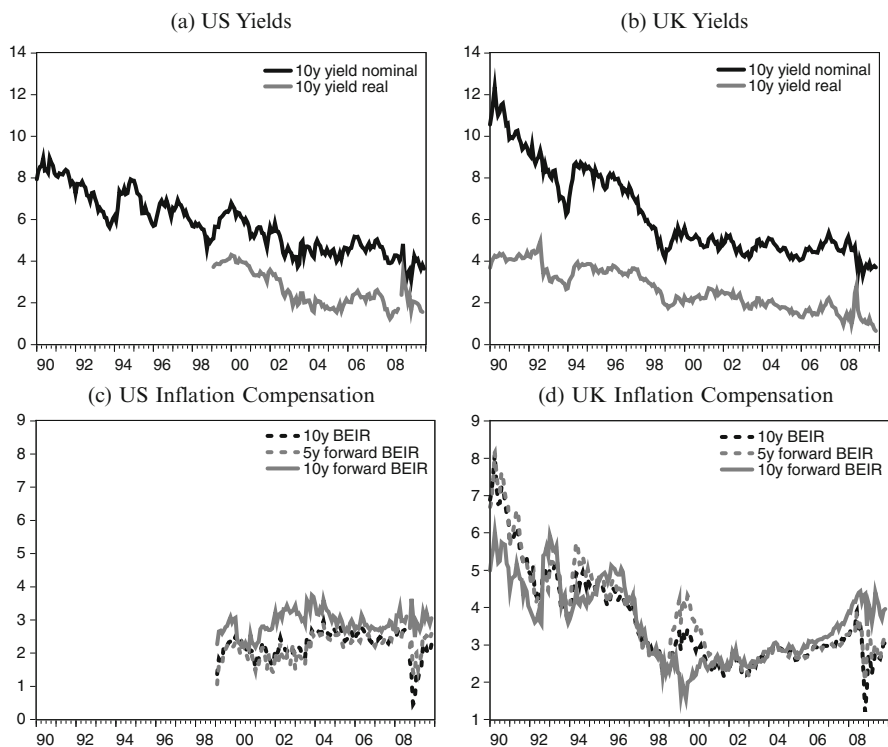


Fig. 6.3 Nominal and real interest rates in the US and the UK

with $\phi_{n,t}^{nom} = \phi_{n,t}^{real} + \zeta_{n,t}$ and $i_{n,t}^{rn} = r_{n,t}^{rn} + E_t \pi_{t,t+n}$. When evaluating (6.23) against (6.20), the break-even inflation rate of respective maturity is composed of average expected inflation and the inflation risk premium that – if significantly large and time-varying – distorts the BEIR as reliable proxy for financial markets' inflation expectations (Hördahl 2008)

$$BEIR_{n,t} = E_t \pi_{t,t+n} + \zeta_{n,t}. \quad (6.24)$$

To assess the role of inflation risk, Figs. 6.3a–b display the recent behavior of US and UK long-term nominal and real interest rates with the two markets being the most established markets for inflation-index bond securities.⁴⁷ The time series

⁴⁷Data for the US are taken from the research database of the Federal Reserve System. The daily nominal and real term structure is calculated according to the Nelson-Siegel-Svensson method and both curves are converted to end-of-month data. Gürkaynak et al. (2006b) and Gürkaynak et al. (2008) are the respective research papers explaining the fitting techniques in detail. Monthly nominal and real interest rates for the UK are available directly from the database of the Bank of England. They are calculated using a cubic spline method.

are restricted from January 1990 to November 2009, due to the relative short data availability of US real interest rates; moreover, estimated inflation risk premia for the later analysis are available only as of 1990.

Starting with the US, interest rates show a considerable decline in both nominal and real terms. From 1999 which marks the beginning of publishing inflation-indexed bond prices, interest rates fell from their peaks in 2000 to historical low levels. The significant decline in nominal interest rates is a phenomenon mostly driven by an equivalent fall of long-term real interest rates – the latter standing at 4% in 2000 and reaching levels of almost 1% in the midyear of 2008. The abnormally strong hike since then (with a recovery in early 2009) is the result of high volatility in bond markets, especially in the 10-year real interest rate. It reflects market factors not driven by expected short-term real rates or interest-rate risk premia. Flight-to-liquidity flows during the financial turmoil in 2008 led to a massive demand for long-term nominal bonds; whereas inflation-indexed bonds suffered from a surge in liquidity premia as a reflex of investors preferences (Hördahl 2009). This process has been further enhanced by institutional factors, in particular by heavily leveraged investors being forced to quickly unwind positions with considerable effects on real bond prices.

Similar results can be found for the UK where real interest rates are available for the entire sample period including the two key monetary policy events, i.e. the first announced inflation target in 1992 and the given independent of the BoE in 1997 with a reformulation of the numerical inflation target. The 10-year inflation-protected interest rate fell from 5% in the beginning of the 1990s to 1% in mid-2008 which has been accompanied by a rectified decrease in its nominal pendant. The inverse dynamic of the nominal and real interest rate in 2008 originates from the same sources as in the US so that these movements should be interpreted with caution when it comes to decomposing nominal interest rates.

Figures 6.3c–d show the history of selected break-even inflation rates for the two countries. The 10-year US spot BEIR was roughly between 1.5% and 2.0% during the first years of the decade and stabilized at a level of about 2.5% until 2008 in the run-up of distortions in financial markets. For the UK, the corresponding spot BEIR has been highest throughout the 1990s and stabilized at 2% until the year 2002 before gradually increasing up to almost 4% in 2008. When using implied forward rates for the calculation of forward BEIRs, differences to the spot BEIRs can be detected. For instance, the 10-year forward inflation compensation is on average slightly higher for the US and it moved in opposite direction to the spot BEIR between 2000 and 2004. For the UK, the 10-year forward break-even rate has experienced a much stronger and prolonged increase in recent years. Soaring average BEIRs deriving from spot rates is then the result of an increase in medium-term to long-term inflation compensation rather than a short-term phenomenon.

Whether such developments are the reflex of changing long-term inflation expectations or of time-varying inflation risk premia is of essential importance for policy makers in order to assess the markets' view about the central bank's intention to maintain price stability over the long run. The decomposition of BEIRs typically

relies on a joint model of the nominal and real yield curve where nominal and real bonds are priced internally consistent so that trading strategies relying on these prices do not allow for risk-free profit opportunities (Hördahl 2008).⁴⁸ Estimated models differ with respect to the explicit use of inflation-indexed securities, the incorporation of survey data, the inclusion of observable macroeconomic variables and the model specification. The latter deals with the research question of applying a pure econometric model or of building a structural macroeconomic one.⁴⁹

Figure 6.4 shows inflation expectations and financial risk premia derived from two types of models. The first model relies on Hördahl (2008) and Hördahl and Tristani (2009). It covers estimates for the US and the euro area where interest rates are modeled through the macro-finance lens. In this respect, macroeconomic dynamics, the nominal as well as the real yield curve are jointly estimated.⁵⁰ The second model of Joyce et al. (2009a) is applied to the UK term structure; it follows the “pure” finance strategy of describing bond pricing dynamics by unobservable, latent factors and by the change in the RPI price index.⁵¹

The US 10-year average inflation expectation is plotted in Fig. 6.4a. Given the break-even inflation rate and the estimated inflation risk premia, it equals the 10-year spot BEIR less the inflation risk premium. With small deviations between 2001 and 2002, total inflation compensation and inflation expectations tend to be close to each other. Indeed, changes in the BEIR are more or less entirely reflected in changes in inflation expectations so that inflation perceptions are well approximated by BEIR dynamics. Turning to the various term premia concepts embedded in real and nominal bond yields, the total US nominal term premium has substantially declined over the sample period, a feature also found in Chap. 4.2.1 or in Kim and Wright (2005). This development was most profound during the “conundrum” period of 2004/2005 and it was the driving force of very low long-term nominal interest rates (Greenspan 2005). Being aware of the fact that the nominal term premium is the sum of the real term premium and the inflation risk premium, total risk can be likewise disentangled. Though the 10-year inflation risk premium is slightly positive on average (14 bps), it turned negative in the years 2001–2003 where inflation significantly fell and deflation scares had been on the top agenda of policy makers and market participants; investors, thus, became less

⁴⁸The arbitrage-free environment refers to the elimination of all risk-free opportunities between the nominal and real term structure and along the respective term structures with different maturities.

⁴⁹It is possible to jointly estimate the real and nominal yield curve without any reference to observable indexed bond prices. By applying the inflation rate in the estimation, real bond yields can be computed entirely consistent with the no-arbitrage model setup. For selected studies with varying model specifications see Buraschi and Jiltsov (2005), Durham (2006), Garcia and van Rixtel (2007), Ang et al. (2008), Chernov and Mueller (2008), D’Amico et al. (2008b), Christensen et al. (2008), Adrian and Wu (2009) or Joyce et al. (2009a).

⁵⁰The macroeconomy is assumed to follow the hybrid New-Keynesian view with an empirically justifiable leads and lags structure. I am indebted to Oreste Tristani (ECB) for sharing 10-year estimated risk premia with me.

⁵¹I deeply thank Mike Joyce (BoE) for providing the estimated times series on forward risk premia.

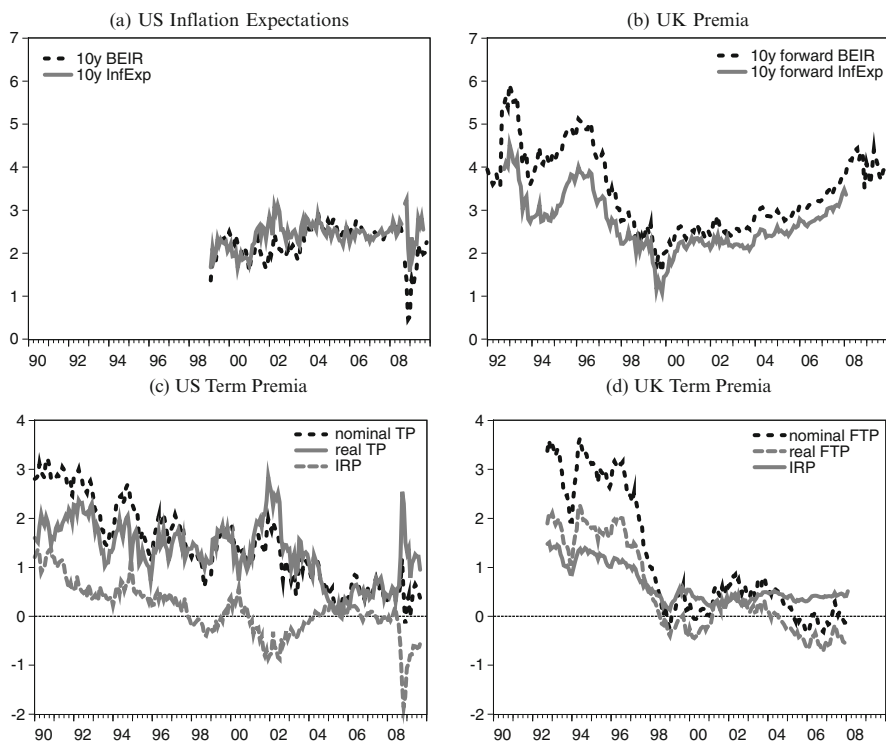


Fig. 6.4 Inflation expectations and term premia in the US and the UK

concerned about inflation risk (Hördahl 2008).⁵² Instead, for the US, most variation in the nominal term premium is attributed to shifts in the real term premium. Real rate uncertainty pops up in appropriate changes in inflation-index interest rates and partly attributes to the synchronization of long-term nominal and real interest rates.

Similar results hold for the estimated UK model of Joyce et al. (2009a). Changes in the 10-year forward BEIR for the most part go hand in hand with changes in expected inflation so that BEIRs represent a reliable proxy for expectational changes – though the absolute level between the two measures considerably differs until 1997 due to the presence of higher inflation risk. In this respect, the BoE's given independence in 1997 plays a crucial role for the dynamics of the estimated long-term inflation risk premium; the independence has led to a sharp decline of the inflation risk premium by over 70 bps where falling long-term inflation expectations supported the slump in forward break-even rates. At the same time, real rate risk

⁵²The findings are robust to different models and sample periods (D'Amico et al. 2008b).

premium fell by even more and stayed on average at negative numbers.⁵³ Since 1998, most of the UK nominal term premium is due to compensation for real risk and inflation risk premia stabilized at a low level of 50 bps for the 10-year forward horizon. From the period of 2003 onwards, long-term forward inflation expectations significantly increased again to a level of around 3%.

For the sake of completeness, studies on break-even inflation rates and inflation risk premia have recently become available for the euro area (Garcia and Werner 2010; Hördahl and Tristani 2009). The analysis on BEIRs and inflation expectations depict that due to higher inflation risk premia, model-implied inflation expectations are significantly below the corresponding break-even inflation rates and more stable than its directly observable proxy. Moreover, the average long-term inflation risk premium mainly accounts for the size of the nominal term premium; whereas the real component appears to be less volatile. These results stand partly in contrast to the findings in the US where inflation expectations can roughly approximated by BEIRs and average nominal term premia are largely determined by real risk.

All in all, the availability of inflation-index bond securities greatly enhances the use of bond instruments for the purpose of extracting crucial information for policy makers and the private sector. It seems that the inflation risk premium is rather more stable than a highly volatile figure ranging between 0% and 1%. Against this background, some caution is needed when interpreting the level of break-even inflation rates as an adequate measure of inflation expectations, in particular for the euro area. Rather, variations in those rates mainly reflect changes in inflation expectations. Monitoring these dynamics enable central banks to gauge insights into their own policy effectiveness and on markets' changing views about the long-term inflation outlook.

⁵³Besides the role of the independence of the BoE, Joyce et al. (2009a) explain the large fall in the level of real term premia by an increased pension-fund demand for inflation-indexed securities against the background of the 1995 Pensions Act which became effective in 1997.

Part III
Financial Stability and Monetary Policy

Chapter 7

Financial Risk and Boom-Bust Cycles

7.1 Traditional Transmission Channels

The monetary transmission is one of the most studied fields of monetary economics from both a theoretical and an empirical perspective. Understanding how monetary impulses are translated into changes in aggregate expenditures is of essential importance for the evaluation of the monetary policy stance. Monetary policy must be conducted with an accurate assessment of the timing and effect of policy changes on prices and quantities on an aggregate level (Mishkin 1995). The evolution and identification of the multiple channels of monetary transmission at work has changed with the emergence of varying conceptional frameworks which have been applied to analyze monetary policy. For instance, the early debate on the propagation effects between Keynesian and Monetarist models has been dominated with the question on how changes in the money supply affect the yields of a (imperfectly substitutable) set of assets. Whereas in the simple IS-LM model the relevant distinction was based on money and a “representative” long-term bond, the monetarist approach highlighted the importance of wealth and substitution effects between various domestic and foreign assets, which are affected by changes in the money supply and have induced changes in investment and consumption paths.¹ By the same token, the recognition that the central bank uses the short-term interest rate as the operating instrument for achieving its policy goals may alter the logic of monetary transmission compared to the older view according to which money supply was treated as exogenous and viewed as under the control of the monetary authority. With an interest-rate setting monetary policy, the stock of money adjusts endogenously and arises from the private sector’s money demand and the bank’s willingness to provide loans (Meyer 2001).

¹ See, for example, Hicks (1937), Friedman (1959) or Meltzer (1995).

In recent years, a consensus view about propagation effects has emerged. It is categorized into the “money view” and the “credit view” of monetary transmission.² The former operates under the premise of perfect financial markets and compromises the effects of real interest rates on consumption and investment decisions; whereas the latter explicitly allows for frictions in credit and financial markets that alter real magnitudes in the economy.

Within the traditional *interest rate channel*, the emphasis is on long-term real interest rates rather than on nominal short-term interest rates. The Expectations Hypothesis of the term structure represents the key link for bond prices of different maturities. Given the rigidity of prices or wages, changes in long-term nominal bond yields also lead to movements in their real counterparts. The demand for durable consumption as well as fixed and residential investment is said to be negatively related to long-term real interest rates through the concept of the user cost of capital (Jorgenson 1963). When monetary policy increases its policy rate, long-term real interest rates likewise tend to increase in the presence of a rising future real rate environment; the user cost of capital rises and the demand for capital assets falls. Similarly, Tobin’s *q* theory of investment can be brought forward to describe the inverse relation between real interest rates and the change in investment dynamics (Tobin 1969). If the market value of installed capital divided by the replacement cost of installed capital is greater than one, firms will increase investment spending.³

Tobin’s *q* is closely related to the *monetarist channel* of monetary transmission. Here, what matters is the universe of relative asset prices and real wealth (Meltzer 1995). The composition of consumption, investment and other asset holdings relies on a generalized choice problem (portfolio approach). Changes in the short-term interest rate induce changes in the re-allocation of assets in the portfolio since relative returns vary. A fall in the short-term interest rate makes bond holdings less attractive relative to equities, thereby causing equity prices to rise. If the market value of firms can be approximated by its share prices, Tobin’s *q* increases and, thus, investment spending picks up. An alternative propagation effect within the monetarist view comes from wealth effects. Consumption spending is determined by lifetime resources of consumers that co-vary with the value of the assets hold in the portfolio.⁴ Lower interest rates drive up prices of stocks, homes and other assets; the increase in total wealth will, in turn, stimulate aggregate consumption (Ando and Modigliani 1963).

Both channels are highly relevant in estimated models of monetary transmission across a variety of countries. Monetary policy shocks contribute to changes in

²In the following, it is abstracted from the exchange rate channel. It belongs to the “money view” where a reduction in domestic interest rates generally lowers the return of domestic assets relative to foreign assets; it triggers a currency depreciation and makes domestic goods cheaper than foreign goods. The effect on aggregate demand comes through an increase in net exports.

³Tobin’s *q* theory can be linked to the neoclassical concept of the user cost of capital; indeed, Hayashi (1982) shows that both investment theories are identical.

⁴Lifetime resources are made up of financial, real and human capital.

long-term interest rates and other asset prices, thereby influencing aggregate demand through direct cost effects and indirect wealth effects.⁵ However, the transmission in these mainstream models relies on the proposition of perfect financial markets; the financial structure is irrelevant for spending decisions. In such a frictionless economy, all consumption and investment projects with a positive net present value are financed, irrespective of the balance sheet positions of economic agents. The “credit view” augments the more traditional channels by considering explicitly how changes in financial conditions influence the effectiveness of monetary policy (Bernanke and Gertler 1995). Borrowers and lenders in financial markets may be faced with asymmetric information and transactions costs that may drive a wedge between the opportunity cost of internal funding and external funding – the external finance premium. It arises from the principal-agent problem that exists between lenders and borrowers. Agency costs occur when lenders are unable to ensure that borrowers act in the lender’s best interest (Hall 2001). They imbed the costs associated with monitoring of the borrowers’ project outcomes and the cost of distortions in the borrowers’ behavior that results from moral hazard and adverse selection.⁶

The *balance sheet channel* (broad credit channel) works through the credit quality of borrowers. Lenders search for signals about the riskiness of loan contracts by evaluating borrowers’ balance sheet positions which imbed information about available liquid assets and marketable collaterals. The external finance premium imposed on external funds and paid by borrowers depends inversely on the net worth of borrowers. If borrowers are willing to contribute to the financing of their projects or if they are able to post an substantial amount of collaterals to back their debt exposures, the conflict of interest between lenders and borrowers may decline. Consequently, the supply of external funds increases and the external finance premium shrinks (Bernanke and Gertler 1989). Moreover, it is assumed that the net worth is typically procyclical during the business cycle so that a fall in the short-term interest rate boosts up asset prices through valuation effects and cash flows through increased sales and profits. Collateral values rise and the external finance premium is then countercyclical with amplifying effects on credit availability and, hence, on investment and consumption dynamics. This element adds to the dynamic response of the economy in case of monetary policy shocks and has come to be known as the “financial accelerator” of monetary transmission.⁷

⁵See, for example, Boivin et al. (2008) and Boivin et al. (2009).

⁶Monitoring costs on part of lenders are typically incurred on the borrower’s repayment schedule so that the latter does not has an incentive to understate the project outcomes. In order to reduce these monitoring costs, Townsend (1979) shows that it is optimal to monitor if the borrower reports a default in order to verify the state. In this respect, it could even be possible that borrowers ration credit if lenders find it unprofitable to attract borrowers with riskier projects despite the possibility of increasing the loan rate (Stiglitz and Weiss 1981).

⁷See Kiyotaki and Moore (1997), Carlstrom and Fuerst (1997), Holmström and Tirole (1997) or Bernanke et al. (1999). Most recently, different variants of balance sheet channels have been

Within the *bank lending channel* (narrow credit channel), banks play a special role in the economy because they are able to reduce the cost of monitoring information which arises when resolving incentive problems between borrowers and lenders (Diamond 1984). A premise that allows the channel to work is that monetary policy affects the supply of loans in an environment where (1) banks find it hard and more expensive to obtain extra funding for credit granting and (2) borrowers can not perfectly substitute between bank loans and other sources of funding (Oliner and Rudebusch 1995). The traditional view on bank lending regards the link between monetary policy and bank deposits as one where monetary policy is implemented through changes in reserves with banks being subject to minimum reserve requirements on their deposits. However, this mechanism is misplaced in a monetary policy regime in which there is no exogenous quantitative constraint on the supply of money and credit. Banks are able to create deposits that are the means of non-bank payments. Bank lending deals with the creation of bank deposits which in turn depends on the demand for and the willingness of banks to extend loans.

Monetary policy changes the opportunity cost of holding deposits and the cost of funding for banks. An increase in the policy refinancing rate triggers price- and quantitative tensions in the interbank market which induces banks to attract deposit and non-deposit sources of funding via higher rate offers.⁸ With banking characterized by a maturity mismatch between asset and liabilities as well as by fixed-rate contracts of assets, banks' profitability shrinks. If banks are not able to perfectly pass on the increased funding costs to the existing stock of assets, or if they do not have enough liquidity buffer, they will adjust or they will be forced to adjust the composition of bank assets, in particular in terms of higher borrowing rates for new loans and a reduction in total loan supply.⁹ This is so, because monetary policy, then, affects banks' capital positions which are subject to minimum regulatory requirements and credit-ratings based target ratios. A shortage of bank capital can trigger a cutback in the loan supply as external financing of lending becomes more costly, in particular for low capitalized institutions (den Heuvel 2002; Gambacorta 2005; Disyatat 2010a). Consequently, the constraint in the bank-lending channel does not come from deposits, but from bank capital. If the reduction in loan supply is not accompanied by a counterbalance of enhanced access to alternative funding sources on capital markets, the private sector loses borrowing capacities and will likely cut down expenditures. This holds in particular for small- and medium-sized firms and individuals which can not switch to security markets and issue bonds.

discussed from the perspective of simple monetary rules and optimal monetary policy (Cúrdia and Woodford 2008; Fiore and Tristani 2009).

⁸In this respect, Disyatat (2010a, 9) remarks that "while traditional [bank lending] models assume that a monetary tightening leads to a shortage of liquidity for banks, the presumption here is that it leads to a disproportionate rise in the price of funding liquidity."

⁹This pattern has been laid out by Bernanke and Blinder (1988), Kashyap and Stein (2000) or Diamond and Rajan (2006). Banks should also be able to sell assets in order to adjust the balance sheet. Although relative liquid banks can withdraw liquid asset to protect their loan portfolios, this strategy is not possible for less liquid and small financial institutions.

The emergence of synthetic and true-sale securitization may have altered the workings and the effectiveness of the bank lending channel.¹⁰ With securitization, banks have increasingly switched from “originate and hold” to a strategy of “originate and distribute”. It implies that a large number of loans becomes marketable, liquid and in principle available outside the banking sector (Franke and Krahnen 2009). Banks may obtain additional funding liquidity since they are able to sell part of their loan portfolio to investors which can then be used for additional loan granting. This may increase the total loan supply without increasing the banks’ balance sheets. Moreover, by removing loans from their balance sheets, banks may improve their capital positions and may obtain regulatory capital reliefs through an increased credit risk transfer. Empirical evidence points out that securitization reduces the importance of the bank lending channel because banks’ funding needs can be reduced in the event of monetary tightening and capital reliefs allow further increases in supplied lending (Altunbas et al. 2009b).

7.2 The Risk-Taking Channel of Monetary Transmission

7.2.1 *Classification and Definition*

The financial crisis, starting in 2007, has refocused the role of financial risk spreads and more generally the impact of risk premia on the monetary transmission mechanism. Against this background, ECB president Trichet (2008) throws into question the connection between macroeconomic performance and financial risk: “Does the Great Moderation determine the quantum of risk traded in the markets? Or does the price of risk determined in the financial markets contribute and partly determine those fluctuations. [...] A complete understanding of these relations would bring invaluable insights for policy considerations. [...] Unfortunately, my conclusion is that the state of our knowledge is not advanced enough to draw definite conclusions about the nature and the directions of influences between risk pricing and the macroeconomy.”

The academic literature has only recently picked up the theme in a more systematic way. The *risk-taking channel* of monetary transmission describes the link between monetary policy and the perception and pricing of risk by economic agents, in particular by financial intermediaries (Borio and Zhu 2008).¹¹ It acknowledges the finance-related view where asset prices rely on the no-arbitrage framework of pricing assets as the discounted stream of future payments. The discount factors compose the risk-neutral interest rates augmented by risk premia that depend on the

¹⁰See ECB (2008b) for an overview of the motives and forms of securitization.

¹¹Short reviews of the risk-taking channel can be found in ECB (2008a, 2009a); Gambacorta (2009).

riskiness of the corresponding assets. Monetary-policy induced changes in short-term interest rates affect asset valuations along three dimensions: (1) prospects for future cash flow payments, (2) the risk-neutral discount factor and (3) the required risk compensation on part of investors. As long as monetary policy has an impact on the pricing of risk, further amplifying effects can emerge in the monetary transmission mechanism.

The impact of monetary policy on valuations, incomes and cash flows has also been discussed within the balance sheet channel in which increases in the net worth of borrowers reduce constraints on external funding. However, this channel limits the perspective of how to capture the concept of risk-taking on a more general level. The focus within the traditional credit view is on the impact of financial frictions in the borrowing sector. Expected default probabilities are counter-cyclical to the net worth of borrowers. In the financial accelerator model of Bernanke et al. (1999), lenders are assumed to be risk-neutral. The charged external finance premium is derived from equating expected payments from lending with the risk-free rate. In contrast, financial asset-pricing models imbed risk pricing as a process where uncertain future cash flows and returns induce risk-averse investors to demand higher expected returns than the risk-free return. Along similar lines, the bank lending channel concentrates on the role of banks in a risk-neutral environment. Banks' balance sheet positions matter because they constrain the availability of funding loan portfolios. Again, explicit risk-taking on part of financial intermediaries over the course of the business cycle is not necessary in order to make changes in the loan supply effective. The crucial issue is, however, whether monetary policy can trigger a financial cycle characterized by an inverse relationship between risk premia and short-term interest rates. If the risk-taking channel is effective, monetary policy actions lead to shifts in the supply of credit through changes in the risk appetite of the financial system and overall risk perceptions.

The emphasis of the propagation effects of risk is not new in the academic literature. Minsky's "financial instability hypothesis" centers financial institutions' tolerance for taking risky positions in financial markets.¹² Risk-taking evolves over time and it affects finance conditions for both firms and financial intermediaries. The level of physical and financial investment is derived from Keynes' two-price system where the demand price of investment and the supply price of capital are adjusted by lender and borrower risk who heavily depend on the degree of external financing and the leverage of the financial system (Minsky 1975). The economy is, thus, characterized by complex borrowing and lending relations with time-varying margins of safety. Economic conditions such as a steep yield curve – eventually initiated by monetary policy – and healthy hedge-financed market participants favor an environment of increased risk-taking by reducing the margins of safety embedded

¹²See Minsky (1975, 1982, 2004). Fisher's debt deflation and Kindleberger's description of booms and busts fit neatly into Minsky's description of the importance of the financial structure for economic activity (Fisher 1933; Kindleberger 1995).

in investment decisions.¹³ In this respect, banks play a special role since they “make on the carry”, thereby exploiting profit opportunities by extending their maturity mismatch of assets and liabilities on balance sheets and by financing long-term, illiquid assets with short-term, rolling-over debt.¹⁴

The risk-taking channel builds upon these basic mechanisms and tries to disentangle the various effects of monetary policy through risk tolerance and risk perception to economic activity. It relies on different methodological tools and makes use of the theory of financial intermediation, risk management practices and the general equilibrium approach of modern monetary economics with financial frictions. Against this background, three main characteristics can be classified: (1) the impact of low policy rates on changing risk compensation, in particular for financial intermediaries, (2) the emphasis on the supply of credit and the existence of financial frictions in the lending sector rather than in the borrowing sector and (3) the arrangement of the monetary policy regime in terms of objectives, transparency and communication policies. All factors can contribute to amplifying dynamics in the monetary transmission and may produce potential non-local effects by aggravating each other to distinct boom-bust cycles (Borio 2003).

7.2.2 Risk-Taking, Financial Intermediaries and the Role of the Short-Term Interest Rate

In order to grasp the risk-taking channel, the elaboration is based on the premise of a short-term policy rate cut. It forestalls the views of Sect. 7.4.1 according to which low policy rates may have contributed to and aggravated the financial crisis starting in 2007. Each of the following effects highlight certain aspects of risk-taking; they should not be read as being separated or even disconnected. Each aspect rather emphasizes one link between monetary policy, asset prices and risk attitudes for economic agents, especially financial intermediaries.

7.2.2.1 Asset Pricing Models and Portfolio Choice

Low levels of short-term interest rates may promote the attractiveness of riskier assets for economic agents in asset allocation. According to portfolio theory, the demand for assets is to be set in the context of the portfolio profile. Agents allocate

¹³The term “hedge-financed” is used differently compared to the more modern view of the management of risky hedge funds. Here, Minsky (1982) refers to the cash flow position of borrowers: they can meet cash flow commitments (loan rate and redemption payments) out of current business cash flow streams.

¹⁴Minsky (2008, 211) emphasis that “[w]ith such a rate pattern, one can make on the carry by financing positions in capital assets by long- and short-term debts, and positions in long-term financial assets by short-term, presumably liquid, debts.”

their wealth and invest in a set of assets that optimizes the respective risk-return structure of their total portfolio. Within the mean-variance analysis framework, changes in the risk-free interest rate have significant effects on asset prices and the optimal portfolio weights attached to the risky asset portfolio and the other portfolio consisting solely of the risk-free asset (Markowitz 1952). The two-fund separation theorem, whose solution was first derived by Tobin (1958), states that investors will choose to hold risky assets in the same relative proportion given by the optimal tangency portfolio provided that investors have identical beliefs regarding the asset return's joint probability distribution and that there are no limits of lending and borrowing at the risk-free rate.¹⁵ Investors only differ in the amount of total wealth invested in the optimal risky tangent portfolio and in the risk-free asset; this choice depends on individual utility functions, in particular on the degree of individual risk-aversion (Pennacchi 2008). If the short-term policy rate proxies the risk-less interest rate, a fall in that rate reduces the total expected portfolio return. Since investors minimize the portfolio variance subject to a given portfolio return, the demand for the risky portfolio and, thus, for the riskier assets increases at the cost of a higher portfolio variance. Simultaneously, given the new optimal balanced portfolio structure, the mean-variance framework predicts that additional assets are added to the total portfolio in terms of a higher weight relative to the other assets at relatively higher prices and lower returns; required risk compensation and the excess return of these assets over the risk-free rate decreases.

The effects of short-term interest rates on risk-taking can be further analyzed within optimal household behavior of a consumption-based asset pricing model. It can be shown that risk aversion of households (investors) moves inversely with changes in short-term interest rates. In contrast to standard utility specifications with constant relative risk aversion, the explicit separation of risk aversion from the intertemporal elasticity of substitution or the introduction of certain types of habit persistence in the consumption utility allows a prominent role of time-varying risk premia when clarifying the links between consumption dynamics and asset prices.¹⁶ With habit formation, risk aversion varies with the level of consumption relative to a habit level or subsistence level. The latter can be interpreted as past average realized consumption levels. Given the fact that lowering interest rates typically boost consumption, risk appetite is higher the more today's consumption levels exceed the habit level. In this way, required risk compensation for holding risky assets, such as equities or long-term bond instruments, should fall. The macro-finance literature on modeling stock prices and bond returns recently has improved in incorporating time-varying risk premia into standard DSGE models. They find evidence that a more accommodative monetary policy *ceteris paribus* leads to rising assets prices which can be attributed to a higher surplus consumption ratio and, thus,

¹⁵For a market equilibrium to hold, the tangency portfolio must coincide with the market portfolio of all risky assets. This is the assumption made by the Capital Asset Pricing model (CAPM).

¹⁶See Epstein and Zin (1991) and Campbell and Cochrane (1999).

to lower investors' risk aversion that translates into falling required excess returns in bond and stock markets.¹⁷

7.2.2.2 Sticky Return Targets as Promoter of Searching for Yields

Optimal portfolio selection can demonstrate why a fall in the short-term interest rate may promote enhanced risk-taking if investors seek for return targets with minimum portfolio variance. Alternatively, investors could also maximize returns given a desired degree of risk. A cut in the policy rate would then trigger an adjustment in the target portfolio return with no change in the portfolio risk exposure. It is therefore ambiguous whether portfolio returns or the amount of risk change in response to interest-rate changes. For the risk-taking channel to work within the portfolio framework, investors must hesitate, be unwilling or even be unable to adjust their desired return targets to changing market conditions with a positive effect on risk appetite.

The behavioral finance approach to asset pricing provides an informative basis on such target rigidity. It can be shown that "narrow framing" contribute to a situation in which investors make decisions that do not capture nominal return dynamics within a broader spectrum that consists of more complex factors that determine wealth.¹⁸ Against this background, some kind of money illusion may induce investors to disregard the fact that low interest rates may be the result of lower inflation expectations and realized inflation. Investors should care about real rather than nominal return targets. Even if nominal returns perpetually fall below some specified nominal target return, the realized real return could coincide with the real target return so that there would be no need to increase risk appetite for the purpose of boosting nominal returns (Akerlof and Shiller 2009). Sticky nominal returns can be further pushed on by investors' efforts to meet nominal returns that had been realized in previous periods. This "status quo bias" becomes evident when a monetary policy induced reduction of low-risk government bond returns promotes investors to shift into higher yielding asset instruments such as corporate bonds or derivative-based securities (BIS 2004; Rajan 2005).

The search for yields in an environment of low short-term and long-term government bond interest rates may also reflect contractual, institutional and regulatory arrangements. For instance, life insurance and pension funds typically manage their assets with respect to nominal liabilities that are often characterized by predefined

¹⁷See, for example, Lyrio et al. (2006), Wachter (2006), Piazzesi and Schneider (2006), Rudebusch and Swanson (2008a), Rudebusch and Swanson (2008b).

¹⁸Asset pricing models rely on assumptions about investor preferences. The vast majority of models apply the von Neumann/Morgenstern approach of describing expected utility. A gamble is merged with pre-existing bets to evaluate whether it represents a worthwhile addition to total wealth. Narrow framing is different because it describes the tendency to treat single gambles separately from other gambles that determine total wealth. Utility is derived directly from the gamble and not indirectly via its contribution to total wealth (Barberis and Thaler 2003).

long-term fixed rates. In some countries, these fixed rates are based on regulatory constraints that provide lenders with a guaranteed minimum return. Similar findings hold for compensation contracts of asset fund managers whose compensation are linked to annual returns in excess of a minimum nominal return (Rajan 2005). Whenever the risk-free rate is high, compensation will be high with only little risk-taking. In contrast, if the risk-less rate is low, managers must take on additional risk in order to exceed the minimum target return. Raising funds by borrowing against this low rate can further boost returns, however by increasing risk through a higher leverage.

7.2.2.3 The Impact on Asset Valuations and Risk Perceptions

Asset prices are an integral building block of the balance sheet channel of monetary transmission. They determine collateral values of firms which affect the external finance premium imposed by borrowers. Ease monetary policy could decrease the riskiness of assets, in particular of stocks, by falling interest-rate costs or by strengthening of borrower's balance sheets.¹⁹ These dynamics can be further amplified in terms of financial procyclicality if asset valuations endogenously change overall risk perceptions and risk tolerance. There are at least three reasons why a monetary-policy induced increase in asset prices is associated with higher risk-taking (Brunnermeier et al. 2009). Firstly, empirical evidence points to the observation that in a rising asset market environment, conditional volatilities of the underlying assets tend to decrease. Higher stock prices are accompanied by lower debt-to-equity ratios thereby reducing the riskiness of stocks which manifests itself in lower volatilities. This effect could alter investor's risk perceptions due to lower expected default probabilities and pay-off volatilities (Gambacorta 2009).

Secondly, common risk management practices can encourage risk-taking when the applied risk measures are mainly estimated in a backward-looking fashion by using past data and the sample period is very short. This is in particular eminent in value-at-risk (VaR) strategies which reflect a binding constraint on investors' portfolio choice. The value at risk for an asset portfolio describes a threshold value for the maximal loss of the marked-to-market portfolio with a given probability over a specified time horizon. Danielsson et al. (2004) show that under such a portfolio risk constraint, a fall in the short-term interest rate followed by a rise in asset prices endogenously amplify these fluctuations. Thereby, market participants' actions feed into to the market dynamics in terms of lower asset price volatility. This effect translates into a relief of the risk constraint with a falling VaR so that further

¹⁹Bernanke and Kuttner (2005) find that stock markets' reaction to monetary policy surprises comes from the effects of policy rate changes on expected excess returns. They argue that monetary policy shocks and the equity risk premium are related through the (perceived) riskiness of stocks or on investors' risk aversion. Furthermore, Kurov (2010) points out that the stock market reaction to monetary policy surprises is asymmetric depending on investor sentiment and overall market conditions (bull vs. bear market).

risk-taking is promoted in portfolio selection even if investors' risk aversion remains constant (Danielsson et al. 2009).²⁰

Finally, similar to the balance sheet channel, informational frictions are lower in an environment of falling interest rates and increasing asset prices. Instead of identifying information asymmetries concerning project outcomes of potential borrowers, this last reasoning attaches riskiness to the collateral value in various funding operations of the lending sector. For example, investors may become less careful about the acceptance of assets as collateral in asset-backed funding transactions when rising asset prices make them less information sensitive; the perceived danger decreases in case a commercial paper seller offers only "bad" assets as collateral (Brunnermeier and Pedersen 2009).

7.2.2.4 Financial Intermediation, Maturity Mismatch and Leverage Targeting

The financial intermediaries' main source of profitability comes from its delegated task of funding long-term illiquid assets with short-term liabilities. Since interest rates on long-term loan contracts are typically higher than those on short-term debt, the marginal profitability of banks by an additional unit of loan depends on the maturity mismatch between assets and liabilities. Within the bank lending channel, the supply of credit is affected by the term spread of interest rates. If short-term interest rates are falling and the spread widens, financial intermediaries earn more profits through an increased interest rate margin and will then seek to expand the supply of credit for the non-financial sector. The slope of the yield curve, therefore, does not only influence the demand of credit through its informational value for future short-term interest rates and, thus, for determining the demand price of capital; it also represents a main determinant of banks' capacity and willingness to grant credit.

These effects are further amplified in a financial system where an increasing number of financial intermediaries rely on the marked-to-market accounting framework of managing balance sheets (Adrian and Shin 2009a). Due to its maturity mismatch, movements in the short-term interest rate (and in the slope of the yield curve) also have valuation effects on balance sheets since assets appear to be more sensitive to changes in short-term interest than liabilities. With such a pattern, the active management of balance sheets takes center stage for the determination of the price of risk in the economy and the dynamics of balance sheet quantities of the financial sector. The main feature of financial intermediaries is that they adjust their

²⁰From a regulatory perspective, risk management rules prior to the financial crisis of 2007 (Basel II) to shield the financial system are counterproductive since they actually expedite endogenous risk and promote higher procyclicality. Bank capital requirements for the purpose of cushioning the underlying VaR are relaxed in periods of booms and low perceptions of risk (Brunnermeier et al. 2009).

balance sheets in response to a changing environment, especially to capital market conditions and the size of equity.²¹

When short-term interest rates are lowered, asset prices tend to increase; since financial intermediaries' balance sheets are marked-to-market, the latter generally becomes stronger and manifests itself in higher asset valuations and an appreciation of the total equity position. The side effect lies in an erosion of leverage defined as the ratio of total assets to equity. Banks and financial intermediaries then hold surplus capital in the sense that they are able to over-cushion the maximal estimated value of loss of assets as measured for instance by the VaR technique.²² They try to expand their balance sheets in order to economize their surplus capital. The liability side increases through additional short-term debt accumulation; whereas on the asset side, the financial sector searches for further potential borrowers. If the "good" projects of borrowers are funded first and the set of such borrowers is limited, the willingness to lend triggers a decline in lending standards and financial risk premia. According to Adrian and Shin (2008b), such active management focuses on a desired leverage target. It could even be possible for leverage to become pro-cyclical when greater demand for assets puts upward pressure on prices with a positive feedback effect in which stronger balance sheets further promote rising asset demand.

Figure 7.1 illustrates the effect of a monetary-policy induced increase of asset prices held by the financial sector. It is assumed that only the asset side is exposed to the effects of valuation and that debt is not affected by asset price changes. This is a simplification because in a marked-to-market system the liability side should also respond to changes in valuations but these effects should be smaller than asset side adjustments due to the maturity mismatch. An increase in asset prices, thus, is entirely translated into higher equity by a capital gain. Since the financial intermediary is leveraged, the relative increase in equity is much higher than the increase in assets. The middle balance sheet mirrors the pure valuation effect of rising prices. With surplus capital and an erosion of leverage, financial intermediaries take on additional short-term debt and hold additional units of risky securities. In such an environment of increased profitability, financial intermediaries' creditors are willing to grant short-term debt since they regard actual leverage as below acceptable levels, as indicated, for instance, by the VaR constraint.²³ Adrian and Shin (2009a) show that the price of risk and, thus, risk premia of securities are decreasing with the value of assets held. Risk premia are particularly low when the size of the leveraged sector is high, a condition which induces banks to increase the supply of credit. Notice that while required risk premia are compressed when balance sheet quantities expand, actual risk may indeed rise due to the search of potential borrowers with lower credit quality.

²¹The following comments rely heavily on Adrian and Shin (2008b, 2009a,b,c).

²²It is usually assumed that a bank aims to adjust its balance sheet so that its economic equity meets total VaR. It can be justified from an optimal contracting problem and from empirical observation (Shin 2008).

²³Alternatively, leverage constraints can be determined through regulatory capital requirements.

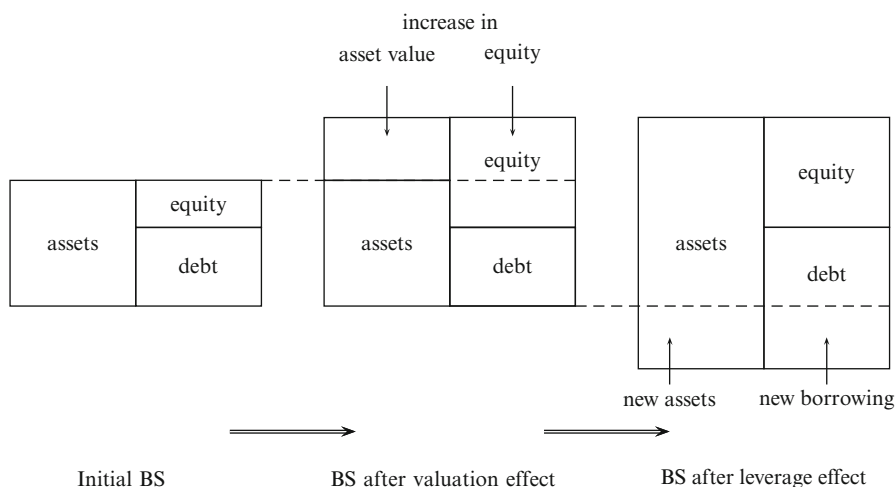


Fig. 7.1 Balance sheet management of financial intermediaries and leverage effect
(Source: Adrian and Shin (2009a))

Funding and market liquidity are closely connected to aggregate balance sheet dynamics. As outlined in Chap. 4.4, funding liquidity describes the ease with which investors can obtain funding on short notice from potential lenders. The leverage ratio is thereby inversely related to the permissible haircut in secured lending.²⁴ An increased leverage of the financial sector pushes haircuts of secured lending down because financial intermediaries and their creditors are willing to lend with lower margins of safety in order to expand their balance sheets (et vice versa). In this context, aggregate liquidity can be regarded as the rate of growth of aggregate balance sheets; it measures how hard the financial sector “searches for borrowers” and is “awash in liquidity” (Adrian and Shin 2009b). A broad set of assets may then become more marketable when their prices are additionally pushed up by falling liquidity premia. With falling liquidity premia, further risk-taking is promoted by facilitating position-taking and the additional relaxation in funding and market liquidity. Consequently, there is a self-reinforcing process between liquidity, leverage and increased risk-taking.

7.2.3 Empirical Evidence

Turning to the empirical evidence of the risk-taking channel at work, at an intuitive economic level, it is common sense to regard the macroeconomic environment as a driver of risk tolerance and risk perceptions. A stylized view would be to

²⁴Leverage most generally is defined as the ratio of total assets to equity. The haircut in secured lending is defined as the percentage amount of equity (cash) a borrower has to bring in order to finance total assets; it is the ratio of equity to total assets.

characterize asset prices as procyclical but risk premia as highly countercyclical. In a recession, economic agents become more risk-averse and are more reluctant to hold riskier assets such as long-term governments bonds, corporate bonds or stocks. On the contrary, during high output, income and employment growth, risk premia are compressed since agents expect a more favorable income and consumption stream thereby accepting higher risk exposures when investing in risky portfolios. The dynamics of risk premia then provide positive impetus for economic activity as laid out in the Sect. 7.2.2.

Indeed, such a pattern can be derived from estimated time-varying term premia on default-free government bond rates as has been extensively presented in Chap. 4.2.2 of this work. Hereby, in the early period of a recovery, when the short-term policy rate is still held down, long-term interest-rate risk premia come down. At the end of the economic upswing, monetary policy tightens, but term premia remain low until the tightening cycle translates into a slowing macroeconomic activity (Cochrane and Piazzesi 2008). Moreover, there is empirical evidence that (unanticipated) policy rate cuts are associated with lower term premia in the term structure of interest rates (Lyrio et al. 2006).

Turning to further risk concepts, Fig. 7.2 shows some descriptive evidence on the dynamics of selected credit risk and risk perception indicators for the US and the euro area. They cover the period of 2002–2006 when monetary policy was regarded to be highly accommodative compared to historical levels and policy rates derived from the Taylor rule (see Chap. 7.4.1 and Taylor (2008a)).²⁵ If the risk-taking channel is effective, these measures should experience a significant decline prior to the crisis of 2007. By analyzing the background of the financial crisis, Goodhart (2008a) similarly makes the under-pricing of risk a driving force of the build-up of financial imbalances prior to the crisis. Moreover, he argues that the crisis was a foreseen crisis by central banks and financial institutions against the background of the financial market developments.

Figure 7.2a displays the spread between BBB- and AAA-rated firms in the US and the euro area. As suspected, this spread contracted starting at the end of 2002 and mirroring lower market prices of credit risk demanded by investors in an environment of very low short-term interest rates.²⁶

²⁵Data are taken from Datastream, ECB, Fed and Merrill Lynch. Corporate bond spreads are defined as the difference between corporate bond rates paid by BBB- and AAA-rated corporate institutions; they are measured in basis points. Lending Standards are specified as the net percentage of banks contributing to tightening standards to large and medium firms in the US and to all firms in the euro area. Expected Default Probabilities are due to the Fitch PRBY Default Index with a maturity of 5 years. Default rates are calculated by dividing the volume of defaulted bonds by the average principal volume outstanding for the time period under observation; the index is measured in basis points. The measure of implied volatility is derived from the Chicago Board of Trade Generic 1st TY future for the US and from the Eurex Generix 1st RX future for the euro area.

²⁶It should be noted, however, that in principle fluctuations in yield spreads can either be due to an increased risk appetite (market prices of risk) or due to a fall in underlying actual credit risk

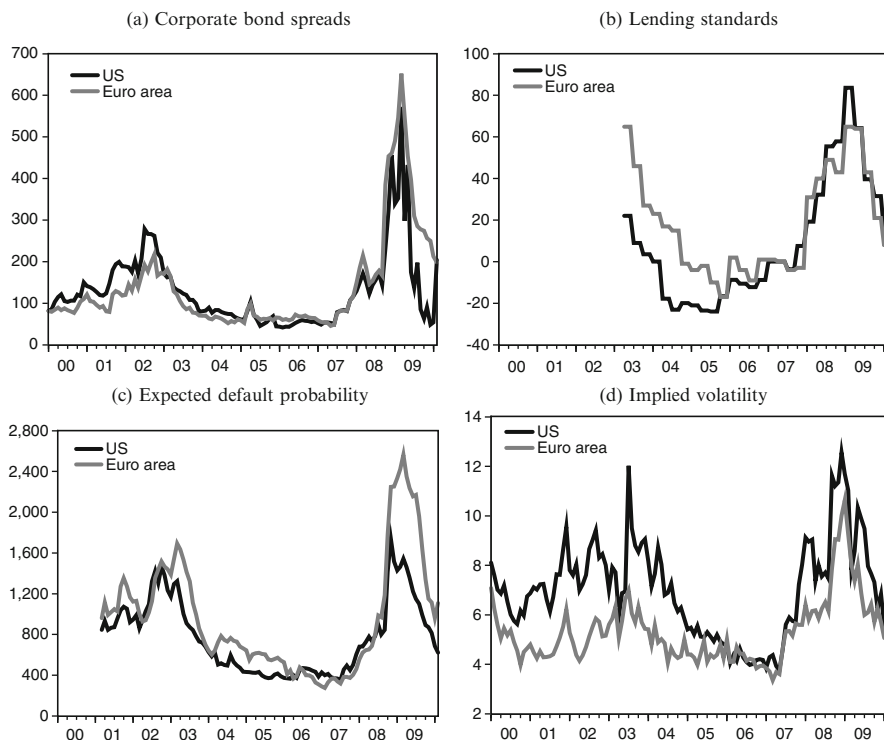


Fig. 7.2 Credit risk, lending standards and market volatility

Credit risk in general exhibits a strong link to macroeconomic activity. Based on a series of econometric studies, Amato (2005) and Amato and Luisi (2006) show that corporate bond spreads as well as credit default swaps (CDS) are negatively correlated with output growth. Since default risk tends to increase in economic slumps, such business cycle risk is priced in spreads and CDSs. They also find a strong relationship to the overall monetary policy stance as measured by a real interest-rate gap concept.²⁷ As risk appetite tends to increase during economic upswings, the real rate gap varies positively with the price of credit risk. In this respect, Gilchrist et al. (2009) point out that corporate bond spreads primarily reflect risk appetite of financial intermediaries. They provide information beyond what is already incorporated in other asset prices indicators such as stock market movements. The spreads exhibit a significant predictive content for future real

(with given market prices of risk). Following Bundesbank (2005b), it can be mostly assumed that in general, time-varying yield spreads are to some extent the expression of changing risk attitudes.

²⁷In addition, Bundesbank (2005a) conducts a VAR analysis; it shows that an increase in the 3-month Euribor rate is followed by an increase in corporate spreads where lower-rated firms' interest-rate costs react more sensitive and stronger to short-rate interest rate changes.

economic activity and may be seen as a reliable indicator of whether monetary policy actions translate into more favorable external finance conditions for non-financial corporations.

Turning to bank lending surveys, Fig. 7.2b reports credit conditions for the private sector. The lines stand for the percentage of the number of banks that tightened credit standards in the current quarter. Again, credit conditions are found to be procyclical so that they are eased during economic recovery leading to amplifying effects in the monetary transmission. This pattern implies an increased risk-taking of the banking sector since banks are more willing to grant credit to the private sector. This comes from both reduced risk perceptions regarding credit risk and increased risk tolerance (DeBondt et al. 2010). In this respect, the level of and the change in short-term interest rates affect lending standards; banks ease access to credit through price and non-price factors when rates are low. Moreover, low levels of short-term interest rates induce investors to “search for higher-yield assets”, in particular for securitized assets. Banks, then, have an incentive to ease their credit standards in order to supply these assets through loan pooling and true-sale securitization.

Figures 7.2c–d summarize the descriptive evidence. Risk perceptions as measured by expected default frequencies of financial and non-financial corporations as well as implied bond market volatility show a common stochastic trend of sharp reductions as of 2003 and non-linear increases triggered by the financial crisis in the third quarter of 2007. This common trend is also detected in the time-series properties of corporate bond spreads and bank lending standards.²⁸ The empirical overview indicates changes in risk perceptions through a monetary policy impact on asset valuations and on greater risk appetite of the financial sector as expressed in a search for high yield assets. This process pushes prices up and brings risk spreads down.

For the sake of robustness of these findings and to gain further insights into the risk-taking channel, a number of studies have recently analyzed the impact of monetary policy on risk-taking by financial intermediaries with the help of microeconomic data.²⁹ The stance of monetary policy is measured via several proxies among them is the short-term policy rate, the slope of the yield curve, Taylor rule residuals or the estimated real interest-rate gap, i.e. the difference between the actual short-term real rate and an estimated natural rate. Banks’ exposure to risk is mainly estimated through the expected default frequency of the individual banks. This variable is seen as reliable forward-looking indicator to credit risk.

The findings support the theoretical discussion about the risk-taking channel. Low interest rates, in the first place, promote a reduction of the probability of default of outstanding loans in banks’ loan portfolio. A lowering of the short-term interest

²⁸ A principal component analysis reveals that the first principle component accounts for almost 84% in the US and for 87% in the euro area of the comovements of selected variables in Fig. 7.2.

²⁹ See Jiménez et al. (2009), Ioannidou et al. (2009), Altunbas et al. (2009a), Manganelli and Wolswijk (2009), Altunbas et al. (2010), Maddaloni and Peydro (2010).

rate leads to a fall in the borrowers' default probability as cash-flow prospects of borrowers increase. In the medium term, however, banks take on additional risk due to the search for yield by granting more risky loans and by relaxing lending standards. Moreover, loan pricing is also directly influenced by monetary policy actions. When the policy rate falls, banks charge lower loan rates, in particular to borrowers exposed to higher default risk. This holds in particular for those banks being characterized by lower capital ratios whose re-financing costs are higher and that have a greater incentive to grant more risky loans due to profitability considerations. Most importantly, the real interest-rate gap and the Taylor residual is the main determinant of changes in banks' credit risk. If the real policy rate is below its natural counterpart, banks do take on more risk. Moreover, a high term spread allows for higher profits of the banking sector due to higher net interest-rate margins generated by maturity transformation. It, therefore, reduces the expected default risk of individual banks in the short-term but increases risk in the long-term.

The risk-taking channel identifies the role of the supply of credit and the balance sheet conditions of lenders as a driving determinant in the monetary transmission. It gives a more complete picture compared to the bank-lending channel that is supposed to operate under general risk-neutrality. In a market-based financial system, the most important marginal supplier of credit is no longer a representative commercial bank: Along the same lines, the most important marginal funding source comes not through additional deposits subject to reserve requirements. With the growing importance of market-based institutions for the economy-wide supply of credit, there has been a focus shift from bank-based institutions as the dominant supplier of credit to financial institutions, which finance themselves mostly through short-term debt such as repurchase agreements and commercial papers, which are heavily involved in the process of securitization (Adrian and Shin 2009b). These financial institutions include among others broker-dealer entities, ABS issuer and finance companies; they are regarded as representing a barometer of overall funding conditions in capital markets. These marked-to-market institutions share one common feature, i.e. they operate with procyclical leverage as laid out in the previous Sect. 7.2.2. If the duration of assets held on the balance sheets is longer than the duration of liabilities, the asset side reacts with more sensitivity to changes in the risk-free interest rate and the policy rate, respectively. With increased marked-to-market equity, financial intermediaries will take on more risk by expanding balance sheets through an enhanced supply of credit and an increased leverage. Following Adrian and Shin (2010), such behavior can be documented when plotting quarterly total asset growth against quarterly changes in leverage of selected financial institutions as in Fig. 7.3. Leverage is here defined as the ratio of total assets to equity. For the US, the balance sheet figures are for security broker dealers and for Germany, MFIs are taken into account.³⁰ Indeed, there is a positive relationship between changes in asset growth (balance sheet growth) and

³⁰The data set is pulled from the Board of Governors, flows of funds and from the Bundesbank, banking statistics.

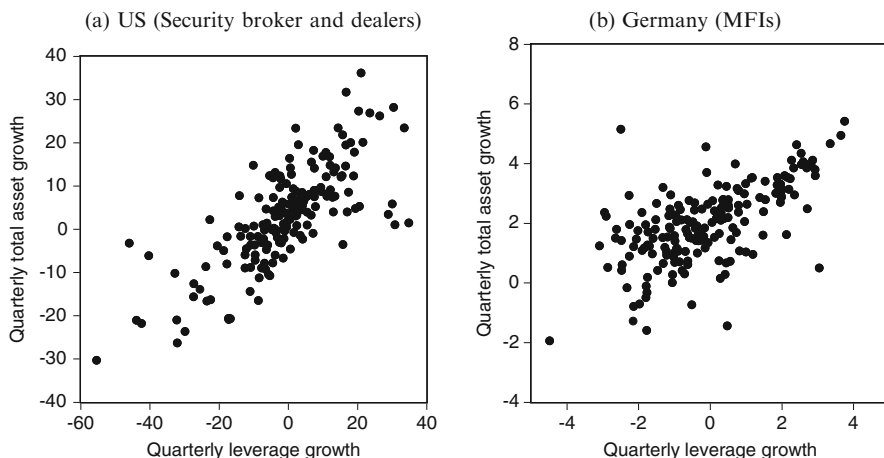


Fig. 7.3 Leverage dynamics in the US and Germany

changes in leverage. It implies that most variation of leverage comes from expanding or contracting the balance sheet via short-term debt adjustments rather than by adjusting equity. Leverage and asset growth is thereby much higher for security brokers and dealers as they are not subject to constrained capital requirements commercial banks (MFIs) are faced with. Still, procyclical balance sheets can be identified in bank-based institutions though the effects are significantly smaller in absolute terms.

The monetary policy impact of changes in policy rates on financial leverage should be higher, the more the term spread of interest rates is affected by policy movements. According to both the bank-lending and risk-taking view of monetary-policy transmission, credit supply should pick up speed, the higher the yield spread turns out to be. It increases the profitability of the financial sector through higher interest-rate margins and it induces lenders to take on more risk as marked-to-market gains relax risk constraints and leveraged institutions can increase their holdings of risky assets and risky loans exposures. Consequently, there is a direct link from short-term interest rates to the supply of credit and, thus, to macroeconomic dynamics as long as changes in policy rates are associated with changes in yield spread measures. Figure 7.4 clearly shows that there is an almost one-to-one negative correlation between annualized changes of the policy rate and the term spread so that monetary policy does exhibit control over changes in the term spread via the policy rate.³¹ The term spread, in turn, has significant impact on future balance sheet dynamics and leverage growth. If the spread widens, net interest-rate

³¹The term spread is measured as the difference between the 3-month interest rate and the 10-year government bond yield. The sample period for the US is from 1982:1–2009:4 and for the euro area from 1999:1–2010:1.

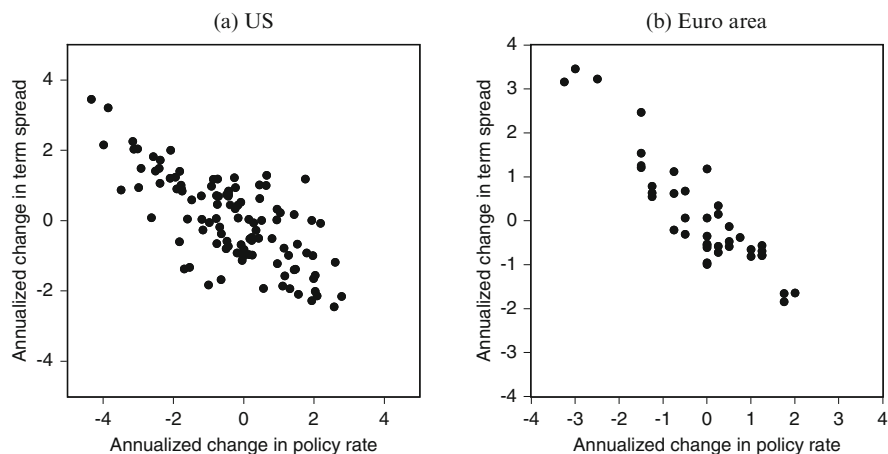


Fig. 7.4 The impact of policy changes on the term spread

margins increase, asset growth on intermediary balance sheets pick up and future GDP increases (Adrian et al. 2010a). This pattern is accompanied by a heightened risk appetite of the financial sector since balance sheet dynamics contain strong predictive content for future excess returns for a wide range of asset classes so that risk premia are determined by quantity adjustments: equity and bond returns are compressed whenever balance sheets expand (Adrian et al. 2010b). Finally, econometric evidence suggests a causal logic running from spread changes to asset growth which translates into higher macroeconomic activity and lower risk premia. The latter amplifies the effects on macroeconomic outcomes as it promotes lower funding costs for both the financial and the real sector.

7.3 The Impact of the Monetary Policy Strategy on Risk Tolerance

7.3.1 *Shaping Risk Premia in Monetary Policy Regimes*

The level and changes of short-term interest rates are regarded to be key factors for the risk-taking channel to work. Both factors exhibit an essential influence on risk perceptions and overall risk tolerance on financial markets. Going one step further, the entire monetary strategy dealing with the set of objectives, transparency, communication and its implementation can be significant for how market participants deal with the various sources of macroeconomics uncertainty. Along the same lines, monetary policy itself can be the originator of uncertainty in financial markets if the conduct of monetary policy is not well understood or the monetary policy regime is arranged so as to produce high volatility in goods, labor and financial markets.

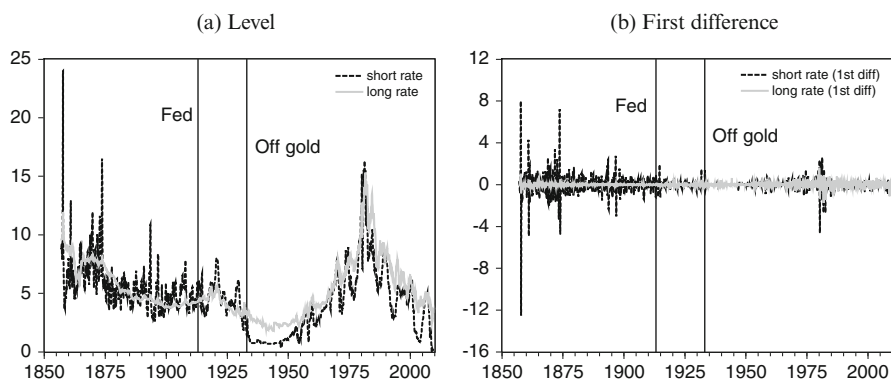


Fig. 7.5 US interest rates and monetary policy regimes

To give an impression how different currency systems provide impetus for yield curve movements, Fig. 7.5 plots representative US short- and long-term interest rates over the sample period 1857:1–2010:2 in levels and in the month-to-month change.³² It is eye-catching that during the gold standard, long-term interest rates were on average lower than short-term interest rates mirroring two important characteristics of the monetary arrangement. Firstly, the trend in long-term interest rates should be seen as a reflex of a steady price level with both periods of inflation and deflation.³³ According to standard asset-pricing theory, inflation compensation over the life of the long-term nominal bonds should on average be low to the same degree if the change in the expected price level is approximately zero. Secondly, the high volatility in short-term interest rates was the result of the inability on part of the government to iron out large swings due to seasonal factors, business cycle fluctuations and periodical bank runs.³⁴ With the establishment of the Federal Reserve Board in 1913 and the abolishment of the gold standard in 1933, the picture for interest rates changed dramatically. On average, long-term interest rates exceeded short-term interest rates with the latter being much less volatile than its long-term counterpart. What the FED did was the creation of a short-term, risk-free interest rate by stabilizing the very short-term money market with its open-market operations.³⁵ It therefore took away the risk associated with changing

³²The data before 1953 are taken from the NBER Macro History Database; the short term interest rate is a 3-month commercial paper rate traded in New York and the long-term interest rate represents US yields on railroad bonds up to 1919 and US yields on long-term governments bonds since then. From 1953 on, data are provided by the Federal Reserve Board.

³³Historical price levels are for instance provided by Robert J. Shiller on his webpage www.econ.yale.edu/shiller.

³⁴The evolution of modern central banking and the switch to interest-rate policy is summarized in Spahn (2001a) or Meltzer (2005).

³⁵Carroll (2009) makes the same point by using one-month interbanking rates for the sample period. With these rates, the reduced volatility in the short-term interest rate is even more obvious.

second moments of the underlying interest rate (but not the uncertainty about its future level). At the same time, the central bank switched to a regime of pursuing a nominal anchor, i.e. a positive inflation rate. However, at least until the mid-1980s, this attempt culminated in highly time-varying long-term inflation expectations. Consequently, the yield curve was on average downward sloping during the gold standard mirroring negative term premia on long-term bonds and it was upward sloping in the post-war period.³⁶

Turning to the current monetary policy regime with central banks around the world setting a short-term policy rate, further features of the monetary policy design may influence the risk-taking behavior of market participants. The extensive literature on macroeconomic volatility and financial market volatility documents a tight correlation between these two concepts. Both have experienced a significant and secular decline starting in the mid-1980s, a phenomenon entitled as the “Great Moderation” on a global level (Stock and Watson 2003b). This trend has been accompanied by central banks becoming more independent and more transparent, using the channel of communication and adopting some variants of inflation targeting. The bottom line of these developments is an increased predictability of monetary policy, in particular for the targeted inflation rate and the likely future path of short-term interest rates. Furthermore, committing to future policy moves conditional on today’s information set may reduce market uncertainty and induce agents to demand a lower risk award for bearing long-term inflation and output risks imbedded in long-term securities. This effect should be stronger the more a central bank aims at removing the conditionality of its future actions in its communication policies through guiding financial markets with the help of “code words”.³⁷ Consequently, the efforts undertaken by policy makers to guide private sector expectations may have contributed to the observable trend decline in risk premia – the “transparency effect” of monetary policy on risk-taking (Borio and Zhu 2008).

Indeed, as it will be demonstrated in the following Sect. 7.3.2, nominal bond risk premia and inflation risk premia are significantly lower under a monetary policy regime where the central bank credibly commits to price stability and, thus, is able to better anchor inflation expectations of the private sector whenever shocks hit the economy. But also the set-up of the monetary authority’s objective function may influence required risk compensation in macroeconomic models. The objective typically deals with the minimization of a loss function in terms of the variances of inflation and output. Depending on the weights attached to inflation and output

³⁶In referring to common business practices of using the maturity mismatch, Cochrane (2008b) remarks that “a pre-1932 hedge fund would *sell* long-term debt and *buy* commercial paper”. Moreover, he shows that even under an anchored inflation regime with a constant expected inflation, the variance of the real value of a nominal bond is much more sensitive to the variance of inflation than the one of a nominal bond in a regime in which the expected price level is constant.

³⁷If a central bank announces that it will tighten policy at a “measured pace”, this coding will not only hint to the expected future policy path, but it may also contribute to lower interest-rate risk priced in long-term bonds.

variability, the equilibrium dynamics of inflation and output significantly change. Under strict inflation targeting, the standard deviation of inflation is lower than under a flexible inflation targeting with a positive weight on output variability. Such policy preferences may come at the cost of higher equilibrium output variation depending on a discretionary or committing central bank regime. Since real and nominal bond risk premia are determined by comovements between output and the short-term interest rate on the one hand and between output growth and inflation on the other hand, the size of risk premia should differ depending on the central bank's preferences. Söderlind (2008) finds within a real business cycle model with sticky wages and a money growth rule that in the face of supply shocks, strict inflation targeting tends to increase real risk premia and decreases the reward of risk on nominal assets.

Monetary policy may also produce moral hazard in the financial industry which may encourage risk-taking. Such behavior can be analyzed as a typical insurance problem where a central bank gives liquidity assistance to money markets and/or particular financial institutions through regular and non-standard, longer-term liquidity provisions in times of crisis. If an insurer insulates risk in economic decision-making, agents may take on more risk because they may perceive that they will not have to bear the burden of losses as risks materialize. If monetary policy stands ready to intervene in case of large liquidity shocks and market participants know that (unlimited) refinancing is obtainable, they may engage in more risky and possibly more illiquid projects.³⁸ This may also hold for other sources of large downside risks in terms of bad economic outcomes if the central bank (predictably) eases monetary policy in case such events hit the economy.³⁹

The “Greenspan put” which states that a central bank can “mitigate the fallout when it occurs and, hopefully, ease the transition to the next expansion”, is one recent example for the possibility of engendering the kind of market dynamics that make financial booms and busts more likely to occur (Greenspan 2002). If monetary policy announces ex-ante low short-term interest rates in times of financial strain, market participants anticipate these low levels. The effect of such an anticipation is that financial institutions will more likely take on more risk by borrowing short-term

³⁸This perspective is central in the debate how to implement lender of last resort policies in order to guarantee the functioning of money markets. The changing views on the lender of last resort in the history of monetary economics are summarized in Knittel et al. (2006).

³⁹There is an analogy to the benefits and costs of deposit insurance. Due to maturity and liquidity transformation, any deposit-taking intermediary is intrinsically exposed to funding liquidity risk. A situation can arise that induces (patient) depositors to withdraw their funds early when they expect other depositors to withdraw. A bank run occurs with forced liquidation of the intermediary's assets. With deposit insurance, (patient) depositors do not initiate bank runs because they expect deposits to be safe (Diamond and Dybvig 1983). However, with deposit insurance, a bank has an incentive to become as risky and large as possible to maximize expected returns and failures absorbed by taxpayers occur with a positive probability (Kareken and Wallace 1978).

and financing long-term illiquid assets. Making on the carry along the yield curve under a liquidity-oriented interventionist policy regime is then profitable and “safe” for an extended period of time (Diamond and Rajan 2009; Cao and Illing 2008). Moreover, strategic complementarities exist since monetary policy not only rescues those institutions who are in need of low interest rates but it creates externalities in terms of higher interest-rate vulnerability for the whole financial sector: it is unwise to be in a minority of institutions exposed to whatever shock; the central bank may be reluctant to accept the costs associated with a lowering of the policy rate. However, the more financial institutions are sensitively exposed to interest-rate risk by building up leverage and engaging in maturity transforming, the more a central bank is obliged to be active if it aims at restoring market and funding liquidity (Farhi and Tirole 2009a,b).

To be clear, the establishment of a lender of last resort is welfare-improving compared to an unsecured banking regime. As Solow (1982, 206) points out, liquidity insurance reduces the probability of bank runs and liquidity crisis; it can be taken “[...] for granted that a national [...] lender of last resort would be instructed to permit no panic that it was able to prevent”. The welfare costs associated with distorted incentives of such types of insurance are non-negligible but they may not exceed the disastrous consequences of a liquidity crisis. However, it should be recognized that increased deregulation of financial markets and the emergence of non-bank financial intermediaries have significantly contributed to wide-scale leverage and maturity mismatch. Monetary policy may have contributed to these developments not only through the very long period of low policy rates in the wake of the Dotcom bubble crash but also via its announcements of cutting interest rates as soon as there were large downside risks to the economy.

7.3.2 *Optimal Monetary Policy and Bond Risk Premia*

This subsection presents a simple New-Keynesian macroeconomic baseline model with an optimizing central bank in spirit of Clarida et al. (1999) and McCallum and Nelson (2004a,b). It explores explicit solutions for bond prices in order to draw statements about the relation between optimal monetary policy and bond risk premia. A similar, but slightly modified model has been recently set up by Palomino (2010). In the following, the baseline model is extended by applying it to both real as well as nominal bonds and to the n -period case.

7.3.2.1 **Discretion vs. Commitment**

The model is specified with a policy-makers objective function at time $t = 0$ which minimizes

$$L = E_0 \sum_{t=0}^{\infty} \beta^t L_t \quad \text{with} \quad L_t = \pi_t^2 + \omega y_t^2 \quad (7.1)$$

subject to

$$\pi_t = \alpha y_t + \beta E_t[\pi_{t+1}] + u_t \quad (7.2)$$

where $\omega > 0$ is the relative importance of output-gap (y_t) deviations in the central bank's preferences.⁴⁰ The target levels for inflation and the output gap are normalized to zero. The minimization constraint is given by a forward-looking price setting equation with $\beta < 1$ and $\alpha > 0$. The cost-push shock⁴¹ follows a simple AR(1) process

$$u_t = \rho u_{t-1} + \sigma \epsilon_t$$

with $\epsilon_t \sim N(0, 1)$. If the monetary authority does not regard the impact of its policies on inflation expectations and the autoregressive character of the shock process in its decision-making, i.e. it takes $F = \beta E_t[\pi_{t+1}] + u_t$ as given, it conducts monetary policy under discretion. It is a sort of sequential optimization since it makes every decision optimal each period without committing itself to any future actions (Gali 2008). Monetary policy is then assumed to choose the values of π_t and y_t in order to minimize the period losses. The optimality condition becomes

$$\pi_t = -\frac{\omega}{\alpha} y_t \quad t = 0, 1, 2, \dots \quad (7.3)$$

In case of inflationary pressure originating from a shock to the inflation equation, monetary policy must drive current output below its equilibrium level so as to generate a negative output gap. This “leaning against the wind strategy” is pursued up to the point where the first-order condition is satisfied. It requires that the central bank keeps the output gap negative as long as inflation is positive.

If, however, the monetary authority takes into account that its current actions alter future outcomes, it follows a commitment strategy under which it relies on a policy plan. The latter allows to credibly commit to some future policy at all possible dates. It holds that the central bank is assumed to choose a sequence $\{\pi_t, y_t\}_{t=0}^{\infty}$ with a periodical optimality condition

$$\pi_t = -\frac{\omega}{\alpha} y_t - \frac{\omega}{\alpha} y_{t-1} \quad t = 1, 2, \dots \quad (7.4)$$

⁴⁰A detailed derivation of the model equations is provided in Appendix E.

⁴¹The terminologies cost-push shock, supply shock or inflation shock are used simultaneously in this Section.

The timeless perspective of credible commitment abstracts from the first-order condition in $t = 0$ to avoid time-inconsistency (Woodford 2003). When comparing the optimality conditions of discretion and commitment under the timeless perspective, monetary policy reacts to changes in the output gap under commitment; whereas under discretion, it targets the level of the output gap. The commitment solution improves the short-run output/inflation trade-off since the monetary authority commits to a history-dependent policy in the future, and it is, thus, able to spread the effects of shocks over time. Because the central bank responds to the lagged output gap, commitment to inertia results in the possibility to influence expected future inflation. Each observable shock induces market participants to expect a monetary policy tightening that reduces output and inflation in the near future as well as in the initial period. Thus, the lagged dependence in the policy rule arises, since the central bank actively manages expectations.

In order to evaluate the outcomes under the different targeting rules, McCallum and Nelson (2004b) derive the equilibrium paths for inflation and output under discretion and commitment. They show that the MSV solution to the targeting rule under discretion (d) and commitment (c) represents the unique rational expectations equilibrium. The solutions are

$$\begin{aligned}\pi_t^d &= au_t \\ y_t^d &= bu_t \\ \pi_t^c &= cy_{t-1} + du_t \\ y_t^c &= ey_{t-1} + fu_t\end{aligned}\tag{7.5}$$

with the complex parameter restrictions

$$\begin{aligned}a &= \frac{\omega}{\omega(1 - \beta\rho) + \alpha^2} & b &= \frac{-\alpha}{\omega(1 - \beta\rho) + \alpha^2} & c &= \frac{\omega(1 - \delta)}{\alpha} \\ d &= \frac{1}{\Omega - \beta(\rho + \delta)} & e &= \delta & f &= \frac{-\alpha}{\omega(\Omega - \beta(\rho + \delta))}\end{aligned}$$

Moreover, it holds that $\Omega = 1 + \beta + \frac{\alpha^2}{\omega}$ and $\delta = \frac{\Omega - \sqrt{\Omega^2 - 4\beta}}{2\beta}$.

The macroeconomic implications of a policy under discretion and commitment can be understood by inspecting the equilibrium process for the output gap and inflation. Firstly, the expected unconditionally inflation under both discretion and commitment is zero so that there is no inflation bias.⁴² Secondly, though in both regimes inflation variability is proportional to a cost-push shock, the commitment regime reduces the sensitivity of inflation to this shock in the initial period. The parameter loading on u_t is, in general, smaller for π_t^c . Only if prices are perfectly

⁴²This holds under the assumption that potential output is normalized to zero and the output-gap target is also zero.

flexible and/or the central banks behaves as an “inflation nutter”, discretion is equivalent to the timeless perspective of committing to future policy actions (Sauer 2007). Likewise, the reaction of the output gap is smaller in the initial period under commitment. For policy evaluation, a stabilization bias associated with discretionary policy occurs since the attempt of stabilizing the output gap in the medium-term does not take into account the benefits that result from allowing larger deviations of the output gap in the future. The effects can be documented in a higher tradeoff between inflation and the output gap in the presence of supply shocks where output is overstabilized and inflation behaves too volatile (Gali 2008).

7.3.2.2 The Macro-Finance Linkage and Optimal Policy

The starting point of the analysis of risk compensation is an assumption about the determination of the pricing kernel. As outlined in Chap. 2.1.3, it can be derived in a consumption-based framework where it is driven by marginal utility of consumption. Aggregate demand in the simplest New-Keynesian model is derived from the households optimality condition. Typically, a representative household maximizes the objective function with respect to a flow budget constraint including nominal as well as real assets. Such a real financial asset can be interpreted as an inflation-indexed bond. The Euler equation represents the first-order condition for optimal consumption, given the returns of financial assets with which asset accumulation and consumption smoothing becomes tractable (for a textbook treatment see Woodford 2003; Gali 2008; Walsh 2003). The stochastic discount factor, i.e. the price of a one-period bond, is then explicitly specified within the familiar Euler equation.

In what follows, it is assumed that periodic utility takes the form

$$U(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma}$$

so that the pricing kernel for real as well as nominal bonds between period t and $t + 1$ becomes

$$M_{t+1} = \beta \left(\frac{U'(Y_{t+1})}{U'(Y_t)} \right)$$

$$M_{t+1}^{\$} = \beta \left(\frac{U'(Y_{t+1})}{U'(Y_t)} \right) \left(\frac{P_t}{P_{t+1}} \right)$$

for $Y_t = C_t$.⁴³ Such a representation of the “demand” side of the economy results in the log-linearized basic New-Keynesian aggregate demand function with

⁴³The baseline model abstracts from aggregate demand components like investment, exports or government purchases so that the goods market clearing condition is simply given by $Y_t = C_t$.

$$y_t = E_t y_{t+1} - \gamma^{-1} (i_t - E_t[\pi_{t+1}] - \zeta) \quad (7.6)$$

and $\zeta = -\log \beta$. From these equations, it is straightforward to compute the equilibrium one-period risk-free interest rate, both in nominal as well as in real terms

$$r_{1,t} = -\log(E_t[M_{t+1}])$$

$$i_{1,t} = -\log(E_t[M_{t+1}^{\$}]).$$

The continuously-compounded real and nominal short-term interest rates are related to the marginal rate of substitution of consumption. The stochastic discount factors in log form are

$$m_{t+1} = \log \beta - \gamma(y_{t+1} - y_t)$$

$$m_{t+1}^{\$} = m_{t+1} - \pi_{t+1}.$$

By substituting the equilibrium paths for output and inflation of (7.5) into the pricing kernel, the solution for discretion is

$$\begin{aligned} m_{t+1}^d &= \log \beta - \gamma b(\rho - 1)u_t - \gamma b\sigma\epsilon_{t+1} \\ &= \log \beta - \gamma \left(\frac{\alpha(1 - \rho)}{\omega(1 - \beta\rho) + \alpha^2} \right) u_t + \frac{\gamma\alpha}{\omega(1 - \beta\rho) + \alpha^2} \sigma\epsilon_{t+1} \end{aligned} \quad (7.7)$$

$$\begin{aligned} m_{t+1}^{\$,d} &= \log \beta - (\gamma b(\rho - 1) + a\rho)u_t - (\gamma b + a)\sigma\epsilon_{t+1} \\ &= \log \beta - \left(\frac{\omega\rho + \gamma\alpha(1 - \rho)}{\omega(1 - \beta\rho) + \alpha^2} \right) u_t + \frac{\gamma\alpha - \omega}{\omega(1 - \beta\rho) + \alpha^2} \sigma\epsilon_{t+1}. \end{aligned} \quad (7.8)$$

Under commitment, the SDFs follow

$$\begin{aligned} m_{t+1}^c &= \log \beta - \gamma(e - 1)y_t - \gamma f\rho u_t - \gamma f\sigma\epsilon_{t+1} \\ &= \log \beta - \gamma(\delta - 1)y_t + \frac{\rho\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} u_t + \frac{\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} \sigma\epsilon_{t+1} \end{aligned} \quad (7.9)$$

$$\begin{aligned} m_{t+1}^{\$,c} &= \log \beta - (\gamma(e - 1) + c)y_t - (\gamma f\rho + d\rho)u_t - (\gamma f + d)\sigma\epsilon_{t+1} \\ &= \log \beta - \frac{(\gamma\alpha - \omega)(\delta - 1)}{\alpha} y_t + \rho \left[\frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))} \right] u_t \\ &\quad + \frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))} \sigma\epsilon_{t+1}. \end{aligned} \quad (7.10)$$

From the equilibrium solutions for the SDFs, it is straightforward to compute the equilibrium paths for the one-period real and nominal interest rate both under

commitment and discretion.⁴⁴ In a log-normal model with above specification, pricing one-period bonds implies

$$E_t[m_{t+1}] + 0.5\text{Var}_t[m_{t+1}] = -r_t \quad (7.11)$$

$$E_t[m_{t+1}^\$] + 0.5\text{Var}_t[m_{t+1}^\$] = -i_t. \quad (7.12)$$

The exact solutions are provided in Appendix E. It is important to recognize that the interest rates derived above describe the equilibrium behavior under the two policy regimes. They do *not* resemble a feedback rule from a “nominal anchor” where the interest rate instrument reacts to observable current and lagged endogenous variables characterizing the state of the economy (McCallum 1981). For instance, the real interest rate under commitment moves inversely to inflationary pressure in the initial period which might give rise to indeterminacy of the system.⁴⁵ Therefore, it is inappropriate to compare such an interest-rate characterization with a Taylor rule since the former is not an instrument rule per se but already an equilibrium solution for short-term interest rates. In this regard, Gali (2008) and Woodford (1999a) show that instrument rules consistent with the equilibrium interest rate paths guarantee a unique non-explosive equilibrium. Recently, Gallmeyer et al. (2005) and Gallmeyer et al. (2007) have combined equilibrium short-rate paths with different monetary reaction functions. They show that in order to restore equilibrium in bond markets, the reaction coefficients can not be chosen independently. They are restricted by the parameters of the nominal pricing kernel.

The structural model gives answers to the question how macroeconomic risk is priced into bonds. The risk compensation in a pricing model can be split up into the quantity of risk and price of risk. While the quantity of risk varies from asset to asset, the price of risk is the same for all assets depending on the structural parameters. The market price of risk, λ , per unit of supply shock is represented as the loading before the shock. It represents the negative of the standard deviation of the pricing kernel $\lambda = -\sigma_m$. The underlying structural model generates constant market prices of risk since economic agents do not adjust risk aversion according

⁴⁴The derivation of the one-period short rate actually follows the same method as in the standard optimal policy literature (Woodford 2003; Evans and Honkapohja 2006). Here, rational expectations of inflation and output are inserted in the aggregate demand function and solved for the interest rate. The only difference is that the specification above accepts the finance view of log-normal returns so that interest rates are augmented by Jensen’s inequality.

⁴⁵Evans and Honkapohja (2006) find that such an interest-rate characterization is not determinate under specific parameter constellation and not learnable for all structural parameter values. Instead, they propose an expectations-based reaction function where the central bank does not substitute out for what the expectations ought to be. On similar grounds, Cochrane (2007a) discusses determinacy properties of New-Keynesian models and makes us aware of the fact that the logic within the models are the opposite to the more “old”-Keynesian models. The marginal utility approach of macroeconomic modeling reveals that high inflation goes along with low output growth and high real interest rates coincides with high output growth; consequently, the propagation channels are different compared to Keynesian models as in Svensson (1997).

to the state of the economy. Moreover, the inflation and output gap driving the pricing kernel exhibit constant variances so that the volatility of the state variables is likewise constant. Risk premia are, thus, constant through time and the Expectations Hypothesis of the term structure holds though risk premia are embedded in longer-term bonds. Monetary policy, then, does not only influence the expected future path of risk-adjusted short-term interest rates, but also affects long-term bonds through a risk-premium channel.

In the discretion regime, the sign of the real market price of risk is always negative. In absolute terms, it is increasing with the degree of risk aversion and with the sensitivity of inflation to the output gap. Moreover, as the preference parameter to the output gap ω approaches zero, market prices of risk increase. Additionally, the nominal price of risk is affected by the shock on inflation. The net effect on the sign of the nominal risk price depends on whether the effect of the shock on the output gap is greater or smaller than the effect on inflation. As long as the effect on the output gap $\gamma\alpha$ is greater than on inflation ω , the market price is negative. With increasing output preference, $\lambda^{\$,d}$ can turn out to be positive. Under commitment, the real risk price is negative and it increases with rising risk aversion and α . The same holds for the real part in the nominal market price of risk. A central bank that puts more weight on output gap stabilization triggers two opposite effects, i.e. it decreases the real risk, but gives rise to an heightened nominal risk compensation.

When comparing the loadings in the discretion and commitment case, market prices of risk under commitment are always smaller than under the discretion regime in absolute values. Under commitment, inflation and output are less sensitive to the underlying supply shock so that the required risk reward is not as high for the commitment solution. It holds that

$$\begin{aligned} \frac{\lambda^c}{\lambda^d} &= \frac{\omega(1 - \beta\rho) + \alpha^2}{\omega(\Omega - \beta(\rho + \delta))} \\ &= \frac{\omega(1 - \beta\rho) + \alpha^2}{\omega(1 - \beta\rho) + \alpha^2 + \omega(\beta - \beta\delta)} < 1 \end{aligned}$$

for $\delta < 1$.⁴⁶

7.3.2.3 Implications for Bond Risk Premia

The effect of the policy regime on risk premia is first analyzed for a two-period bond. The price of a two-period bond satisfies

$$P_{2,t} = E_t[M_{t+1}P_{1,t+1}].$$

⁴⁶Sauer (2007) shows that δ is always $0 < \delta < 1$ in order to have stability and to fulfill the MSV criterion.

With $p_{2,t} = -2r_{2,t}$, $E_t m_{t+1} = -r_t - 0.5 \text{var}_t(m_{t+1})$ and log-normality, the pricing equation becomes

$$\begin{aligned} p_{2,t} &= E_t[m_{t+1} + p_{1,t+1}] + \frac{1}{2} \text{var}_t(m_{t+1} + p_{1,t+1}) \\ -2r_{2,t} &= -r_t - E_t r_{t+1} - \frac{1}{2} \text{var}_t(r_{t+1}) - \text{cov}_t(m_{t+1}, r_{t+1}). \end{aligned} \quad (7.13)$$

The expected excess return of the 2-period bond is $E_t x r_{2,t+1} = E_t r r_{2,t+1} - r_t = 2r_{2,t} - E_t r_{t+1} - r_t$.⁴⁷ From the pricing equation above, it is defined as

$$[E_t r r_{t+1} - r_t] - \frac{1}{2} \text{var}_t(r_{t+1}) = \text{cov}_t(m_{t+1}, r_{t+1}). \quad (7.14)$$

If the SDF (output growth) is positively (negatively) correlated with the expected short-term interest rate in $t + 1$, then a risk-averse investor expects to suffer from return losses of holding a two-period bond from which she wants to receive compensation. The reason is that the two-period bond is less attractive than the one-period bond because its payoff is low in bad times when marginal utility is high. The covariance term captures the required compensation for risk embedded in two-period bonds.

Under discretion, the covariance term is

$$\begin{aligned} \text{cov}_t(m_{t+1}^d, r_{t+1}) &= \left(\frac{\alpha \gamma}{\omega(1 - \beta \rho) + \alpha^2} \right)^2 (1 - \rho) \sigma^2 \\ \text{cov}_t(m_{t+1}^{s,d}, i_{t+1}) &= \frac{(\gamma \alpha - \omega)(\gamma \alpha (1 - \rho) + \omega \rho)}{(\omega(1 - \beta \rho) + \alpha^2)^2} \sigma^2 \end{aligned}$$

Under commitment the conditional covariances follow

$$\begin{aligned} \text{cov}_t(m_{t+1}^c, r_{t+1}) &= \left(\frac{\alpha \gamma}{\omega(\Omega - \beta(\rho + \delta))} \right)^2 (1 - \delta - \rho) \sigma^2 \\ \text{cov}_t(m_{t+1}^{s,c}, i_{t+1}) &= \left(\frac{\alpha \gamma - \omega}{\omega(\Omega - \beta(\rho + \delta))} \right)^2 (1 - \delta - \rho) \sigma^2. \end{aligned}$$

A comparison of the two regimes reveals that an increased credibility, in terms of shifting from a discretionary policy to a policy under commitment, decreases the magnitude of overall risk premia. This comes as a result of a lower impact of the shock on inflation and output as well as of diminishing volatility of inflation and output so that the inflation/output tradeoff can be reduced.

⁴⁷In order to avoid confusion, the expected two-period return is defined as $E_t r r_{2,t+1} = p_{1,t+1} - p_{2,t} = -r_t + 2r_{2,t}$.

In the discretion regime, risk premia on a two-period real bond are positive if $\rho < 1$. As long as there is a marginal utility effect of the supply shock, the risk premium is non-zero. Since the central bank operates via aggregate demand in order to fight inflation, a compensation for this real risk is required by agents. With an increased output preference, the risk premium falls. This happens because monetary policy reduces the volatility of output and makes consumption smooth which, in turn, reduces the risk compensation for holding a real two-period bond. In the limiting case where the shock to the economy follows a random walk, the risk premium becomes zero as there is no effect on marginal utility.

For nominal bonds, a similar condition holds where in addition $\gamma\alpha$ needs to exceed the parameter value ω . An increasing ω has two countervailing effects on overall risk premia. First, since any nominal bond is also exposed to real risks, the effect of a higher ω should follow along the same lines as for a real bond, i.e. the risk premium tends to fall. Second, with increasing ω , the shock impact on inflation and its variability rises. This generates an obvious inflation-output tradeoff (captured by $\omega\rho$). Whenever there is a negative correlation between inflation and output, the compensation for nominal risk increases.

Under commitment, two-period real and nominal bonds carry positive risk premia if $\delta + \rho < 1$. Consequently, the lower the persistence in the output gap and the lower the autocorrelation of the shock term, the higher risk premia tend to be. When comparing 2-period nominal and real bonds, the risk premium on nominal bonds falls relative to real bonds with rising ω because a higher output preference results in an interest-rate reaction that leads to a less pronounced positive correlation between $m_{t+1}^{$.c}$ and i_{t+1} . Moreover, a supply shock always contributes to a positive risk compensation through the direct effect on inflation ($\omega\rho$).

What is also evident from the simple 2-period analysis is that it is perfectly possible that risk premia can be negative. Economically, if a central bank is doing a very good job of stabilizing expectations, investors might not require positive compensations for holding assets “exposed to the central bank’s objectives”. Quite on the contrary, instruments affected by these risk factors (inflation and output gap) might represent a hedge for marginal utility and then imply a negative premium. If the monetary authority conducts a commitment strategy, deviations in the output gap and in inflation from their respective targets persist beyond the life of the initial shock. With such a response, it manages to reduce the inflation/output tradeoff in the initial period when the shock occurs. Consequently, market prices of risk are lower and corresponding risk premia in the initial period fall. The latter can even become negative so that investors prefer investing in a two-period bond compared to the one-period risk-free interest rate because the two-period bond represents a hedge to them. It offers a high return in real terms exactly when it is needed by economic agents, i.e. when marginal utility in $t + 1$ is high and short-term interest rates are low.

As outlined in Chap. 2.1.4.3, it is straightforward to extract the inflation risk premium from the pricing formulas. Recall, that the one-period nominal interest rate is the sum of the one-period real interest rate, expected one-period inflation, an inflation risk premium and an adjustment term:

$$i_t - r_t - E_t[\pi_{t+1}] = -\gamma cov_t(\Delta y_{t+1}, \pi_{t+1}) - \frac{1}{2} var_t(\pi_{t+1}) \quad (7.15)$$

The identity says that if inflation and the output move in opposite direction, the conditional covariance is negative, and, thus, imposes a positive risk premium on the nominal bond. Economically, if low output growth and a high marginal utility go hand in hand with high inflation, nominal bonds are less desirable relative to real bonds, as inflation is high in those states where nominal bonds suffer from real return losses. A central bank affects the inflation risk premium through its choice of its preference parameter ω . To see this, the inflation premium under discretion and commitment is given by

$$\begin{aligned} -\gamma cov_t(\Delta y_{t+1}, \pi_{t+1})^d &= \gamma \frac{\alpha \omega}{(\omega(1 - \beta \rho) + \alpha^2)^2} \sigma^2 \\ -\gamma cov_t(\Delta y_{t+1}, \pi_{t+1})^c &= \gamma \frac{\alpha \omega}{[\omega(\Omega - \beta(\rho + \delta))]^2} \sigma^2. \end{aligned}$$

The model specification allows for no negative inflation risk premium in case of supply shocks for one-period bonds. This is self-evident from the structural set-up, since a supply shock always introduces a negative correlation between the output gap and inflation dynamics in the initial period when the supply shock hits the economy. An increased preference in terms of ω has the net effect that it increases the inflation risk premium so that the negative correlation between the change in expected output and inflation widens. Finally, the inflation risk premium is always lower under commitment than under discretion due to the lower inflation/output tradeoff when the shock hits the economy. In particular, the economy is less vulnerable to supply side shocks since the sensitivity of output and inflation is lower under commitment. The compensation for being exposed to this risk is then also lower.

7.3.2.4 The Term Structure of Risk Premia

Within the affine term structure representation of Chap. 3.5, the analysis above can be extended to the n -period case of term premia. In this respect, the relevant state variables of the simple model can be stacked into a vector X_t . A simple VAR(1) process describes the equilibrium dynamics

$$X_t = \mu + \phi X_{t-1} + \Psi^\top \Sigma \varepsilon_t \quad (7.16)$$

where $\Psi^\top$ is a matrix of structural parameters and $\Sigma \Sigma^\top$ is the variance-covariance matrix of the stochastic shocks with $\varepsilon_t \sim i.i.d. N(0, I)$. As outlined in Appendix E, interest rates follow

$$\begin{aligned}
i_{n,t} &= -n^{-1} \log(P_{n,t}) \\
&= n^{-1} (-A_n - B_n^\top X_t) \\
&= a_n + b_n^\top X_t
\end{aligned} \tag{7.17}$$

with $a_n = -A_n/n$, $b_n = -B_n/n$ and the recursive loadings are defined as

$$A_n = A_{n-1} + B_{n-1}^\top (\mu - \Psi^\top \Sigma \Sigma^\top \lambda_0) + \frac{1}{2} B_{n-1}^\top \Sigma_\psi B_{n-1} - \delta_0 \tag{7.18}$$

$$B_n^\top = B_{n-1}^\top (\phi - \Psi^\top \Sigma \Sigma^\top \lambda_1) - \delta_1 \tag{7.19}$$

where $\Sigma_\psi = \Psi^\top \Sigma \Sigma^\top \Psi$ with $A_1 = -\delta_0$ and $B_1 = -\delta_1$.

This simple model can be transformed into the discrete-time affine class of term structure models by imposing restrictions on the parameters. Under discretion, the only relevant state variable is the time-series property of the supply shock which is transformed into the pricing kernel via the output gap and inflation. Under commitment, the state space needs to be augmented by the evolution of the output gap in the rational expectations equilibrium. Appendix E provides a derivation of the restrictions on $\mu, \phi, \Psi, \Sigma, \lambda, \delta_0$ and δ_1 for both optimal monetary policy under discretion and commitment. With this closed-form solution, expected excess returns can be easily computed. They are defined as $E_t x r_{n,t+1} = p_{n-1,t+1} - p_{n,t} - i_{t,1}$. Based on the definition of the yield curve, they can be written as

$$E_t [x r_{n,t+1}] = B_{n-1}^\top \Sigma_\psi \lambda_0 - 0.5 B_{n-1}^\top \Sigma_\psi B_{n-1}. \tag{7.20}$$

It is also possible to calculate inflation risk premia of relevant maturities. With the model dynamics represented in the VAR(1) form, inflation expectations can be computed up to the maturity spectrum of the term structure. For that purpose, the shock equation, inflation and output can be stacked into a vector $Z_t = [u_t, \pi_t, y_t]^\top$ so that conditional expectations of the system for the periods up to $t+n$ at the initial shock period take the form

$$E_t [Z_{t+n}] = \Lambda^n Z_t$$

with $Z_t = \mathcal{E} \varepsilon_t$ and the coefficient matrices chosen appropriately to describe the law of motion. The term structure of instantaneous inflation risk premia⁴⁸ are calculated according to

$$r \pi_{t+n}^{d,c} = i_{n,t}^{d,c} - r_{n,t}^{d,c} - E_t [\bar{\pi}_{t+n}^{d,c}] \tag{7.21}$$

with $E_t [\bar{\pi}_{t+n}^{d,c}] = n^{-1} \sum_{i=1}^n E_t [\pi_{t+i}^{d,c}]$ denoting average expected inflation over the life of the bonds.

⁴⁸The Jensen inequality term is excluded from the risk premium computation.

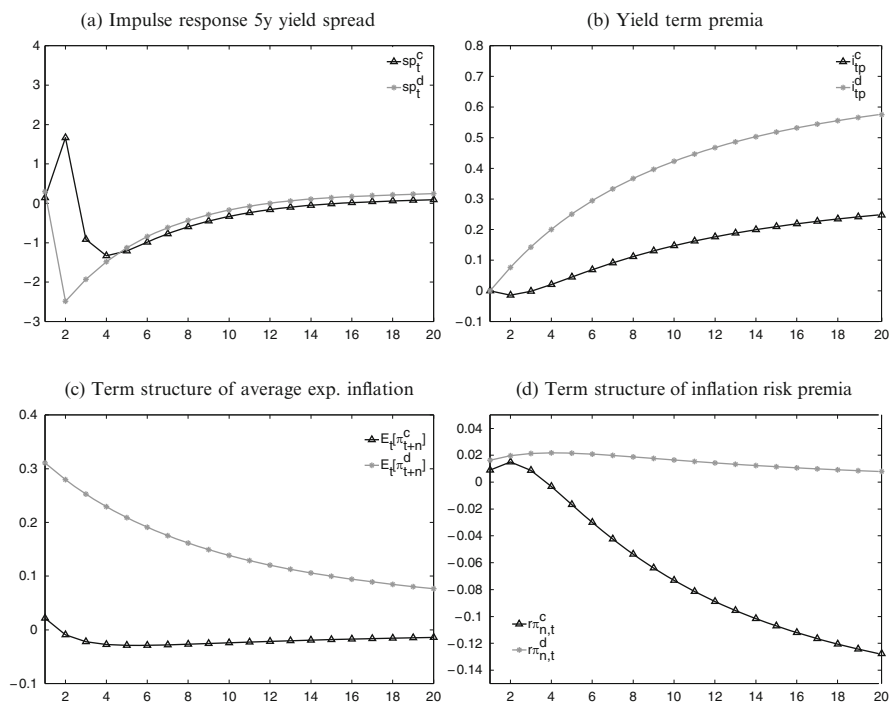


Fig. 7.6 The term structure of risk premia and effects of monetary policy

Model-implied risk premia and the basic behavior of interest rates are plotted in Fig. 7.6.⁴⁹ Figure 7.6a reveals that the dynamics of the yield spread differ considerably under the two regimes. Under discretion, the response of the spread is slightly higher and moves in opposite direction than under commitment. Since the short-term interest rate rises in the course of the restrictive monetary policy response, the spread becomes negative and long-term interest rates rise less than the short rate. The commitment solution of optimal monetary policy makes clear that it is possible under reasonable parameter values that in an environment of inflationary pressure, the term spread widens. This is not the result of an increase in long-term interest rates; it rather reflects a significant fall in the short-term interest rate if monetary policy credibly commits to future interest-rate moves. Again, the dynamic response of the short-term interest rate under commitment does not guarantee uniqueness and learnability of the macroeconomic system as explained earlier in this Section.

⁴⁹The parameter values are chosen as $\beta = 0.99$, $\alpha = 0.17$, $\omega = (.5/4)^2 = 0.0156$, $\rho = 0.8$ and $\gamma = 4$. See McCallum and Nelson (2004b) and Gali (2008).

Figure 7.6b gives an idea of the sign and the height of nominal yield term premia.⁵⁰ Risk premia under both regimes are upward-sloping so that the yield curve is characterized by the normal shape, i.e. a positive slope along the maturity spectrum. Notice, that unconditional and conditional risk premia coincide since the latter do not depend on a changing economy. It is eye-catching that the lack of credibility of discretionary policy leads to higher term premia being demanded by economic agents; it increases the compensation for risk in nominal bonds.

To shed further light on the implications of optimal monetary policy on financial risk premia, Fig. 7.6c plots the average expected inflation with forecast horizon n in the period the shock hits the economy. As expected by the RE solution for inflation, average inflation expectations under discretion are much higher and more persistent than under commitment. This effect shifts up the nominal term structure under discretion; whereas, according to the commitment solution, inflation expectations become even slightly negative and the nominal yield curve is scaled down relative to the real term structure of interest rates. The different expectations pattern is also translated into a lower nominal risk price. As highlighted in Fig. 7.6d, inflation risk compensation is significantly smaller under commitment, since the initial increase in inflation is followed by a period of disinflation with an average expected inflation equal to zero. Such inflation risk premia patterns imply that the real risk premium is greater than the nominal risk premium under commitment. Indeed, when plotting impulse response functions for inflation and the output gap (not reported here), both variables co-vary positively from the fourth quarter onwards. Nominal bonds with longer maturity then provide a good hedge for economic agents so that required risk compensation of inflationary risk is negative.

Despite its attractiveness of representing the economy within such a simple form, the model's financial asset pricing implications are sensitive to the numerical parameter values of the economy. The benchmark specification generates a nominal yield curve that is, on average, upward sloping as it is observed in international yield curve data (Backus et al. 1998). To some extent, the empirical evidence on real interest rates reaches different conclusions. The unconditional mean of the slope of the real yield curve is slightly negative and mirrors negative real risk premia (Piazzesi and Schneider 2006; Ang et al. 2008). This fact can hardly be reproduced by the model since an inflationary shock is considered to produce real output risk for which agents want to get compensated. With increasing ω , real risk premia relatively fall, but they are still positive so that the real yield curve cannot monotonically fall. Long-term real term premia can become negative, if holding long-term bonds provides a hedge against the underlying structural shocks. This fact can be overcome in a general equilibrium setting in which economic agents are not only exposed to inflationary risk and optimal monetary policy targeting rules but also to (persistent)

⁵⁰They are calculated by setting the risk price vector equal to zero and by subtracting Jensen's inequality to get a measure for risk-neutral yields. The yield term premium is then defined as the difference between actual yields and risk-neutral yields.

aggregate demand shocks. In a series of papers, Palomino (2009, 2010) shows that preference shocks driven by consumption-dependent stochastic risk aversion, and a pure taste shock to agent's utility, allow to produce both a downward as well as an upward sloping real yield curve.⁵¹ Finally, the inflation risk premium in the model is higher at the short end of the yield curve since the presented model does not provide any long-term nominal attractor or shifting endpoint. The analysis of yield curve movements in Chap. 5.3 has shown that such slow-moving risk factors are important in order to fit the long-end of the yield curve and to generate risk premia with appropriate magnitude.

7.3.3 Risk Premia in the New-Keynesian Model Economy

Chap. 5.3 has dealt with term structure implications of New-Keynesian Macroeconomics. The model has first been log-linearized around the steady state and asset pricing implications have been derived by applying the log-normal approach to the pricing kernel. Such a modeling strategy produces constant market prices of risk and, thus, constant risk premia. This subsection shortly presents evidence on the excess return loadings of the macroeconomic shocks and of the net effect on yield term premia.

Figure 7.7 plots nominal excess returns for a given macroeconomic shock and yield term premia for selected policy rule specifications. Figures 7.7a–b display the benchmark case with inflation and output coefficients set to $\rho = 1.5$ and $\chi = 0.5$. Excess returns can be decomposed into the instantaneous contribution of a shock to total one-period excess returns for bonds with maturity n . Excess returns are positive whenever the conditional covariance between the pricing kernel and future short-term interest rates is positive.⁵² For instance, a positive innovation in the output shock triggers a rise in the current short rate. Simultaneously, it also raises long-term interest rates and lowers its bond price. Since a positive output shock leads to higher marginal utility, the stochastic discount factor also goes up so that it is negatively correlated with long-term bond prices. As a consequence, bond holders require a positive return compensation for holding long-term bonds. The same mechanism holds in case of monetary policy shocks and innovations to the natural rate output.

In contrast, a cost-push shock to the price setting equation raises the inflation rate which, while it lowers the stochastic discount factor, but it also lowers bond prices and raises long-term interest rates. Investors then regard long-term bond instruments

⁵¹Notice, that preference shocks can be interpreted as demand shocks since they show up in the natural rate specification. Shocks to the natural rate are compressed in an exogenous demand shock vector (Gali 2008).

⁵²Alternatively, excess returns are greater than zero if the conditional covariance between the pricing kernel and (1) the future expected sequence of pricing kernels is negative or (2) the future bond prices is negative.

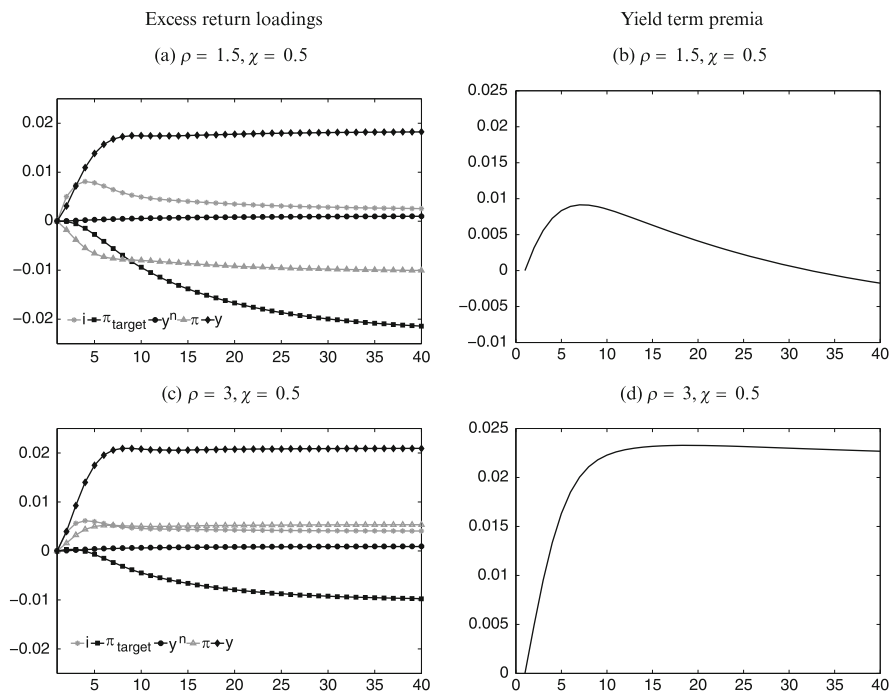


Fig. 7.7 Excess return loadings and risk premia in NK-model

as a hedge against cost-push shocks and require a negative risk compensation. Similarly, an increase in the inflation target triggers a fall in the current short rate, it raises the stochastic discount factor and it lowers long-term interest rates so that excess returns for one unit of inflation target shocks are negative (see also Ichiue (2005); Hördahl et al. (2006) for these results). The yield term premium is calculated by taking the difference between yields and risk-neutral yields. It is positive and increasing up to yields with maturity of ten quarters and then it is monotonically decreasing and becomes slightly negative for long-term bonds due to the dominance of the inflation target at longer maturities.

The choice of policy responsiveness to inflation and output alters the sign and height of excess returns loadings and the term structure of yield term premia. To see this effect, Figs. 7.7c–d display risk premia implications for a stronger response of monetary policy to inflationary pressure by setting $\rho = 3$. The main change comes from a positive loading of the cost-push shock. A more aggressive central bank brings about expected return uncertainty of long-term bonds since the stochastic discount factor becomes negatively correlated with long-term bond prices. Still, long-term bond instruments provide a hedge to inflation target risk but the size effect is smaller than for the benchmark calibration. As a result, yield term premia are increasing with time to maturity and do not move back as is the case with a less aggressive monetary authority.

One considerable drawback in the above New-Keynesian analysis is that other sources of risk, i.e. liquidity and default risk are not captured. Risk premia are purely derived from aggregate consumption risk and optimal household behavior in a frictionless financial market economy. The introduction of financing costs enables to model default and liquidity risk premia. Goodhart et al. (2006, 2009) and Espinoza et al. (2009) show that the aggregate consumption risk approach, as is found in standard New-Keynesian models, is not the most important risk component in asset valuations. With default and cash-in-advance constraints in a heterogeneous agent model, there is a proper demand for liquidity with interest rates mirroring a “loss reserve payment protecting the agent against default.” Uncertainty about aggregate liquidity in future states induces agents to command a premium that generates higher long-term interest rates and a positive yield spread in equilibrium.

In the standard model, financial intermediation activity with its amplification effects is likewise not included. However, as has been laid out within the credit view and its re-emphasis on the financial intermediary sector and risk-taking attitudes, financial frictions and financial shocks may be additional if not the main driving forces of financial risk premia and, obviously, macroeconomic outcomes. Advances have been made in incorporating liquidity, default and heterogeneity aspects into general equilibrium modeling. Frictions in both the borrowing and the lending sector allow for the modeling of multiple interest rates and credit spreads that act as amplifiers of real shocks and as a source of financial shocks to the economy. The difference between the rate paid to depositors and the borrowing rate can be regarded as the risk premium demanded by the intermediary sector. It can arise from various sources. Constraints in the borrowing sector are due to the principal-agent problem between borrowers and lenders where the external finance premium varies in accordance with the net worth of collaterals (Kiyotaki and Moore 1997; Bernanke et al. 1999; Christiano et al. 2007). Within the lending sector, the costly “loan production approach” assumes that intermediaries have increasing marginal costs in the volume of lending due to originating and servicing loans and managing loan portfolios in the presence of the fixity of some factors (Cúrdia and Woodford 2008; Goodfriend and McCallum 2007). Another way of generating endogenous credit spreads and risk premia is to implement capital and funding constraints in the intermediary sector which are more binding in the recession phase than in the boom phase. These include the “value-at-risk” constraint sketched out by Adrian and Shin (2009a) and the “deposit-diverting” idea of Gertler and Kiyotaki (2010). Since funding of additional loan granting is partly obtained from the wholesale market, funding shocks within this market give rise to (non-linear) liquidity spirals as described by Brunnermeier and Pedersen (2009) and which have been recently incorporated into general equilibrium models by Brunnermeier and Sannikov (2010) and Gertler and Kiyotaki (2010); endogenous leverage dynamics, together with the standard financial accelerator of Bernanke et al. (1999), have been applied to a DSGE model by Angeloni et al. (2010).

7.4 Challenges for Monetary Policy

7.4.1 *The Debate on “Too Low for Too Long” in the Pre-Crisis Period 2002–2006*

“Could monetary policy have helped prevent the financial crisis” that began in August 2007? This question was raised by ECB Governor Bini-Smaghi (2010) in a speech about the lessons to be drawn from the crisis. To him, “[...] a case can be made that interest rates should have been higher than was the case *before* the crisis.” Not only European policy makers admit that, at least to some extent, monetary policy played some role. FRSB Kansas City president Hoenig (2010) remarks that “[...] experience has shown that, despite good intentions, maintaining excessively low interest rates for a lengthy period runs the risk of creating new kinds of asset misallocations, more volatile and higher long-run inflation, and more unemployment.” To be precise, the position is not to regard monetary policy as the single cause. The insight rather emphasizes the plurality of factors that, together with unsound monetary policy, produced the financial crisis – among them faulty regulation, intransparent financial product innovations, the growing importance of shadow banking and global imbalances.

In order to address the issue of whether central banks have set interest rates too low for too long in the period 2002–2006, a review of macroeconomic conditions is necessary in order to draw conclusions about the monetary policy stance during this period. Major industrialized countries suffered from a recession following the dot-com crash in 2000, as well as geo-political shocks including the terrorist attacks on September 11, 2001 and the Iraq war in 2003. Particularly in the US, according to Bernanke (2010), fears of a jobless recovery and an unwelcome scenario of deflationary pressure induced the FED to aggressively cut the policy rate to 1% and to keep the rate at this low level until the tightening cycle was implemented in mid-2004. Easing of monetary conditions was also the primary focus in the euro area, Canada and the United Kingdom. At the same time, activity in the housing market picked up considerably, first and foremost in the US, Spain and the UK. Strong dynamics in asset prices could also be identified in other asset markets, in particular in stock and bond markets. Therefore, decline in returns was a phenomenon not attached to one specific market but it was apparent on a general level and for a broad set of asset classes.⁵³

There are several ways in which to judge the overall monetary policy stance during this period. One first crude measure is to calculate a real short-term rate as the difference between the 3-month interest rate and inflation.⁵⁴ Real short

⁵³For an empirical overview of the coincidence of low nominal short-term interest rates and rising asset prices for OECD countries see Ahrend et al. (2008).

⁵⁴The nominal short rate is the 3-month interest rates and inflation is the CPI all items index. Data are pulled from FED Fred and the ECB.

rates were particularly low in 2002–2004 in the US and the euro area (Fig. 7.8). Against the background of the macroeconomic environment, monetary policy was accommodative: for the US, the calculated real rate stayed at negative levels for over 40% of the time in the years 2000–2009, a number not even reached in the turbulent years of the 1970s when inflation skyrocketed and output volatility was high (not reported here).

Another approach is to conduct a counterfactual analysis for monetary policy and to apply simple rule-based policy reaction functions to the period 2002–2006. The Taylor rule can be used as a benchmark rule where the interest rate derived from this normative perspective can be compared with actual short-rate dynamics. In other words, one can ask how the short rate would have behaved had a central bank followed the Taylor rule specification. As noted previously, Taylor (1993) showed that his rule performed well in the 1980s and early 1990s when inflation was successfully brought down accompanied by robust economic growth in many OECD countries. Taylor specified his rule to be 1.5 times inflation plus 0.5 times the output gap, plus 1. The underlying assumptions were an average, or “natural” real rate of 2% and an inflation target of roughly 2%. Though applying such a standard Taylor rule to the US and the euro area, it becomes

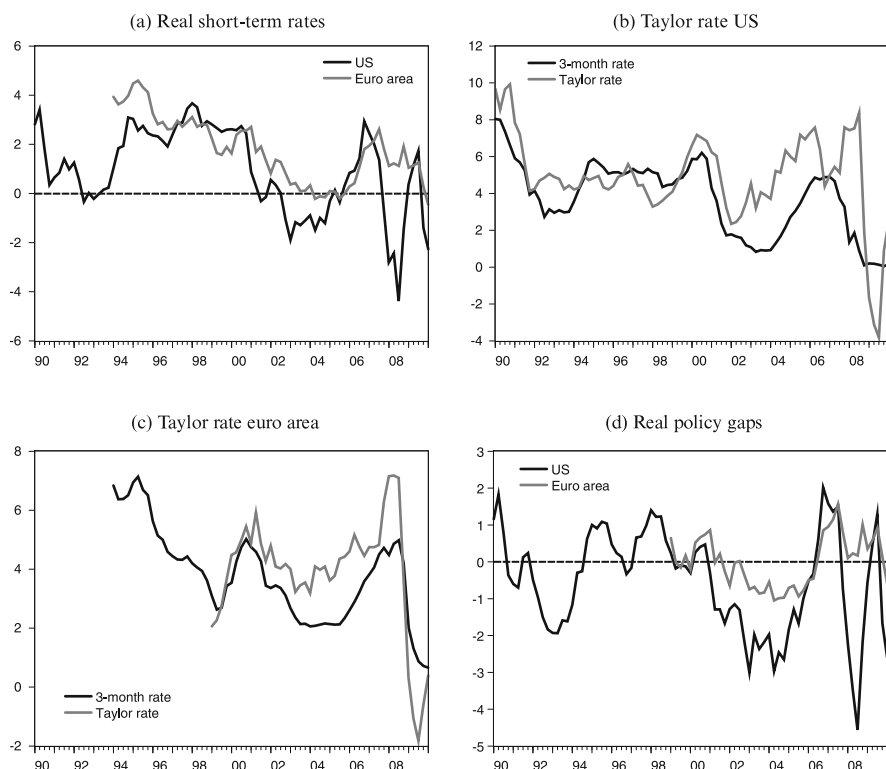


Fig. 7.8 Real policy gaps

evident that there was a large deviation from the Taylor rule starting in 2004; it persisted until the beginning of 2007 when short-term interest rates continued to rise.⁵⁵ Actual nominal short rates were significantly below the prescribed levels recommended by Taylor's specifications. The maximum spread was 4% in the fourth quarter 2004 for the US and 2% in the second quarter 2004 in the euro area.

There are a number of caveats to use the Taylor rule as operational guidance for monetary policy. It is well-known that the choice of the price index can considerably alter the deviations between actual rates and normative rates. Depending on the use of headline inflation or core inflation, the height and the volatility of Taylor rates may differ.⁵⁶ The benchmark rate calculation in Fig. 7.8 is conducted with inflation including seasonal and volatile price index components, mainly energy as well as food prices. However, when checking for robustness of the persistence of the highly accommodative period 2002–2006 by using core inflation, the results basically remain unchanged though the gap between the Taylor rate and the actual rate in both currency areas is a bit smaller (not reported here).⁵⁷ Furthermore, since monetary policy should be forward-looking, some Taylor rule specifications do not depend on currently observed inflation but rather on inflation expectations. The latter can be based on the central bank's own inflation projections or on private sector forecasts. Orphanides and Wieland (2008) show that a projection-based rule relative to the above outcome-based rule improves the fit with actual short-rate dynamics for the period 2002–2003 but it is likewise not possible to reproduce the federal funds rate behavior in the years 2004–2005. But even if one (reasonably) takes it for granted that the FED uses its own inflation projection in current interest-rate decisions, a major shortcoming occurs, i.e. the Fed's forecasts of future inflation were persistently too low during the period of consideration which might have contributed to the lower target interest rate (Taylor 2010). Indeed, an interest-rate reaction function that incorporates private sector forecasts of inflation also signal higher short-term rates as actually observed.

The original and estimated versions of Taylor rules share further difficulties in terms of the observability and estimation of the neutral rate and the output gap in real time. It is well-known that ex-post revisions of potential output considerably differ from real-time estimates that can produce serious mistakes in the monetary policy formulation (Orphanides 2001). Moreover, it seems reasonable to assume that the underlying natural or equilibrium real interest rate is a time-varying variable strongly affected by productivity growth rather than a constant term entering the Taylor rule (Laubach and Williams 2003). Evaluating monetary policy must then

⁵⁵For the calculation, headline inflation (CPI index all items) has been used. The output gap is measured as the percentage deviation of actual real GDP from potential real GDP. The short rate is the 3-month interest rate. Data sources: Fed Fred, ECB and OECD.

⁵⁶For a discussion about the use of different price indices see Mishkin (2007).

⁵⁷For the US, the core PCE inflation rates is taken as benchmark for core inflation and for the euro area, the HCPI less energy and food components serves as proxy for core inflation.

take into account the real-time problems policy makers are faced with and the time-varying nature of the equilibrium real rate.

In the following, it is verified if the assessment of a too accommodative monetary policy still holds when controlling for a time-varying natural rate of interest. The stance of monetary policy can be regarded as expansionary if the short-term real interest rate lies lower than the benchmark natural rate (Wicksell 1898). The latter is calculated as the average real short-term interest rate plus the annual growth rate in potential output less its long-term average. The cyclical behavior of the natural rate is, thus, approximated by potential output growth which in turn should be predominately influenced by productivity growth (Gambacorta 2009).⁵⁸ Figure 7.8d displays the real policy gap defined as the difference between the real short-term interest rate and the natural rate. Again, policy rates have been unusually low and returned towards a neutral level but not before the second quarter of 2006. This pattern predominantly holds for US monetary policy whose deviations appear much stronger than in the euro area. The US real rate gap stayed negative for 22 quarters during the period 2001–2006, a duration never reached before in the post-war era. Although, according to NBER, the 1990 and 2001 economic contraction phases each endured no longer than two quarters, the real policy gap in the early 1990s has been negative for only 15 quarters. The character of a highly expansionary US monetary policy can be also seen in cumulated negative real rate gaps during the two periods: from 2001–2006, they added up to 40% and from 1990–1994 to only 17%. Surely, this description must be judged against the background of a different macroeconomic environment, in particular the threat of deflationary pressure in the US economy in the years 2002–2003 (Bernanke 2002). The FED was therefore well-advised to aggressively cut its policy rate to 1%. However, the preferred inflation measure, PCE inflation, stabilized again in the third quarter of 2002 at 2%; and, as of 2003, the output gap recovered.⁵⁹ It is exactly for this reason why the conventional Taylor rule recommended policy-rate increases starting in 2002–2003, and not because core inflation was above a specified target level, say 2%, but the (still) negative inflation and output gap narrowed which called for rate adjustments upwards.⁶⁰ For this reason, Greenspan's interpretation of using the Taylor as a measure for the monetary policy stance is incorrect when he writes that "[...] the Taylor rule also gave a false signal for policy to stabilize the core PCE price" (Greenspan 2010, 42).

All in all, the assessment above suggests that monetary policy was unusually accommodative in the period following the dotcom crash. Policy rates remained at

⁵⁸ Average real rates and average growth in potential output are from 1980:1–2009:4 in the US and from 1999:1–2009:4 in the euro area. The generated time series broadly matches natural rate dynamics which are estimated within structural models such as those of Belke and Klose (2010).

⁵⁹ Other inflation measures such as the CPI likewise stabilized around the same level. Only the core PCI index fell in the beginning of 2003 from 2% to 1.4%, but recovered at 1.9% in the end of 2003.

⁶⁰ Moreover, it is questionable to assume a 2% inflation target for core inflation. This would bring about much larger and more volatile dynamics in headline inflation (Mishkin 2007).

levels too low and for too long when measured against conventional benchmark rates. A final issue during this period was that the FED announced in 2003 that it was likely to stay accommodative for a considerable period of time. Such communication might have fostered the market view of low policy rates in the future thereby promoting low long-term interest rates and excessive risk-taking. Break-even and survey measures of inflation *did* pick up in July 2003 and inflation *was* increasing in the euro area and the US (Clarida 2010). Headline inflation in the US rose to 4.5% (CPI inflation even to more than 5.5%) in 2008 and core inflation picked up to 2.6%. In the euro area, overall inflation reached 3.8% in 2008. Therefore, it seems maintainable to characterize monetary policy as too expansionary in the run-up of these inflation figures.

Closely related to this debate is the question of whether loose monetary policy contributed to the surge in housing prices – mainly in the US – during this period. According to Bernanke (2010), strong US housing investment, going along with above-average house price growth, was already a market development beginning shortly before and during the year 2000 when the 30-year mortgage rate dropped from its mid-2000 peak although monetary policy started its easing cycle in January 2001. However, the most rapid price gains occurred after 2002 when the Taylor and real rate gap were by a large amount in the negative corridor. The timing of events suggests that loose monetary policy and excessive house-price appreciation were positively correlated.

By lowering the policy rate, monetary policy affects housing dynamics mainly through its effect on mortgage rates and expected house price appreciations. If the short-term interest rate is lowered, long-term interest rates and mortgage rates also tend to decline so that the user cost of capital falls and housing demand picks up. Moreover, provided that the short rate remains low for a prolonged period, markets may expect a real increase in the value of houses which induces current demand to rise. Empirical evidence on the transmission mechanism of monetary policy on house prices is rather mixed. Dokko et al. (2009) find that for the US, monetary policy only accounted for a small portion of housing prices based on historical data until the beginning of the excessive housing boom in 2002. Even if it is assumed that the FED would have followed the Taylor rule, house prices would not have significantly performed with lower dynamics in terms of both growth and level. In contrast, Jarocinski and Smets (2008) and Taylor (2009) find an inconcealable effect of loose monetary policy on housing activity when including the latest housing boom in the empirical analysis. The former note that “[t]here is also evidence that monetary policy has significant effects on housing investment and house prices and that easy monetary policy designed to stave off perceived risks of deflation in 2002–2004 has contributed to the boom in the housing market in 2004 and 2005” (p. 362). This result can be explained by the fact that – opposed to the 1980s and 1990s – an increasing number of mortgage applications was based on adjustable rate contracts so that these rates are more closely linked to short-term interest rates. A positive link between negative policy gaps and increases in residential property prices is also identified across OECD countries by Ahrend et al. (2008).

But even if this causal link remains controversial from an empirical point of view, the Section on the risk-taking channel demonstrates that interest-rate setting and the monetary policy strategy does affect the pricing of a whole range of assets and it does alter the dynamics of credit supply schedules.

7.4.2 *Financial Intermediaries, the Yield Curve and Credit Boom-Bust Cycles*

The phenomenon of business cycle moderation beginning in the 1980s with low inflation and output volatility can be understood as a reflex of manifold developments. Favorable supply conditions, in terms of excess labor supply on a global scale and an enhanced credibility of conducting monetary policy, contributed to a low-inflation environment with low and stable inflation expectations as well as low interest rates. This moderation supported the abolishment of stop-and-go policies which translated into lower overall macroeconomic uncertainty. The side effect could be documented in a trend decline of risk compensation on asset markets that was driven by both an increased risk appetite and a shrinking quantity of risk (see Chap. 2.1.4.1 and Sect. 7.3.1). The progress of stabilizing goods and labor markets, thus, has promoted increased risk-taking on the part of financial market participants which culminated in *abrupt* asset price reversals in the wake of increasing financial imbalances (Borio and Lowe 2002; Trichet 2008). The point is that the establishment of an effective anti-inflationary monetary policy regime may lead to an environment where signs of unsustainable macroeconomic dynamics do not first show up in rising inflation or excess goods demand pressure but, rather, in a build up of financial imbalances – the “paradox of credibility” (Borio 2006).

There is a trade-off between monetary and financial stability. Price stability is not a sufficient condition for financial stability as claimed by the “monetarist” view of the interlinkages between goods and asset markets (Schwartz 1995; Issing 2003); nor does price stability necessarily promote financial stability (Bordo and Wheelock 1998). To be clear, monetary stability decreases the level of uncertainty in the economy thereby reducing the likelihood of mispricing due to uncertain cash flow prospects and narrowing of information asymmetries. It is common sense to assume that price stability translates into lower asset price volatility which, in the end, can foster economic growth, at least ideally. However, as extensively and previously explained, price stability is like a Janus head for financial markets since it can push up financial imbalances due to higher risk-taking. With low risk perceptions, risk tolerance accumulates in the expansion phase of the business and financial cycle while risk materializes at the peak of the financial cycle when recessionary developments crop up. Ironically, while markets infer low risk in the boom phase, risk is actually increasing “in the background”. Market participants suffer from the time-dimension aspect of risk assessments, in particular from assessing the non-diversifiable, systemic wide risk that manifests itself in an endogenous and a

highly non-linear way.⁶¹ Such behavior is the result of both imperfect knowledge of the systemic nature of individuals' actions,⁶² and incentive gaps that induce financial institutions to take actions that may result in undesirable outcomes with negative adverse effects for other financial institutions and the economy as a whole (e.g. triggered by information asymmetries and liquidity spirals or by the insurance guarantee of public institutions in case system-wide risks materialize).

This pattern produces high procyclicality in financial markets with amplifying affects on credit and asset growth as well as in both the boom and the bust phases. Procyclical financial market developments are natural to business cycles as the valuation of current and future cash flow streams of tradable assets are positively related to the level of economic activity. The simultaneous build-up of financial imbalances is the side effect of such amplifying affects; they are in high gear when three factors pick up speed in financial markets: the size of balance sheet growth, leverage and the degree of maturity transformation on the part of financial intermediaries. As has been analyzed previously, the ability to expand the size of financial intermediaries' exposures comes from capital constraints.

In the boom phase, a relief in the capital position through an increase in asset prices or an improved net interest rate margin generates overcapacity. The attempt to extend the size of intermediation activity goes hand in hand with increased leverage and a reduction in measured risk. Most importantly, for a given fixed pool of outside funding from depositors (households) and a low stock of liquid assets, bloating balance sheets can be only achieved through borrowing and lending from each other *within* the financial sector (Shin 2010).⁶³ It massively contributes to the interconnectedness of financial institutions and makes them heavily reliant on wholesale market conditions (Haldane 2009; ECB 2010c). Funding liquidity,

⁶¹Following DeBandt and Hartmann (2002), a financial shock that affects the whole economy is labeled as "systematic" since it is not diversifiable. This shock may produce a "systemic" event in financial markets where simultaneously a large number of financial institutions are hit by the shock (e.g. a macro shock). Systemic events can also be the result of limited or even idiosyncratic shocks to one or a small number of institutions which spread out via contagion (domino effects, loss and liquidity spirals). Finally, systemic events are more likely to appear when stronger imbalances have built up over time and then unravel abruptly.

⁶²For instance, Caballero and Krishnamurthy (2008) find that the process of increased intermediation chains make the financial system too complex. Agents are not capable of detecting all the correlations in order to optimally diversify portfolios. In such an environment, disruptions trigger panic and fundamental uncertainty. This uncertainty is heavily pronounced for untested financial innovations that lead agents to question their worldview.

⁶³Such funding activity is closely related to the emergence of the wholesale interbanking markets across industrialized countries. The first experiences with wholesale funding on an international level took place in the late 1960s with the fast growing euro-dollar market. Traditional banking relies on the available stock of retail deposits for funding liquidity whose size was only partially or not at all in the bankers' control. The flexibility of extending or contracting loan activity mainly originated from the stock of liquid asset (government securities). With a deep wholesale market, holding of liquid buffers became inefficient from an institution's point of view so that marginal funding took place in the interbanking market. Asset market liquidity had been replaced by funding liquidity for the margin of freedom of additional credit granting (Goodhart 2010).

leverage and intermediation chains are endogenous, self-reinforcing concepts during the course of the financial cycle.

A systematic or idiosyncratic shock to one or more financial institutions gets the bust phase going. It is important to note that such a shock may be the “inevitable” result of the endogenous nature of expanding balance sheets. Increased risk measures and losses in asset prices make the capital constraint quickly binding, thereby forcing them to deleverage in order to justifiably shed acceptable capital ratios.⁶⁴ Due to the interconnectedness and dependence on the wholesale market, for each individual institution, the dry-up in funding liquidity appears like a bank run with other banks withdrawing funding opportunities. A reinforcing loop process of falling market and funding liquidity, with increasing margin haircuts in the wholesale market, forces financial institutions to take fire-sale actions on their asset portfolios with liquidity suddenly disappearing (see Chap. 4.4). Prices of assets then become heavily disconnected from their “fundamental” values as determined by expected cash flows; in times of liquidity distress, prices mirror solely the degree of market liquidity. Falling asset prices then trigger a shrinkage in capital positions and can manifest in a profound solvency crisis for the financial sector as a whole and severe negative spill-over effects for the real economy. The higher the leverage with low capital and the longer the maturity transformation chain, the more exposed financial institutions are to funding liquidity risk, in particular haircut funding and short-term debt roll-over risk.⁶⁵

Financial instability is most pronounced in times of a high maturity mismatch of assets and liabilities; and the degree of maturity mismatch, in turn, mostly depends on interest rate spreads between long-term, illiquid assets and short-term debt. The yield curve provides a causal mechanism for financial sector dynamics and the spill-over to real economic activity. Typically, the term spread is regarded as a pure information variable in the transmission mechanism where a positive spread points to higher future short-term interest rates, strong (expected) credit demand and rising inflation expectations (Chap. 5.1.1). An inverted curve, instead, reflects falling future short-term interest rates and low aggregate demand. The yield curve matters since it is the long-term interest rate that alters aggregate credit demand schedules. If taking the role of financial intermediaries in the transmission process as a significant driver, the term spread directly influences the marginal profitability of financial businesses and gives rise to expanding or contracting financial balance sheets and overall credit. Moreover, term-spread induced changes in the supply of credit further

⁶⁴In order to fulfill their capital constraints, banks can either sell assets or they can raise new capital. According to Myers (1977) and Kashyap et al. (2010), the “debt overhang” problems makes banks reluctant to raise new capital because new capital would be immediately siphoned off by the more senior creditors. Therefore, in the shareholders’ interest, banks shrink assets with the social costs of liquidity dry-ups and credit crunches rather to raise new capital that would not produce such negative welfare effects.

⁶⁵Brunnermeier (2009) notices that in the run-up to the financial crisis of 2007, investment banks had to re-finance one-quarter of total assets via overnight repo contracts. Historical support for the three criteria, size, leverage and maturity mismatch can be found in Schularick and Taylor (2009).

influence risk premia demanded by financial institutions which provide a further amplifying effect on real activity. In a market-based financial system, it is not (only) the extent to which the long rate reacts to short rate changes, with its implications for credit demand, but it is the term spread with its impact on the supply of credit to households and corporations that dominates the quantity of credit. For a tightening cycle to become effective, policy rate increases do not exclusively depend on its “lever” effect on the long end of the yield curve. A flattening of the yield curve is sufficient to alter financial intermediaries profitability and bank lending. Paradoxically, given stable long-term interest rates, restrictive monetary policy might be even more effective since short-term changes one-by-one translate into term spread changes.⁶⁶

Monetary policy matters directly. It affects financial intermediation activity, balance sheet growth, leverage dynamics and the pricing of risk. The total quantity of credit is increasingly driven by endogenous shifts in the credit supply function – the risk-taking channel of monetary policy. It enhances credit granting to the private sector (and within the financial system): $\Delta i \rightarrow \Delta C r^d, \Delta C r^s \rightsquigarrow \Delta M$. Once a policy innovation hits the financial sector, further endogenous shifts of the supply curve in the course of the financial cycle increase the availability of credit. The amplification mechanism works through the leverage and the balance sheet quantities of the financial sector. In order to measure financial sector activity, the focus should be on these balance sheet quantities. Funding of increasing quantities of financial assets (including credit) in a market-based financial system takes place by a rising share of “non-monetary” liabilities such as repos and commercial papers. Therefore, aggregate money growth gives a blurred picture as an indicator of overall financial activity (denoted by $\rightsquigarrow \Delta M$). This holds in particular for the narrow, high-powered money concept M1; it does not imbed these marketable instruments of the banking sector. Moreover, this pattern is more pronounced, the larger is the share of non-bank financial intermediaries such as broker-dealers, ABS issuers, hedge funds and money market mutual funds (shadow banks). With increased (true-sale) securitization, a significant part of total loans to the private sector is transferred from banks’ balance sheets to special purpose vehicles (SPVs) that pool the loans and issue asset-backed securities. From an accounting perspective, this shrinks banks’ outstanding amount of credit so that total loans to the economy should be higher than reported outstanding loans of the banking sector. At least conceptually, as SPVs need to fund their activities, SPVs holdings of loans can either lead to a decrease of SPVs’ deposits at bank institutions or they can contribute to an increase of deposits in case SPVs first have to borrow in order to acquire loans. In any case, the money multiplier (and money demand), then, becomes a fragile concept so that

⁶⁶The story of the interest-rate conundrum of 2004–2005 then can be regarded from a different perspective. Whereas the common view suggests that the restrictive fed funds cycle did not become effective since long-term interest rates did not pick up, the alternative view would claim that monetary policy was effective working through the financial sector and triggering a decline in credit growth (Adrian et al. 2010a).

a monetary aggregate insufficiently pictures credit, liquidity and funding dynamics in the economy. Firstly, it addresses only the liability side of deposit-taking banks with no consideration of market-based refinancing; this could be resolved by analyzing broad money (M3) which at least involves marketable instruments of the commercial banking sector. Secondly, no monetary aggregate adequately captures the whole financial intermediary sector and its financial intermediation activity. The switch to, and the analysis of, the asset side of financial intermediaries may improve the insights derived from balance sheet dynamics.⁶⁷ Indeed, historical evidence suggests that the growth rate of total credit is the best predictor of future financial instability and it outperforms any broad money concept (Schularick and Taylor 2009).

Figure 7.9 reviews key balance sheet variables of the financial intermediary sector for the euro area. According to the European System of Accounts (ESA), the financial sector is subdivided into the (a) central bank, (b) other monetary and financial institutions, (c) other financial intermediaries (OFIs), (d) financial auxiliaries and (e) insurance companies and pension funds (ICPFs). Group (a) and (b) form together the monetary and financial institutions sector (MFIs).⁶⁸ When comparing euro area financial intermediaries, currently (2010:Q1) 62% of total assets are held by MFIs, 12% by ICPFs and 26% by OFIs. Figures 7.9a, b plot annualized growth rates of MFIs total asset growth, total loan growth and loan growth to the private sector (households and non-financial corporations) vis-à-vis M1 growth and M3 growth, respectively. As suspected, while the relationship between banking activity and M1 appears rather loose, M3 tracks asset and loan growth much more accurately. The latter seems to be a reliable indicator of banking activity in the euro area. Due to securitization, ECB (2008b) estimates of total loan growth to the private sector lies approximately 1% higher than private loan growth reported by MFI statistics. Moreover, since credit to the private sector also entails marketable debt-securities, it is reasonable to take total asset growth (loans and debt securities) as a benchmark variable for total credit dynamics in the economy. This holds all the more because MFIs also hold debt securities issued by OFIs generated by the securitization process.⁶⁹

Figures 7.9c, d depict balance sheet dynamics of the various groups within the financial intermediary sector. Asset growth is procyclical as documented in stylized

⁶⁷ Along similar lines, the instability of the money-income relationship (money demand function) has been one of the motives of central banks for abandoning monetary targeting and downgrading money as an information variable for predicting future inflation (Friedman 1988). To what extent money can be still used as information variable mainly depends on improvements in statistical methods and refinements of money-demand equations (Fischer et al. 2006; Assenmacher-Wesche and Gerlach 2006).

⁶⁸ MFIs incorporate credit institutions, money market funds, and other financial deposit-taking institutions. OFIs come closest to the concept of “shadow banks” with securitization vehicles and dealers belonging to the latter group. For an analysis of the OFI sector see ECB (2010c).

⁶⁹ Indeed, ECB (2008b) reports that the amount of true-sale securitization corresponds to MFIs debt security purchases issued by OFIs.

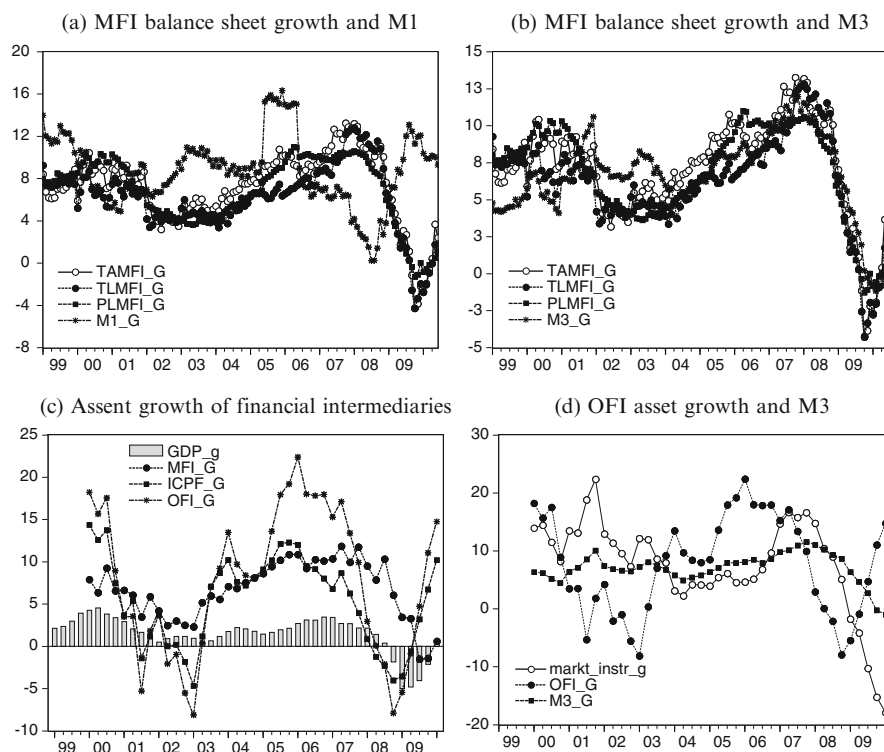


Fig. 7.9 Euro area financial sector activity and monetary aggregates

(Note: TAMFI: total assets of MFI, TLMFI: total loans of MFI to euro area residents, PLMFI: private loans of MFI to households and non-financial corporations, M2: narrow money, M3: broad money. The variable *markt_instr* contains repos, money market funds units and debt securities up to 2 years (including commercial papers); it equals the difference between M3 and M2. Data are taken from the monthly aggregate balance sheet statistic of Monetary and Financial Institutions (MFIs) excluding the Eurosystem provided by ECB. Financial intermediaries' activity imbeds total assets of MFIs, insurance companies and pension funds (ICPFs) and other financial intermediaries (OFIs). Quarterly data are taken from the Euro Area Accounts statistic provided by ECB.)

facts about business and financial cycles (Kindleberger 1995). Euro area balance sheet dynamics of OFIs are much more pronounced than of the MFI sector; they show strong amplitudes during the financial cycle. Moreover, OFIs balance sheet growth precedes the MFIs counterpart and tends to anticipate GDP growth (*GDP_g*) much better than overall banking activity. The latter result can also be found in US data where asset growth of broker-dealers helps to predict future GDP growth after controlling for banking activity (Adrian and Shin 2009a, 5): "Our results point to key differences between banking as traditionally conceived and the market-based banking system that has become increasingly influential in charting the course of economic events." Similarly, M3 growth is incapable of replicating the non-bank financial intermediaries' activity over the sample period since it does not contain

information about the market-based funding of the OFIs group of the financial sector. Since OFIs' balance sheet are mostly marked to market, they may be seen as an indicator of funding conditions in financial markets. Stronger cyclicity of the banking sector can be identified when inspecting only the marketable instruments of the M3 components. Again, they tend to follow OFI dynamics with a lag (2004–2007) or tend to move in opposite direction (2001–2003: portfolio shifts into M3 assets due to heightened uncertainty).

Summing up, the link between prices and quantities in a market-based financial system can be restored by the interconnection between monetary policy and balance sheet growth of the financial sector. Paradoxically, the theoretical and empirical analysis on boom-bust cycles stands in contrast to the implications of the quantity theory. Meanwhile the quantity theory predicts that a rise in the stock of money (quantity) is followed by goods price inflation, the boom-bust cycle approach predicts that a rise in financial sector balance sheets (quantity) is followed by disinflation or even deflation of consumption goods. It is rather asset price inflation and excessive credit dynamics that seed the roots for future severe economic downturns. The origin and the excrescence of financial cycles result in the following proposition: “Contrary to the common view that monetary policy and policies towards financial stability should be seen separately, they are inseparable. Monetary policy should be better coordinated with policies towards financial stability.” (Adrian and Shin 2009b, 605).

7.4.3 Macroprudential Policy and Implications for Central Banking

Output losses are large when a systemic financial event hits the economy. “Balance-sheet and credit booms gone bust” is a recurrent pattern. Macroeconomic policy has learned from pre-war experience that implementing countercyclical monetary and fiscal policy is an effective instrument for fighting slack in aggregate demand. However, the changing nature of financial intermediation, with its emphasis on increased intermediation chains and higher leverage, has made the economic system more vulnerable to such events and quantitatively has made financial shocks much larger with negative feedback effects for the real sector.⁷⁰ As a consequence, output and income reductions in the aftermath of financial crisis remain high. It provides a rationale for ex-ante policy measures for the purpose of preventing the materialization of systemic risk.

“Macroprudential” policy addresses the financial system’s resilience and evolving imbalances for the objective of limiting systemic risk. In contrast to microprudential regulation with its focus on individual financial institutions’ healthiness, it implements regulatory and supervisory arrangements from a system-wide

⁷⁰For a historical record, see Schularick and Taylor (2009); Reinhart and Rogoff (2009).

perspective. It acknowledges the insights that risk evolves endogenously within the financial sector where time-varying correlations and interconnectedness are the cornerstones of looming systemic risk; they produce high externalities for the macroeconomy (Borio 2003).⁷¹ In this respect, the macroprudential approach has two dimensions with distinct ratiocinations. The first dimension focuses on the evolution of risk over time and its implications for procyclicality and amplification effects of the financial system. The second dimension deals with cross-sectional aspects that describe the interlinkages between financial institutions, how aggregate common risk is distributed among them and how this common source of risk makes them vulnerable to joint failures (Borio 2010).

Each dimension calls for different macroprudential instruments depending on the criteria attached to the instruments. For instance, in the time dimension, if the criteria coincides with the resilience of the financial system, an instrument could ensure that liquidity and equity buffers increase in the boom phase and are high enough so that they can be drawn down in the cyclical downturn. However, if the criterion lies in the preemptive mitigation of financial instabilities, in terms of balance sheet growth and build-up of risk, the instrument should be designed to act as a “speed limit” in a countercyclical way. The cross-section calls for instruments that detect the systemic significance of individual institutions. It can support the identification of individual’s contribution to overall systemic risk and it can serve as a measure to impose charges in order to internalize the costs associated with systemic events. Additionally, such measures help to classify financial institutions as individually systemic (too-big-to fail), systemic part of a herd or non-systemic (Brunnermeier et al. 2009; Adrian and Brunnermeier 2009).

The design of the macroprudential instrument depends on whether to implement measures on the supply side (financial institutions’ balance sheets) or the demand side (borrowing sector) of credit. Depending on the sources of systemic risk, these instruments can be applied as a scissor strategy for ensuring resilience and for serving as retaliatory actions to fight evolving imbalances. Finally, macroprudential regulation should be aimed at all financial intermediaries and not only towards the banking sector. The financial turmoil of 2007 has shown that the interlinkages of a vast group of intermediaries has contributed to the bust in financial and goods markets (Brunnermeier 2009).

In order to achieve the objective of macroprudential regulation, the identification and the assessment of systemic risks lies at the heart of the effectiveness of operational instruments. These instruments can be fixed over time or they can be adjusted in response to movements in systemic risk measures. Fixed limit

⁷¹In 1979, the term “macroprudential” was first used by the chairman of the Basel Committee on Banking Supervision. He said that “micro-economic problems (which were of concern to the Committee) began to merge into macro-economic problems (which were not) at the point where micro-prudential problems became what could be called macro-prudential ones. The Committee had a justifiable concern with macro-prudential problems and it was the link between those and macro-economic ones which formed the boundary of the Committee’s interest”, cited from Clement (2010, 60).

instruments can nonetheless act as automatic stabilizers over the course of the financial cycle to the extent that they are more binding in a boom than in a bust. Early-warning signal models can be used to derive indicators about the probability of systemic events in the future on which time-varying instruments can be based. The latter allow to take ex-ante measures in order to prevent or at least to reduce the materialization of systemic risk.⁷² Empirical studies on such signals work out a common ground for optimal indicators. They show that a combination of “unusual” credit (to the private sector) growth with increasing debt ratios and high asset and balance sheet growth precede banking and financial crisis.⁷³ Concerning the interconnectedness of financial institutions, measures based on the overall systemic risk conditional on a particular institution being in financial distress reveal the marginal contribution of this institution to overall risk by imposing charges. Moreover, they can support the forward-looking aspect of crisis prevention by forecasting systemic risk contribution into the future to address the procyclicality of balance sheet management.⁷⁴ To what extent the justifiable critique of the capability of such indicators in signalling evolving imbalances is accounted for, is closely associated with the discussion whether a central should respond to asset price misalignments; this issue is, therefore, discussed later in this Section when the role of the monetary authority for financial stability is explored.

Table 7.1 summarizes selected measures of macroprudential regulation with a categorization of the aim of macroprudential measures. They are related to the robustness and resilience of the system (R) as well as to the direct impact on evolving imbalance (FI). Moreover, measures can be taken at the balance sheet side of financial institutions (S) or at the borrowing sector (D). Moderate procyclical fluctuations in leverage and balance sheet size is achieved through time-varying capital targets including caps on maximum permitted leverage and minimum capital requirements on risk-weighted assets (RWA).⁷⁵ Capital charges can be further established to generate buffers in the boom phase that can be drawn down in the bust phase. Charges on capital and liquidity positions are important since it is not

⁷²ECB (2010a) generalizes four approaches how to identify and assess systemic risk and financial imbalances: (1) coincident models for identification of the current state of financial stability, (2) early-warning models, (3) macro stress-testing models and (4) contagion and spillover models.

⁷³See Borio and Lowe (2004), Borio and Drehmann (2009), Alessi and Detken (2009), Gerdesmeier et al. (2009), ECB (2010b). The term “unusual” is typically related to some measures of equilibrium levels. The latter can be estimated with the help of structural economic and statistical models or through the calculation of long-term averages. In this respect, joint indicators based on credit, equity and property information perform best and the inclusion of a global perspective in terms of these indicators improve on the prediction power and on the reduction of false alarms.

⁷⁴Adrian and Brunnermeier (2009) construct a $CoVar^i$ variable which is calculated as the VaR of the whole financial sector conditional on institution i being in distress. $\Delta CoVar^i$ denotes the difference between $CoVar^i$ and financial sector VaR; whereas $\Delta CoVar^{ij}$ displays the increase in risk of institution i conditional on risk on institution j .

⁷⁵Gersbach and Hahn (2010) propose a systematic, rule-based aggregate equity ratio of the banking sector that is related to the current state of money and credit.

Table 7.1 Macroprudential measures of the financial sector

Measures	Instruments	Objective	Credit
Definition of capital	Eligibility criteria for high quality capital	R	S
Leverage ratio	Minimum ratio of capital over total assets	R,FI	S
Counterparty credit risk	Strengthened and time-varying capital requirements for counterparty credit risk exposures (haircuts)	R,FI	S
Minimum capital requirements	(Time-varying) risk-weighted capital ratio	R,FI	S
Capital buffer	(Time-varying) capital buffer and restrictions on dividend and bonuses payments	R,FI	S
Contingent instruments	Conditional convertibles debt instruments to strengthen capital in times of distress	R	S
Provisioning	Moving from the incurred loss to the expected-loss over the cycle approach of loan provisioning	R,FI	S
Stable funding ratio	Loan-to-deposit ratio	R	S
Liquidity coverage ratio	High-quality liquid to total asset ratio	R	S
Moral hazard and externalities	Risk-based private insurance schemes; recovery and resolution plans for large and complex institutions; individual capital surcharges or Pigovian tax according to the contribution to overall systemic risk (liquidity and default risk); mark-to-funding accounting for assets that are long-term funded	R, FI	S
Collateral issues	Time-varying loan-to-value ratios in real estate but also in corporate and security markets	FI	D
Transparency	Centralized clearing house arrangements for derivative markets	R	S,D

Note: The objective refers to the resilience (R) and the direct impact on financial imbalances (FI). Instruments can be aimed at credit supply (S) or credit demand (D). Measures and instruments are taken from Brunnermeier et al. (2009), ECB (2010c), BIS (2010) and Goodhart (2010).

only the quality (riskiness) of financial institutions' asset side that matters but also how asset positions are related to the maturity structure of their corresponding funding including leverage and maturity mismatch.⁷⁶ Fixed limit ratios such as funding ratios and liquidity coverage ratios support the reduction of liquidity risk in the course of the financial cycle. Another way to reduce procyclicality is to operate directly on equity through forward-looking provisioning on additional loan granting or via a Pigovian tax according to the marginal contribution of systemic risk when an institutions' balance sheet expands. Both instruments reduce the equity level in the boom phase through less profits compared to the non-regulated case so that for any given leverage target, this goes hand in hand with lower total asset

⁷⁶Most interestingly, the discussion on the benefits and costs of deposit insurance culminated in the conclusion that intermediaries must hold much more capital to protect against risky activities (Kareken and Wallace 1978). The discussion took place in the late 1970s and early 1980s (!) when capital ratios were yet significantly higher than before the financial crisis of 2007 (Goodhart (2010) reports on the historical evolution of capital ratios in the US).

holdings. At the same time, tax revenues can be put into an institutions' resolution fund so that they do not imply a net transfer away from the financial sector (Shin 2010).

Taking on a more structural view on macroprudential regulation, the switch from mark-to-market to a framework of mark-to-funding accounting improves the resilience of the financial system and reduces excessive fluctuations of asset positions (see for the following explanation Brunnermeier et al. 2009). The idea is to value an asset according to the funding capacity of the asset holder and not according to the intention of the holder. If a 20-year asset is funded via short-term debt, no matter what the intention, the asset should be valued at market prices. However, if the asset is funded with a 10-year debt instrument, the asset should be valued at the average price over the next ten years so that high-frequency market price volatility does not manifest itself in asset valuations. Consequently, those institutions being funded by long-term debt may not be forced to sell assets with a high fair value in the medium- to long term in times of fire-sale downward liquidity spirals.

High risk-taking is pronounced for financial institutions labeled as too-big-to-fail and too-interconnected-to-fail. These institutions presume that governmental and supervisory bodies would not let any of these market participants fail since the costs associated with "gone concern" are far too high (see Sect. 7.3.1). In order to reduce such distorted incentives, besides regulatory liquidity and capital buffers, an effective ex-post measure lies in the development of recovery and resolution schemes (living wills) in joint work by institutions and regulators that can contribute to a reduction in the impact of financial distress (ECB 2010d). In particular, resolution plans should regularly update the identification of payment and funding structures to which the financial institution is connected. Simultaneously, they should fix a sequence of operations of how to disconnect the institutions from the financial system in an orderly manner. Such plans allow for a reduction of moral hazard and could promote lower risk-taking due to a strengthened perception of shareholder responsibility.

One last set of regulatory efforts deals with the resilience of the financial system concerning the market structure. First, systemic markets should be arranged as a centralized clearing house where payment structures, network effects and the spill-over of counterparty risk can be more efficiently handled. It allows for an adjustment of the conditions of trading and for the ability to identify cross exposures of a variety of market participants. Second, as emphasized by Shin (2010, 16), "[...] long intermediation chains carry costs in terms of greater amplitude of fluctuations in the boom bust cycle of leverage and balance sheet size. Shorter intermediation chains carry benefits for the stability of the financial system." The longer the chain, the more short-term is the funding of (shadow) credit intermediation. From a macroprudential perspective, giving incentives to shorten these chains can contribute to the robustness of the financial system. Covered (mortgage) bonds with recourse against the issuing bank, such as *Pfandbriefe*, is an instrument of how to bring ultimate lender and borrowers more directly together and to shorten intermediation chains (intermediation).

It is evident that macroprudential regulation does not operate within a “vacuum” in which instruments become effective without any influence of other policy measures. In particular, liquidity provision and lender of last resort actions are integral functions of a central bank; they support the effectiveness of macroprudential oversight and control during the course of the financial cycle. With respect to the implementation of *macroprudential* regulation, these tasks should be conducted by the central bank as a supervisor of financial stability. Both from an economic and juridical perspective, there are sound arguments in favor for this approach. Maintaining financial stability and fostering financial developments was at the very heart of the early days of central banking. The Bagehot rule, according to which the monetary authority should lend freely at a high rate against good collateral for individual institutions, or the emergency liquidity assistance to markets are guiding principles of how to cope with financial distress. There is essential interest for a central bank to promote financial stability, since imbalances represent a severe threat to macroeconomic outcomes. Indeed, most charters of central banks directly or implicitly recognize the monetary authorities’ responsibility in striving toward financial stability (Ferguson 2003). Since enhancing the resilience of the financial system is inevitably connected to liquidity management, the obvious institution assigned with this objective is the central bank. As Goodhart (2010) argues, “[i]t would cause massive complications if liquidity management remained the sole province of the Central Bank while a separate financial stability authority was to be established without any command over liquidity management. [...] [T]he financial stability authority has to be given command over liquidity management [...] [and] would have command over the Central Bank balance sheet. Indeed, the financial authority would then, de facto, become the true Central Bank.”

Furthermore, a central bank is already equipped with various tools for detecting and assessing financial risk against the background of evaluating the outlook of inflation and output for the stance of monetary policy. Letting macroprudential regulation be a further instrument on part of the central bank is adequate also from an informational perspective. Finally, and of most relevance, if prudential policy is adjusted to financial conditions, the monetary policy transmission is altered depending on the degree to which instruments bind in terms of capital and liquidity constraints. The effectiveness of monetary policy actions is then closely related to regulation thereby calling for increased coordination. It is conceivable that the financial system is under stress; whereas inflation pressure hits the economy. Under this scenario, prudential regulation should be loosened but monetary policy tightens in order to reduce aggregate demand and keep inflation expectations on track. On the contrary, a regulation could be tightened that reduces the availability of credit so that monetary policy might be obliged to lower interest rates to give stimulus to credit demand in order to achieve the objective of price stability. The scenarios do not imply that there is a de facto policy conflict; however, there is little experience with such circumstances so that a close collaboration is necessary for the purpose of complementing each other. If decisions on monetary policy and macroprudential regulation would be made by two institutions, there is the threat of a “push-me,

pull-you”⁷⁷ outcome that can at least be reduced if both policies are conducted by one institution.

Despite the various efforts of implementing measures and instruments of macro-prudential policies, it would be naive to assume that regulation can *per se* assure financial stability. It can actually do harm if wrongly lashed. Financial market participants try to relax their budget constraint in terms of findings ways to circumvent regulation-imposed capital and liquidity constraints through regulatory arbitrage over time.⁷⁸ It includes enhanced financial activity in those entities that reduce regulatory costs in terms of capital and liquidity requirements. The question arises whether further policy measures, such as the interest-rate setting of the central bank, can support regulatory actions in particular against the background of the direct effects of policy rate changes on financial intermediary activity.

7.4.4 Addressing Financial Instability from a Monetary Policy Perspective

There is a long standing debate on how monetary policy should cope with asset price dynamics. Two opposing views compete with each other over whether a central bank should react to (detected) asset price misalignments. One view claims that monetary policy should react to such bubbles only in so far as they have an impact on current and expected output growth and inflation. The second view requires the monetary authority to respond to asset prices over and above the effects on output and inflation in the near time, since evolving financial imbalances through asset price bubbles increase the likelihood of a bust that can do harm to the economy in terms of high real output costs.⁷⁹ The debate is centered around two discussion lines, one tackling asset prices within a flexible inflation targeting framework, and the other dealing with implementation difficulties.

Within the flexible inflation targeting framework, various work has analyzed the effects of asset prices and the economy. Under conventional assumptions with a loss function embedding inflation and output deviations as well as a transmission mechanism with aggregate demand affected by asset price growth, any optimal policy response to asset prices can be only motivated through an impact on the

⁷⁷The terminology is due to Bean et al. (2010).

⁷⁸For instance, prior to the financial crisis in 2007, European banks were subject to risk-weighted capital requirements but not to a cap on leverage. Consequently, banks increased leverage by adding alleged “highly rated” asset-backed securities which did not contribute to an increase in risk-weighted assets. On the contrary, US commercial banks were faced with a leverage ratio but not to capital requirements based on risk-weighted assets. Therefore, they filled up their portfolios with more risky securities (Goodhart 2010).

⁷⁹See for modeling results and literature reviews Bernanke and Gertler (1999), Bernanke and Gertler (2001), Cecchetti et al. (2000), Cecchetti et al. (2002), Filardo (2004), Posen (2006), Roubini (2006), Disyatat (2010b).

final target variables. It can be shown that the short rate directly reacts to the bubble component of the asset pricing equation and it indirectly responds to asset prices through its weights on inflation and output gaps (Disyatat 2010a). It also implies that monetary policy ought to “lean against the wind” of cyclical non-fundamental asset price fluctuations. Asset prices matter because they contain information about future output gap and inflation like forecasts about the future output gap and inflation themselves. If financial imbalances build up yet do not provide quantitative information for the inflation outlook and disruptions emerge in the distant future, “cleaning-up” by lower interest rates and liquidity assistance is the most effective way in dealing with both the emergence of imbalances and financial distress. Consequently, there is no trade-off between price stability and financial stability, and it becomes clear that the analysis based on standard assumptions falls short of adequately capturing the endogeneity of financial activity.

The more relevant research question is whether a rationale exists to react to evolving financial imbalances even if they do not imply any direct impact on future goal variables within the policy-relevant time horizon; but, these imbalances may sow the seeds for disruptions to the economy in the distant future.⁸⁰ In this respect, Shirakawa (2010, 27) asks: “How should monetary policy be conducted in an environment in which all the symptoms of the economy except for inflation signal a need for policy tightening: asset prices are rising, credit and leverage are increasing, maturity mismatch is widening, and the economy is overheating, while only inflation remains low and stable?” This proactive, pre-emptive view relies on the premise that the duration of imbalances heighten the risk of sudden reversals. Most importantly, in contrast to the standard New-Keynesian transmission mechanism of the inflation targeting approach, imbalances do not occur as exogenous shocks but they build up in the boom phase which depends on the stance of monetary policy. While the debate in the early 2000s has concentrated (solely) on asset price bubbles, risk-taking mechanisms focus on the market-based financial system as a whole. The risk-taking channel highlights the endogenous nature of credit, balance sheet, risk and asset price dynamics as well as the non-linear disruptions when risk materializes. Therefore, it is not (only) the course of asset prices that matters; rather the underlying financial/credit cycle determines the nature and the degree of systemic risk.⁸¹ Under the proactive view, monetary policy is used to lower the probability of economic busts in the economy by means of fighting the procyclicality of financial intermediaries and by countersteering the attempt to extend balance sheets in an unsustainable way. Short-term interest rates are increased whenever misalignment in private balance sheets and/or on financial markets are detected. These hikes

⁸⁰For this perspective, see Bordo and Jeanne (2002), Borio and Lowe (2002), White (2006).

⁸¹Weber (2008, 3) notes that “[t]he debate about monetary policy and financial markets is too often slanted to the question on how to deal with asset price bubbles. [...] In my opinion, the view of monetary policy and asset prices is too narrow. A more fruitful debate on appropriate monetary policy reactions to developments on financial markets would be possible if the focus were redirected from financial bubbles to the issue of procyclicality.”

act as insurance against asset price reversals at the costs of short-term output losses (Berger et al. 2007). Consequently, an inter-temporal trade-off exists between preserving future financial stability and current price stability so that monetary policy must evaluate current and future losses of a set of policy options.

To derive optimal monetary policy implications, the “transparency effect” and the “paradox of credibility” of Sects. (7.3.1) and (7.4.2) need to be incorporated into a macroeconomic model and inflation targeting framework, respectively. Both approaches deal with the mechanisms how financial imbalances evolve despite no visible signs of unsustainable inflation and output dynamics. However, macroeconomic models that are capable of reproducing such phenomena hardly exist. Most monetary models with financial frictions, an intermediary sector and financial shocks produce outcomes in the financial sector that directly affect current output and inflation dynamics. For instance, Cúrdia and Woodford (2008, 65) admit that their model has “[...] nothing to say about the issue of how monetary policy decisions should take into account financial stability concerns – either possible consequences of interest-rate decisions for systemic risks to the financial sector [...] or possible consequences of interest-rate policy for risk-taking behavior [...] since we simply abstract from such concerns in our reduced-form model of the financial sector.” These models outline policy implications for a situation in which financial shocks and disruptions in banking wholesale markets are already realized. Policy measures aimed at crisis management can be evaluated but measures based on crisis prevention can not be addressed. Besides traditional interest-rate setting, the former includes unconventional policy instruments based on a central bank’s balance sheet (liquidity management) such as changes in reserve balances over and above those operations that make the short-term policy rate effective, changes in the composition of private to public debt securities and explicit engagements in particular private debt segments.⁸² To what extent these policies can and should also be used for prevention purposes and whether they are effective within a model with financial intermediary activity is discussed later. The point is rather that with mentioned conventional models, the pre-emptive approach can not be evaluated from an optimal monetary policy perspective. But what these models show is that with financial frictions and an intermediary sector, monetary policy is yet advised to change the intercept of a conventional Taylor rule in response to a varying credit spread by a way of increasing the intercept whenever the spread shrinks (et vice versa). Such policy response is welfare-improving since it acknowledges that the relevant long-term interest rate is not only effected by the risk-neutral version of the Expectations Hypothesis but also by variations in the credit risk premium.⁸³

⁸²See for an overview and the effectiveness of unconventional monetary policy Cúrdia and Woodford (2009a), Cúrdia and Woodford (2010), Borio and Disyatat (2009), Gertler and Kiyotaki (2010).

⁸³For a theoretical derivation and a rule-based approach see McCulley and Toloui (2008), Taylor (2008b), Cúrdia and Woodford (2009b), Belke and Klose (2010), Giavazzi and Giovannini (2010).

On a highly stylized level, attempts have been made to derive optimal policy results where financial imbalances at least in part are endogenous to the state of the economy (Bordo and Jeanne 2002; Filardo 2004; Disyatat 2010a). It turns out that it is optimal to set the policy rate explicitly according to the perceived misalignment of financial conditions in a non-linear way, i.e. if imbalances exceed some pre-defined threshold value, monetary policy reacts in terms of raising its policy rate.⁸⁴ In this respect, it is allowed that inflation and output longer deviate from its target level since the adjustment speed appears to be slower. At the same time, a pre-emptive policy implies a “negative inflation bias” in the sense that in an environment of stable goods prices except for signs of financial imbalances, the interest rate increase pushes, on average, inflation below its target level. Still, welfare losses can be minimized if monetary pre-emptively tightens, since the monetary authority takes on an intertemporal perspective of trading-off (small) short-term output and inflation losses with (high) losses in the future. Financial stability can act as an intermediary target within the flexible inflation targeting framework since, in the long run, the trade-off appears as a trade-off between current economic stability and future economic stability. The lengthening of the policy horizon gives systemic financial risk concerns an explicit role in policymaking and, in principle, there is no inconsistency between inflation targeting and the use of monetary policy to counteract imbalances.

One of the main criticisms of leaning against asset price bubbles is centered around the inability to detect deviations of asset prices from their fundamental values (Bernanke and Gertler 1999; Posen 2006). Similarly, it is unlikely that a central bank would be a better judge than the markets of the existence of a bubble since this would imply a superior information set on the part of the central bank compared to private agents, which is a very unlikely scenario. Concerning the latter, this argumentation is flawed since a central bank is not a trader or a direct actor in financial markets with the purpose of profit maximization. Bubbles can knowingly exist and, notwithstanding, there are rationales for investors to ride the bubble and not to bet against asset price increases.⁸⁵ By the same token, asset price misalignments represent a symptom rather than the direct cause of financial imbalances. It is not about correctly identifying the bubble component of a security; the interaction and combination of credit supply (and demand) dynamics produce the procyclicality of the financial system with the observable facts: easing of lending standards, increasing indebtedness, lower risk premia and, of course, rising asset prices (Sect. 7.2.3). The previously stated work on early warning indicators is an improvement in detecting the systemic risk component of

⁸⁴In a similar spirit, Diamond and Rajan (2009, 1) call for higher policy rates in order to fight imbalances: “To offset incentives for banks to make more illiquid loans, authorities may have to commit to raising rates when low, to counter the distortions created by lowering them when high.”

⁸⁵See Brunnermeier (2008) on the literature on bubble models emerging from rational or behavioral agents and from limits to arbitrage; or to put it in Keynes’ words: “Markets can remain irrational longer than you can remain solvent,” cited from Lowenstein (2000, 123).

financial market activity. Additionally, the construction of new financial condition indices support the identification of evolving imbalances.⁸⁶ These models rely on a multitude of financial indicators and circumvent the drawback of focussing on only one particular market segment, in case an asset price bubbles wants to be detected. Clearly, despite valuable model innovations, there is still a considerable degree of uncertainty surrounding the identification process. Such an argument can not, however, be put forward against the proactive view, since standard inflation targeting also operates under similar parameter, data and model uncertainty.⁸⁷ Taken to extremes, it could be argued that, from a risk management perspective, pursuing a financial stability objective is more robust due to the multidimensional approach of addressing financial instability.

Another caveat against the proactive view in the debate is based on the effectiveness of monetary policy to reduce the likelihood of developing misalignments. It has been argued that the increasing of interest rates is not the appropriate tool since an asset price bubble will not burst in response to small changes in interest rates (Greenspan 2002). Again, this argumentation is misplaced in this context since it is not directly about asset prices but about financial intermediaries. As highlighted by Adrian and Shin (2008a, 291), “a difference of a quarter or half percentage in the funding cost may make all the difference between a profitable venture and a loss-making one for leveraged financial intermediaries.” With long intermediations chains, each participant skims the market along the yield curve by utilizing term and credit spreads. At the peak of the financial cycle, financial intermediaries are highly leveraged since such a strategy maximizes the return on equity in an environment in which interest rate spread between assets and liabilities increasingly shrinks over time. Monetary policy exhibits considerable influence on the length and duration of intermediation activity through its variation of the short-term interest rate. It alters both the term spread and the price of risk demanded by market participants.

As a consequence, the yield spread seems to be an effective point of reference for monetary policy. It acts as a reference on how fast and how deep the financial sector “breathes” and is engaged in liquidity transformation. A steep yield curve provides powerful incentives to borrow short-term and lend long-term. The longer the spreads remain high in the expansion phase, the more likely it is that financial instability builds up by means of higher risk-taking. Monetary policy is well advised to flatten the yield curve in a pre-emptive way whenever there are early signs of arising financial imbalances in the economy.⁸⁸ Such policy action feeds through the

⁸⁶See Borio and Lowe (2004), Borio and Drehmann (2009), Alessi and Detken (2009), Gerdesmeier et al. (2009), Hatzius et al. (2010), ECB (2010b).

⁸⁷This holds particularly for the concept of the output gap which can be only measured with strong noise.

⁸⁸This rationale of flattening the yield curve stands in contrast to the findings of Chap. 6.2.1. Whereas in the former Chapter, the focus is on the demand side of credit with the expectational effects of monetary policy, this Section emphasizes the supply side of credit and the bank-lending as well as the risk-taking channel of monetary policy transmission.

whole set of asset prices – via arbitrage and financial intermediary activity. However, a central bank is confronted with a serious dilemma. As stated previously, pre-emptive tightening generates a trade-off between current output losses and future financial reversals; but leaving the yield curve steep for an extended period of time further fuels imbalances and delays possible adjustments in financial markets. This easing of the monetary policy stance may force a central bank to aggressively raise interest rates at the end of the expansion phase, and precipitate a rapid change of funding conditions for financial markets, to which highly leveraged market participants can not prepare in a timely manner. Therefore, a policy of “benign neglect,” in the beginning of the “unsustainable” expansion, aggravates the measured systemic shock component that hits the economy in the run-up to the bust phase. In the long-run, such a strategy would make the bust more costly in terms of higher output losses. Moreover, cleaning-up and massive easing in the bust phase can push the economy into a low interest-rate environment and can sow the seeds for the next crisis due to too much risk-taking. The crisis, in turn, is heavily dependent on low interest rates so as to keep the financial system alive. In this case, rising rates becomes extremely difficult to achieve and monetary policy remains stuck in a low rate equilibrium.⁸⁹ Such a description mirrors the experiences of the last two decades in the US as well as in some European countries.

A situation can arise in which funding (liquidity) conditions and return opportunities are still favorable for financial market participants despite short rate hikes. Expectations of higher short rates in the future might prevent the yield curve to sufficiently flatten. Credit spreads could be enduringly compressed due to low volatility of macroeconomic risk and low perceived default risk. By the same token, margin requirements (haircuts), in secured short-term interbank lending, might not sufficiently change to generate a slow-down in credit dynamics. It would surely be a mistake to attach any role in increasing volatility to monetary policy, by means of inducing a heightened frequency of macroeconomic fluctuations. However, further instruments on behalf of a central bank, besides standard interest rate policies, could support the aim of reducing the procyclicality of the financial system, in particular when short rate changes do not translate into an appropriate change in credit dynamics. The choice for further policy tools could also be justified against the background of the famous “Tinbergen principle” that claims that, in the case of two independent policy objectives, monetary policy should operate with two independent stabilization tools.⁹⁰

⁸⁹Giavazzi and Giovannini (2010) model such a low rate equilibrium and point out that the ex-ante real rate might be too low in “normal” times. White (2009) describes this low-rate trap scenario with the help of macroeconomic performance in the US following the stock market crash in 1987, in the early 1990s, in the early 2000s and currently, in the aftermath of the financial crisis that began in 2007. Indeed, there is a clear trend decline in the policy rate with massive pre-emptive easing in the run-up to financial crisis. After each crisis, the policy rate did not return to its old level.

⁹⁰Applying the Tinbergen principle in this context and calling for two policy instruments, is not convincingly valid. Like for the standard inflation targeting approach, the solution to a problem

Besides the instrument of setting the short-term interest rate, monetary policy exhibits a further instrument, i.e. the management of its balance sheet, which can be carried out quite independently from interest-rate policy. Following Borio and Disyatat (2009), the implementation of monetary policy can be characterized by two elements. The first one refers to the signalling effect of the policy stance that describes the desired and announced policy rate. In principle, since the implementation of monetary policy takes place in the market for bank reserves (central bank balances, base money) and the monetary authority is the monopoly supplier of this asset, the latter can make any level of the policy rate effective by buying and selling any amount of reserves through open market operations. This is the second element of policy implementation which involves the use of the central bank's balance sheet (liquidity management). At first sight, one might suspect, in accordance with the traditional view of monetary policy implementation, that there is an inverse relationship between the targeted interest rate and central bank balances in the context of an interest-rate elastic demand for reserves (Poole 1970).⁹¹ Demand should vary inversely with the benchmark policy rate because this rate resembles the opportunity cost of holding (excess) reserves. However, the levels commercial banks aim to hold are typically chosen to be a minimum level so that the demand is a rather weak interest-elastic or even interest-inelastic demand (Disyatat 2008).

In practice, there is no “liquidity effect” when monetary policy announces a new policy rate; nor does the volume of open market operations increase to make this rate effective (Friedman and Kuttner 2010). These observations can be explained through the combined importance of market expectations and the way in which the money market is organized. If the central bank stands ready to make any rate change effective, and if such possible action is perceived to be credible, market participants in the interbank market will trade near that target without the need of the central bank undertaking open market operations. In case of an interest-inelastic, vertical demand curve, the monetary authority adjusts its supply endogenously to this reserve demand. With a reserve remuneration scheme where reserve holdings give a rate of return that is lower than the policy rate, “[...] once the demand for bank reserves has been met, the central bank can set the overnight rate at whatever level it wishes by signalling the level of the interest rate it would like to see” (Borio and Disyatat 2009, 3). In case demand inversely reacts to the current market rate, the central bank can use upper and lower standing facilities that form a corridor for the market rate, in order to prevent it from not significantly deviating from its target. Finally, the quantity of reserve balances can be fully isolated from the process

with one instrument and multiple targets can be derived in terms of the intended trajectory for any one arbitrarily chosen target. This can hold for the intra-temporal inflation/output trade-off but for the inter-temporal price stability/future financial stability trade-off, too. In any way, the independence of the different objectives depends on the time horizon within monetary policy is supposed to operate.

⁹¹Reserves over and above some outright reserve requirements (excess reserves) are held primarily as a buffer against unexpected payment shocks and due to institutional characteristics.

of interest-rate setting, if the central bank takes advantage of the “floor” system. In this system, the target rate equals the deposit rate (lower standing facility), i.e. the rate paid on reserve balances held on the central bank account. At this rate, the demand curve becomes horizontal because the opportunity cost of holding reserves becomes zero. Any additional supply of base money beyond the (minimum) demand is absorbed by the banking sector at this rate. Balances are held at the central bank’s account and the central bank can supply any amount by way of open-market operations.⁹²

The essence of the argument above is that the central bank is able to choose both the interest rate and the quantity of reserves supplied in a (quite) independently manner. This decoupling gives monetary policy a further instrument at hand, i.e. the management of its balance sheet. As it has been laid out, a situation can arise in which the short-term policy rate is not capable of fighting evolving imbalances on financial markets. Operations based on changes in the central bank balance sheet can affect financial conditions more directly. In this respect, the central bank balance sheet can be altered in terms of its size, composition and risk profile. The main objective is to alter the relative scarcity of various assets in order to influence the market valuation and the required liquidity and risk compensation of those assets. This can be achieved by signalling the central bank’s attempt by means of communication and using its influence on market expectations. Furthermore, direct actions seek to alter the composition of private sector portfolios when there is imperfect substitutability on the asset and liability side of the private balance sheets. Balance sheet policy can take on different forms varying with its impact on the transmission mechanism. It can be classified into (1) quasi-debt management policy, (2) credit policy and (3) bank reserve policy (Borio and Disyatat 2009).

Central bank operations aimed at altering returns, liquidity and interest-rate risk on government bonds belong to the first category. Since the yields on these bonds serve as benchmarks for a multitude of private transactions, the central bank can affect various other asset classes of the private sector. Most importantly, it can support a flattening of the yield curve by buying long-term government securities which makes it possible to cut intermediation dynamics back. If the induced compression of the yield spread exhibits a more pronounced impact on the supply of credit than the implied reduction of the long-term rate on credit demand, overall credit growth should shrink.⁹³ If the credit cycle is mainly demand driven, a central bank is advised to sell long-term governments bonds so that long-term yields rise.

The second category, credit policy, is related to central bank actions that are directly aimed towards targeting private asset market segments. Assets with different

⁹²For a graphical representation of the various forms of implementing monetary policy the reader is referred to Keister et al. (2008), Spahn (2010). The authors also discuss shortcomings of organizing the money market as a floor system. Among other arguments, the loss of information on the state of the money market, as well as the requirement to hold a large stock of public (and private) securities may be regarded as disadvantage.

⁹³See Adrian et al. (2010a) on the empirical dominance of the supply effect over the demand effect.

characteristics, in terms of credit and liquidity risk, are substituted for each other in order to influence the price of risk. Credit policy also comprises the choice of private assets accepted as collateral in secured monetary operations with the central bank. It alters the private sector's demand for those assets and affects their risk and liquidity attributes. The need to collateralize central bank funds then implies a cost for commercial banks if they have to reshuffle portfolios in order to participate in monetary policy operations and to deviate from an otherwise optimal asset choice (Callado-Munoz and Restoy 2010). It makes the refinancing of those assets dependent on the state of collateral policy of the central bank. Such actions are likely to be translated in private collateralized lending operations in wholesale markets and they alter haircut requirements for the financial sector as a whole (Geanakoplos 2010a,b). In this respect, it appears most promising that the criteria for eligibility of collateral assets in open market operations have also a direct effect on the funding conditions for the shadow banking system and system-wide leverage.⁹⁴ It is obvious that credit policy influences relative prices with distributional effects for the economy. The central bank dictum is typically to avoid operations that directly affect capital investment decisions. However, if one acknowledges the adverse distributional outcomes generated by endogenous financial cycles, then credit policy "only" restores the *ex-ante* status-quo.

Finally, independent of its impact on the asset side of its balance sheet, a central bank conducts bank reserve policy if it targets an amount of bank reserves. Changes in the size of available aggregate liquidity (here narrowly defined) are expected to influence credit flows and risk-taking. For instance, it could reduce the supply of reserves for the commercial banking sector thereby sending out a clear signal to cut back the aggregate availability of liquidity (Spahn 2010).⁹⁵ Such action can induce market participants to become more sensitive to funding liquidity risk and may contribute to a re-pricing of financial risk towards more "fundamental" levels. At the same time, if bank reserve policy interacts with quasi-debt management policy, transactions in the government security market can produce desired effects on term and credit spreads. Bank reserve policy, thus, can be used to counteract the modern dogma of money to be fully demand driven where money supply is a completely endogenous concept.

⁹⁴The concept of collateral management is similar to the idea of imposing asset-based reserve requirements (ABBR) where required reserves are held against assets rather than against deposits (Palley 2004, 2007). Along the same lines, ABBR change the banking's asset structure by imposing additional costs on selected assets. The same mechanism holds for imposing capital requirements that increase with the growth of credit collateralized by assets whose prices have unsustainable soared (Schwartz 2003). It has also been argued that regional differing collateral management within the euro area is an effective tool to address imbalances and heterogeneity across member countries (Geiger and Spahn 2007; Brunnermeier 2010).

⁹⁵It is clear that even under the floor system there arise an inverse relationship between the market rate and the supply of reserves in case the latter is cut back significantly and money demand is not fully allotted.

For the purpose of crisis prevention, balance sheet policy can support the pre-emptive approach of fighting financial imbalances. As has been previously argued, financial intermediaries try to relax their budget constraint purported by regulatory measures and interest-rate policy. A central bank can actively use its instruments of liquidity management to provide an additional monetary restriction for the financial sector. Liquidity management offers the primary advantage that it does not need to rely on adequate policy rate moves to exhibit effects on the pricing of assets and overall funding conditions. From a balance of risk consideration, if a central banker is faced with the increased probability of an economic downturn, when identified imbalances are countervailed by pre-emptive policy rate increases, she might be advised to use the various forms of balance sheet policy to address distortions in credit and asset price dynamics. Still, even if those actions are successful in curbing credit dynamics, it remains unclear to which extent the reduced credit supply manifests itself in reduced lending for real (physical) assets or financial assets.

Against this background, the communication of a central bank's objectives and intentions is essential for the effectiveness of the proactive view of addressing financial imbalances. The standard (practical) inflation targeting framework, with its focus on price and output stability in the medium run, suffers from a lack of credibility. A monetary authority finds it difficult to justify any policy action that responds to financial factors that do not affect the inflation and output forecast for a given policy horizon. Disyatat (2010b, 153) remarks that "[i]t is very important, therefore, to ensure that the central bank's communication strategy does not lead to a situation where the maintenance of credibility becomes a constraint on the flexibility of monetary policy." Assigning monetary policy with the explicit objective of financial stability, is an appropriate step to communicate financial stability concerns with the public. Monetary policy must ensure that markets understand the nature of procyclical evolving credit dynamics that can increase the likelihood of the bust in the distant future. Indeed, many central banks are already equipped with communication instruments that sensitize markets to financial developments.⁹⁶ In the run-up to the financial crisis starting in 2007, those institutions have repetitively emphasized the mispricing of risk measures across a set of financial assets and the extraordinarily overextension of financial intermediaries' balance sheets.⁹⁷ However, their hands were tied to react to these facts which in turn induced financial markets to expect a favorable interest-rate environment as long as there are no medium-term threats to inflation. In case monetary policy is also designed towards striving for financial

⁹⁶Many central banks publish financial stability reports on a regular basis; among them are the European Central Bank, Bank of England, Sveriges Riksbank, Bank of Canada or the Bank of Japan.

⁹⁷For instance, the ECB comments in its June (2005, 9) Financial Stability Review: "On the one hand, there has been a broad-based improvement in the capacity of the financial system to absorb adverse disturbances. On the other, financial imbalances are already quite large and could expand further, primarily at the global, but also at the domestic, level."

stability and it announces to pre-emptively respond to the build-up of financial market misalignments, market participants perceive a higher risk of monetary restrictions and consider such likelihood in investment decisions. In the end, the pure announcement effect can bring about desirable adjustment dynamics. If this effect is not strong enough, monetary policy is advised to take pre-emptive actions that include an increase in the short-term interest rate supported, if necessary, by balance sheet policy. For agents to understand such steps, the central bank must communicate its greater tolerance to depart from the inflation target due to the inter-temporal tradeoff between current and future economic stability. Indicating the rationale for policy actions is essential in keeping inflation expectations solidly anchored.

A few years ago, former FED president Alan Greenspan (2004) remarked that “[...] the conduct of monetary policy [...] has come to involve, at its core, crucial elements of risk management. The conceptual framework emphasizes the understanding as much as possible the many sources of risk and uncertainty policymakers face, quantifying those risks when possible, and assessing the costs associated with each of the risks.” This risk management perspective towards monetary policy can be interpreted as a robust approach of dealing with various forms of uncertainty, in particular with events characterized by a difficult to measure probability but with a large impact on the economy (Blinder and Reis 2005). It provides policymakers with the necessary freedom of action (discretion) in the decision-making process to react to developments that can not be captured within the standard time-invariant response to various shocks.⁹⁸ In the context of financial instability and imbalances, the risk management approach recommends a discretionary reaction to financial developments rather than a strictly rule-based reaction.⁹⁹ Monetary policy needs to be timely, decisive and flexible. It includes episodes of pre-emptive and non-gradual policy actions as an appropriate form of risk management with the scope of reducing the likelihood of severe outcomes. It is inapprehensible why, until today, these calls for flexibility in monetary policy are biased toward aggressive easing in times of financial disruptions. The asymmetric reaction, by way of easing the policy stance when the likelihood of near-time adverse events is significant, finds broad consensus among academic and policy circles.¹⁰⁰ However, as previously stated, it could be rather argued that aggressively cutting interest rates in times of financial distress promotes an environment of favorable conditions for subsequent unsustainable financial developments. Effective risk management should then concentrate on pre-emptive tightening and not on pre-emptive easing. This holds in particular against

⁹⁸Typically, optimal monetary policy is evaluated in a linear-quadratic framework in which the magnitude and the direction of macroeconomic risk does not matter; this is so because certainty equivalence holds in these models among them those of Clarida et al. (1999), Giannoni and Woodford (2003b), Svensson and Woodford (2005).

⁹⁹In this respect, Kindleberger (1995, 35) remarks that “[w]hen speculation threatens substantial rises in asset prices, with a possible collapse later, and harm to the financial system [...], monetary authorities confront a dilemma calling for judgement, not cookbook rules of the game.”

¹⁰⁰As a representative see Mishkin (2008, 2010).

the background of the analysis of boom-bust cycles that highlights the need to resist procyclicality within a new macro-financial framework.

The two-pillar strategy of the ECB seems to be a natural starting point for such a framework with two objectives. Whereas the first pillar is directed towards price stability, the formerly monetary pillar could address the financial stability objective (DeGrauwe and Gros 2009; Gali 2010). Indeed, the ECB itself emphasizes the information content of its second pillar for the assessment of risks to financial stability (Issing 2005b; Stark 2010). As there is a short- and medium-term trade-off between both objectives, each pillar can be used independently to analyze upside and downside risks to its objectives. The first pillar includes both the economic and the monetary analysis as recommended by the flexible inflation targeting approach. As long as non-monetary and monetary models contain information about future inflation, they should be imbedded in this pillar. At the same time, such a re-definition of the two-pillar strategy offers the convenient advantage of circumventing the justified critique that money should be not treated differently from any other indicator model to predict inflation (Gerlach 2004). The central bank can clarify that its interest-rate setting is the primary instrument (1) in achieving the macroeconomic objective of price stability, (2) in providing a determinant nominal anchor for the macroeconomy and (3) in restoring equilibrium after some demand and supply shocks. The financial stability pillar, instead, is set up to provide a timely assessment of the financial system's resilience and evolving imbalances for the purpose of limiting systemic risk. It includes those early-warning indicator models, financial condition indexes and network analysis that help to define a threshold for financial imbalances. If this threshold is reached, the central bank reacts in a non-linear, decisive way with its instruments. They include time-varying macroprudential tools beyond fixed instruments and interest-rate as well as balance sheet policy. Finally, the central bank should support its actions with a clear and transparent communication of its intentions.

Chapter 8

Conclusion and Outlook

Combining finance models with monetary macroeconomics helps to understand the link that exists between them. Essentially, the *macro-finance* approach of modeling the yield curve and the macroeconomy allows for the analysis of how macroeconomic variables spread across bond yields of different maturities. For monetary policy, these interactions are of primary interest. Firstly, the relationship between short and long-term interest rates is relevant for the transmission mechanism. The dynamic of New-Keynesian business cycle models is driven by changing expectations about future economic variables. Forward-looking agents form expectations that profoundly influence the current economic outcome. The term structure of interest rates takes on a central role as it imbeds the expectations-driven long-term interest rates that determines aggregate credit demand. Secondly, the term structure reflects private expectations of future macroeconomics developments and provides monetary policy with valuable information about the perceived future monetary policy stance as well as inflation and output expectations.

The central element of a joint modeling strategy is the short-term interest rate. From a macroeconomic perspective, the short rate is the policy rate that is adjusted in accordance with a central bank's policy objectives. It is determined by an additive combination of multiple macroeconomic variables. Within the finance view, the short rate shares the same functional form, i.e. it is represented by a linear combination of latent risk factors. By replacing the finance factors with macroeconomic variables, the no-arbitrage condition on financial markets fixes the unique cross-section structure of interest rates with different maturities, and enables it to identify the macroeconomic factors behind yield curve dynamics, in particular regarding the impact on its level and its shape. A general finding of this part of the analysis in this work is that the explanatory power of macroeconomic shocks for interest rate movements decreases with the time to maturity. Moreover, shocks that are highly persistent generate a significant effect at the long end of the yield curve. Whereas inflationary and output shocks have their greatest impact in describing fluctuations of short-term interest rates, shocks to the perceived inflation target and to the natural output mainly affect long-term bond yields. More lasting impacts on the whole yield curve are generated in an environment in which agents form

expectations in an adaptive learning process where bond securities are priced with subjective beliefs. It provides a strong transmission mechanism for macroeconomic shocks resulting in more persistent effects on bond prices. The modification of the conduct of monetary policy through the inclusion of term structure information into the rule-based interest setting brings about ambivalent results. It is an improvement of welfare in the sense that the volatility of inflation and output can be reduced. However, as such a modification of the policy rule is associated with reacting to private expectations, the danger of self-fulfilling expectations emerges, pushing the macroeconomy toward an indeterminate and explosive solution.

With the absence of arbitrage opportunities, movements in long-term interest rates reflect time-varying risk premia, as demanded by risk-averse agents, rather than the average expected path of future short-term interest rates. Indeed, by applying the affine term structure set-up, the empirical analysis of Government bond rate dynamics for Germany, the US and the UK points out that the size of the yield spread can be mostly attributed to the risk compensation that arises from the uncertain evolution of future interest rates. Term premia can also be the reflex of default risk. With the help of a case study for the euro area government security market, this work has shown the extent to which global and country-specific factors have accounted for observable yield differentials. With the introduction of the Single Currency in 1999, the compression of yield differentials was the result of global risk appetite and various aspects of the Eurosystem's institutional setting. For instance, the collateral framework of open market operations treated all government bonds as eligible assets that can be used as collateral so that fundamentally justifiable default-risk differentials can be suppressed. With the financial crisis and the massive deterioration of national fiscal positions in 2008, market participants started to discriminate between the quality of national fiscal policies by imposing an additional risk compensation. Finally, liquidity risk is a further factor that contributes to overall term premia. Besides the distinction between market and funding liquidity, a stylized model of a liquidity spiral has been presented where increased market-based funding of financial intermediaries generates a correlation between both liquidity concepts. Any tension in market liquidity affects markets participants' funding liquidity. With falling market liquidity, funding becomes more expensive requiring a re-shuffling of the funding structure. A self reinforcing loop process of a loss and margin spiral can create an environment where liquidity suddenly dries up. In this respect, Keynes' theory of liquidity preference can be understood through the lens of financial intermediation and active balance sheet management. An increase in liquidity preference is, thereby, associated with a reduced activity in liquidity and maturity transformation.

The no-arbitrage approach further reveals monetary policy's role in shaping risk premia on financial markets. This work has brought forward the argument in favor of the existence of a broad risk-taking channel that describes the effects of monetary impulses on financial markets and macroeconomic dynamics. Within this channel, it can be explored how the strategy and the conduct of monetary policy influence the risk perception and the risk tolerance of market participants. With the risk premium split into the quantity and the market price of risk, the latter determines the required

compensation for various macroeconomic sources of uncertainty. The analysis of optimal monetary policy, augmented by term structure effects and bond risk premia, comes to the conclusion that the market price of risk for inflationary shocks is smaller in case of monetary policy being conducted under commitment rather than under discretion. Under commitment, inflation and output react less sensibly to the shock process over the time to maturity of the relevant bonds. Along similar lines, increased central bank transparency contributes to an increased predictability of monetary policy actions that reduces market uncertainty and induces market participants to demand a lower risk award. This transparency effect has significantly contributed to the observed trend decline in risk premia in many industrialized countries. Moreover, the risk-taking channel has shown that moral hazard issues arise in the form of excessive risk-taking, if market participants assume that the central bank stands ready to tackle downside risks to the economy by lowering interest rates and providing ample liquidity.

The risk-taking channel explicitly allows for financial frictions and financial shocks in the lending sector. Active balance sheet and risk management on part of financial intermediaries generate high procyclicality in the financial sector. Monetary policy easing triggers higher risk tolerance due to sticky return and leverage targets that culminates in expanding balance sheets, softer lending standards and a compression of credit spreads. These amplifying effects are in high gear when three factors pick up speed: the size of the balance sheet, leverage growth and the degree of maturity transformation. They can produce endogenous boom-bust cycles driven by financial intermediary activity. The degree of financial instability depends on the maturity mismatch between assets and liabilities. Therefore, the steepness of the yield curve is an effective point of reference for the appraisal of evolving imbalances. The longer the spread is high in the expansion phase, the more likely financial instability builds up by means of higher risk-taking – a mechanism also emphasized by the Post-Keynesian model introduced by Hyman P. Minsky to explain financial instability. In essence, the duration of financial imbalances heightens the risk of sudden reversals to which market participants cannot quickly adjust – with an obvious, disastrous real economic outcome.

For the conduct of monetary policy, this work has developed new arguments in favor of pre-emptive tightening to address financial instability. Prior to the financial crisis starting in 2007, the consensus view relied on the “benign neglect” approach according to which monetary policy eases and cleans-up up afterwards once the boom has turned to bust. The logic of the risk-taking channel creates an alternative rationality how to cope with the systemic component of financial risk that builds up in the background of the financial system. It recommends a pre-emptive tightening and, consequently, a non-linear policy rule whenever forward-looking, multi-dimensional risk indicators exceed a historically defined threshold level. Standard critics against this proactive view can be dismissed because it does not rely on the need to identify a particular asset price bubble in one market segment nor is monetary policy ineffective in altering imbalances since, with leveraged institutions, small policy rate changes significantly influence the profitability of liquidity and maturity transformation. Despite the short-run trade-off between price and financial

stability, the assignment of a financial stability objective might help to communicate the policy intentions with the public. In a last step, this work has presented further monetary policy instruments that support the resistance of procyclicality of the financial system. They include macroprudential and balance sheet policy, among them time-varying capital and liquidity buffers as well as the design of collateral management in open market operations. All in all, what these policy measures imply is the effective influence of the monetary budget constraint for the financial sector as a whole.

Current (general equilibrium) models are still silent about how to build a financial sector with evolving financial imbalances and sudden reversals. Most model attempts have little to say about the issue of how monetary policy decisions are able or should take into account financial instability concerns from the perspective of systemic risk, since they analyze macroeconomic and financial dynamics next to the steady state in a log-linear setting. Against the background of this work, the insights discussed may give rise to further research efforts in this direction. The models should include endogenous financial risk, non-linear asset price and liquidity spirals. In times of disruptions, these mechanisms can plunge the system into a crisis state with low economic activity and high uncertainty; they may take the system far away from the local steady state. The associated economic costs may make it possible to incorporate financial imbalances into the formulation of optimal monetary policy. It may allow for the derivation of a policy rule that makes the prescription of pre-emptive actions operational.

Appendix A

Dynamic Optimization

In a dynamic setting, the task is to find the entire path of some choice or control variable $u_{t=0}^{\infty}$ which maximizes the infinite sum of a objective function $r(x_t, u_t)$.¹ Formally, the agent's problem is to solve the objective function of the form

$$\max_{u_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \quad (\text{A.1})$$

subject to the dynamic constraint

$$x_{t+1} = g(x_t, u_t) \quad \forall t = 0, 1, \dots, \infty \quad (\text{A.2})$$

$$x_0 \text{ given.} \quad (\text{A.3})$$

Above, x_t is a state variable at time t and u_t is the control variable. The constraint equation is also called the transition equation since it describes the transition from the state x_t at t to the state x_{t+1} at time $t + 1$. The aim of the dynamic decision problem is to seek a time-invariant policy function (or control equation) which maps the state x_t into the control u_t

$$u_t = h(x_t). \quad (\text{A.4})$$

As a first step to solving the problem, the infinite dynamic optimization has to be transformed into a sequence of two-period problems. The value function $V(x_0)$ at the initial date expresses the value of the objective function at the optimum. That is, the value function gives the maximum value of the objective function when the control variabel is chosen optimally. In general, the value function at the initial date is defined as

¹The solution of the dynamic programming problem basically follows the lines of Ljungqvist and Sargent (2004).

$$V(x_0) = \max_{u_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \right\} \quad (\text{A.5})$$

The restriction is that it is not feasible to know $V(x_0)$ until one has solved the whole problem. Since the problem arises from the initial date until infinity, it is also possible to write

$$V(x_0) = \max_{u_{t=0}^{\infty}} \left\{ r(x_0, u_0) + \sum_{t=1}^{\infty} \beta^t r(x_t, u_t) \right\}$$

which can be modified to

$$V(x_0) = \max_{u_{t=0}^{\infty}} \left\{ r(x_0, u_0) + \sum_{t=0}^{\infty} \beta^{t+1} r(x_{t+1}, u_{t+1}) \right\}. \quad (\text{A.6})$$

It is clearly recognized that the second term of (A.6) can be expressed as being β times the Value of V as defined in (A.5) with the initial condition (x_1, u_1) . Therefore, one could write

$$V(x_0) = \max_{u_{t=0}^{\infty}} \{ r(x_0, u_0) + \beta V(x_1) \}. \quad (\text{A.7})$$

Equation (A.7) highlights the dynamic recursive structure of the decision problem. At each date t , the agent has to deal with the same decision problem, i.e. choosing the control or choice variable that maximizes the current value (which is $\max_u r(x_t, u_t)$) plus the discounted value of the optimum plan from $t + 1$ onwards (which is $\max_u \beta V(x_1)$). Since this problem repeats itself every period and holds for any starting date, (A.7) holds more generally as the Bellman equation

$$V(x_t) = \max_{u_{t=0}^{\infty}} \{ r(x_t, u_t) + \beta V(x_{t+1}) \} \quad (\text{A.8})$$

or in compact representation

$$V(x_t) = \max_{u_{t=0}^{\infty}} \{ r(x_t, u_t) + \beta V[g(x_t, u_t)] \} \quad (\text{A.9})$$

subject to

$$x_{t+1} = g(x_t, u_t) \quad \forall t = 0, 1, \dots, \infty \quad (\text{A.10})$$

$$x_t \text{ given.} \quad (\text{A.11})$$

The functional form of the Bellman equation can be written according to

$$V(x_t) = r(x_t, h(x_t)) + \beta V(g(x_t, h(x_t))) \quad (\text{A.12})$$

When writing the Bellman equation in such a functional relationship, the problem is specified in such a way that today's control variable will unescapably become tomorrow's state variable.

There are a number of methods to derive the policy function. Here the focus is placed on finding the corresponding Euler equations. The Envelope theorem or the Lagrangian method allow solving for the Euler equation. What they both have in common is that they seek to find the first-order condition of the Bellman equation. Taking the partial derivative of (A.9) with respect to the control gives the first-order condition

$$\frac{\partial r}{\partial u_t}[x_t, u_t] + \beta V'[g(x_t, u_t)] \frac{\partial g}{\partial u_t}[x_t, u_t] = 0. \quad (\text{A.13})$$

In a next step, Ljungqvist and Sargent (2004) show that for an interior solution the value function V is differentiable with

$$V'(x_t) = \frac{\partial r}{\partial x_t}[x_t, h(x_t)] + \beta V'(g[x_t, h(x_t)]) \frac{\partial g}{\partial x_t}[x_t, h(x_t)] \quad (\text{A.14})$$

This property of a solution is also called the Benveniste-Scheinkmann condition. In some cases, it is possible to formulate the transition equation in such a way that the first derivative of the equation is $\partial g / \partial x_t = 0$ so that the condition collapses to

$$V'(x_t) = \frac{\partial r}{\partial x_t}[x_t, h(x_t)]. \quad (\text{A.15})$$

Taking this latter equation one step forward yields

$$V'(x_{t+1}) = \frac{\partial r}{\partial x_{t+1}}[x_{t+1}, u_{t+1}] \quad (\text{A.16})$$

which can be substituted into the first-order condition of (A.13) giving rise to the famous Euler equation:

$$\frac{\partial r}{\partial u_t}[x_t, u_t] + \beta \frac{\partial r}{\partial x_{t+1}}[x_{t+1}, u_{t+1}] \frac{\partial g}{\partial u_t}[x_t, u_t] = 0. \quad (\text{A.17})$$

Since most of the economic models deal with uncertainty, the question is whether the solution method, according to the Envelope Theorem, also holds in such an environment. The most natural way to incorporate uncertainty is obtained by the transition equation $g(x_t, u_t, z_t)$ where uncertainty is exogenous and may follow an AR(1) process

$$z_{t+1} = \alpha z_t + \varepsilon_t \quad (\text{A.18})$$

where ε_t is independently and identically distributed. Ljungqvist and Sargent (2004) prove that as long as this shock has these properties, the adequate modification of

non-stochastic method will work as well. In a stochastic setting, the agent faces the following dynamic problem

$$\max_{u_t=0}^{\infty} E_0 \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \quad (\text{A.19})$$

subject to the dynamic constraint

$$x_{t+1} = g(x_t, u_t, z_{t+1}) \quad \forall t = 0, 1, \dots, \infty \quad (\text{A.20})$$

$$x_0 \text{ given.} \quad (\text{A.21})$$

Following the arguments of the non-stochastic case, the Bellman equation can be written according to

$$V(x_t) = \max_{u_t=0}^{\infty} \{r(x_t, u_t) + \beta E_t [V(x_{t+1})]\} \quad (\text{A.22})$$

or

$$V(x_t) = \max_{u_t=0}^{\infty} \{r(x_t, u_t) + \beta E_t [V[g(x_t, u_t, z_{t+1})]]\} \quad (\text{A.23})$$

and in the functional form as

$$V(x_t) = r(x_t, h(x_t)) + \beta E_t [V(g(x_t, h(x_t), z_{t+1}))]. \quad (\text{A.24})$$

Following the lines of equations (A.13)–(A.17) the Euler equation is

$$\frac{\partial r}{\partial u_t}[x_t, u_t] + \beta E_t \left\{ \frac{\partial r}{\partial x_{t+1}}[x_{t+1}, u_{t+1}] \frac{\partial g}{\partial u_t}[x_t, u_t, z_{t+1}] \right\} = 0. \quad (\text{A.25})$$

Appendix B

State-Space Model and Maximum Likelihood Estimation

This appendix introduces the statistical state-space model. It describes the tools that are employed for the estimation of various term structure models (see for subsequent work Harvey 1990; Hamilton 1994; Gouriéroux and Monfort 1997; Lemke 2006).

Structure of the State Space Model

A state-space model is a representation of the joint dynamics of an observable random vector y_t with size $N \times 1$ that can be generally described by an unobservable state vector α_t with size $(r \times 1)$. It consists of a measurement equation and a transition equation. The former governs the evolution of the state vector, the latter specifies the empirical link between the set of observable variables and the state. Such a model is said to be Gaussian if the innovations to the state space are normally distributed. The representation can be written as

$$\alpha_t = c + T\alpha_{t-1} + D\eta_t \quad (\text{B.1})$$

where c is a $N \times 1$ vector T is a $N \times N$ matrix and D is $r \times g$. The measurement equation is given by

$$y_t = b + M\alpha_t + \varepsilon_t \quad (\text{B.2})$$

where M is a $n \times r$ matrix and b is an $N \times 1$ vector. The state innovations ($g \times 1$) and measurement errors ($N \times 1$) are normally distributed with the first two moments given by

$$\begin{pmatrix} \eta_t \\ \varepsilon_t \end{pmatrix} = \mathcal{N} \sim \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} Q & 0 \\ 0 & H \end{pmatrix} \right)$$

so that the disturbances are uncorrelated and independent to each other. The initial conditions are $\alpha_0 \sim (a_0, P_0)$ and $E(\eta_t \alpha_0), E(\varepsilon_t \alpha_0) = 0$.

Kalman Filter

The Kalman filter is an algorithm used to estimate and extract the evolution of the unobservable state variables given the sequence of observable variables y_t via a feedback-control rule. It calculates linear least square forecasts of the state vector on the basis of data observed through time $t - 1$. Optimality is achieved by minimizing the mean squared error (MSE) of the state variables. The best *a priori* estimate of the state variable vector is $E_t(a_t | \mathcal{I}_{t-1})$ and its variance-covariance matrix is $E_t(\Sigma_t | \mathcal{I}_{t-1})$, conditionally on all information $\mathcal{I}$ available at time $t - 1$ where Σ_t is the MSE. Since y_t has not yet been observed, the prediction of the observable variables takes the form of $E_t(\hat{y}_t | \mathcal{I}_{t-1})$ and the error of this forecast is $y_t - \hat{y}_{t|t-1}$ with variance-covariance matrix F_t after the measure has been observed. Continuing from here, the Kalman filter recursively updates the *a priori* estimates of the conditional means and (co)-variances to yield the *a posteriori* estimates to get $E_t(a_t | \mathcal{I}_t)$ and $E_t(\Sigma_t | \mathcal{I}_t)$, respectively.

The Kalman filter algorithm typically starts in **Step 1** with an initialization of the first two moments of the measurement equation

$$a_{0|0} = \bar{a}_0 \quad \Sigma_{0|0} = \bar{\Sigma}_0$$

The system of prediction equations can be summarized in **Step 2** as

$$\begin{aligned} a_{t|t-1} &= c + T a_{t-1|t-1} \\ \Sigma_{t|t-1} &= T \Sigma_{t-1|t-1} T^\top + D Q D^\top \\ \hat{y}_{t|t-1} &= b + M a_{t|t-1} \\ v_t &= \hat{y}_{t|t-1} - y_t \\ F_t &= M \Sigma_{t|t-1} M^\top + H. \end{aligned} \tag{B.3}$$

After Step 2, y_t is now observed and the current value of a_t can be updated. For that purpose, a coefficient matrix K_t (Kalman gain) is introduced with which the difference between the *a posteriori* and the *a priori* estimate of the variance-covariance matrix of the state variables can be minimized. It is a weighting matrix that defines the extend to which the difference between the *a priori* estimate and the observed measure is weighted in the *a posteriori* estimate. It is proportional to the mean squared error of the forecast for the state vector and inversely related to the mean squared error of the observable vector. The higher K_t , the greater is the weight of the observable measure on the *a posteriori* estimate of the state equation. The system of updating equation can be documented in **Step 3** as

$$\begin{aligned}
K_t &= \Sigma_{t|t-1} M^\top F_t^{-1} \\
a_{t|t} &= a_{t|t-1} + K_t (y_t - \hat{y}_{t|t-1}) \\
\Sigma_{t|t} &= \Sigma_{t|t-1} - K_t M \Sigma_{t|t-1}.
\end{aligned}$$

In **Step 4**, this procedure is repeated by setting $t = t + 1$ if $t < T$ and one goes back to Step 2.

The Kalman filter provides the sequence of conditional means and covariances for the relevant conditional distributions. In estimation, the initial mean and covariance of the state variables are calculated as their unconditional equivalents (provided that the transition equation is stationary).¹ Thus, $a_{0|0}$ is chosen as

$$a_{0|0} = (I - T)^{-1}c$$

and the covariance-variance matrix is given in a column vector as

$$\Sigma_{0|0} = \text{vec}[I - (T \otimes T)]^{-1} \text{vec}(Q).$$

Maximum Likelihood Estimation

If the parameters describing the state space are unknown, they can be estimated with maximum likelihood (ML). For a given distributional assumption of the innovations, the ML estimate of an unknown parameter set is the value that maximizes the probability density. For a linear Gaussian model (normality assumption of innovations) with a set of unknown elements stacked in the vector ϑ , the conditional density function of a simple VAR(1) process with $y_t \sim N(\mu, \Omega)$ and dimension N can be written in general as

$$f(y_t | \mathcal{I}_{t-1}; \vartheta) = (2\pi)^{-N/2} |\Omega|^{-1/2} \exp \left[-\frac{1}{2} (y_t - \mu)^\top \Omega^{-1} (y_t - \mu) \right].$$

The joint density function from observation t through T satisfies

$$f(y_T | \mathcal{I}_{T-1}; \vartheta) = \prod_{t=1}^T f(y_t | \mathcal{I}_{t-1}; \vartheta)$$

and the log-likelihood function is given as

$$\ln \mathcal{L}(\vartheta) = -\frac{NT}{2} \log(2\pi) - \frac{T}{2} \log |\Omega| - \frac{1}{2} \sum_{t=1}^T [(y_t - \mu)^\top \Omega^{-1} (y_t - \mu)].$$

¹See Hamilton (1994, p. 378).

The Kalman filter can be used to calculate the sequence of conditional means and (co)-variances so that for above state-space specification, the distribution of y_t conditional on $\mathcal{J}_{t-1}$ is given by (B.3)

$$y_t | \mathcal{J}_{t-1} \sim N(\hat{y}_{t|t-1}, F_t)$$

with $\hat{y}_{t|t-1} = b + Ma_{t|t-1}$. Accordingly, the log-likelihood function becomes

$$\ln \mathcal{L}(\vartheta) = -\frac{NT}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T [\log |F_t| + v_t^\top F_t^{-1} v_t].$$

This function can be maximized with numerical optimization techniques (see Lemke 2006, p.79): (1) choose an initial value for $\vartheta = \vartheta_0$, (2) run the Kalman filter and store the sequences ϑ_t and F_t , (3) use them to compute the log-likelihood and (iv) use an optimization procedure that repeats steps (1)–(3) until a maximizer $\hat{\vartheta}$ has been found.

For a sufficiently large sample size T , the distribution of the estimate $\hat{\vartheta}$ can be approximated as

$$\hat{\vartheta} = N(\vartheta_0, T^{-1}I^{-1})$$

where I is the information matrix and ϑ_0 denotes the true parameter vector. The information matrix can be estimated in two ways. The first way is to calculate the Hessian to get

$$I_H = -\frac{1}{T} \left[\frac{\partial^2 \ln \mathcal{L}(\vartheta)}{\partial \vartheta \partial \vartheta^\top} \right].$$

The second way is based on the outer-product estimate

$$I_{OP} = \frac{1}{T} \sum_{t=1}^T [h(\hat{\vartheta}, \mathcal{J})][h(\hat{\vartheta}, \mathcal{J})]^\top$$

where $h(\cdot)$ denotes the the vector of derivatives evaluated at $\hat{\vartheta}$.

According to Hamilton (1994, Sect. 5.8), the variance matrix for $\hat{\vartheta}$ can be then given as

$$Var(\hat{\vartheta}) = \frac{1}{T} [I_H I_{OP}^{-1} I_H]^{-1}.$$

Standard errors are calculated based on the quasi-maximum likelihood estimator for potentially misspecified models when the underlying data are not generated by the assumed density of the error terms.

Appendix C

Recursive Nature of the Expectations Hypothesis

The use of the pure form of the Expectations Hypothesis can give a simple example of a term structure model. Suppose that the short-term interest rates follows an AR(1) process with

$$i_{1,t} = \mu + \phi i_{1,t-1} + \varepsilon_t \quad (\text{C.1})$$

and $\varepsilon_t \sim N(0, \sigma^2)$. The yield-to-maturity hypothesis in log-form is given by

$$i_{n,t} = \frac{1}{n} \sum_{j=0}^{n-1} E_t[i_{1,t+j}]. \quad (\text{C.2})$$

For a two-period bond, it must hold that

$$i_{2,t} = \frac{1}{2}(E_t[i_{1,t+1}] + i_{1,t}) \quad (\text{C.3})$$

$$= \frac{1}{2}(\mu + \phi i_{1,t} + i_{1,t}) \quad (\text{C.4})$$

$$= \frac{1}{2}(\mu + (1 + \phi)i_{1,t}). \quad (\text{C.5})$$

For a three-period bond, the EH implies

$$i_{3,t} = \frac{1}{3}(E_t[i_{1,t+2}] + E_t[i_{1,t+1}] + i_{1,t}) \quad (\text{C.6})$$

$$= \frac{1}{3}((2 + \phi)\mu + (1 + \phi + \phi^2)i_{1,t}). \quad (\text{C.7})$$

For a four-period bond, the term structure model becomes

$$i_{4,t} = \frac{1}{4}(E_t[i_{1,t+3}] + E_t[i_{1,t+2}] + E_t[i_{1,t+1}] + i_{1,t}) \quad (\text{C.8})$$

$$= \frac{1}{4}((3 + 2\phi + \phi^3)\mu + (1 + \phi + \phi^2 + \phi^3)i_{1,t}). \quad (\text{C.9})$$

The n -period yield then equals

$$i_{n,t} = \frac{1}{n} \left[\sum_{j=1}^n [(n-j)\phi^{j-1}] \mu + \sum_{j=0}^{n-1} \phi^j \right]. \quad (\text{C.10})$$

The financial factor model representation of Chap. (3.4.1) takes the form of

$$i_{n,t} = a_n + b_n X_t \quad (\text{C.11})$$

$$X_t = \mu + \phi X_{t-1} + v_t \quad v_t \sim i.i.d.N(0, \sigma). \quad (\text{C.12})$$

The first equation describes the measurement equation and the second equation provides the law of motion of the state variable (transition equation). For the purpose of representing the pure Expectations Hypothesis, it is not necessary to add a measurement error to the measurement equation. Here, X_t is the short-term interest rate $i_{1,t}$ and the scalars a_n and b_n for $n > 1$ follow

$$b_n = \frac{1}{n} \sum_{j=0}^{n-1} \phi^j = \frac{1 - \phi^n}{n(1 - \phi)} \quad (\text{C.13})$$

$$a_n = \frac{1}{n} \sum_{j=1}^{n-1} b_n^* \mu, \quad \text{where} \quad b_n^* = \sum_{j=0}^{n-1} \phi^j. \quad (\text{C.14})$$

Appendix D

Derivation of Affine Coefficient Loadings

The derivation of the difference equations follows the guess-and-verify strategy similar to the method of undetermined coefficients supposed by McCallum (1983). For convenience, the relevant starting equations are

$$\begin{aligned} X_t &= \mu + \phi X_{t-1} + \Sigma \varepsilon_t \\ P_{n,t} &= E_t^{\mathcal{P}}[M_{t+1} P_{n-1,t+1}] \\ M_{t+1} &= \exp(-i_{1,t} - 0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}) \\ i_{i,t} &= \delta_0 + \delta_1 X_t \end{aligned}$$

Duffie and Kan (1996) guess a solution for bond prices as

$$P_{n,t} = \exp(A_n + B_n X_t).$$

For a one-period bond, it can be easily shown that

$$P_{1,t} = E_t M_{t+1} = \exp(-i_{1,t}) = \exp(-\delta_0 - \delta_1 X_t). \quad (\text{D.1})$$

Matching coefficients yields $A_1 = -\delta_0$ and $B_1 = -\delta_1$. Recursive solving and matching coefficients can also be applied to a n -period bond:

$$\begin{aligned} P_{n,t} &= E_t^{\mathcal{P}}[M_{t+1} P_{n-1,t+1}] \\ &= E_t^{\mathcal{P}}[\exp(-i_{1,t} - 0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}) \exp(A_{n-1} + B_{n-1}^\top X_{t+1})] \\ &= E_t^{\mathcal{P}}[\exp(-\delta_0 - \delta_1 X_t - 0.5\lambda_t^\top \lambda_t - \lambda_t^\top \varepsilon_{t+1}) \exp(A_{n-1} + B_{n-1}^\top (\mu + \phi X_t \\ &\quad + \Sigma \varepsilon_{t+1}))] \\ &= \exp(-\delta_0 - \delta_1 X_t - 0.5\lambda_t^\top \lambda_t + A_{n-1} + B_{n-1}^\top \mu + B_{n-1}^\top \phi X_t) \\ &\quad E_t^{\mathcal{P}}[\exp((B_{n-1}^\top \Sigma - \lambda_t^\top) \varepsilon_{t+1})]. \end{aligned}$$

The sources of uncertainty ε_{t+1} are assumed to be log-normal distributed with mean zero and variance $I_{N \times N}$ so that $E(e^{b\varepsilon}) = \exp(0.5bIb^\top)$. This result makes it possible to modify the expectations term of the recursive solution:

$$\begin{aligned} P_{n,t} &= \exp(-\delta_0 - \delta_1 X_t - 0.5\lambda_t^\top \lambda_t + A_{n-1} + B_{n-1}^\top \mu + B_{n-1}^\top \phi X_t) \dots \\ &\quad \exp((B_{n-1}^\top \Sigma - \lambda_t^\top) \text{var}_t(\varepsilon) (B_{n-1}^\top \Sigma - \lambda_t^\top)^\top) \\ &= \exp(-\delta_0 - \delta_1 X_t - 0.5\lambda_t^\top \lambda_t + A_{n-1} + B_{n-1}^\top \mu + B_{n-1}^\top \phi X_t) \dots \\ &\quad \exp(0.5B_{n-1}^\top \Sigma^\top \Sigma B_{n-1} - B_{n-1}^\top \Sigma \lambda_t + 0.5\lambda_t^\top \lambda_t). \end{aligned}$$

Substituting $\lambda_t = \lambda_0 + \lambda_1 X_t$ and netting out gives

$$\begin{aligned} &= \exp(-\delta_0 - \delta_1 X_t + A_{n-1} + B_{n-1}^\top \mu + B_{n-1}^\top \phi X_t) \dots \\ &\quad 0.5B_{n-1}^\top \Sigma^\top \Sigma B_{n-1} - B_{n-1}^\top \Sigma (\lambda_0 + \lambda_1 X_t) \end{aligned}$$

To this end, matching coefficients yields

$$\begin{aligned} \exp(A_n + B_n X_t) &= \exp(A_{n-1} + B_{n-1}^\top (\mu - \Sigma \lambda_0) + \frac{1}{2} B_{n-1}^\top \Sigma \Sigma^\top B_{n-1} - \delta_0 \\ &\quad + B_{n-1}^\top (\phi - \Sigma \lambda_1) X_t - \delta_1 X_t) \end{aligned}$$

so that scalar A_n and vectors B_n with size $N \times 1$ follow complex difference equations

$$\begin{aligned} A_n &= A_{n-1} + B_{n-1}^\top (\mu - \Sigma \lambda_0) + \frac{1}{2} B_{n-1}^\top \Sigma \Sigma^\top B_{n-1} - \delta_0 \\ B_n^\top &= B_{n-1}^\top (\phi - \Sigma \lambda_1) - \delta_1. \end{aligned}$$

Continuously-compounded interest rates then follow

$$\begin{aligned} i_{n,t} &= -n^{-1} \log(P_{n,t}) \\ &= n^{-1} (-A_n - B_n^\top X_t) \\ &= a_n + b_n^\top X_t \end{aligned}$$

with $a_n = -A_n/n$ and $b_n = -B_n/n$.

It is straightforward to calculate model-implied forward rates. Since forward rates are defined as

$$f_{n,t} = p_{n+1,t} - p_{n,t}$$

they can be easily computed as

$$f_{n,t} = (A_{n+1} - A_n) + (B_{n+1}^\top - B_n^\top) X_t.$$

Risk-neutral yields and forward rates can be defined as those that would prevail if investors did not price risk (λ_t) and all other parameters remain unchanged. The simple recursions for deriving risk-neutral rates can be defined as

$$\begin{aligned} A_n^{rn} &= A_{n-1}^{rn} + B_{n-1}^{rn,\top} \mu + \frac{1}{2} B_{n-1}^{rn,\top} \Sigma \Sigma^\top B_{n-1}^{rn} - \delta_0 \\ B_n^{rn,\top} &= B_{n-1}^{rn,\top} \phi - \delta_1. \end{aligned}$$

Bond risk premia are therefore computed as the difference between the model-implied yields and forwards and its artificial counterparts derived as if investors were risk-neutral.

$$\begin{aligned} \phi_{i_{n,t}} &= i_{n,t} - i_{n,t}^{rn} \\ \phi_{f_{n,t}} &= f_{n,t} - f_{n,t}^{rn}. \end{aligned}$$

If we want to impose the pure form of the expectations hypothesis (PEH), not only are investors insensitive to risk, but interest rates need to be deterministic. This is achieved by setting the variance-covariance matrix of the state variables equal to zero. Expected interest rates then follow

$$E_t[i_{1,t+n}] = f_{n,t}^{rnc}$$

where $f_{n,t}^{rnc}$ is the risk-neutral forward rate minus the convexity effect due to Jensen's inequality.

Appendix E

Optimal Monetary Policy

Discretion vs. Commitment. The central banker's objective function at time time $t = 0$ is given by

$$L = E_0 \sum_{t=0}^{\infty} \beta^t L_t \quad \text{with} \quad L_t = \pi_t^2 + \omega y_t^2 \quad (\text{E.1})$$

subject to

$$\pi_t = \alpha y_t + \beta E_t[\pi_{t+1}] + u_t. \quad (\text{E.2})$$

With discretion, minimizing the loss function subject to the constraints results in the Lagrangian

$$\Lambda_t = \pi_t^2 + \omega y_t^2 - \lambda_t(\pi_t - \beta E_t[\pi_{t+1}] - \alpha y_t - u_t) \quad \forall \quad t = 0, 1, 2, \dots$$

The first-order conditions are

$$\frac{\partial \Lambda_t}{\partial y_t} = 2\omega y_t + \alpha \lambda_t = 0 \quad (\text{E.3})$$

$$\frac{\partial \Lambda_t}{\partial \pi_t} = 2\pi_t - \lambda_t = 0$$

resulting in

$$\pi_t = -\frac{\omega}{\alpha} y_t. \quad (\text{E.4})$$

Under commitment, it holds that the central bank is assumed to choose a sequence $\{\pi_t, y_t\}_{t=0}^{\infty}$ so that the Lagrangian under commitment is given by

$$\Lambda_t = E_t \sum_{i=0}^{\infty} \beta^i [\pi_t^2 + \omega y_t^2 - \lambda_t (\pi_t - \beta \pi_{t+1} - \alpha y_t - u_t)]$$

and the first-order conditions are

$$\frac{\partial \Lambda}{\partial y_t} = 2\omega y_t + \alpha \lambda_t = 0 \quad t = 0, 1, 2, \dots$$

$$\frac{\partial \Lambda}{\partial \pi_t} = 2\pi_t - \lambda_t = 0 \quad t = 0$$

$$\frac{\partial \Lambda}{\partial \pi_t} = 2\pi_t - \lambda_t - \lambda_{t-1} = 0 \quad t = 1, 2, \dots$$

leading to

$$\pi_t = -\frac{\omega}{\alpha} y_t \quad t = 0 \quad \text{and}$$

$$\pi_t = -\frac{\omega}{\alpha} y_t - \frac{\omega}{\alpha} y_{t-1} \quad t = 1, 2, \dots$$

As shown by McCallum and Nelson (2004b), the solutions are

$$\begin{aligned} \pi_t^d &= a u_t \\ y_t^d &= b u_t \\ \pi_t^c &= c y_{t-1} + d u_t \\ y_t^c &= e y_{t-1} + f u_t \end{aligned} \tag{E.5}$$

with the complex parameter restrictions

$$\begin{aligned} a &= \frac{\omega}{\omega(1 - \beta\rho) + \alpha^2} b = \frac{-\alpha}{\omega(1 - \beta\rho) + \alpha^2} c = \frac{\omega(1 - \delta)}{\alpha} \\ d &= \frac{1}{\Omega - \beta(\rho + \delta)} e = \delta f = \frac{-\alpha}{\omega(\Omega - \beta(\rho + \delta))} \end{aligned}$$

Moreover, it holds that $\Omega = 1 + \beta + \frac{\alpha^2}{\omega}$ and $\delta = \frac{\Omega - \sqrt{\Omega^2 - 4\beta}}{2\beta}$.

Pricing Kernel and Short-term Interest Rates. Periodic utility is specified as

$$U(C_t) = \frac{C_t^{1-\gamma}}{1-\gamma}$$

so that the pricing kernel for real as well as nominal bonds between period t and $t + 1$ becomes

$$M_{t+1} = \beta \left(\frac{U'(C_{t+1})}{U'(C_t)} \right)$$

$$M_{t+1}^{\$} = \beta \left(\frac{U'(C_{t+1})}{U'(C_t)} \right) \left(\frac{P_t}{P_{t+1}} \right).$$

With $Y_t = C_t$ and $Y^* = 0, \pi^* = 0$, stochastic discount factors in log form can be also represented as

$$m_{t+1} = \log \beta - \gamma(y_{t+1} - y_t)$$

$$m_{t+1}^{\$} = m_{t+1} - \pi_{t+1}.$$

It is straightforward to compute the equilibrium one-period risk-free interest rate, both in nominal as well as in real terms

$$r_{1,t} = -\log(E_t[M_{t+1}])$$

$$i_{1,t} = -\log(E_t[M_{t+1}^{\$}]).$$

With a log-normal distribution, the short-rate process becomes

$$r_{1,t} = -E_t[m_{t+1}] - 0.5\text{var}_t(m_{t+1}) \quad (\text{E.6})$$

$$i_{1,t} = -E_t[m_{t+1}^{\$}] - 0.5\text{var}_t(m_{t+1}^{\$}). \quad (\text{E.7})$$

Substituting the equilibrium solutions of Eq. (E.5) into the pricing kernel and the short rates, gives the law of motions

$$m_{t+1}^d = \log \beta - \gamma b(\rho - 1)u_t - \gamma b\sigma\varepsilon_{t+1}$$

$$= \log \beta - \gamma \left(\frac{\alpha(1 - \rho)}{\omega(1 - \beta\rho) + \alpha^2} \right) u_t + \frac{\gamma\alpha}{\omega(1 - \beta\rho) + \alpha^2} \sigma\varepsilon_{t+1} \quad (\text{E.8})$$

$$m_{t+1}^{\$,d} = \log \beta - (\gamma b(\rho - 1) + a\rho)u_t - (\gamma b + a)\sigma\varepsilon_{t+1}$$

$$= \log \beta - \left(\frac{\omega\rho + \gamma\alpha(1 - \rho)}{\omega(1 - \beta\rho) + \alpha^2} \right) u_t + \frac{\gamma\alpha - \omega}{\omega(1 - \beta\rho) + \alpha^2} \sigma\varepsilon_{t+1}. \quad (\text{E.9})$$

$$m_{t+1}^c = \log \beta - \gamma(e - 1)y_t - \gamma f\rho u_t - \gamma f\sigma\varepsilon_{t+1}$$

$$= \log \beta - \gamma(\delta - 1)y_t + \frac{\rho\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} u_t + \frac{\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} \sigma\varepsilon_{t+1} \quad (\text{E.10})$$

$$\begin{aligned}
m_{t+1}^{s,c} &= \log \beta - (\gamma(e-1) + c)y_t - (\gamma f \rho + d\rho)u_t - (\gamma f + d)\sigma \varepsilon_{t+1} \\
&= \log \beta - \frac{(\gamma\alpha - \omega)(\delta - 1)}{\alpha} y_t + \rho \left[\frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))} \right] u_t \\
&\quad + \frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))} \sigma \varepsilon_{t+1}.
\end{aligned} \tag{E.11}$$

$$r_t^d = -\log \beta + \frac{\gamma\alpha(1-\rho)}{\omega(1-\beta\rho) + \alpha^2} u_t - \frac{1}{2} \left(\frac{\gamma\alpha}{\omega(1-\beta\rho) + \alpha^2} \right) \sigma^2 \tag{E.12}$$

$$r_t^c = -\log \beta + \gamma(\delta - 1)y_t - \frac{\rho\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} u_t - \frac{1}{2} \left(\frac{\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))} \right) \sigma^2 \tag{E.13}$$

$$i_t^d = -\log \beta + \frac{\omega\rho + \gamma\alpha(1-\rho)}{\omega(1-\beta\rho) + \alpha^2} u_t - \frac{1}{2} \left(\frac{\gamma\alpha - \omega}{\omega(1-\beta\rho) + \alpha^2} \right) \sigma^2 \tag{E.14}$$

$$\begin{aligned}
i_t^c &= -\log \beta + \frac{(\gamma\alpha - \omega)(\delta - 1)}{\alpha} y_t - \frac{\rho(\gamma\alpha - \omega)}{\omega(\Omega - \beta(\rho + \delta))} u_t \\
&\quad - \frac{1}{2} \left(\frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))} \right)^2 \sigma^2.
\end{aligned} \tag{E.15}$$

The Term Structure of Risk Premia. The model can be represented as VAR(1) model with

$$X_t = \mu + \phi X_{t-1} + \Psi^\top \Sigma \varepsilon_t \tag{E.16}$$

where $\Psi^\top$ is a matrix of structural parameters and $\Sigma \Sigma^\top$ is the variance-covariance matrix of the stochastic shocks with $\varepsilon_t \sim i.i.d.N(0, I)$. The VAR form is specified in order to describe the pricing kernel dynamics in equilibrium. Against this background, the affine term structure representation is used in order to draw statements for long-term bond prices. It holds that

$$M_{t+1}^s = \exp(-i_{1,t} - 0.5\lambda_t^\top \lambda_t - \lambda_t^\top \Sigma \varepsilon_{t+1}).$$

with market prices of risk

$$\lambda_t = \lambda_0 + \lambda_1 X_t.$$

and the short rate given by

$$i_{1,t} = \delta_0 + \delta_1^\top X_t.$$

Notice that the term $-0.5\lambda_t^\top \lambda_t$ is a correction term that preserves the linearity of interest rates in the log-normal setting. The variance term is, thus, eliminated from

the short-rate process. Following the discrete-time version of the essentially affine class of term structure models, bond prices follow

$$P_{n,t} = \exp(A_n + B_n^\top X_t)$$

with the recursive equations

$$A_n = A_{n-1} + B_{n-1}^\top (\mu - \Psi^\top \Sigma \Sigma^\top \lambda_0) + \frac{1}{2} B_{n-1}^\top \Sigma \Psi B_{n-1} - \delta_0 \quad (\text{E.17})$$

$$B_n^\top = B_{n-1}^\top (\phi - \Psi^\top \Sigma \Sigma^\top \lambda_1) - \delta_1 \quad (\text{E.18})$$

where $\Sigma_\Psi = \Psi^\top \Sigma \Sigma^\top \Psi$ with $A_1 = -\delta_0$ and $B_1 = -\delta_1$. Since continuously-compounded interest rates are related to the logarithm of bond prices, $i_{n,t}$ is given by

$$\begin{aligned} i_{n,t} &= -n^{-1} \log(P_{n,t}) \\ &= n^{-1} (-A_n - B_n^\top X_t) \\ &= a_n + b_n^\top X_t \end{aligned} \quad (\text{E.19})$$

with $a_n = -A_n/n$ and $b_n = -B_n/n$ (see Appendix D for the solution technique; an analytical example is Palomino (2010)).

The simple model can be stacked into the discrete-time affine class of term structure model by imposing restrictions on the parameters. Showing that under discretion, the only relevant state variables is the time-series property of the supply shock which is transformed into the pricing kernel via the output gap and inflation dynamics. Under commitment, the state space needs to be augmented by the evolution of the output gap in the rational expectations equilibrium.

Proposition E.1. *The equilibrium interest rate characteristics under discretion for real bonds are*

$$\begin{aligned} X_t &= u_t, & \mu &= 0, & \phi &= \rho \\ \Sigma &= \sigma & \delta_0 &= -\log \beta \\ \delta_1 &= \gamma \left(\frac{\alpha(1-\rho)}{\omega(1-\beta\rho) + \alpha^2} \right) & \lambda_0 &= -\frac{\gamma\alpha}{\omega(1-\beta\rho) + \alpha^2} \\ \lambda_1 &= 0 & \Psi^\top &= 1. \end{aligned}$$

For nominal bonds, the affine representations has to be modified according to

$$\begin{aligned} \delta_0^\$ &= -\log \beta \\ \delta_1^\$ &= \frac{\omega\rho + \gamma\alpha(1-\rho)}{\omega(1-\beta\rho) + \alpha^2} & \lambda_0^\$ &= -\frac{\gamma\alpha - \omega}{\omega(1-\beta\rho) + \alpha^2} \end{aligned}$$

Proposition E.2. *The equilibrium interest rate characteristics under commitment for real bonds are*

$$\begin{aligned}
 X_t &= [u_t, y_t]^\top & \mu &= [0, 0]^\top & \phi &= \begin{bmatrix} \frac{-\rho}{\omega(\Omega - \beta(\rho + \delta))} & 0 \\ \delta & \delta \end{bmatrix} \\
 \Sigma &= \begin{bmatrix} \sigma & 0 \\ 0 & 0 \end{bmatrix} & \lambda_0 &= [-\frac{\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))}, 0]^\top \\
 \delta_0 &= -\log \beta & \delta_1 &= [-\frac{-\rho\gamma\alpha}{\omega(\Omega - \beta(\rho + \delta))}, \gamma(\delta - 1)]^\top \\
 \lambda_1 &= 0_{3 \times 3} & \Psi^\top &= \begin{bmatrix} \frac{1}{-\alpha} & 0 \\ \frac{-\alpha}{\omega(\Omega - \beta(\rho + \delta))} & 0 \end{bmatrix}.
 \end{aligned}$$

With nominal bonds, the affine representation has to be modified according to

$$\begin{aligned}
 \delta_0^s &= -\log \beta & \delta_1^s &= [-\rho \frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))}, \frac{(\gamma\alpha - \omega)(\delta - 1)}{\alpha}]^\top \\
 \lambda_0^s &= [-\frac{\gamma\alpha - \omega}{\omega(\Omega - \beta(\rho + \delta))}, 0]^\top.
 \end{aligned}$$

With this closed-form solution, expected excess returns can be easily computed. They are defined as $xr_{n,t+1} = p_{n-1,t+1} - p_{n,t} - i_{i,1}$. Based on the definition of the yield curve, they can be written as

$$E_t xr_{n,t+1} = B_{n-1}^\top \Sigma_\psi \lambda_0 - 0.5 B_{n-1}^\top \Sigma_\psi B_{n-1}^\top. \quad (\text{E.20})$$

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